

BRIEFINGS

black hat

Tiny Chips Big Leaks!

Breaking TrustZone-M with Single-Stepping Attacks

Cristiano Rodrigues, Sandro Pinto, Marton Bogнар, and Jo Van Bulck
Centro ALGORITMI, University of Minho, DistriNet, KU Leuven

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THE FOLLOWING **TECHNICAL PREVIEW** HAS BEEN APPROVED FOR
THE BLACK HAT COMMUNITY
FOR EDUCATIONAL PURPOSES ONLY

NR	NOT RATED
	The Content of This Talk Has Not Been Evaluated

-----BEGIN RSA PRIVATE KEY-----

-----END RSA PRIVATE KEY-----

-----BEGIN RSA PRIVATE KEY-----

*****:

-----END RSA PRIVATE KEY-----

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010111001101010101010100100111010101101011
101000101011001010101111000101010101010110
1010101010101001010101010101010101010111111
010101010101001010101010110001110011010010
10100101010101010110111111111101010010100
101000101010101010101010110101001001010101
010101010100101010100101111111001010101010
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010101010101011010001111010101010101010101
010101110101010010101010110101010101010101
01

-----BEGIN RSA PRIVATE KEY-----

-----END RSA PRIVATE KEY-----

010111001101010101010100100111010101101011
101000101011001010101111000101010101010110
1010101010101001010101010101010101010111111
010101010101001010101010110001110011010010
10100101010101010110111111111101010010100
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010101010100101010100101111111001010101010
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010101010101011010001111010101010101010101
010101110101010010101010110101010101010101
01

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-----END RSA PRIVATE KEY-----

-----BEGIN RSA PRIVATE KEY-----

-----END RSA PRIVATE KEY-----

-----BEGIN RSA PRIVATE KEY-----

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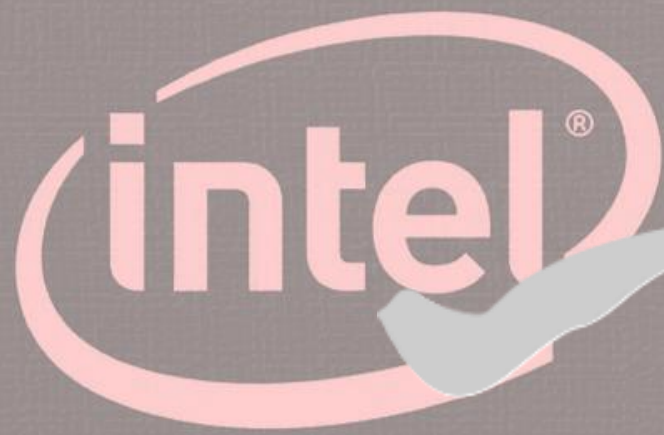
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intel[®]
SGX



intel®
SGX





SGX

SGX-Step

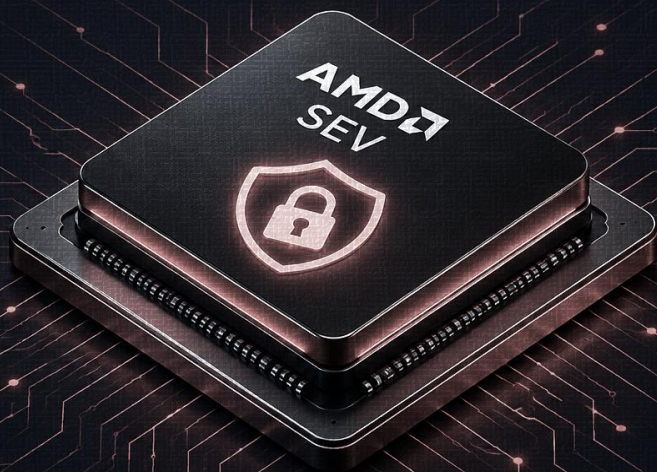




arm
TrustZone[®]



AMD
SEV



intel[®]
TDX



intel[®]
SGX

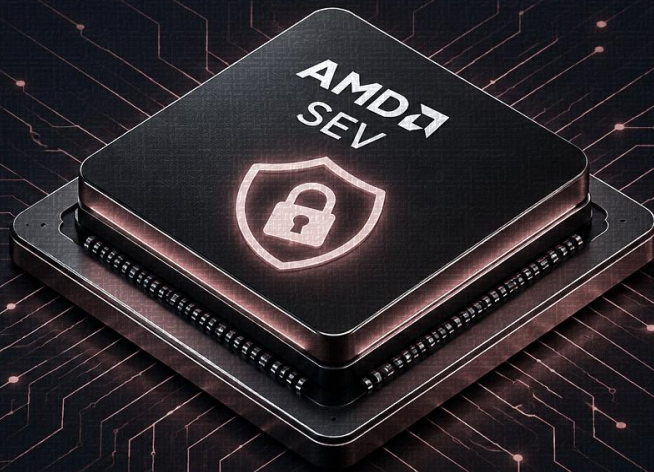


SGX-Step

arm
TrustZone[®]



AMD
SEV



intel[®]
TDX



intel[®]
SGX

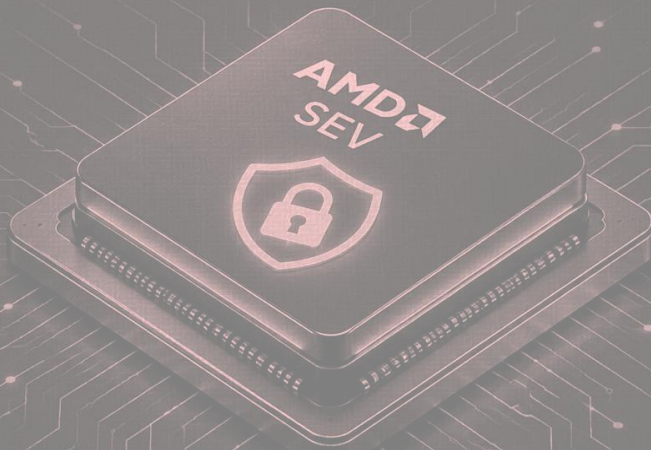


SGX-Step

arm
TrustZone[®]



AMD
SEV



intel[®]
TDX



intel[®]
SGX

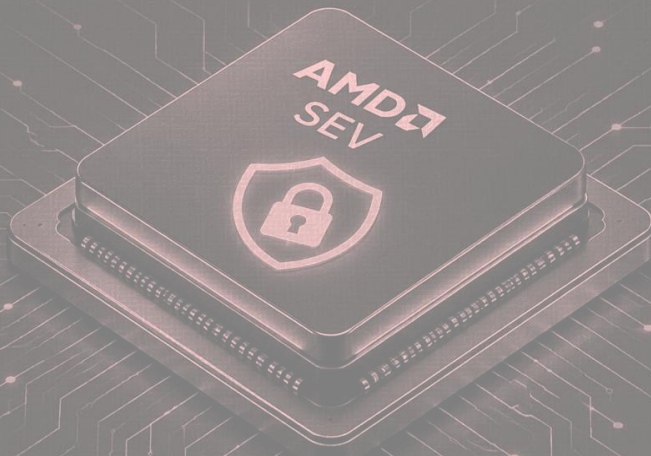


SGX-Step

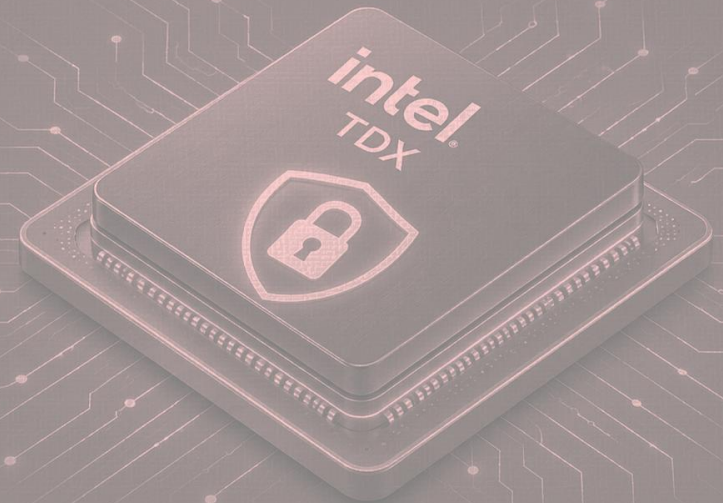
arm
TrustZone[®]



AMD
SEV

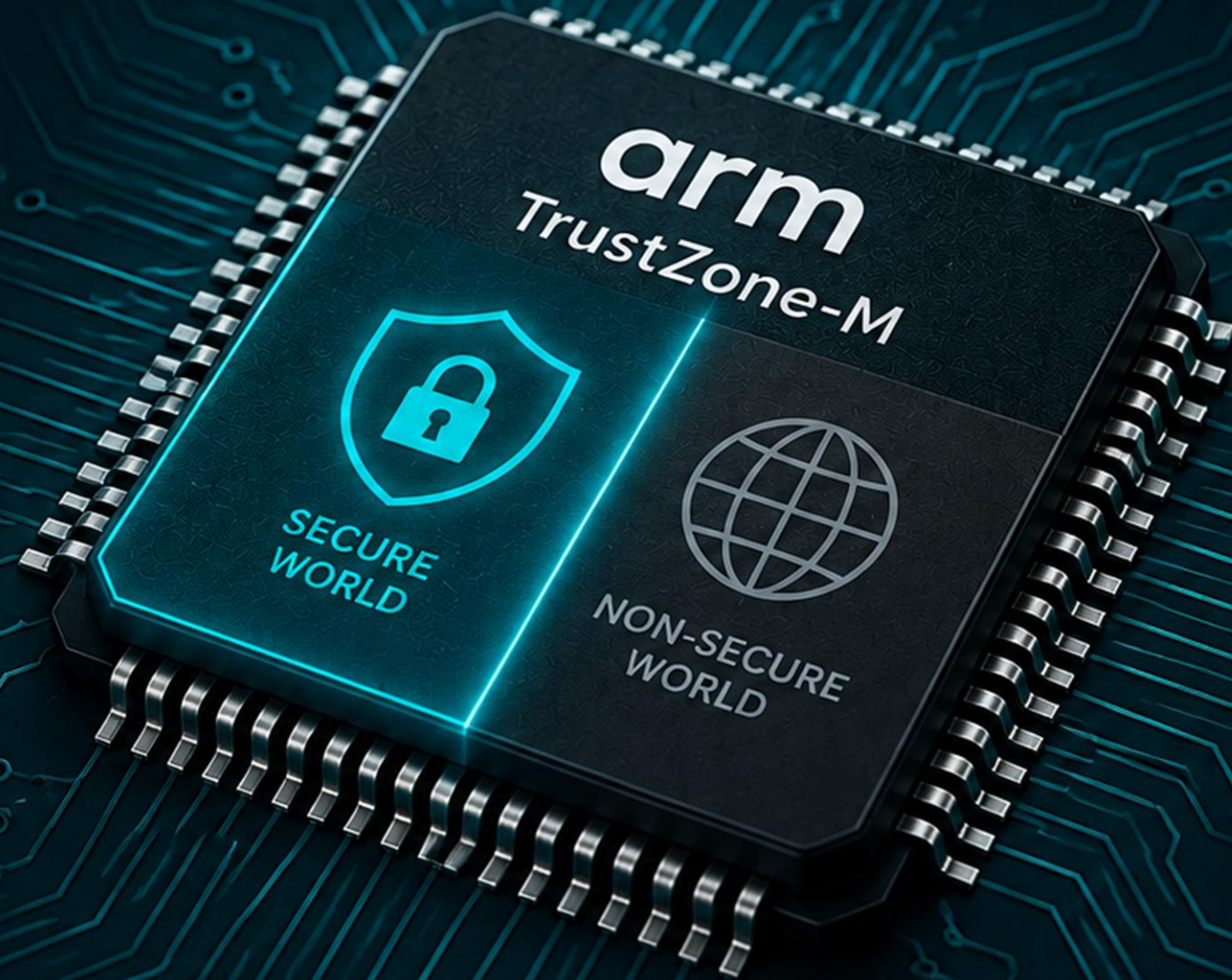


intel[®]
TDX



arm[®]

TRUSTZONE-M



arm[®]

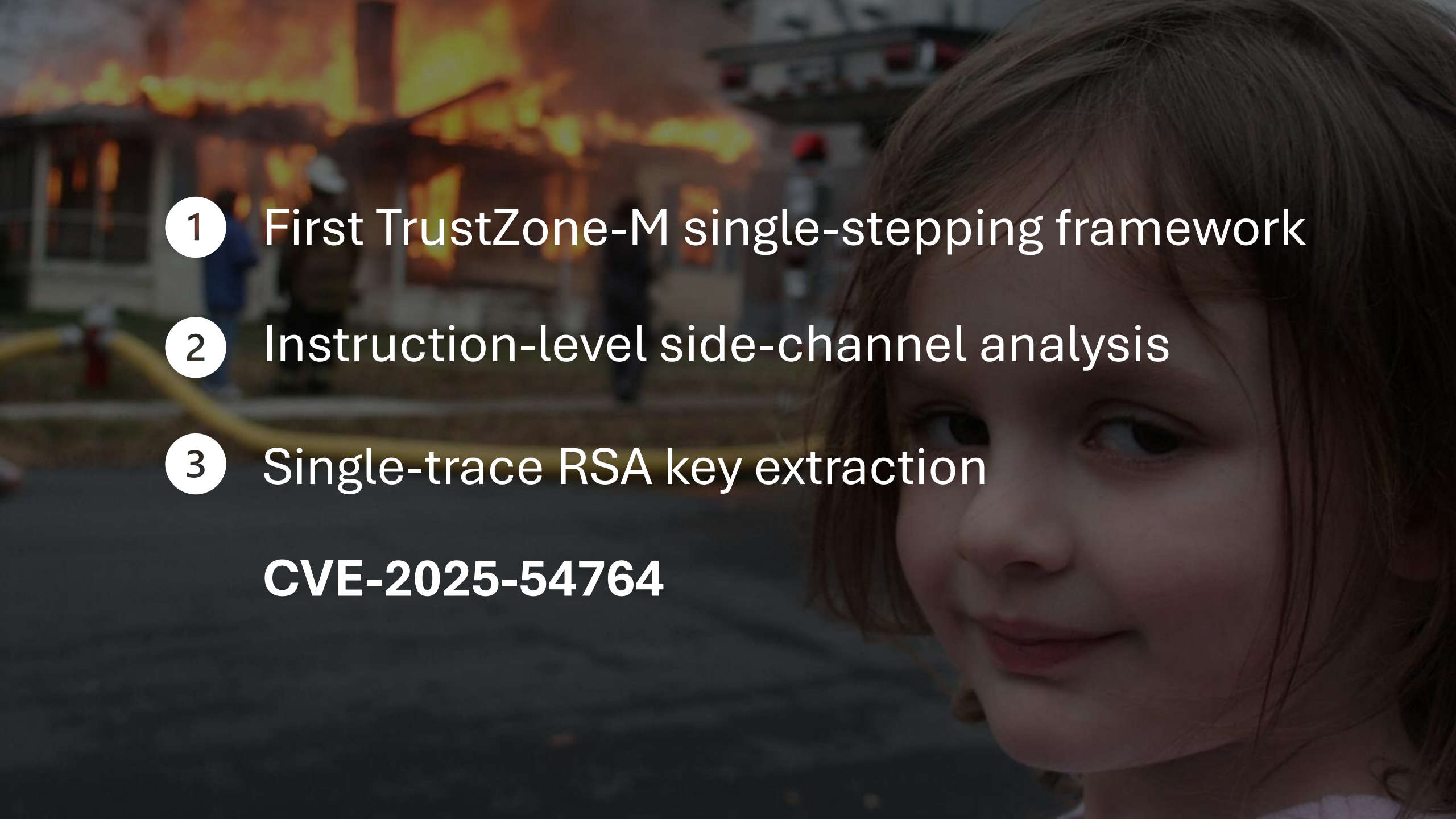
TRUSTZONE-M



arm[®]

TRUSTZONE-M



- 
- 1 First TrustZone-M single-stepping framework
 - 2 Instruction-level side-channel analysis
 - 3 Single-trace RSA key extraction

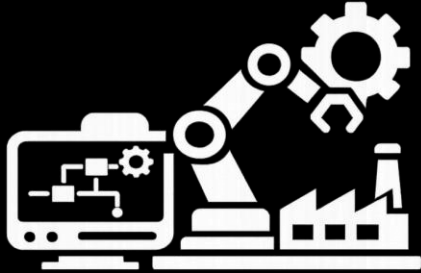
CVE-2025-54764

Tiny chips have become valuable targets

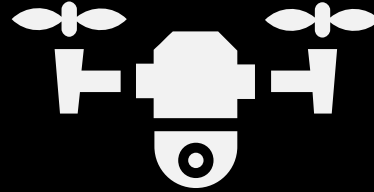
Modern Connected World



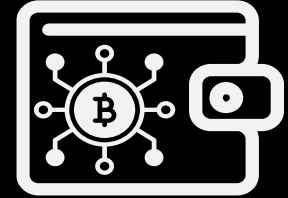
Satellites



Industrial



Drones



Hardware Wallets

Cars



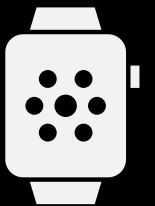
Medical



Appliances



Wearables



Modern Connected World



Satellites



Factories



Drones



Smart Wallets

What powers almost all of them?

Cars



Medical

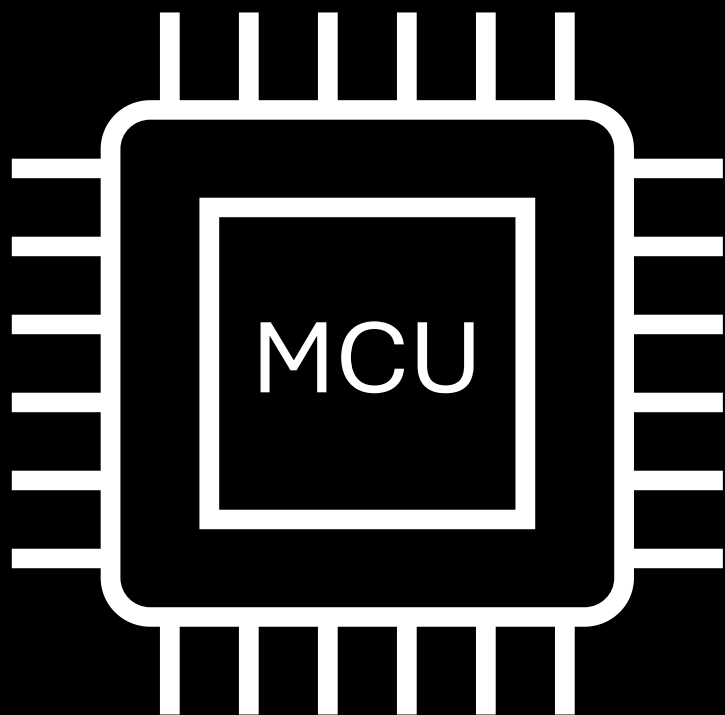


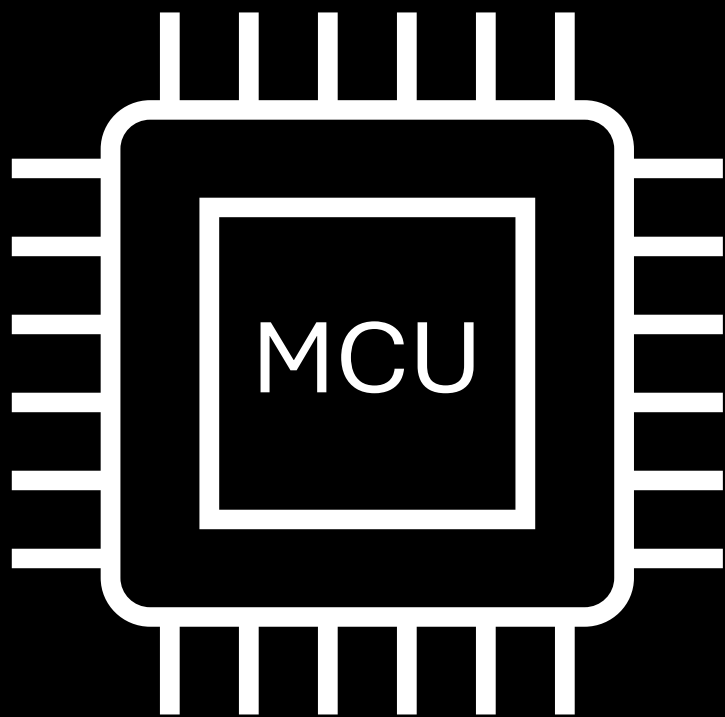
Smart Appliances



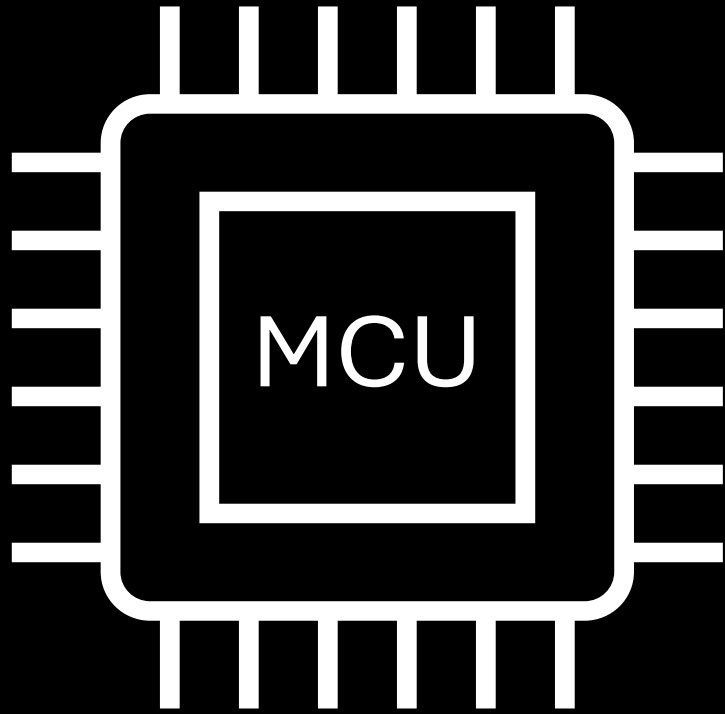
Wearables





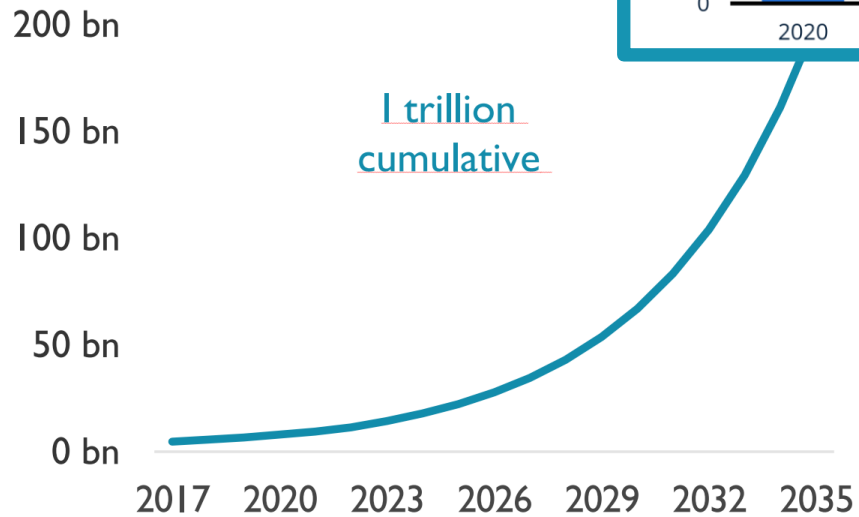


arm

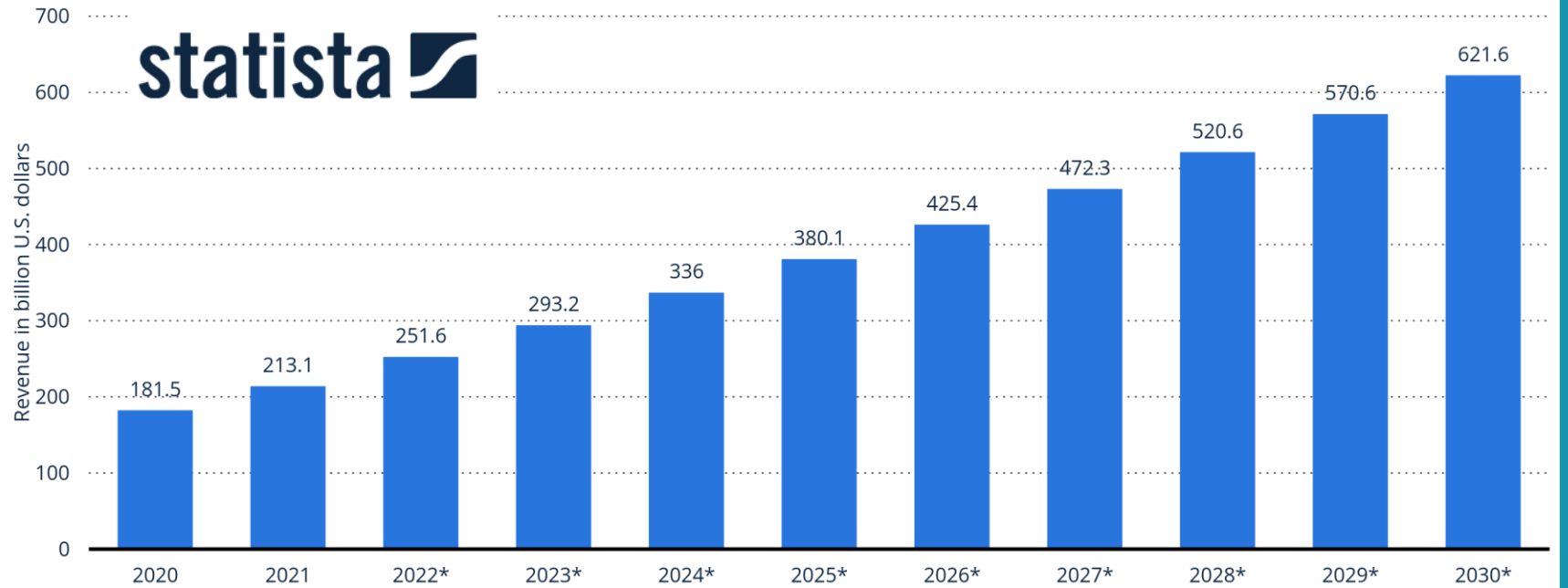


arm
CORTEx-M

Annual Production of IoT devices

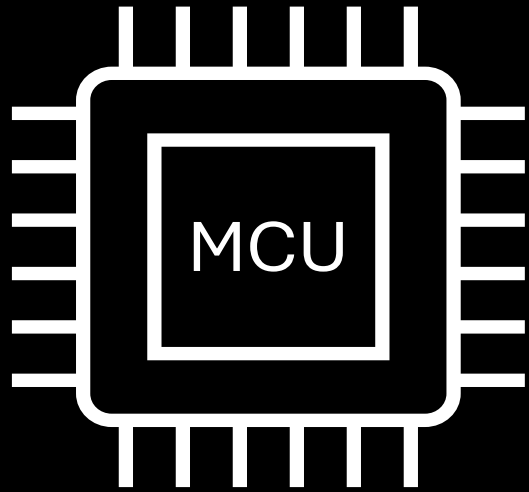


Source: SoftBank and ARM estimates



The route to a trillion devices

The outlook for IoT investment to 2035



arm
CORTEx-M

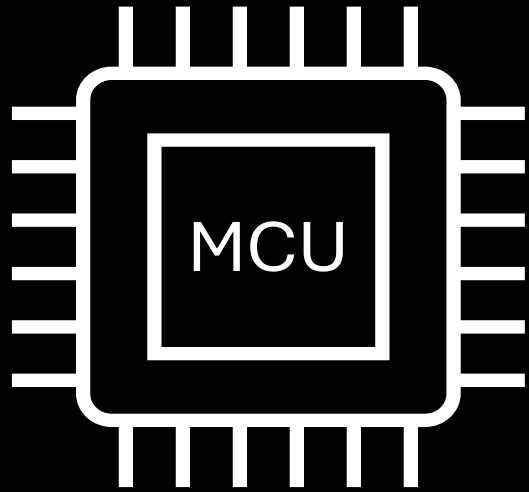
Armv6-M & Armv7-M

Base Architecture

Unprivileged
THREAD

Priv. THREAD

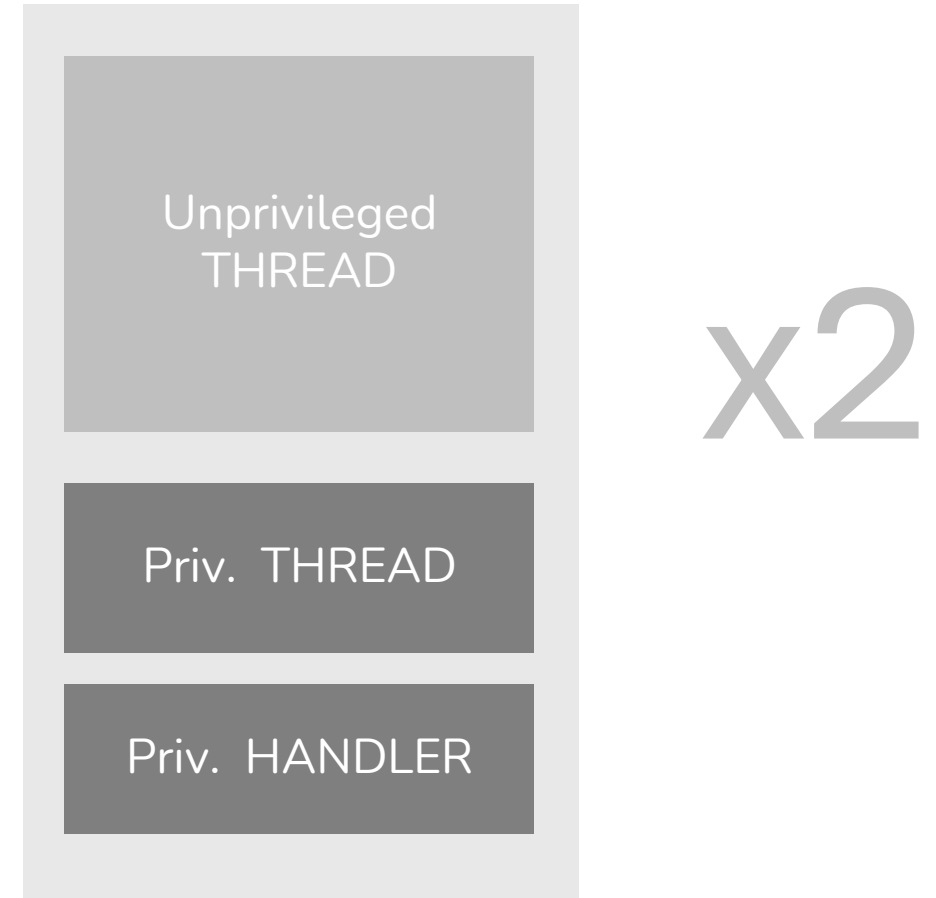
Priv. HANDLER

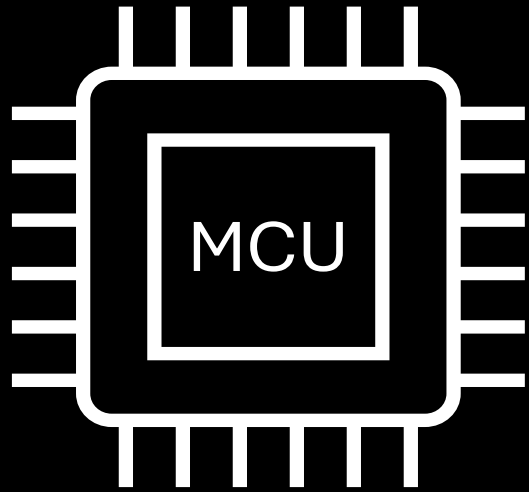


arm
TRUSTZONE-M

Armv8-M

Armv6/7-M Base Architecture

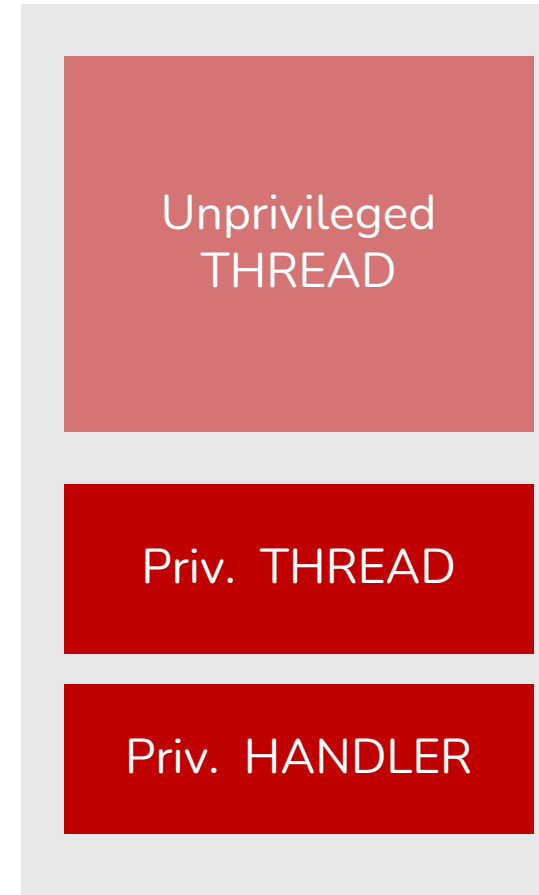




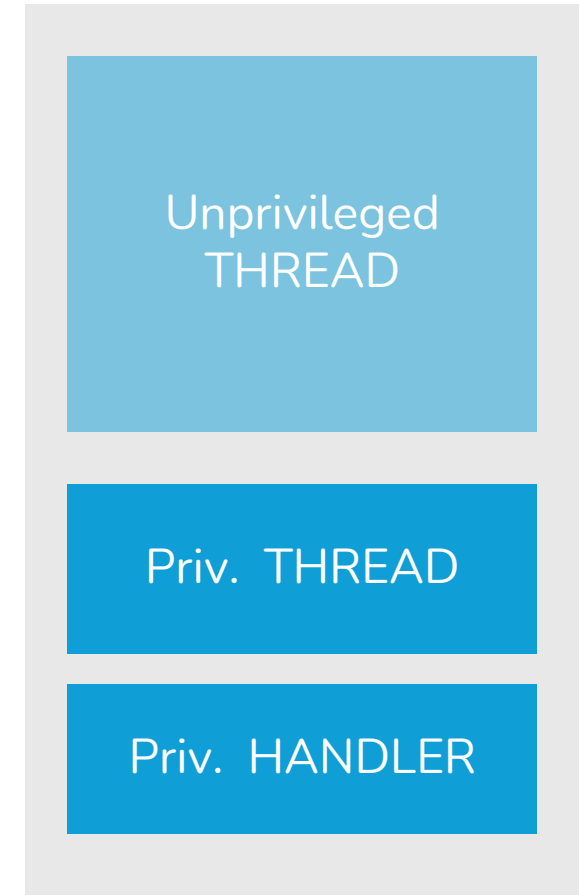
arm
TRUSTZONE-M

Armv8-M

Non-Secure State



Secure State





Hand Me Your Secret, MCU!
Microarchitectural Timing Attacks on Microcontrollers are Practical



Hand Me Your Secret, MCU!
Microarchitectural Timing Attacks on Microcontrollers are Practical

**What the TrustZone-M Doesn't See, the MCU Does Grieve Over:
Lessons Learned from Assessing a Microcontroller TEE**





Hand Me Your Secret, MCU! Microarchitectural Timing Attacks on Microcontrollers are Practical

What the TrustZone-M Doesn't See, the MCU Does Grieve Over:
Lessons Learned from Assessing a Microcontroller TEE

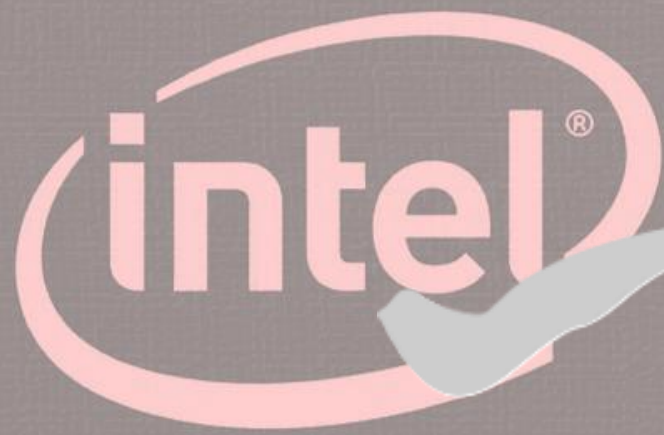


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THE LAST MISSING FRAMEWORK

intel[®]
SGX





SGX

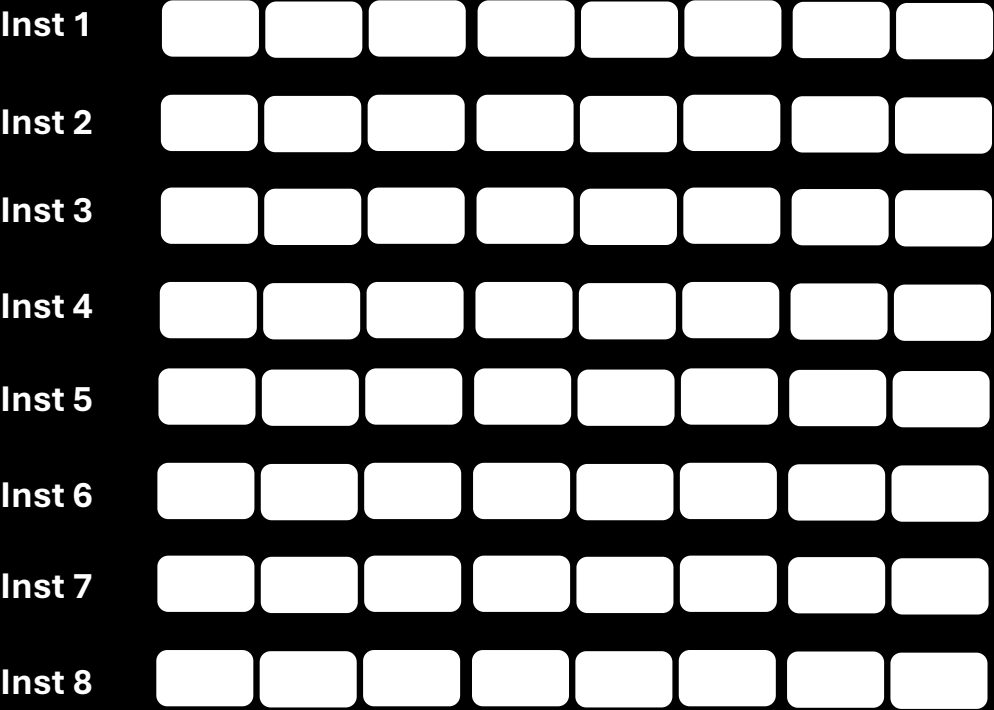
SGX-Step



End-2-End Timing

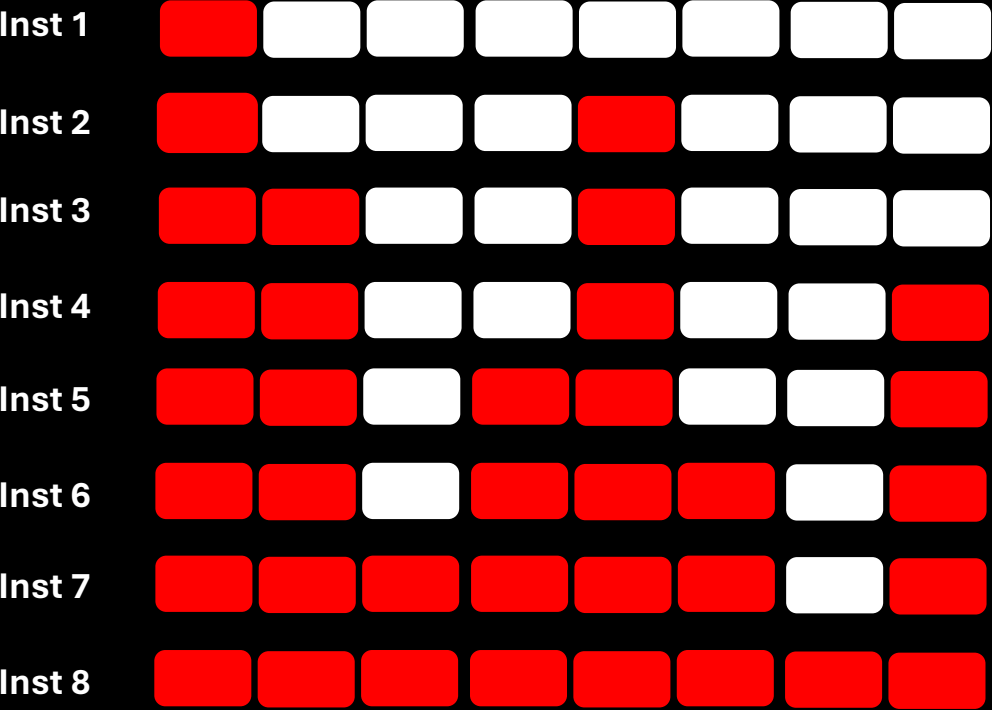
Single-Stepping

End-2-End Timing



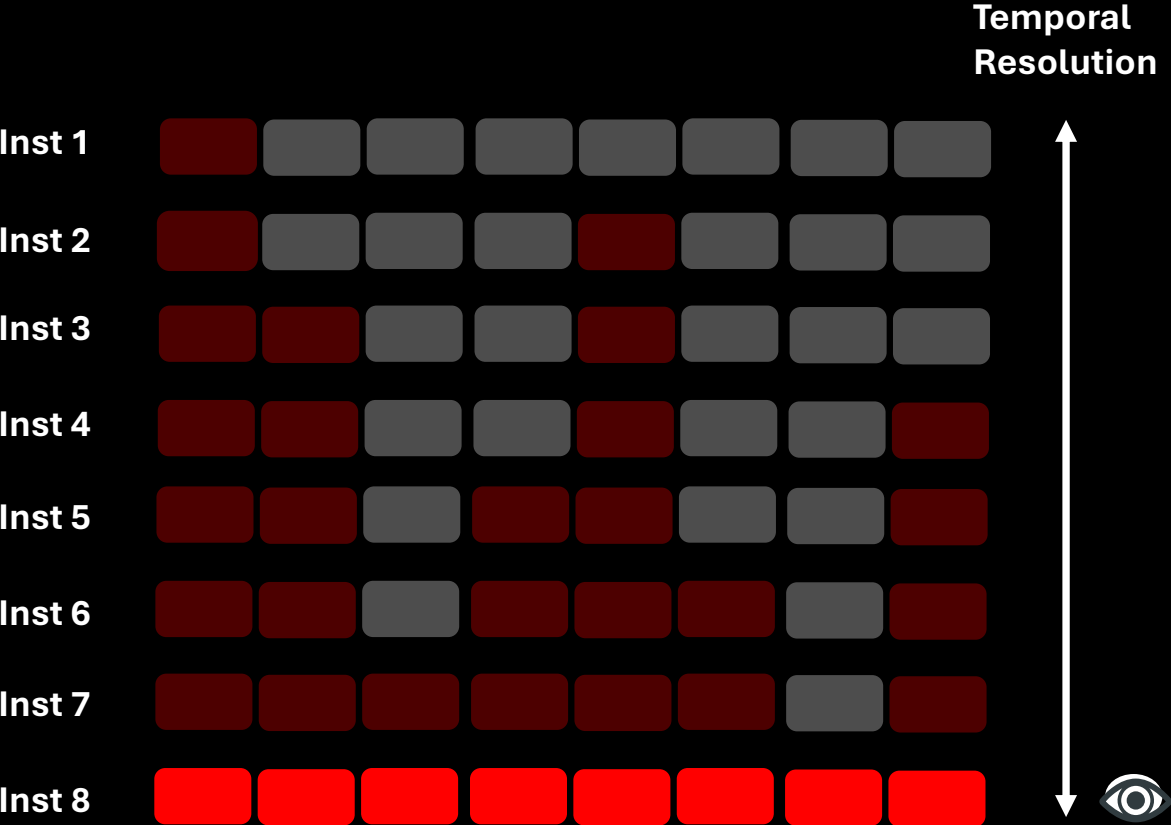
Single-Stepping

End-2-End Timing



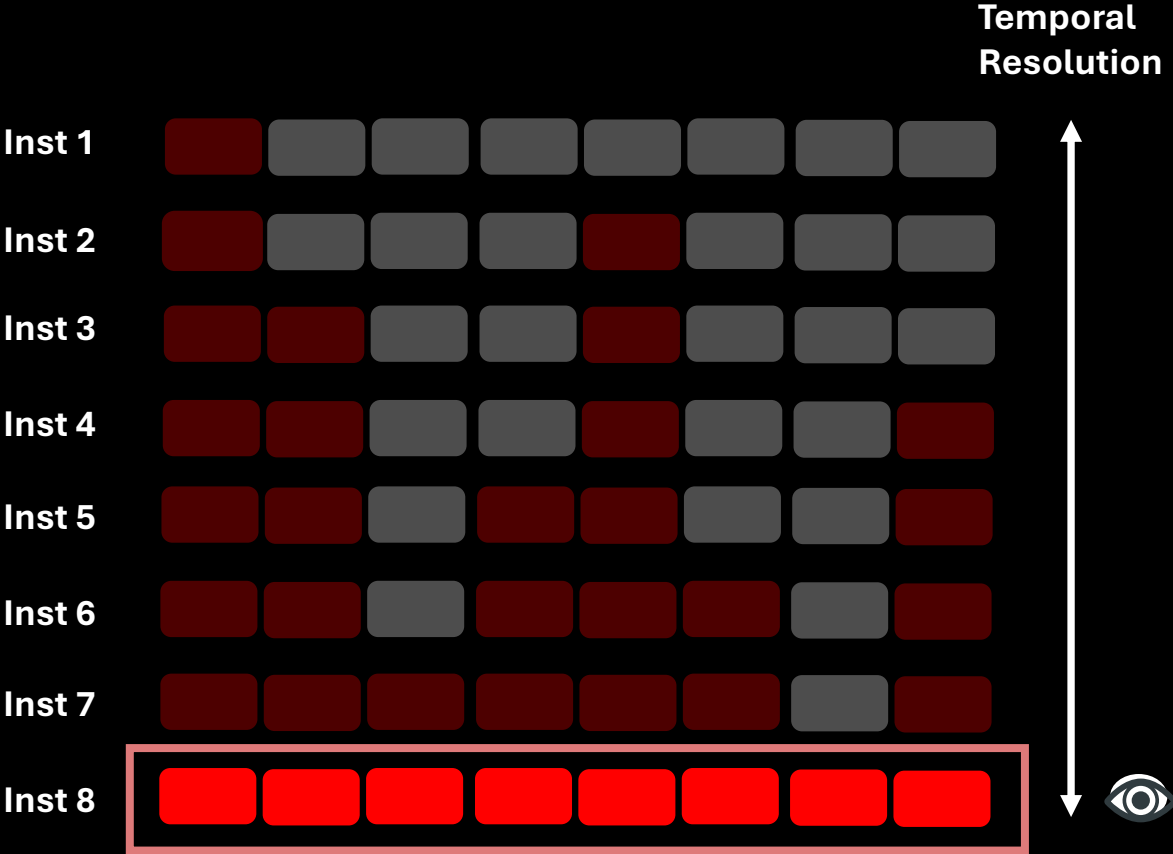
Single-Stepping

End-2-End Timing



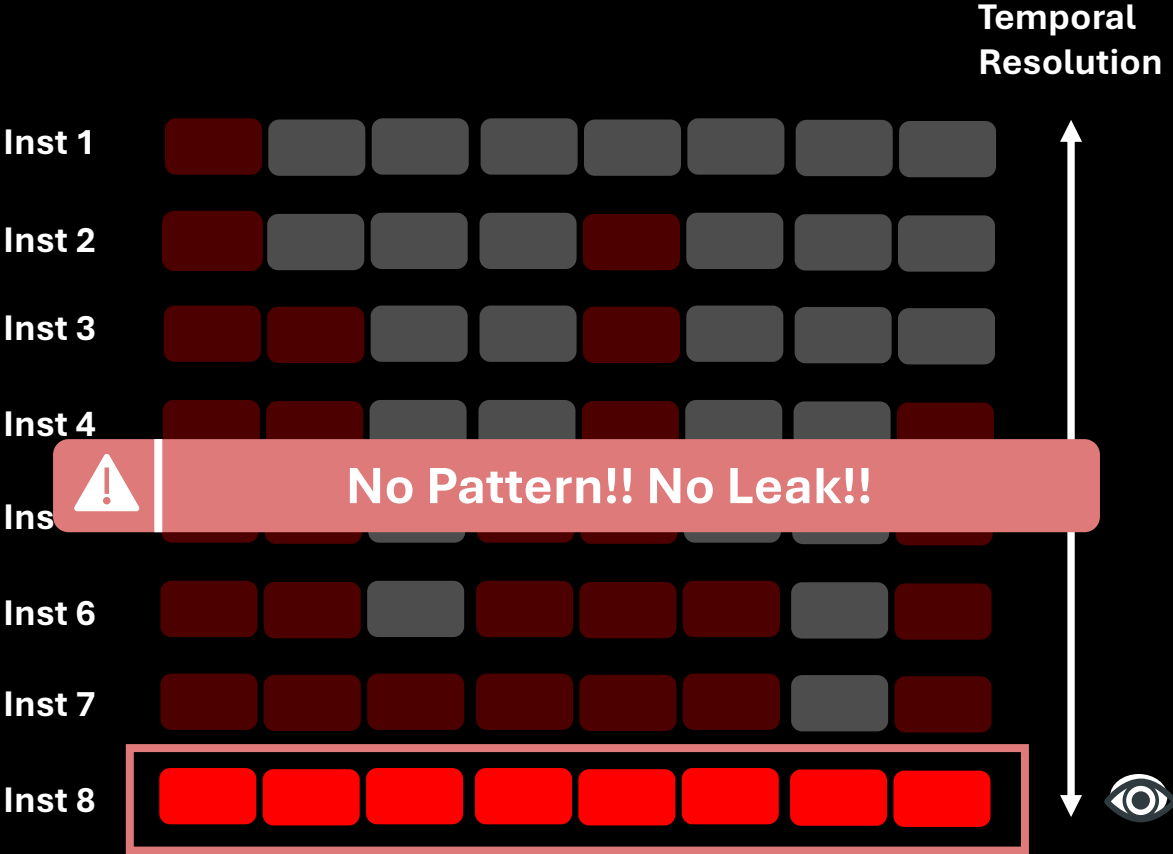
Single-Stepping

End-2-End Timing



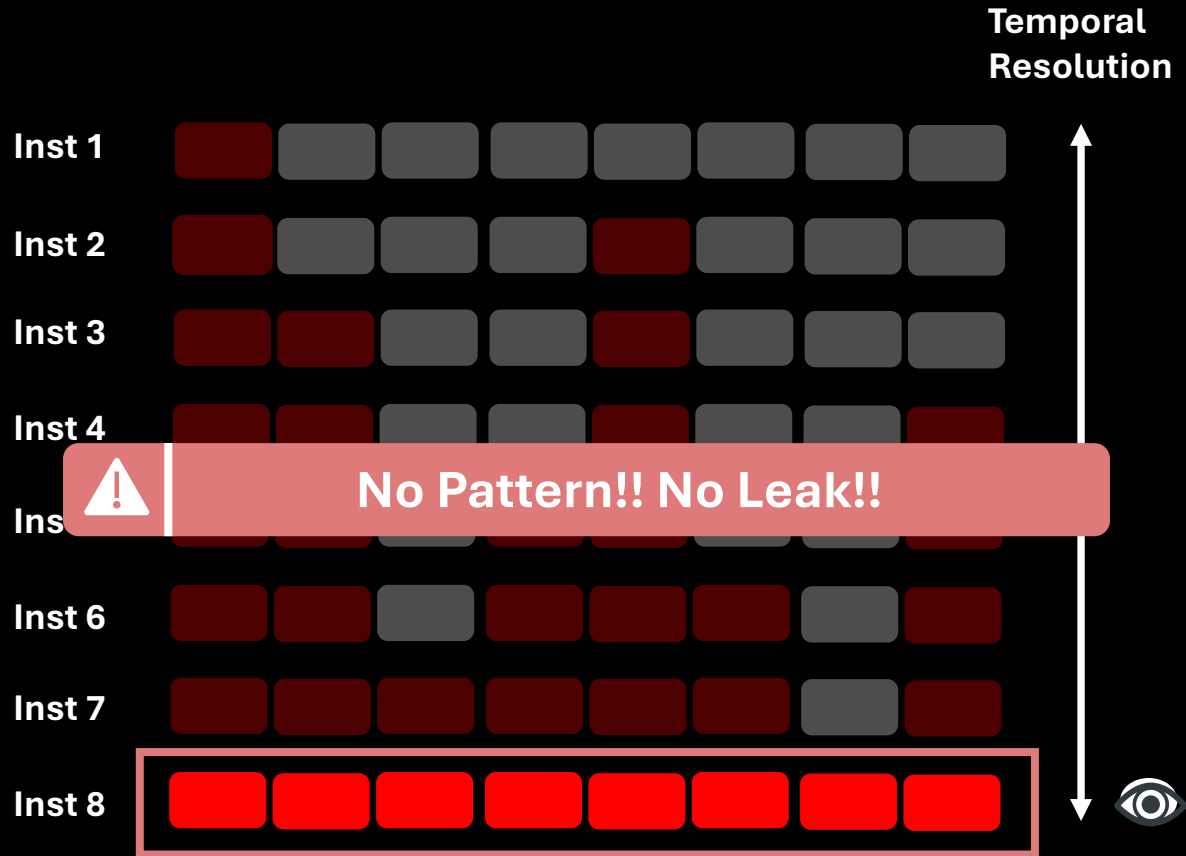
Single-Stepping

End-2-End Timing

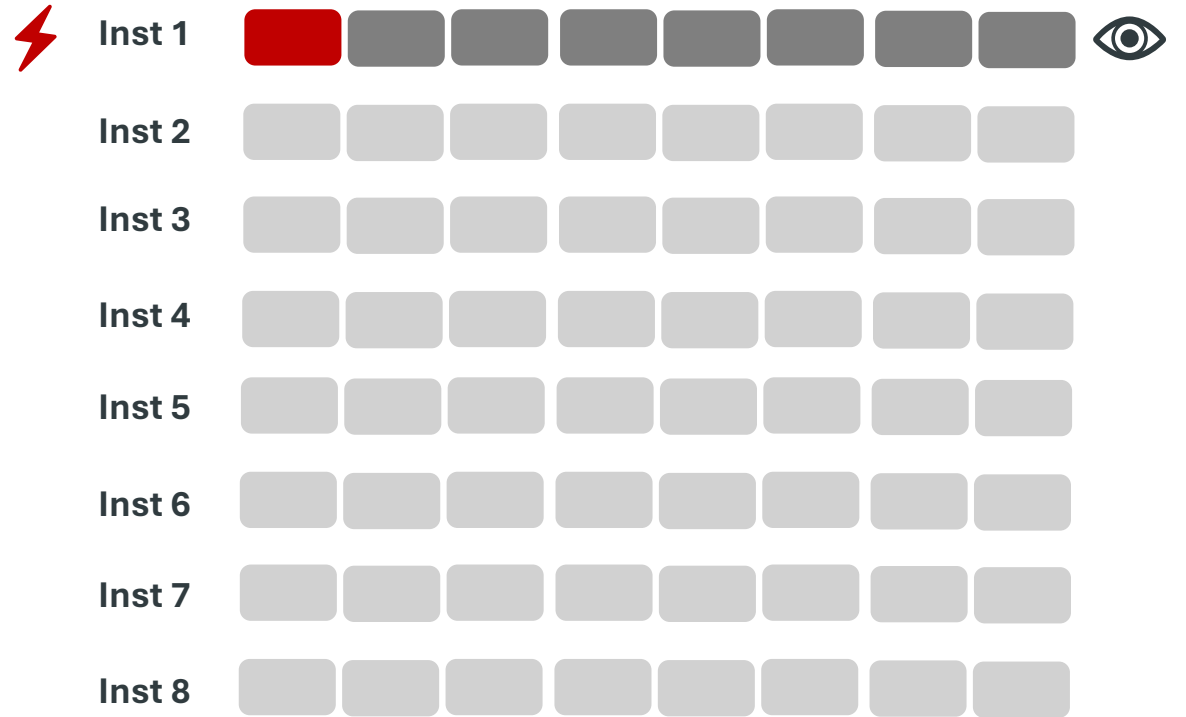


Single-Stepping

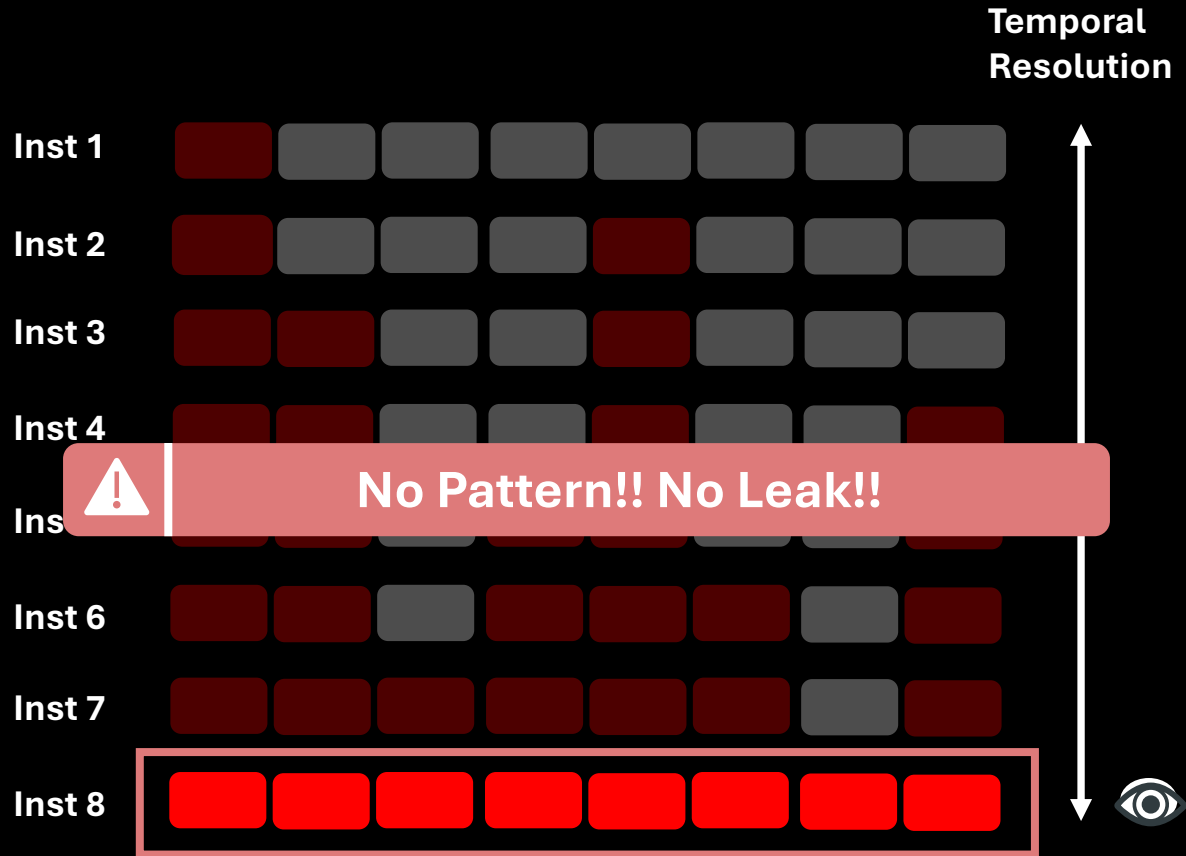
End-2-End Timing



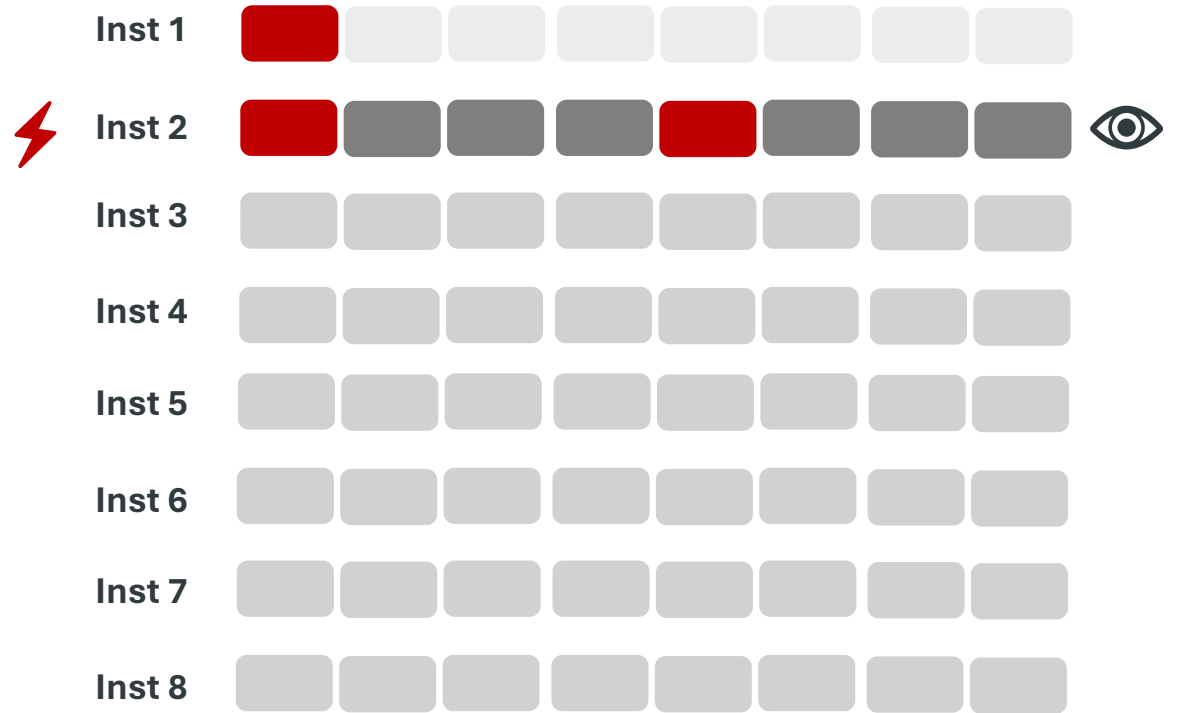
Single-Stepping



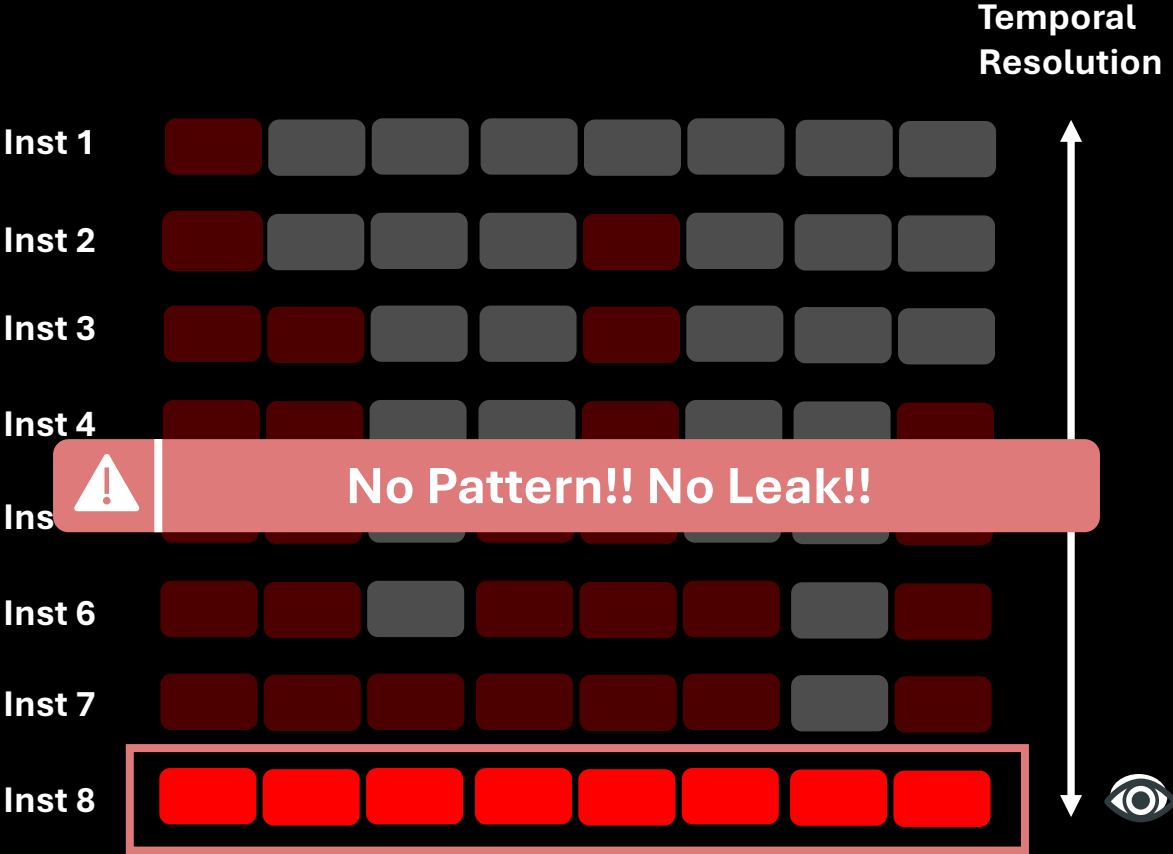
End-2-End Timing



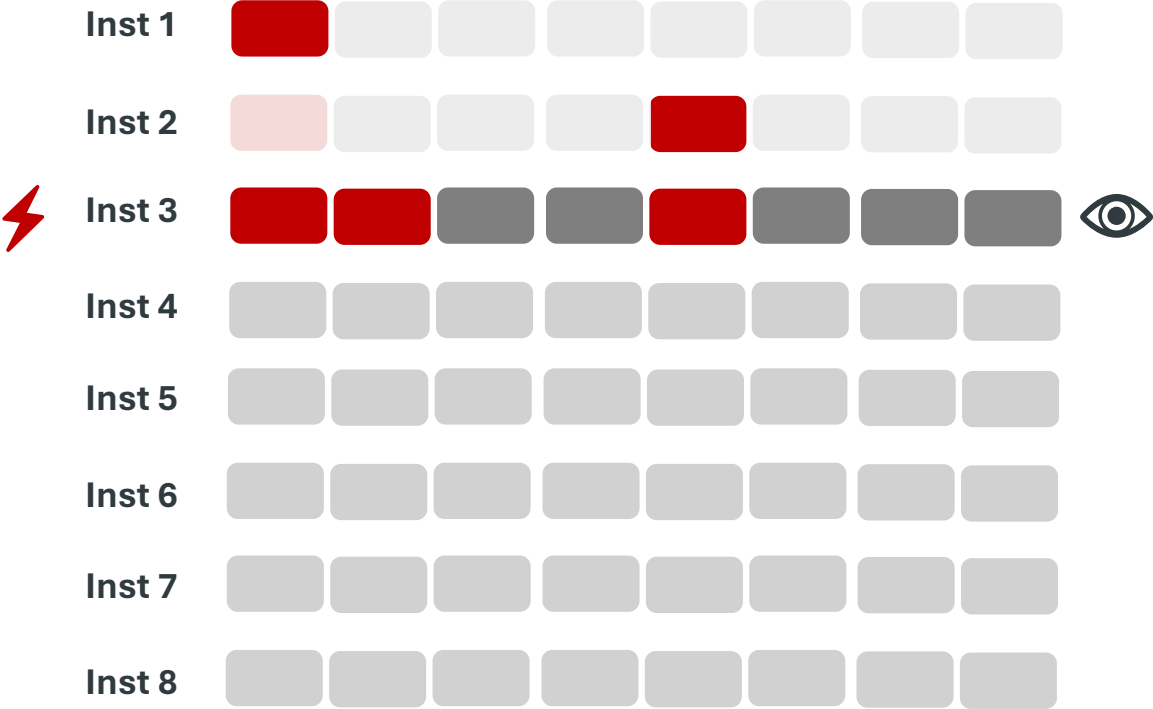
Single-Stepping



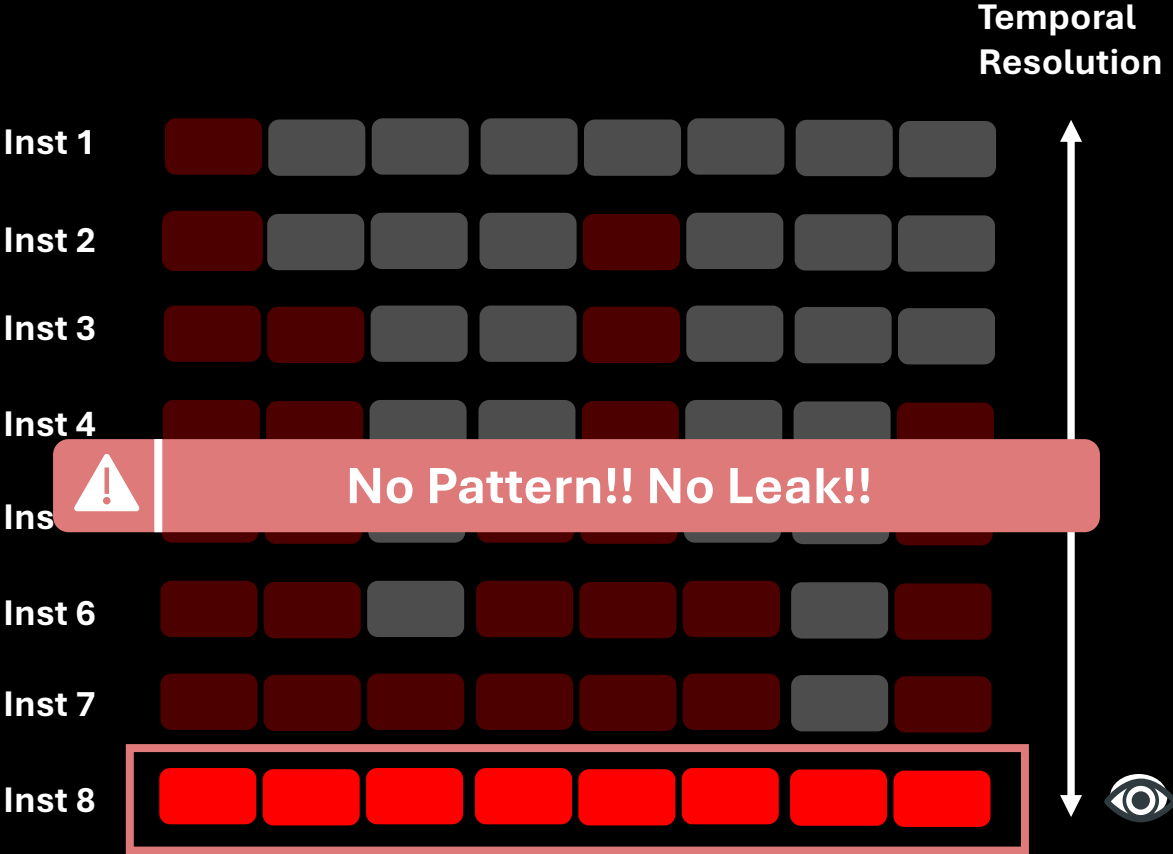
End-2-End Timing



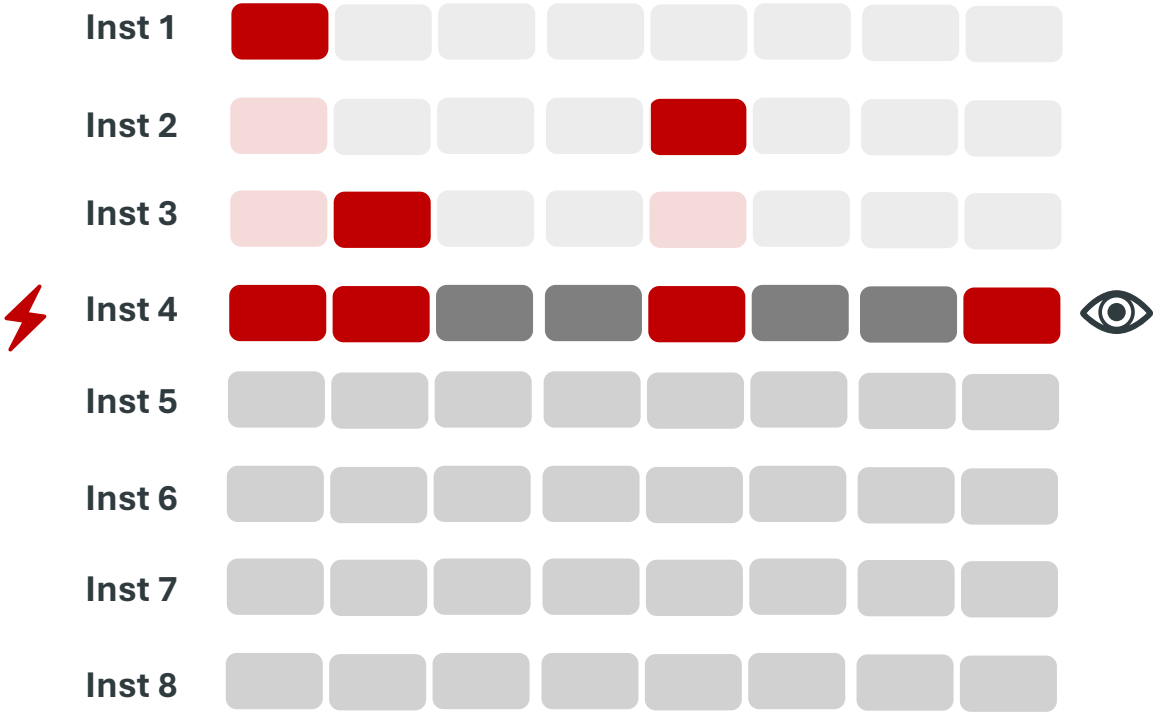
Single-Stepping



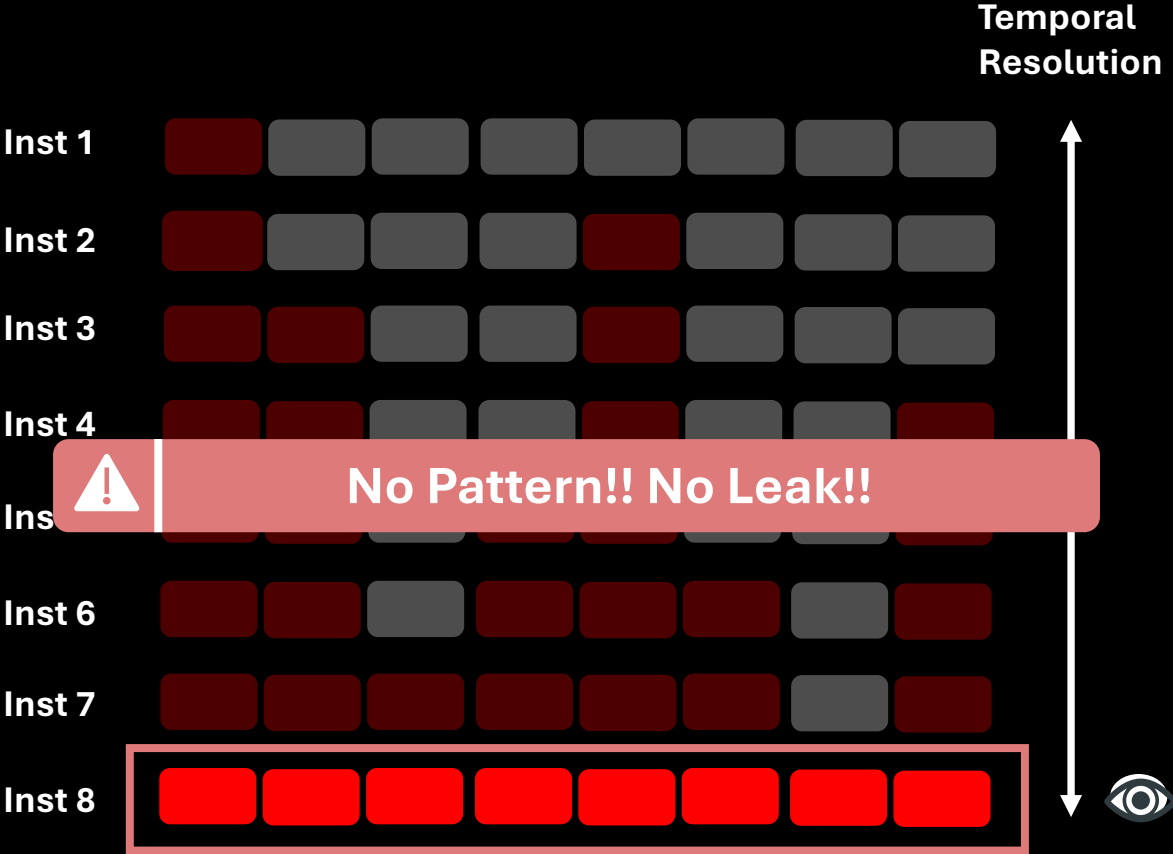
End-2-End Timing



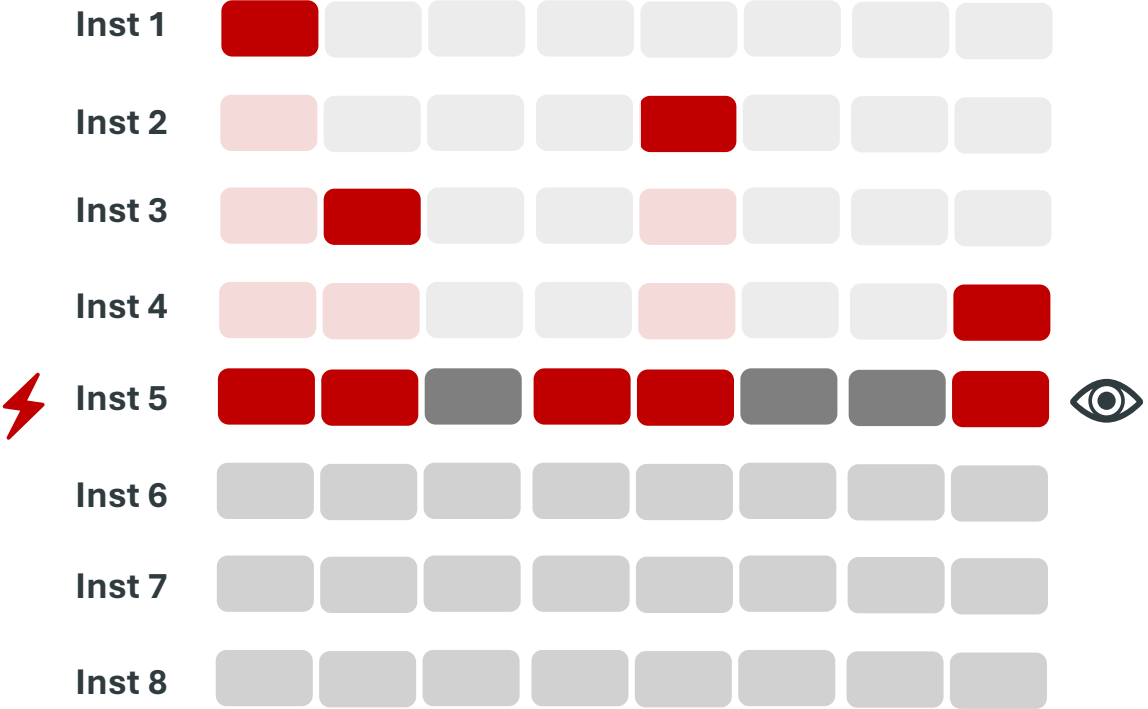
Single-Stepping



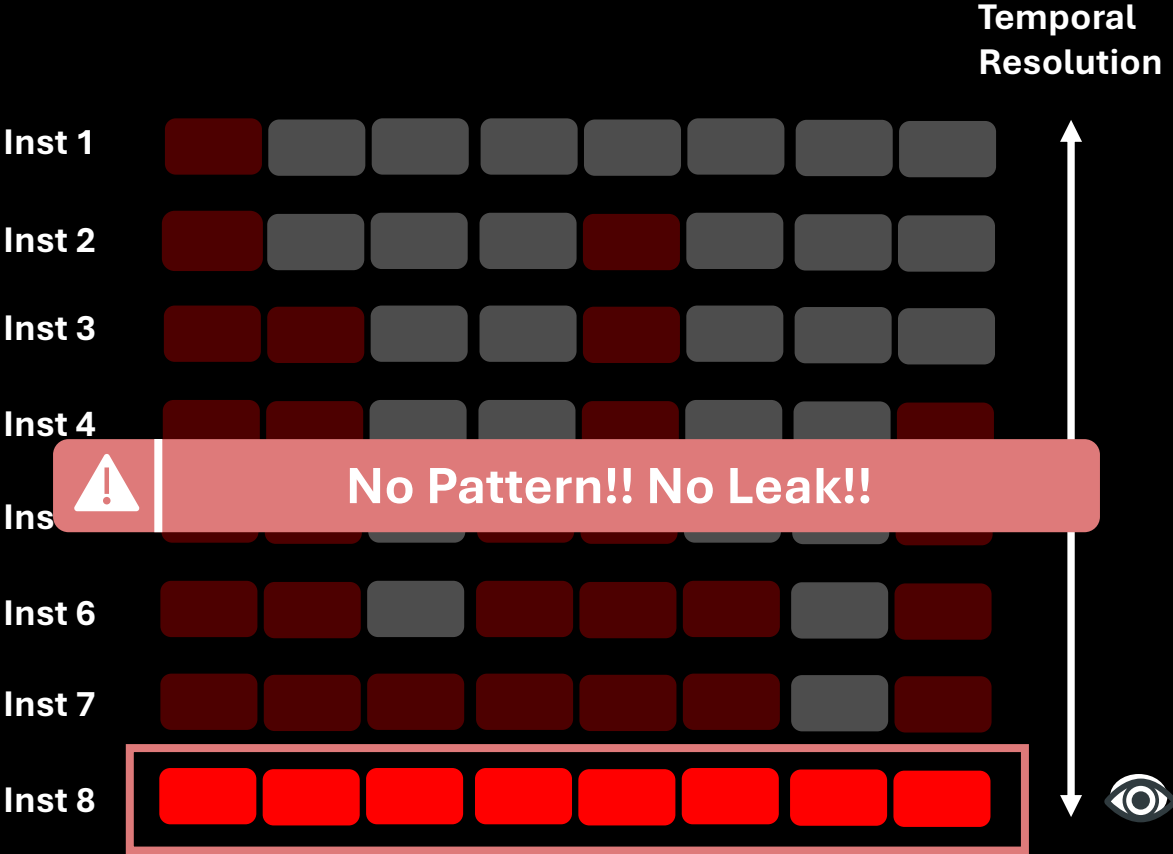
End-2-End Timing



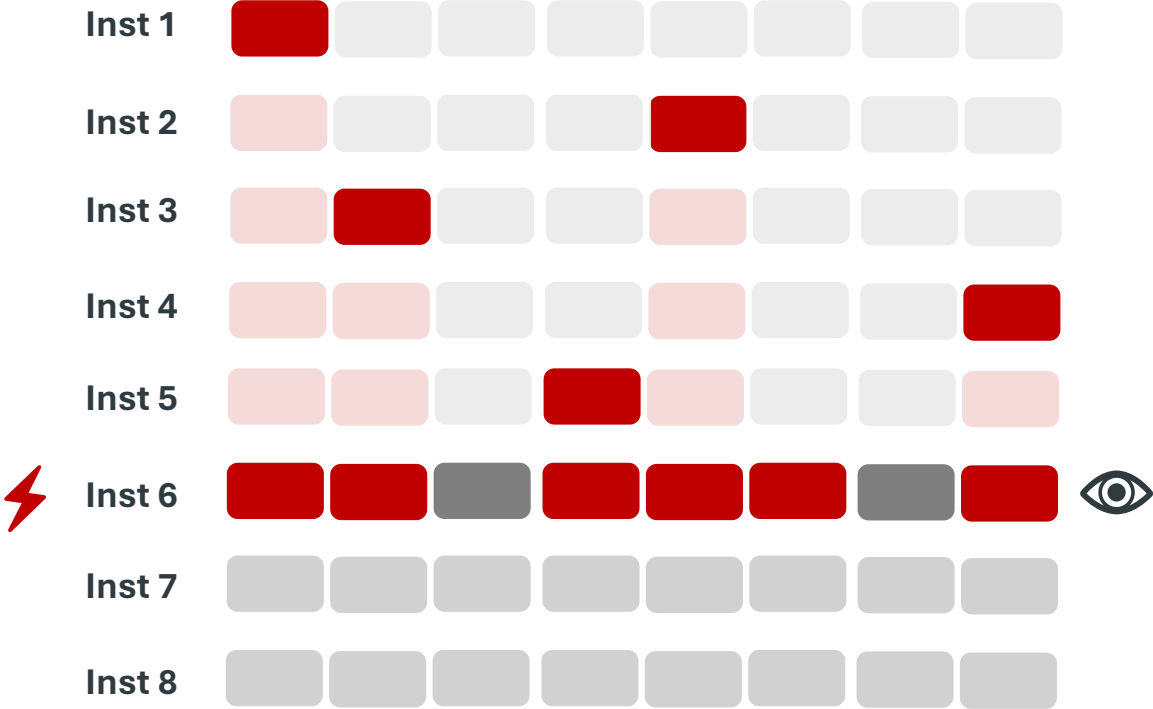
Single-Stepping



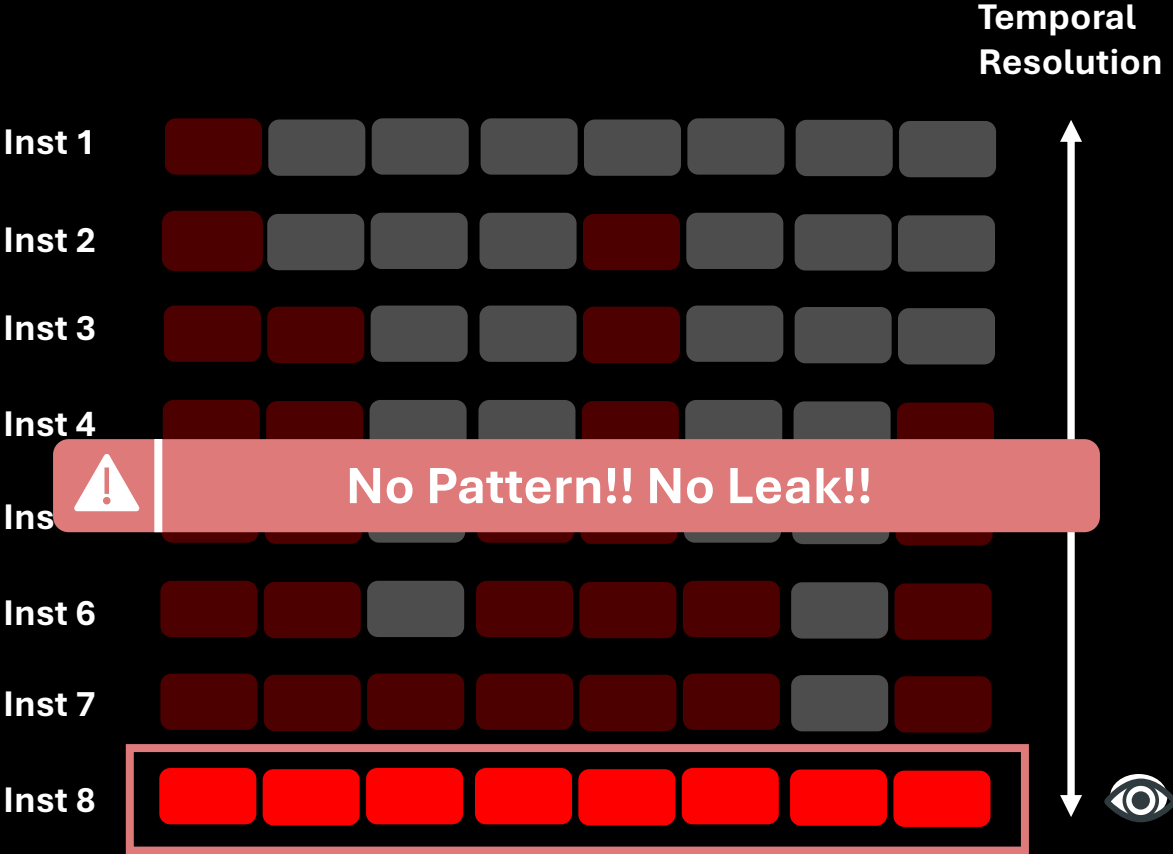
End-2-End Timing



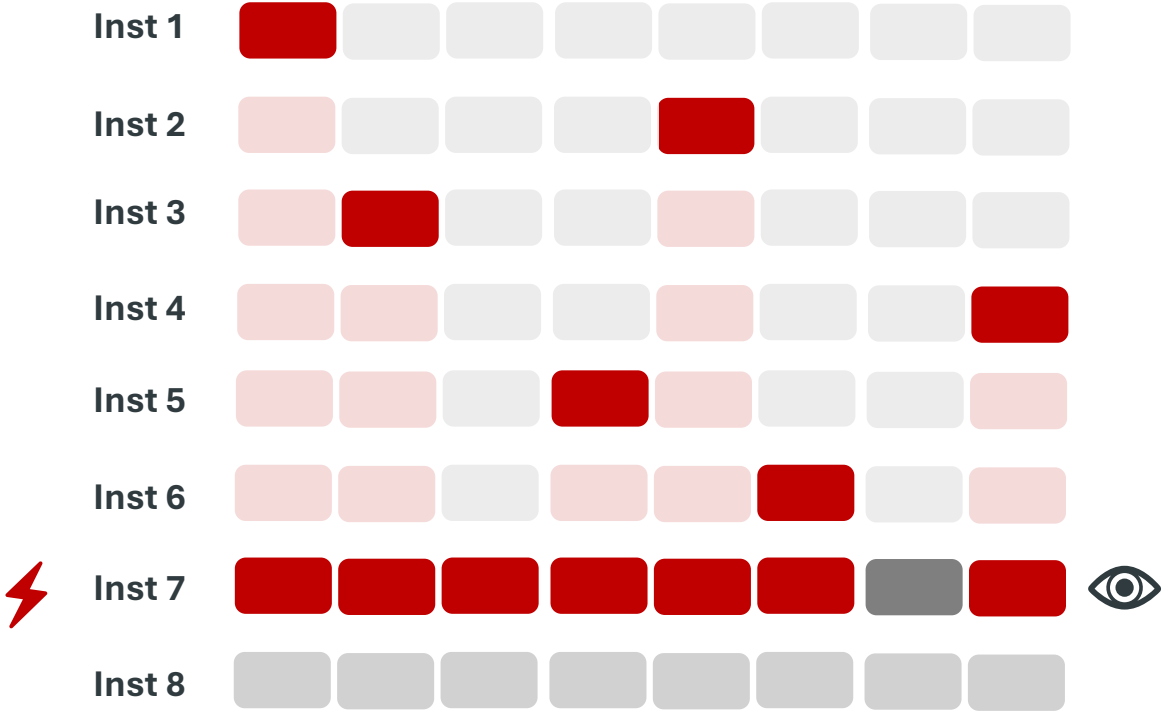
Single-Stepping



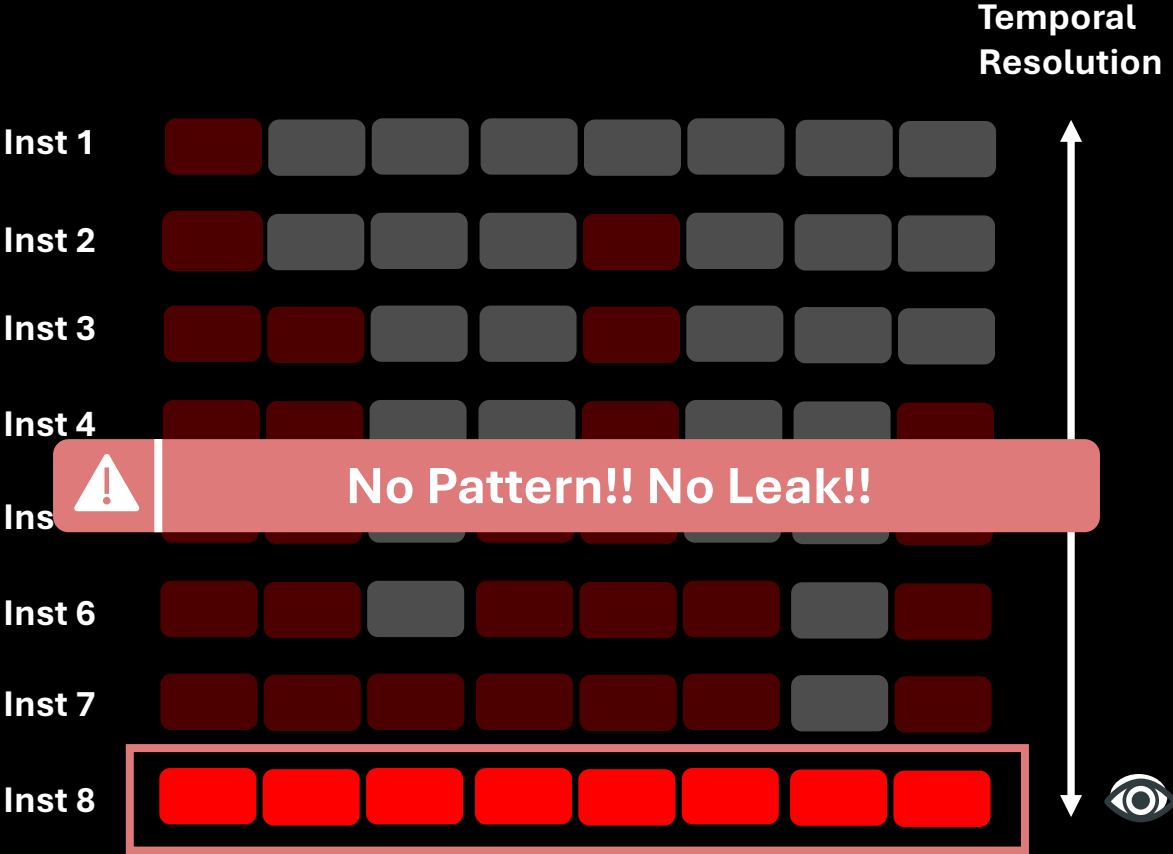
End-2-End Timing



Single-Stepping



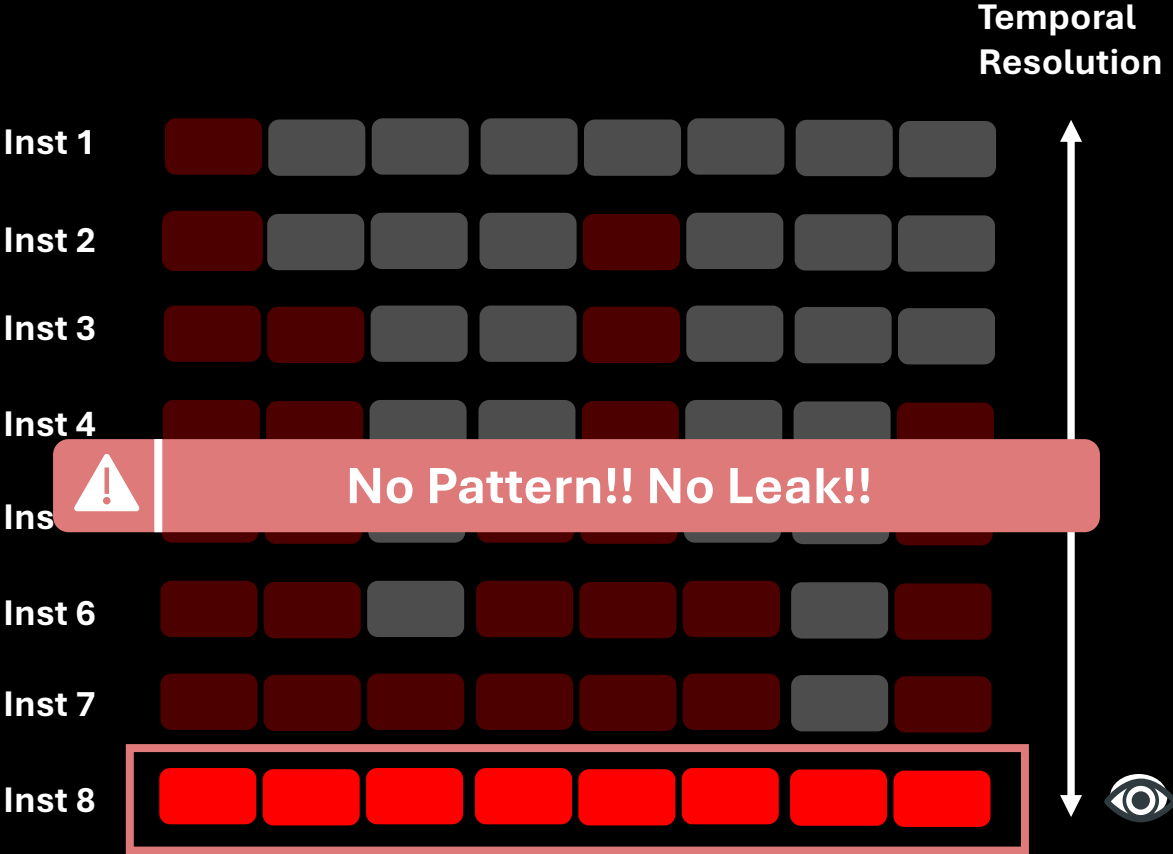
End-2-End Timing



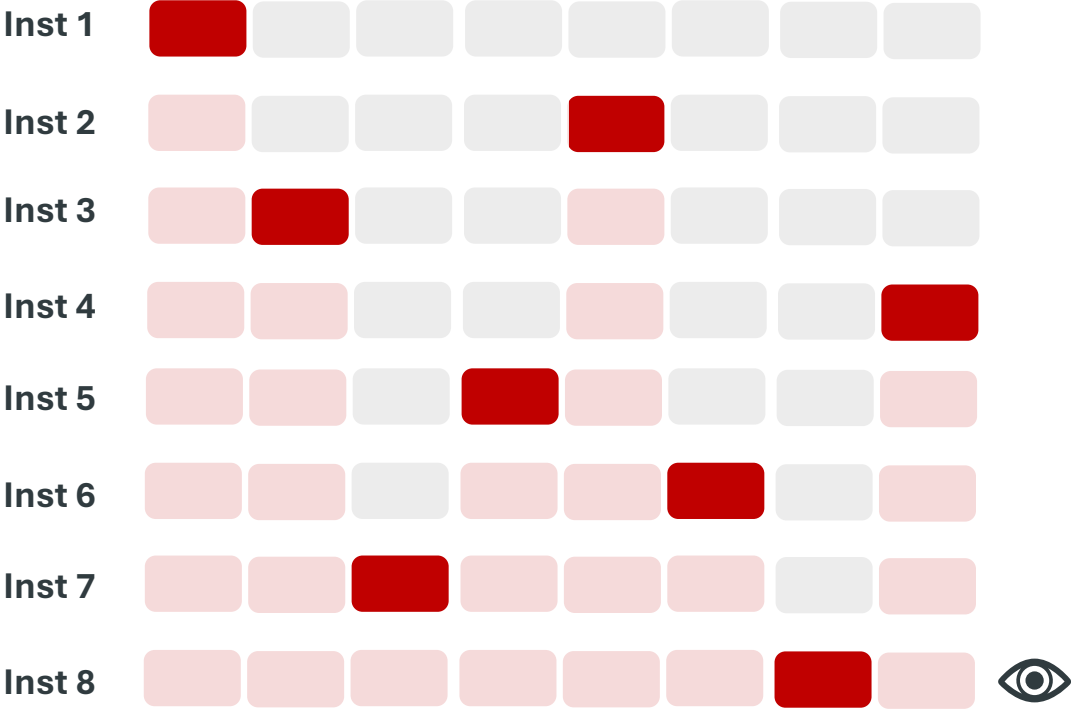
Single-Stepping



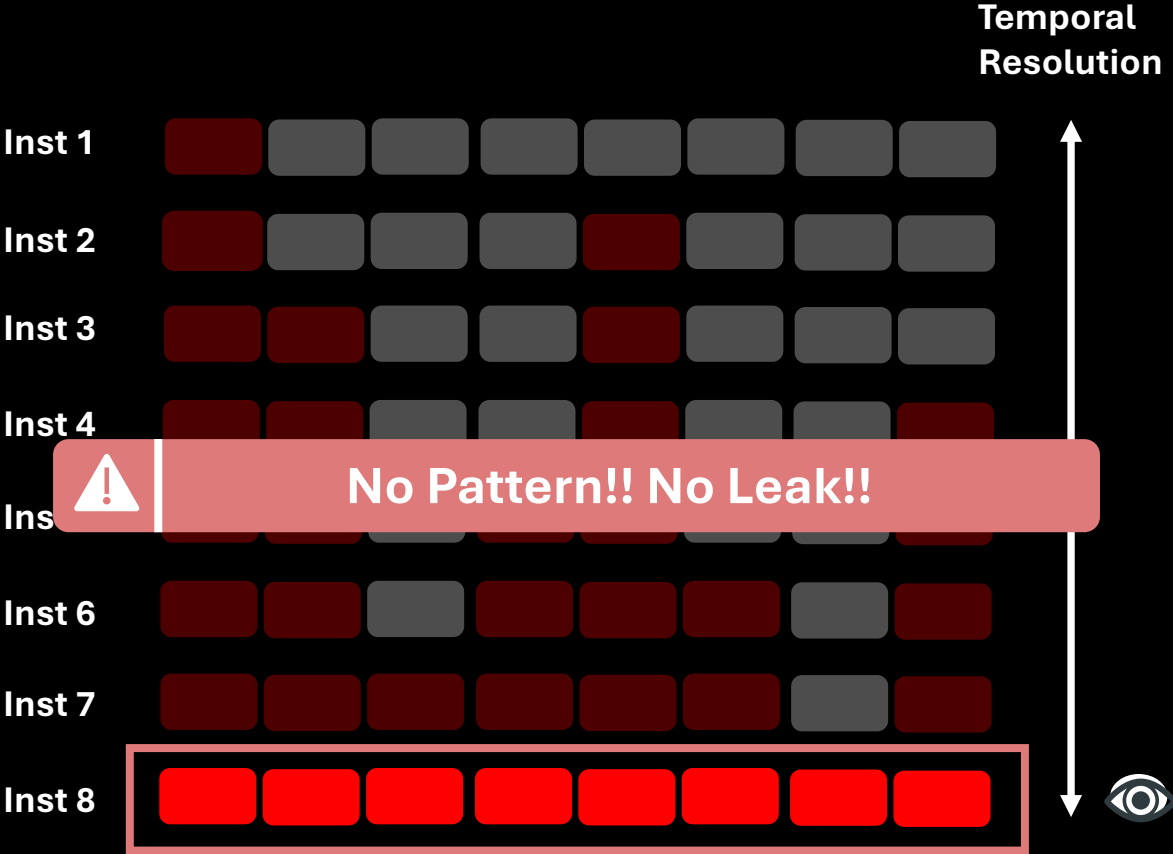
End-2-End Timing



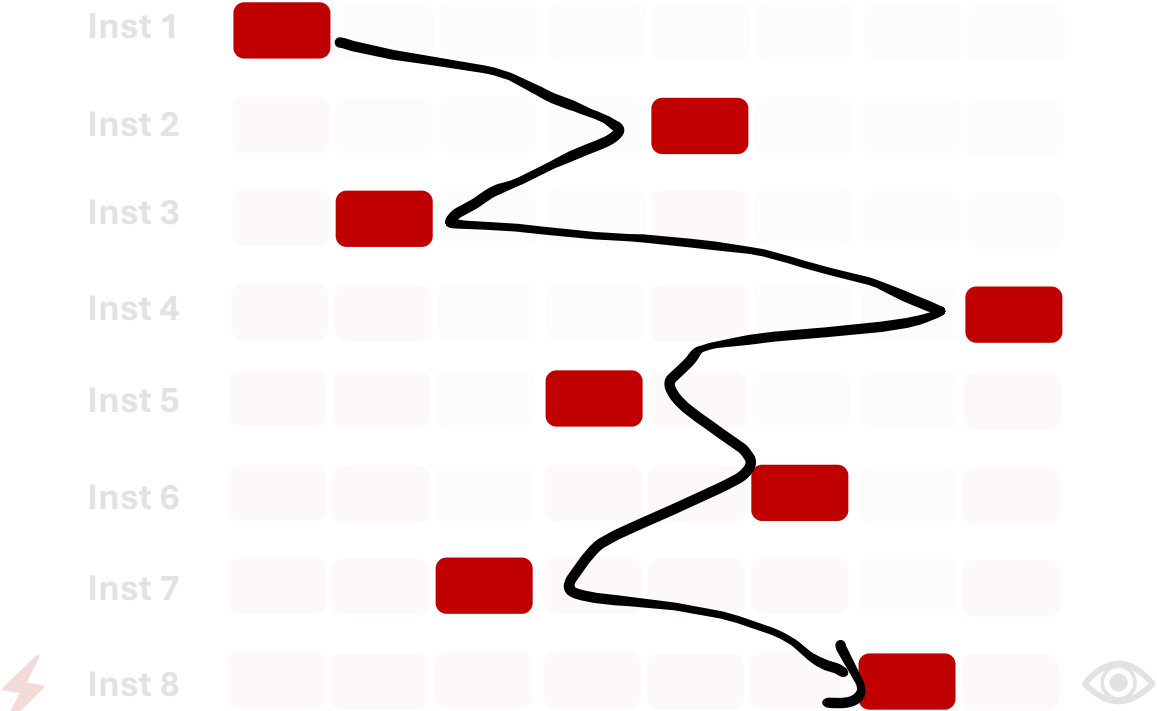
Single-Stepping



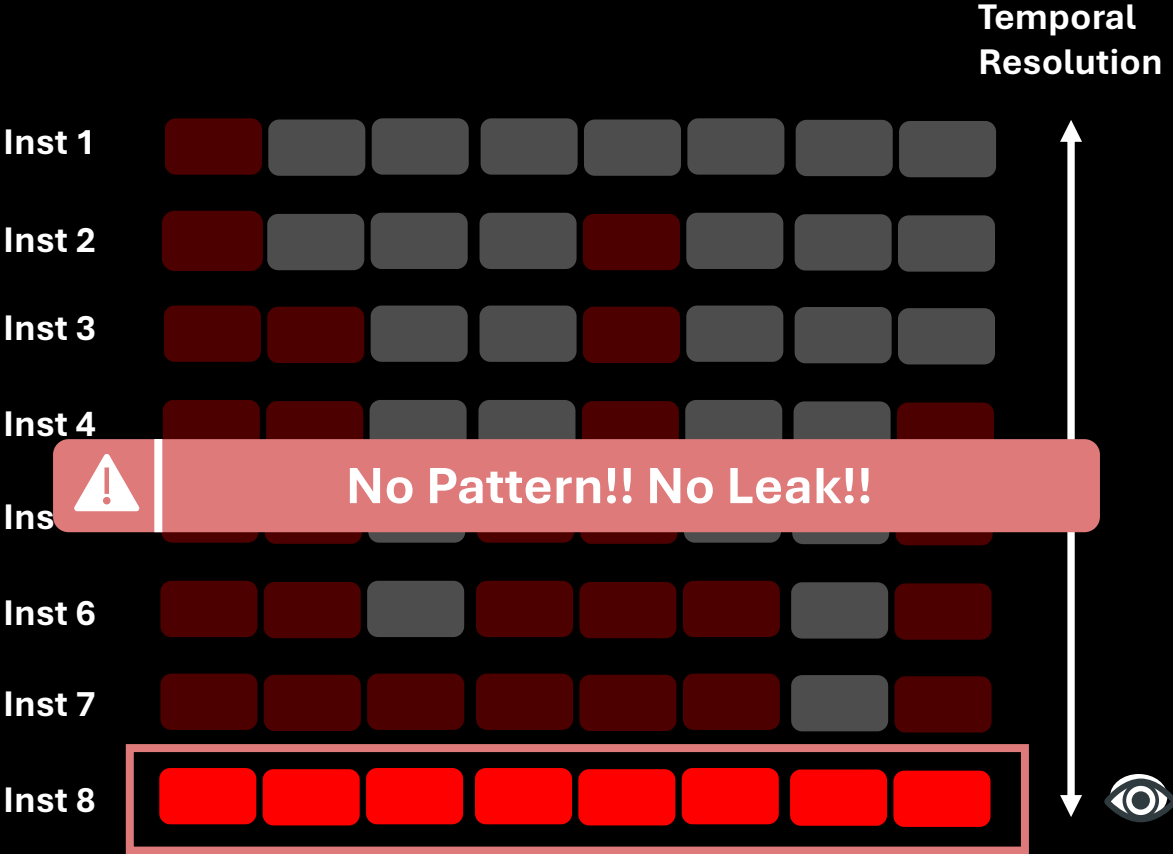
End-2-End Timing



Single-Stepping



End-2-End Timing



Single-Stepping



Single-Step Frameworks



SGX-Step

CacheGrab

Load-Step

SEV-Step

TDXexploit

TDXdown

intel.

SGX

arm

TrustZone-A

arm

TrustZone-A

AMD

SEV

intel.

TDX

intel.

TDX

Single-Step Frameworks



SGX-Step

CacheGrab

Load-Step

SEV-Step

TDXexploit

TDXdown



intel

SGX

arm

TrustZone-A

arm

TrustZone-A

AMD

SEV

intel

TDX

intel

TDX

TRUSTZONE-M

The Spark: we noticed something weird

General Assumption

instruction being interrupted. Alternatively, ARM Cortex M0 processors [5] abandon multi-cycle instructions to handle any pending interrupt immediately. While such processors are immune to the IRQ latency timing attacks described in this paper, they remain

Jo Van Bulck et al, "Nemesis: Studying Microarchitectural Timing Leaks in Rudimentary CPU Interrupt Logic".

General Assumption

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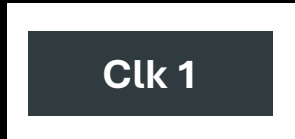
Jo Van Bulck et al, "Nemesis: Studying Microarchitectural Timing Leaks in Rudimentary CPU Interrupt Logic".

Instruction
MOV

Instruction
UDIV

Instruction

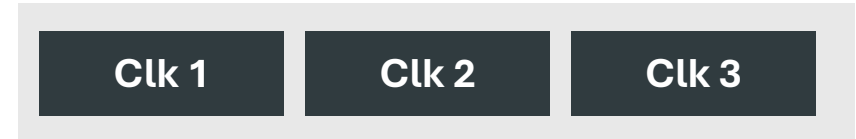
MOV



1 Clock cycle

Instruction

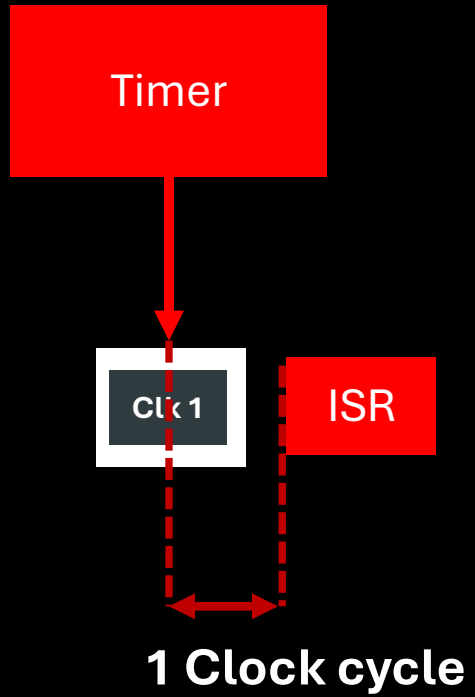
UDIV



3 Clock cycles

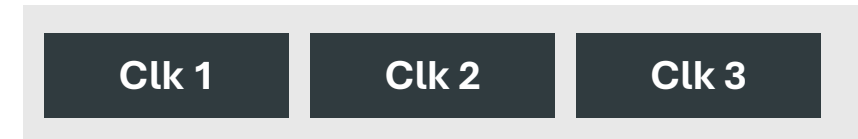
Instruction

MOV



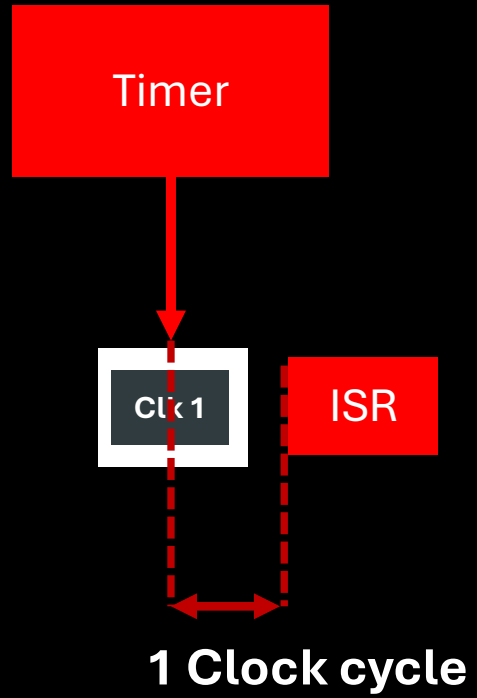
Instruction

UDIV

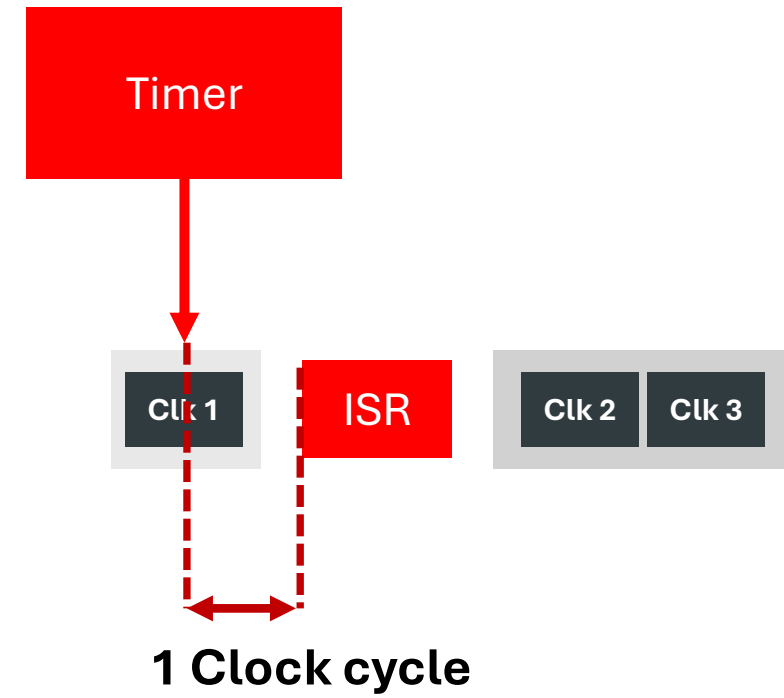


3 Clock cycles

Instruction MOV



Instruction UDIV

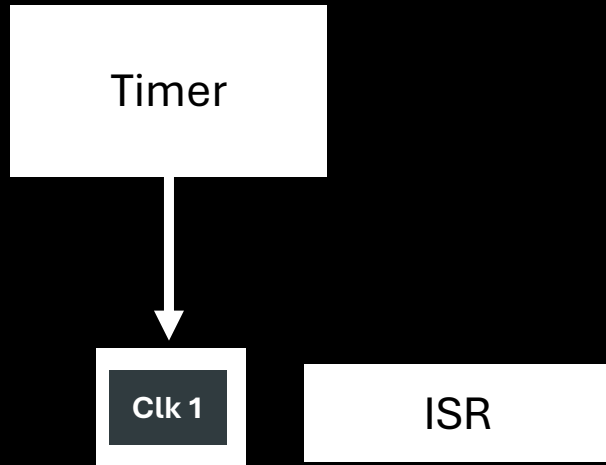


Instruction
MOV

Instruction
UDIV

Instruction

MOV

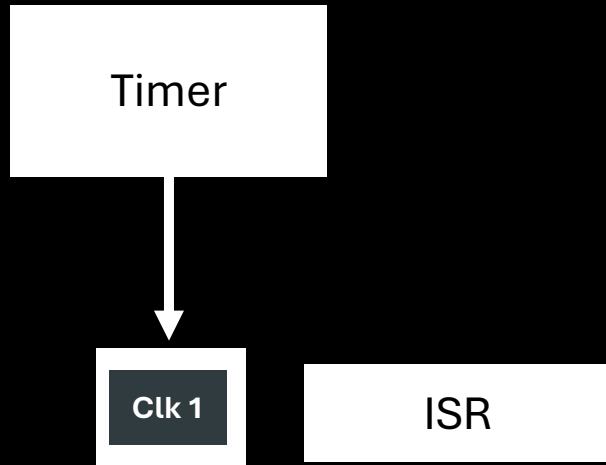


Instruction

UDIV

Instruction

MOV

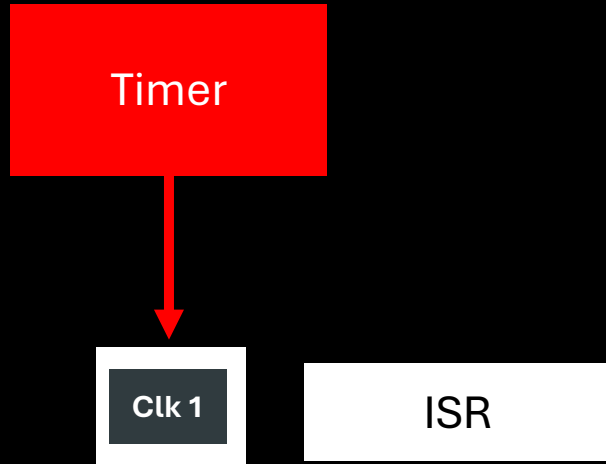


Instruction

UDIV

Instruction

MOV

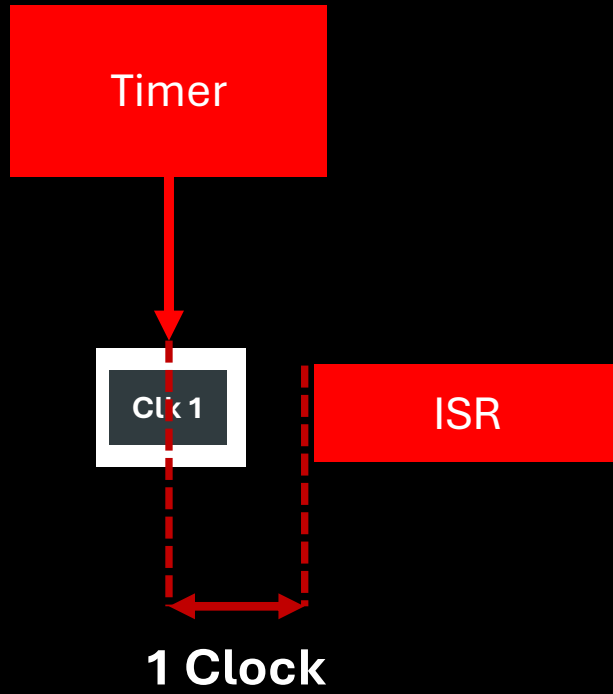


Instruction

UDIV

Instruction

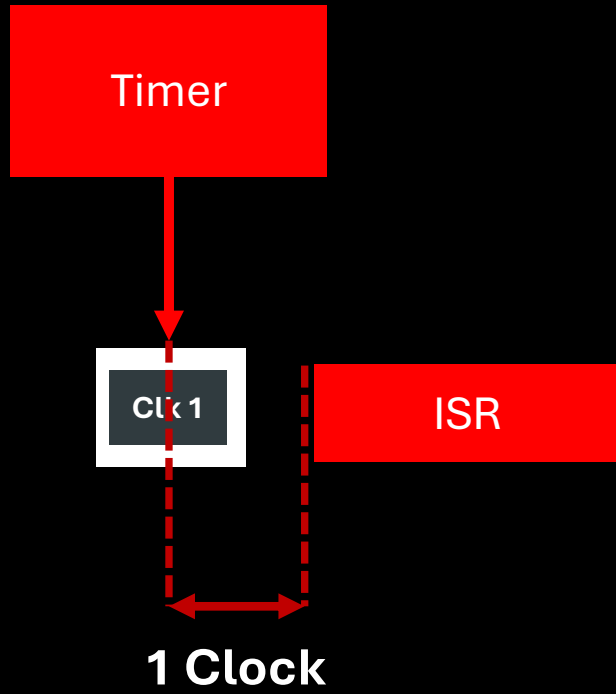
MOV



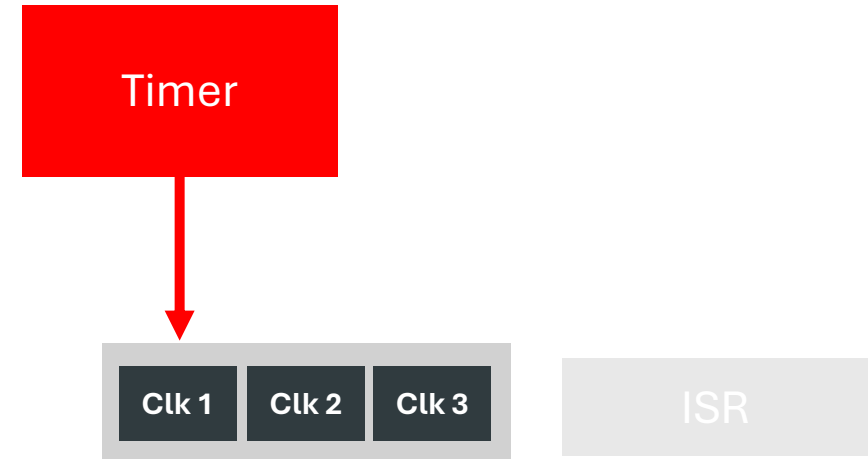
Instruction

UDIV

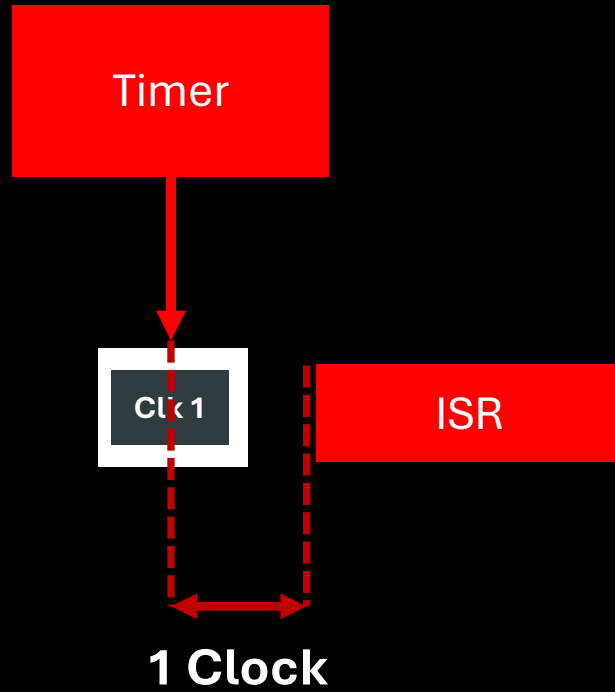
Instruction MOV



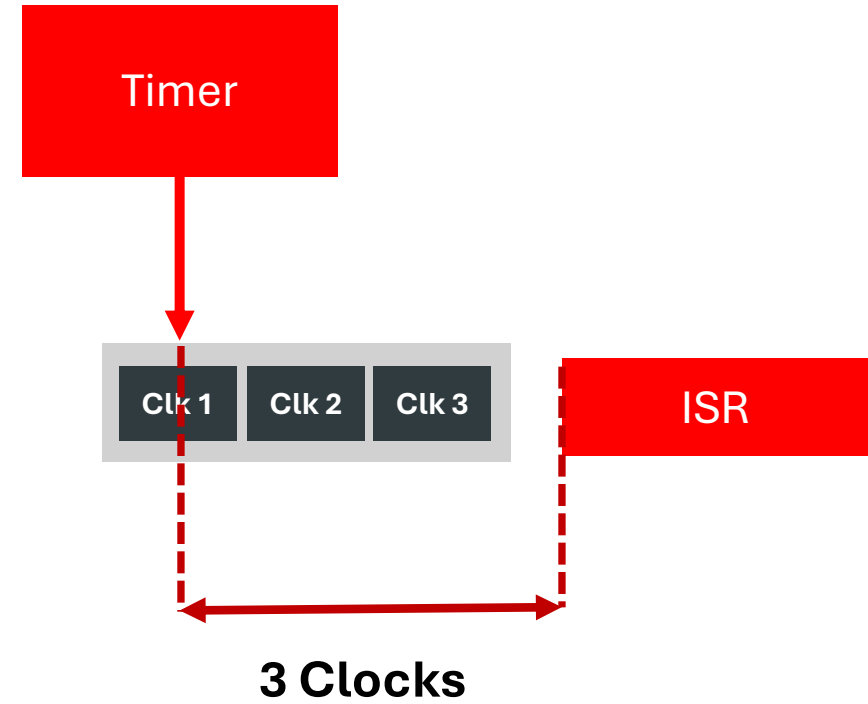
Instruction UDIV



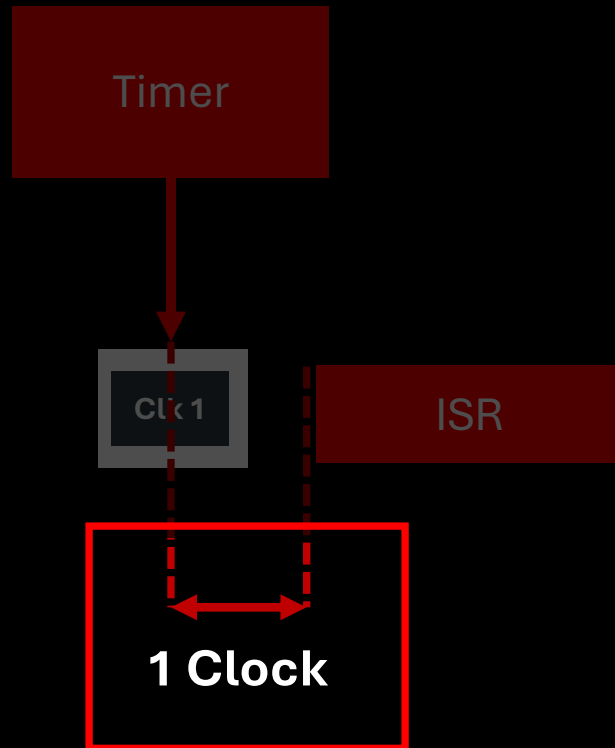
Instruction MOV



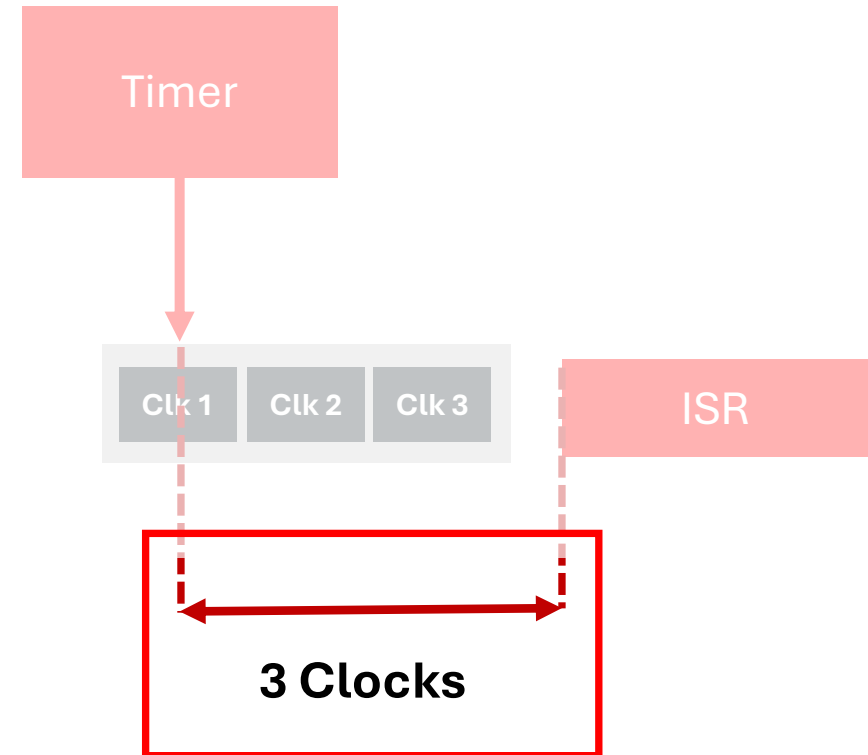
Instruction UDIV



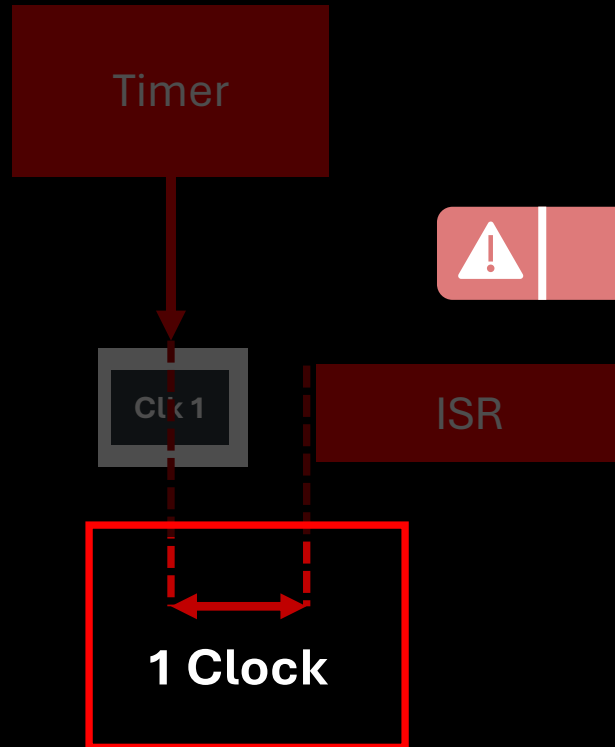
Instruction MOV



Instruction UDIV

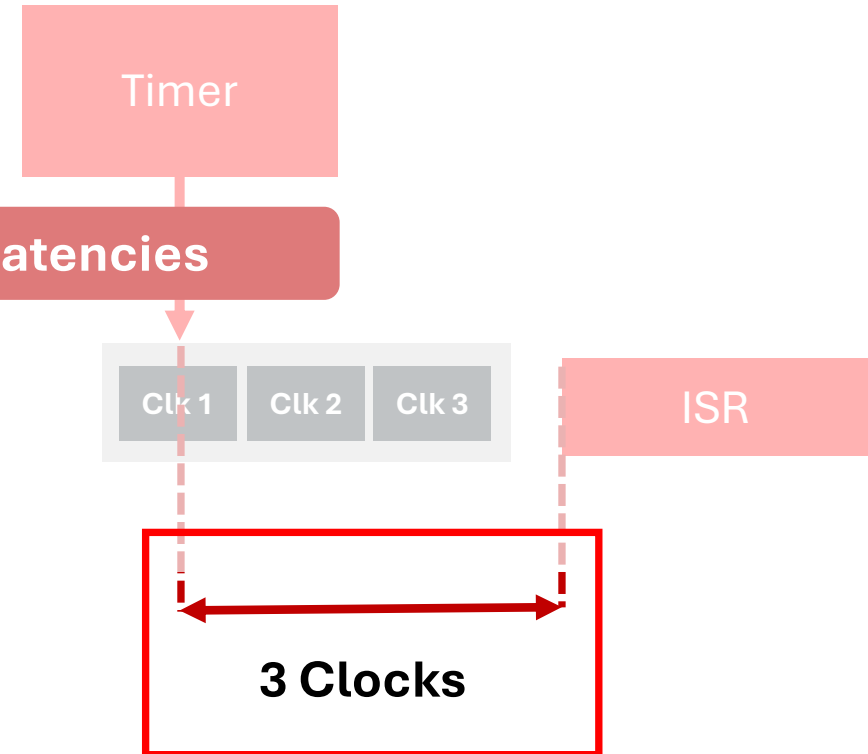


Instruction MOV

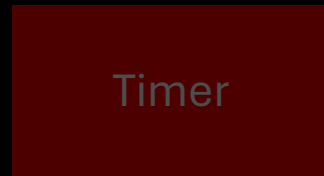


Different Interrupt Latencies

Instruction UDIV



Instruction MOV



1 Clock

Instruction UDIV



3 Clocks

! Different Interrupt Latencies

! LEAK!

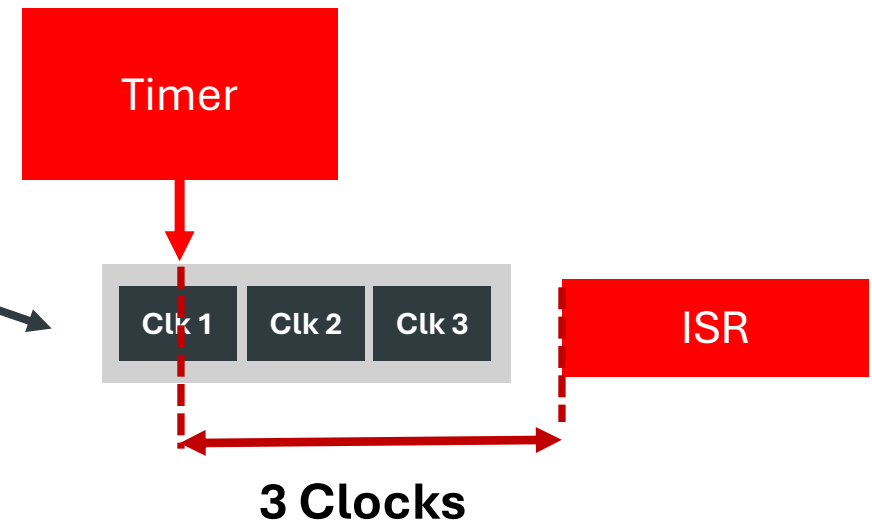
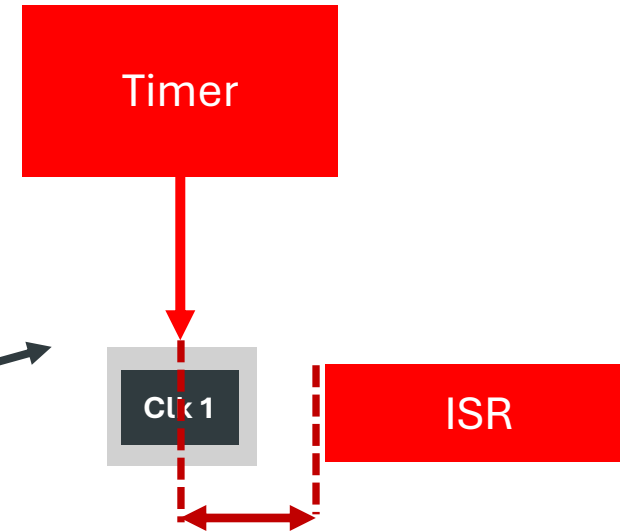


If (S=1)

Instruction 1

else

Instruction 2



The Bumpy Road to Achieve Single-Stepping

Single-Stepping

Instruction

Instruction

Instruction

Instruction

Instruction

Debug

Instruction

Instruction

Instruction

Instruction

Instruction

Production

Single-Stepping

Instruction

Instruction

Instruction

Instruction

Instruction

Debug

Instruction

Instruction

Instruction

Instruction

Instruction

Production

Single-Stepping

Instruction

Instruction

Instruction

Instruction

Instruction

Debug

Instruction

Instruction

Instruction

Instruction

Instruction

Production

Single-Stepping

Instruction

Instruction

Instruction

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Debug

Instruction

Instruction

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Instruction

Instruction

Production

Single-Stepping

Instruction

Instruction

Instruction

Instruction

Instruction

Debug

Instruction

Instruction

Instruction

Instruction

Instruction

Production

Single-Stepping

Instruction

Instruction

Instruction

Instruction

Instruction

Debug

Instruction

Instruction

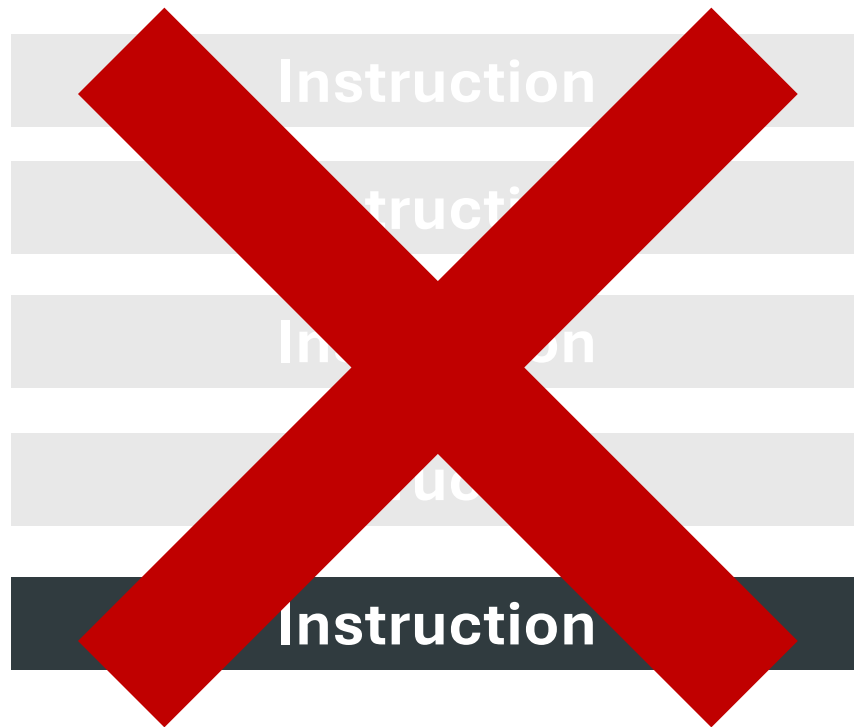
Instruction

Instruction

Instruction

Production

Single-Stepping

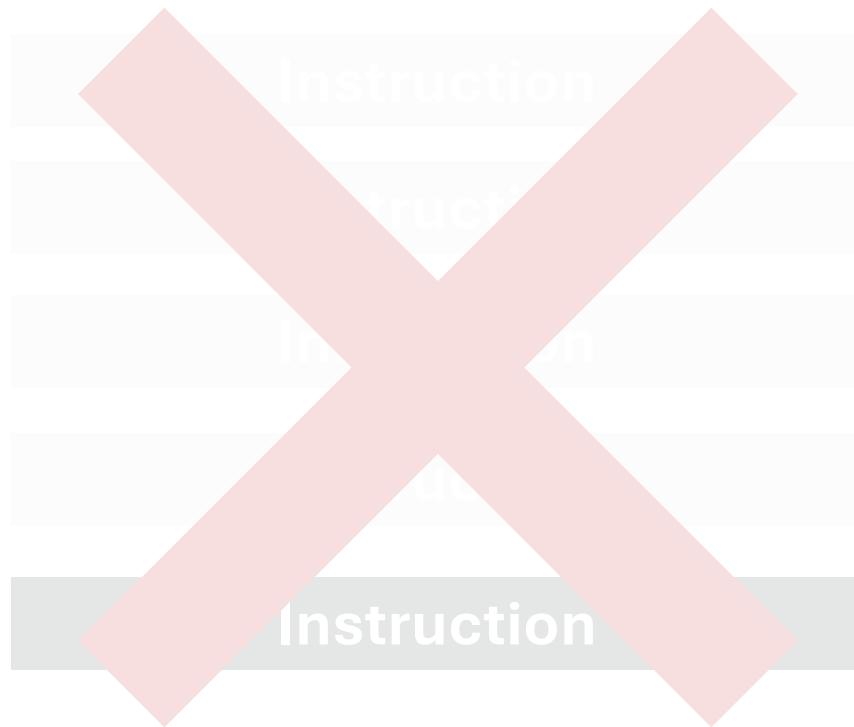


Debug

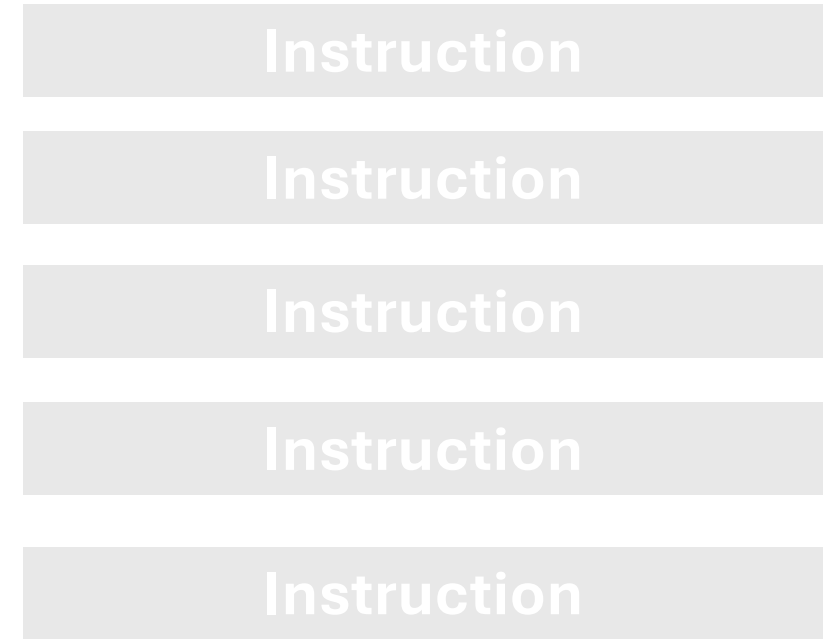


Production

Single-Stepping

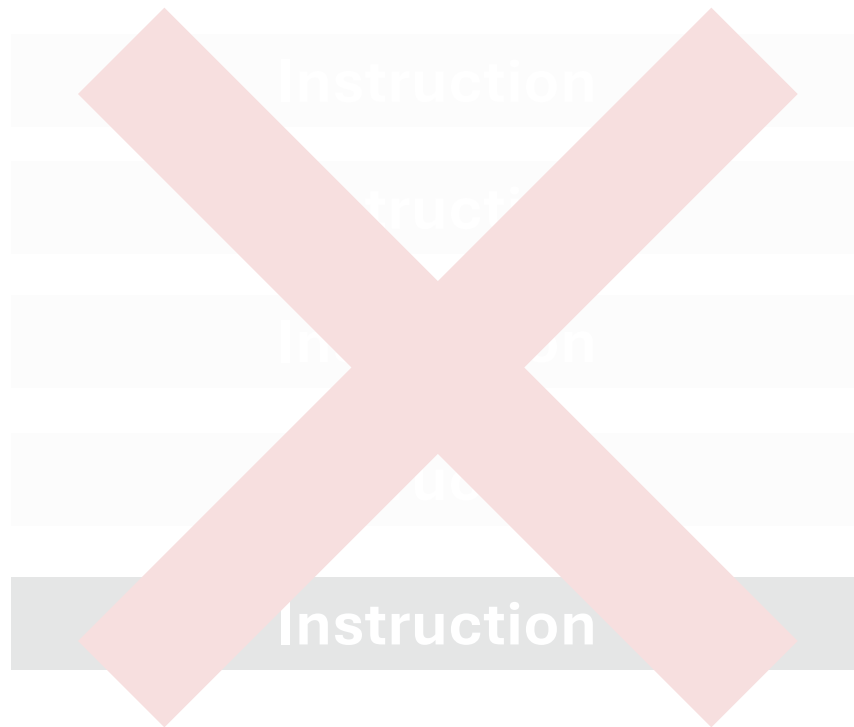


Debug

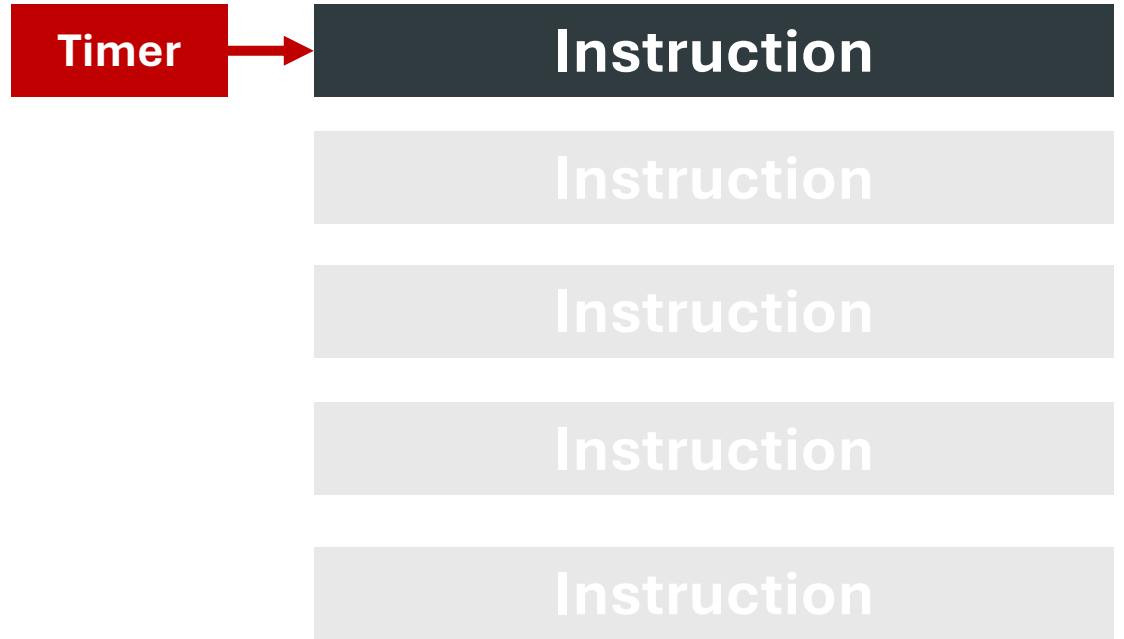


Production

Single-Stepping

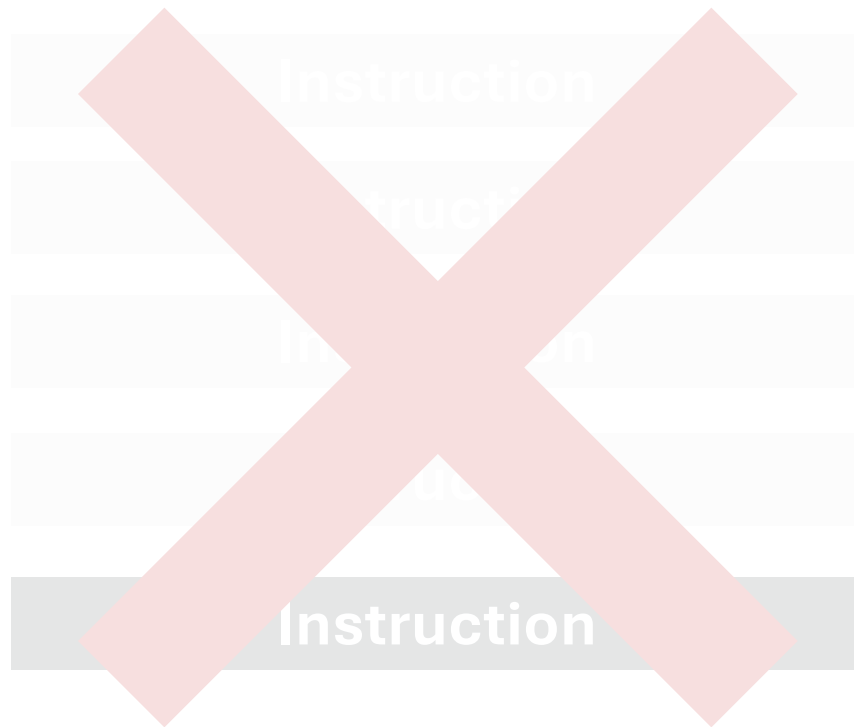


Debug

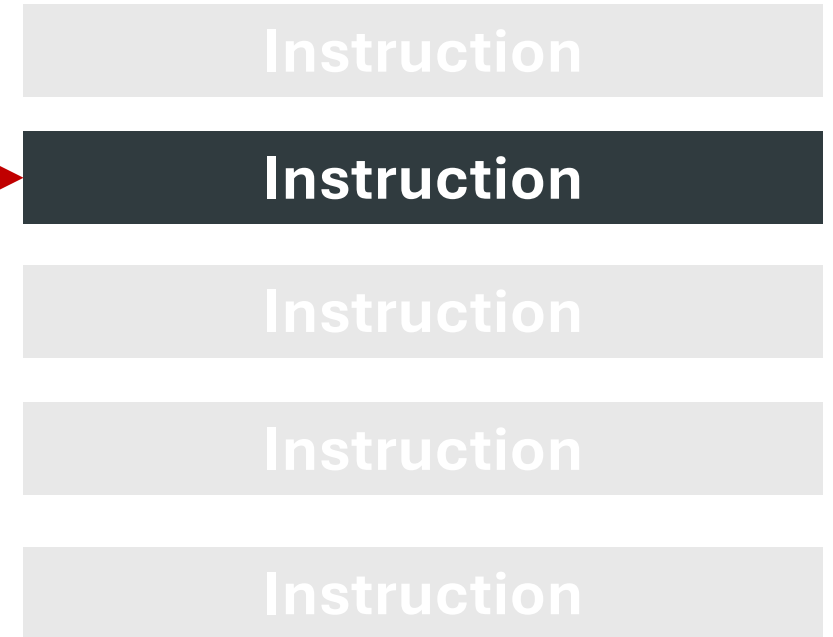


Production

Single-Stepping

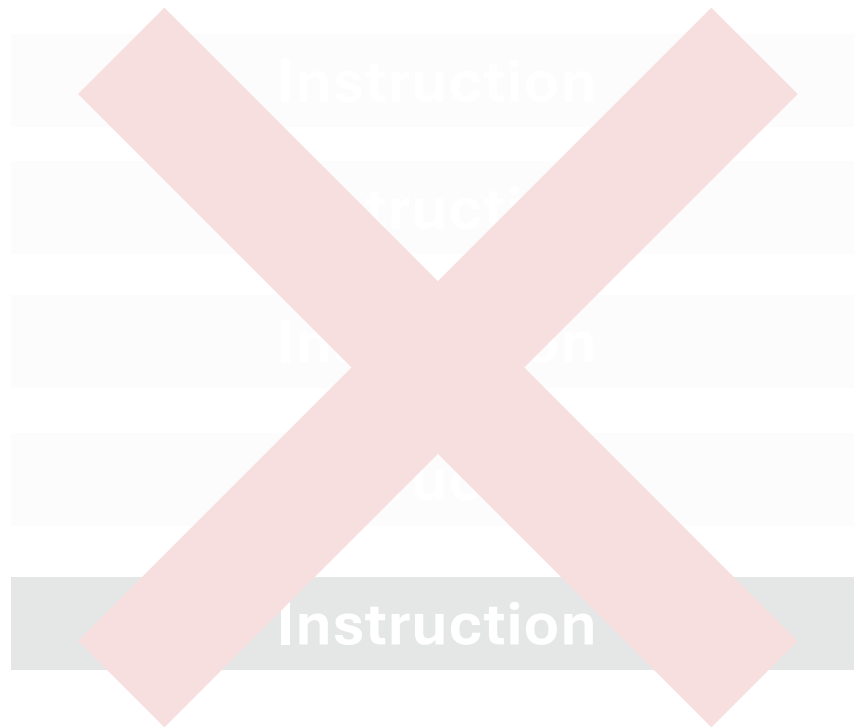


Debug

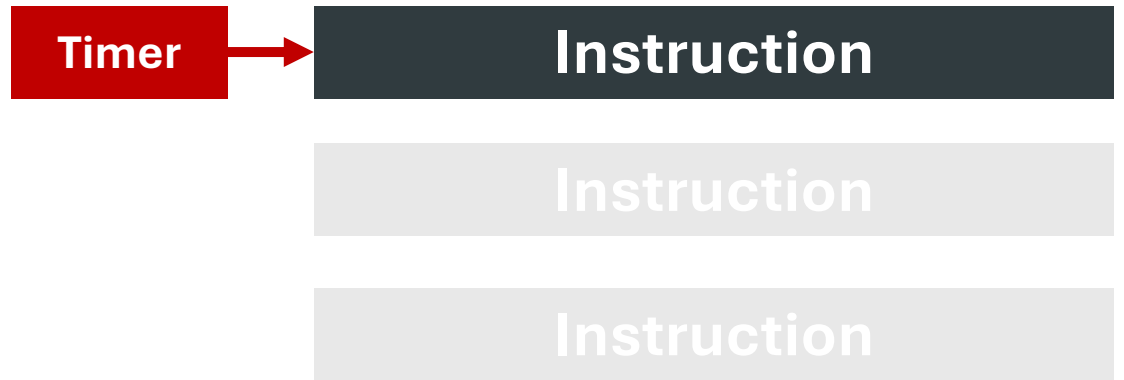


Production

Single-Stepping

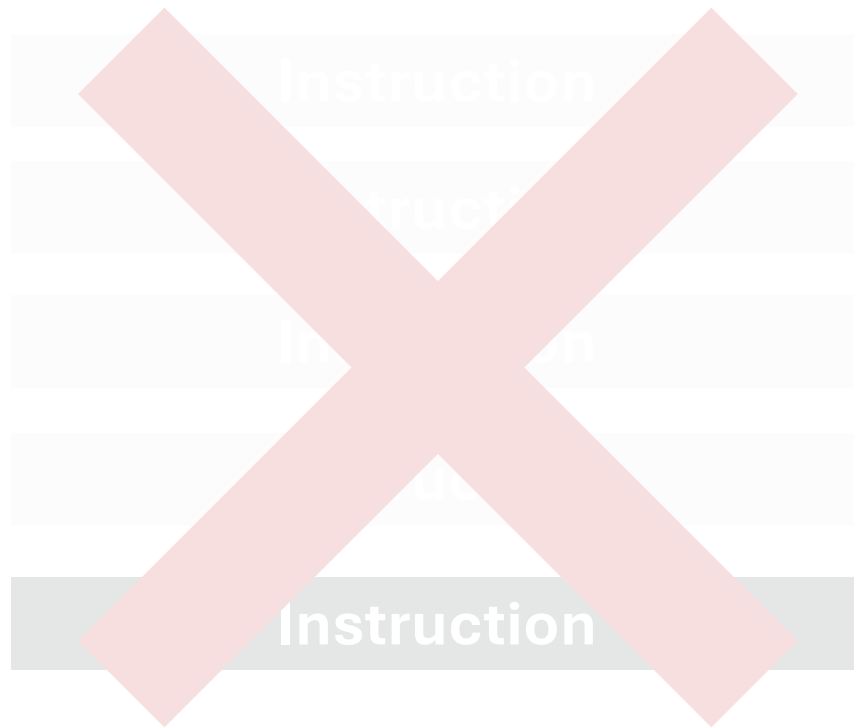


Debug

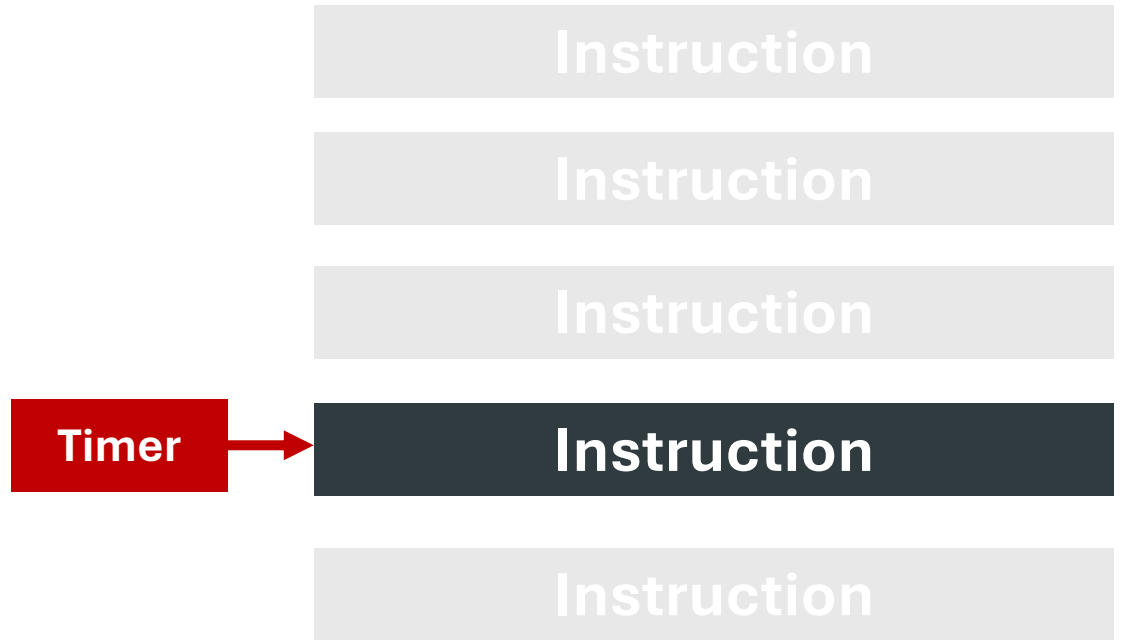


Production

Single-Stepping

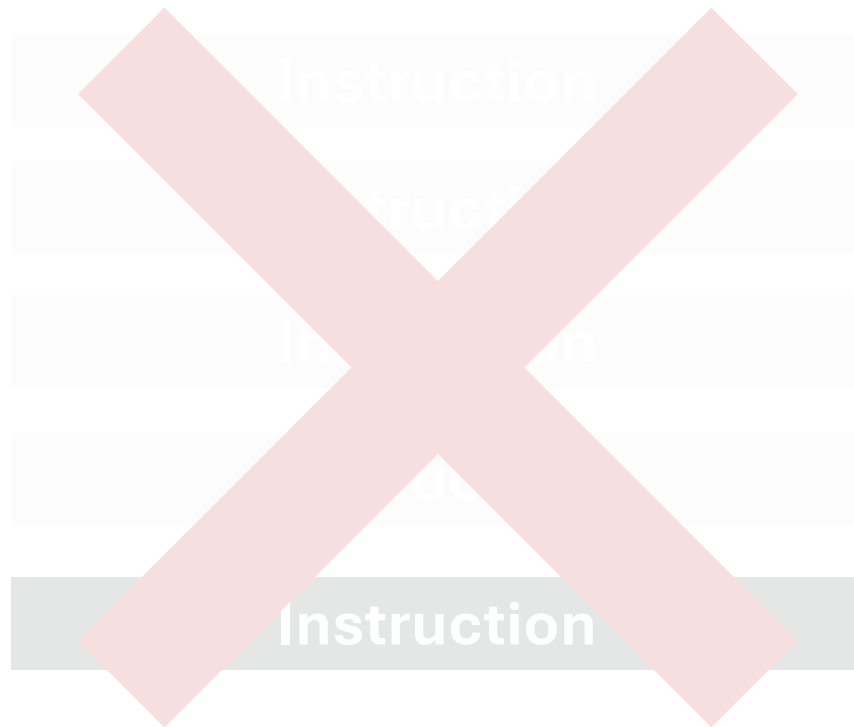


Debug

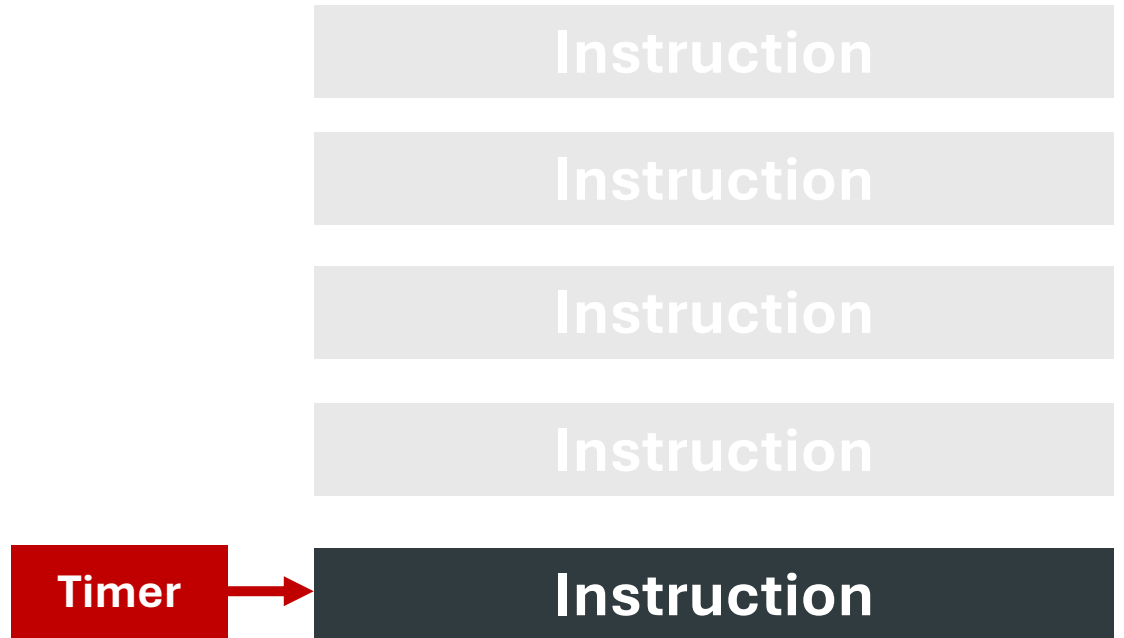


Production

Single-Stepping



Debug



Production

Timer

Context Switch

Inst

Inst

Inst

Inst

Timer

Context Switch

Inst

Inst

Inst

Inst

Timer

Context Switch

Inst

Inst

Inst

Inst

Timer

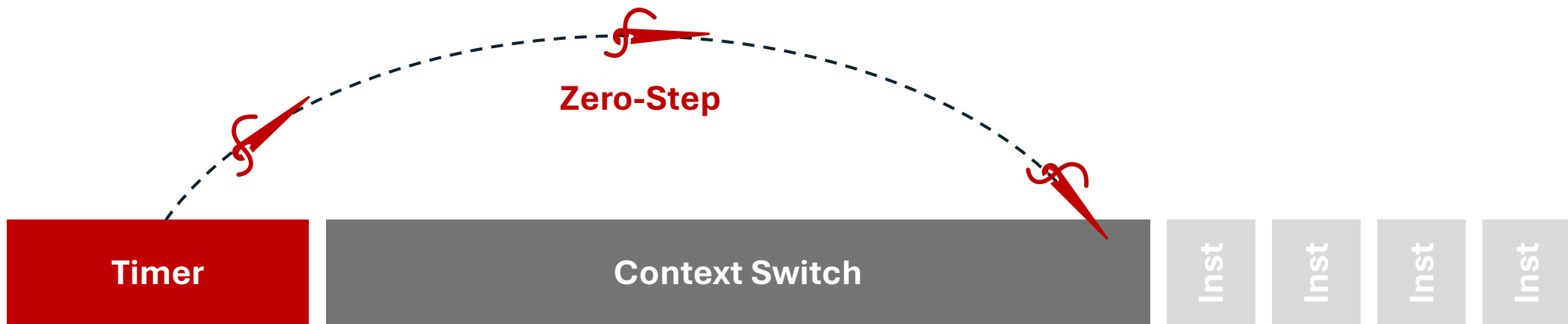
Context Switch

Inst

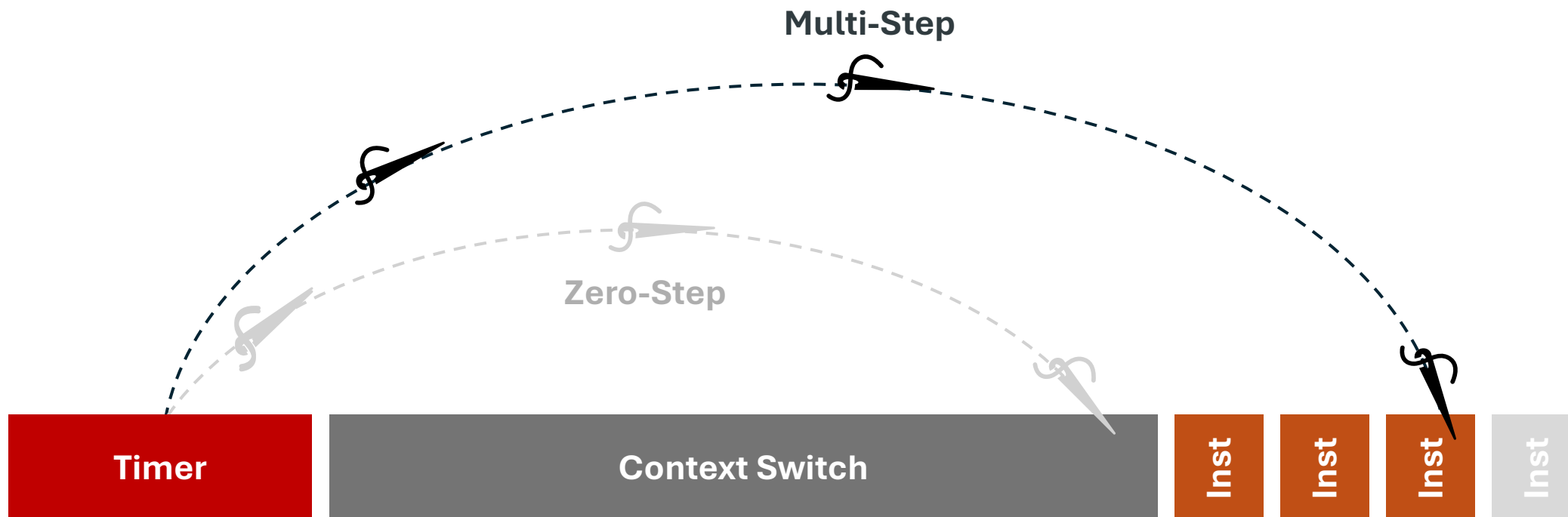
Inst

Inst

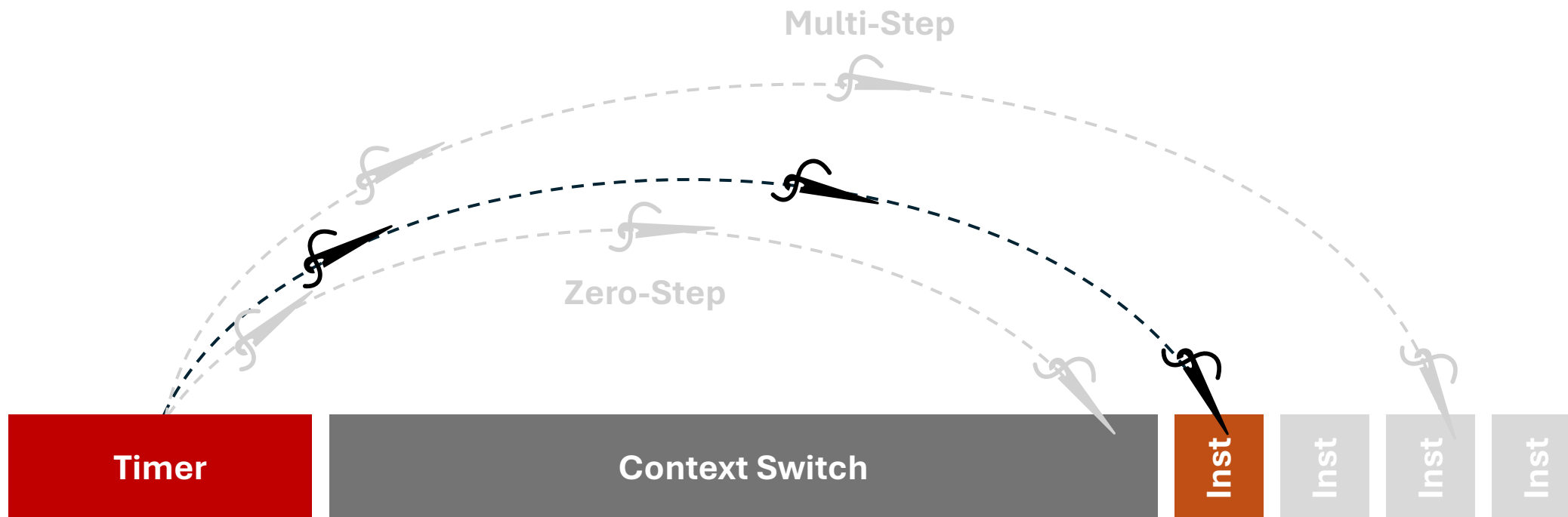
Inst



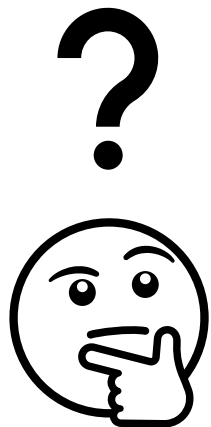
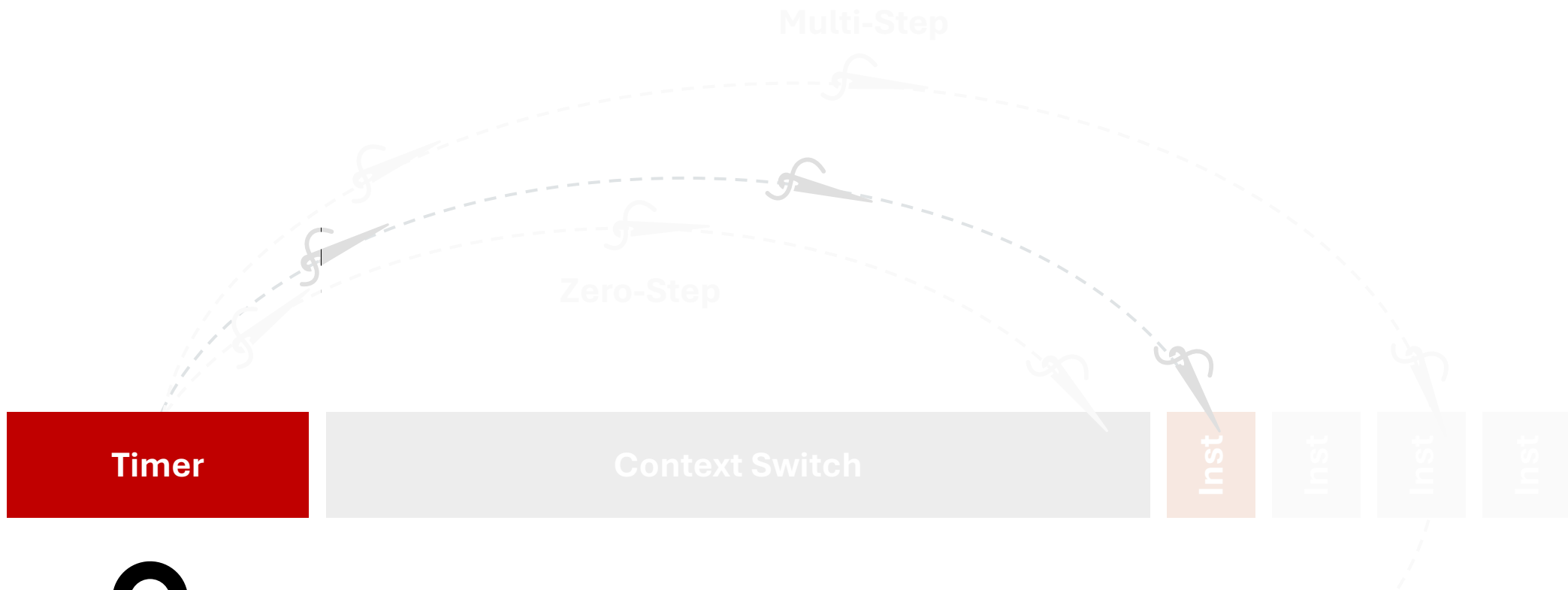
⚠ Too Early!



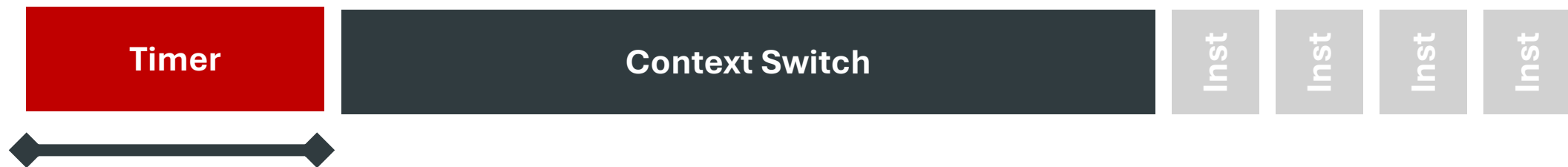
 **Too Late!**

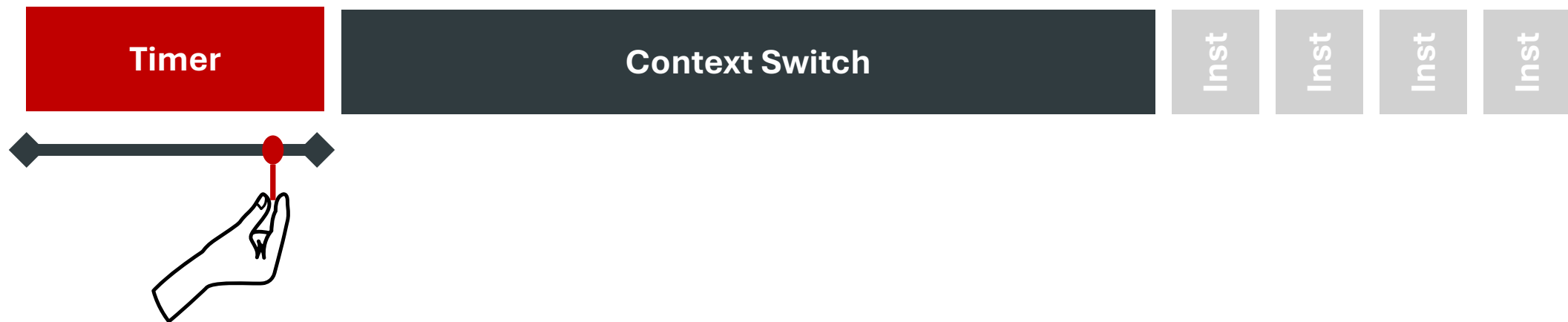


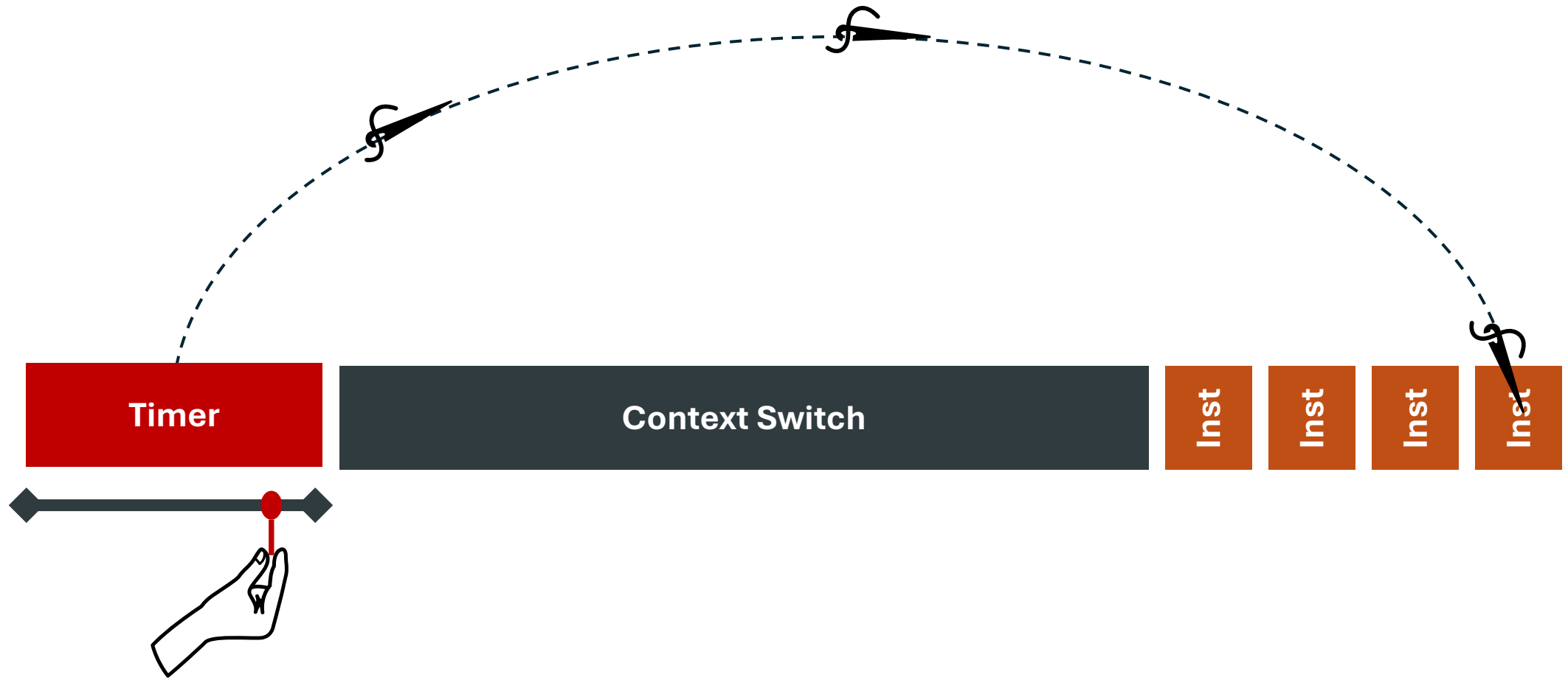
✓ **Single-Step!**

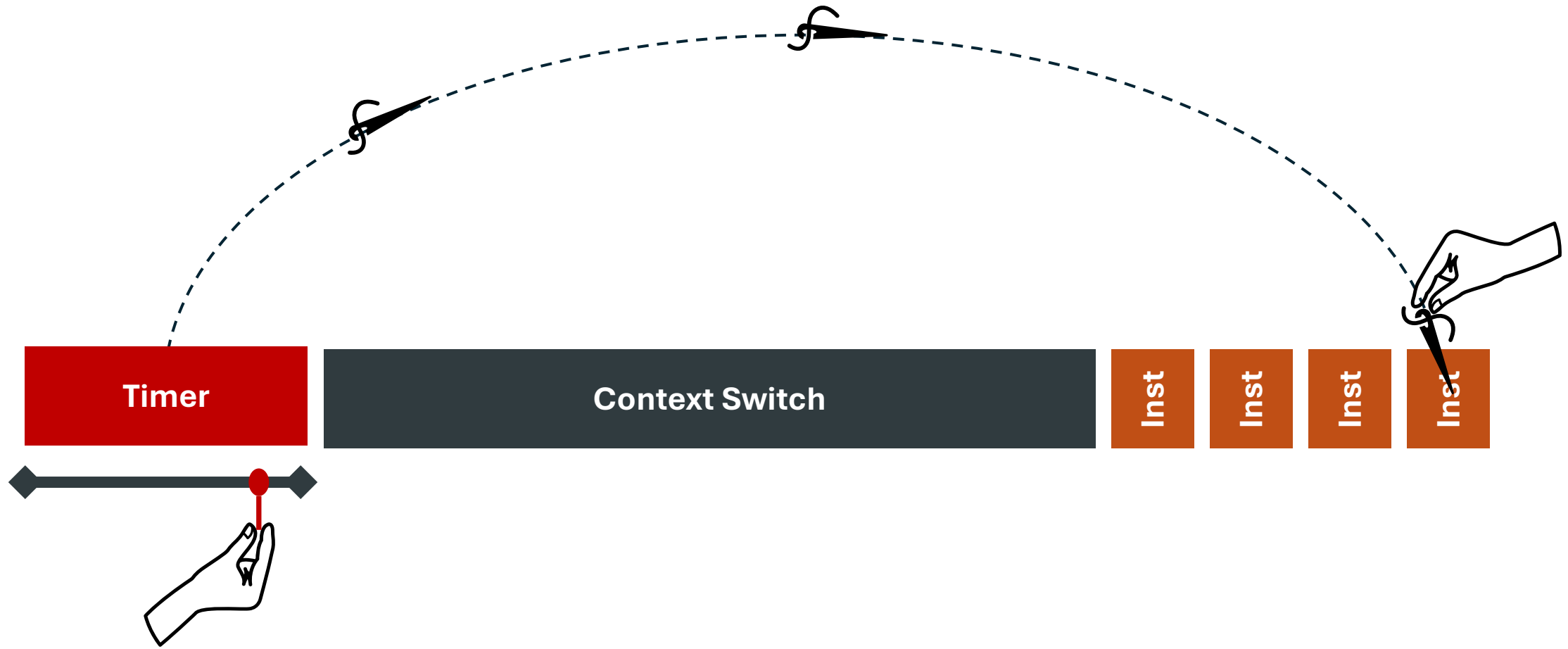


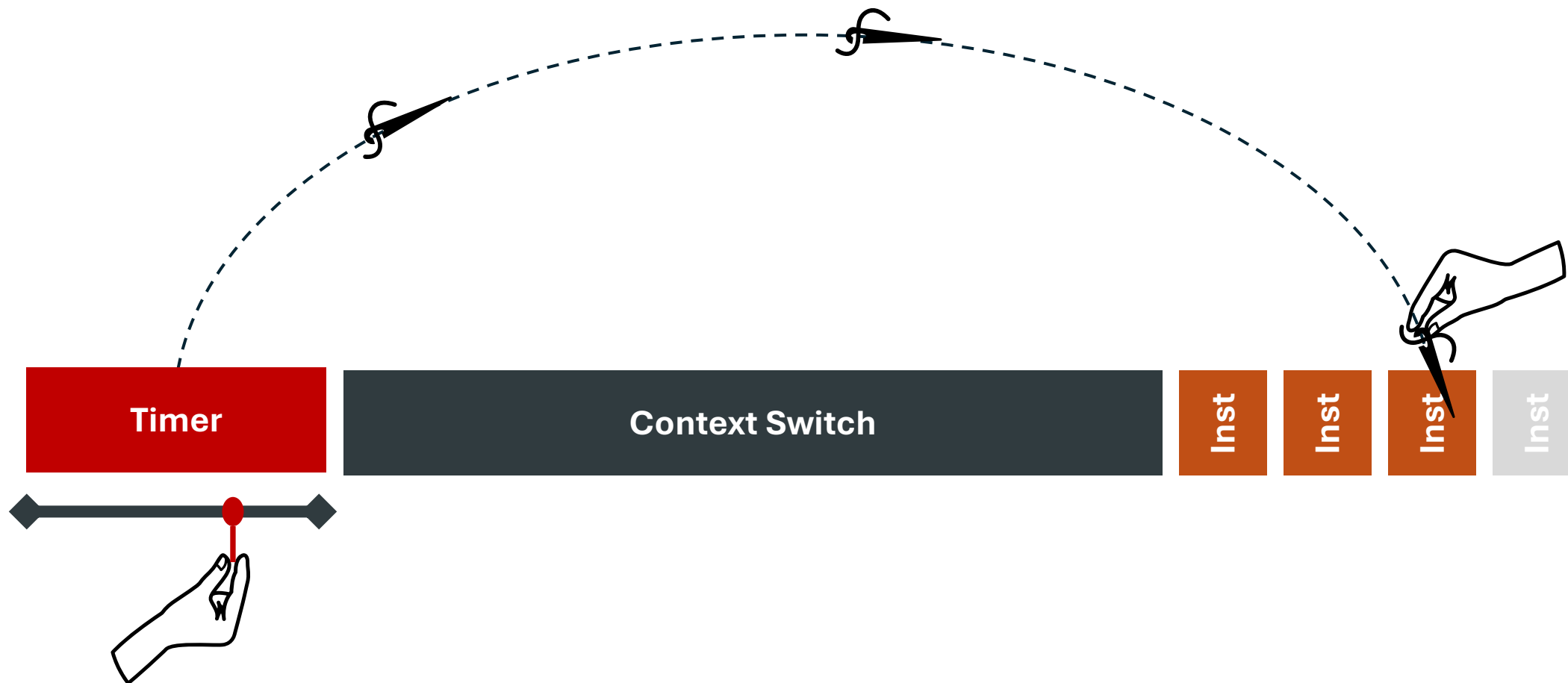
✓ Single-Step!

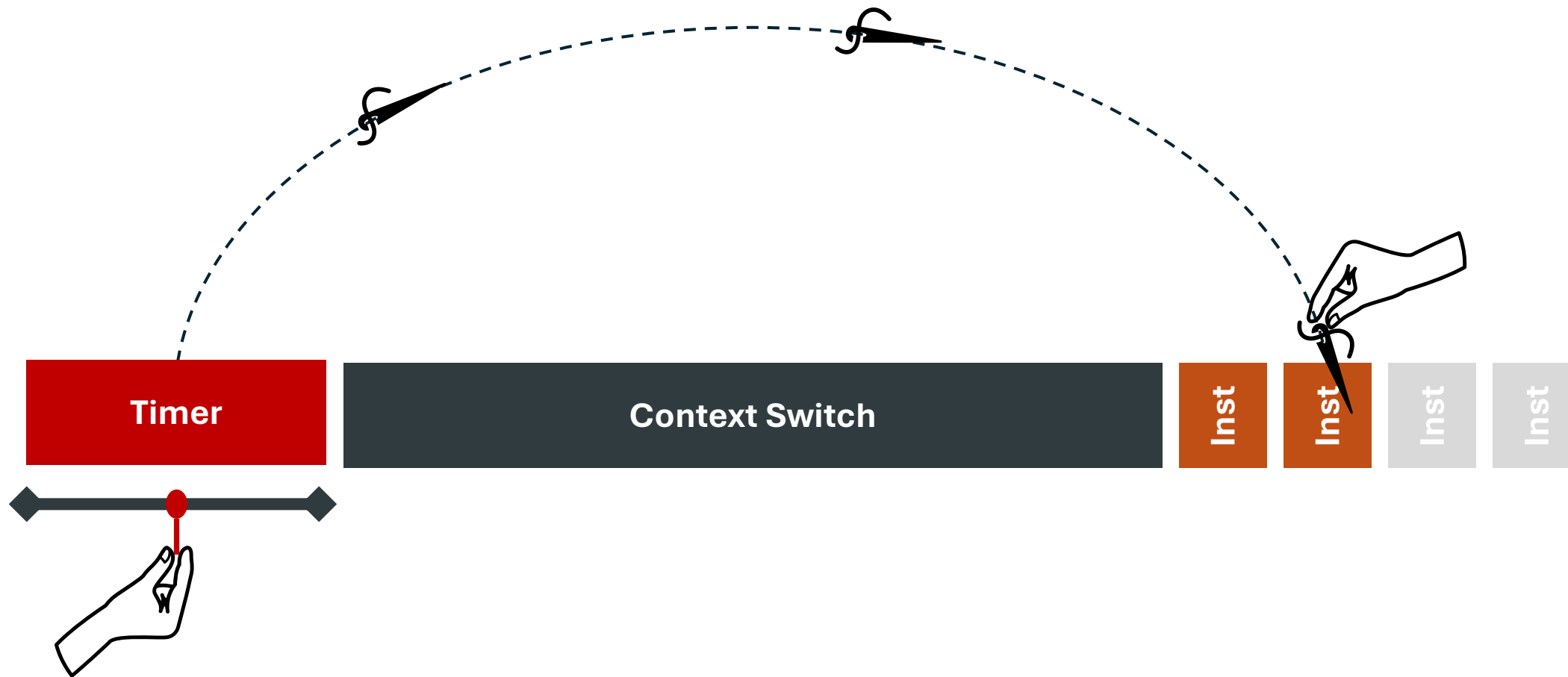


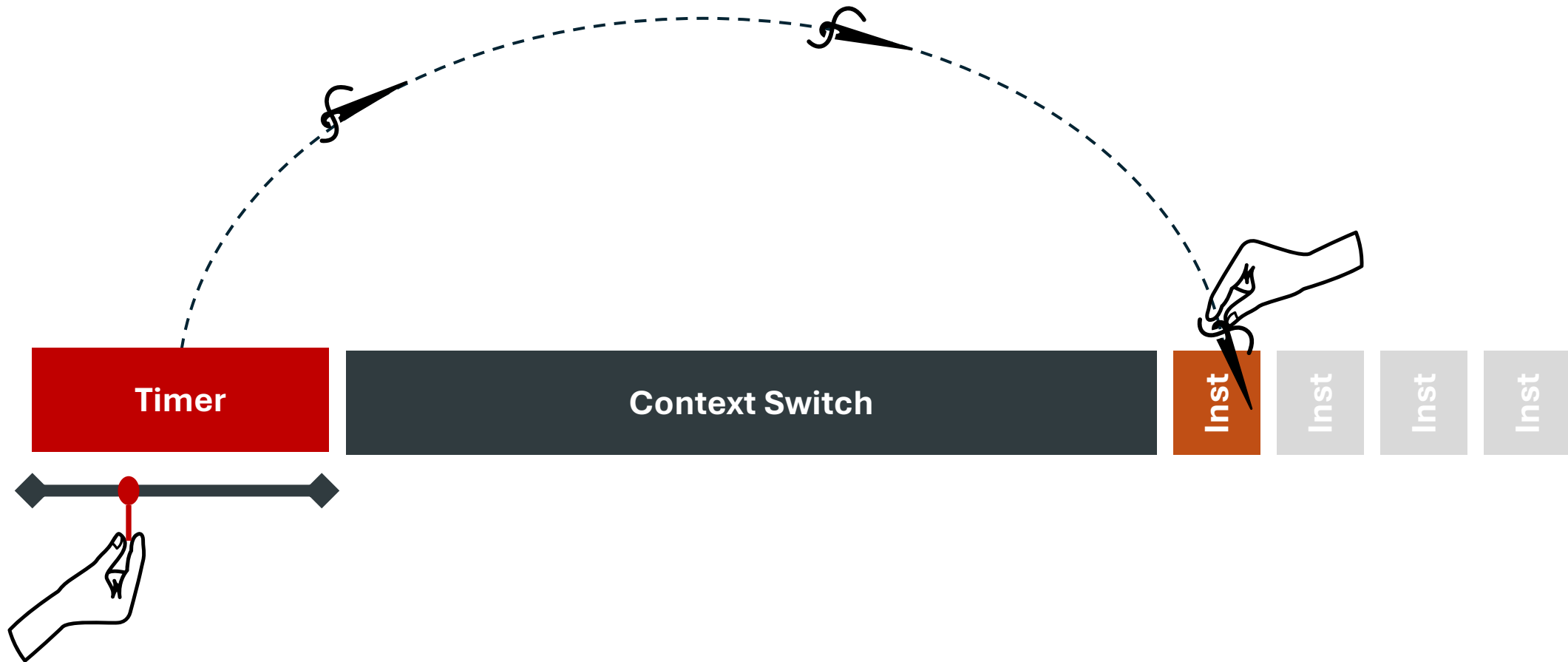


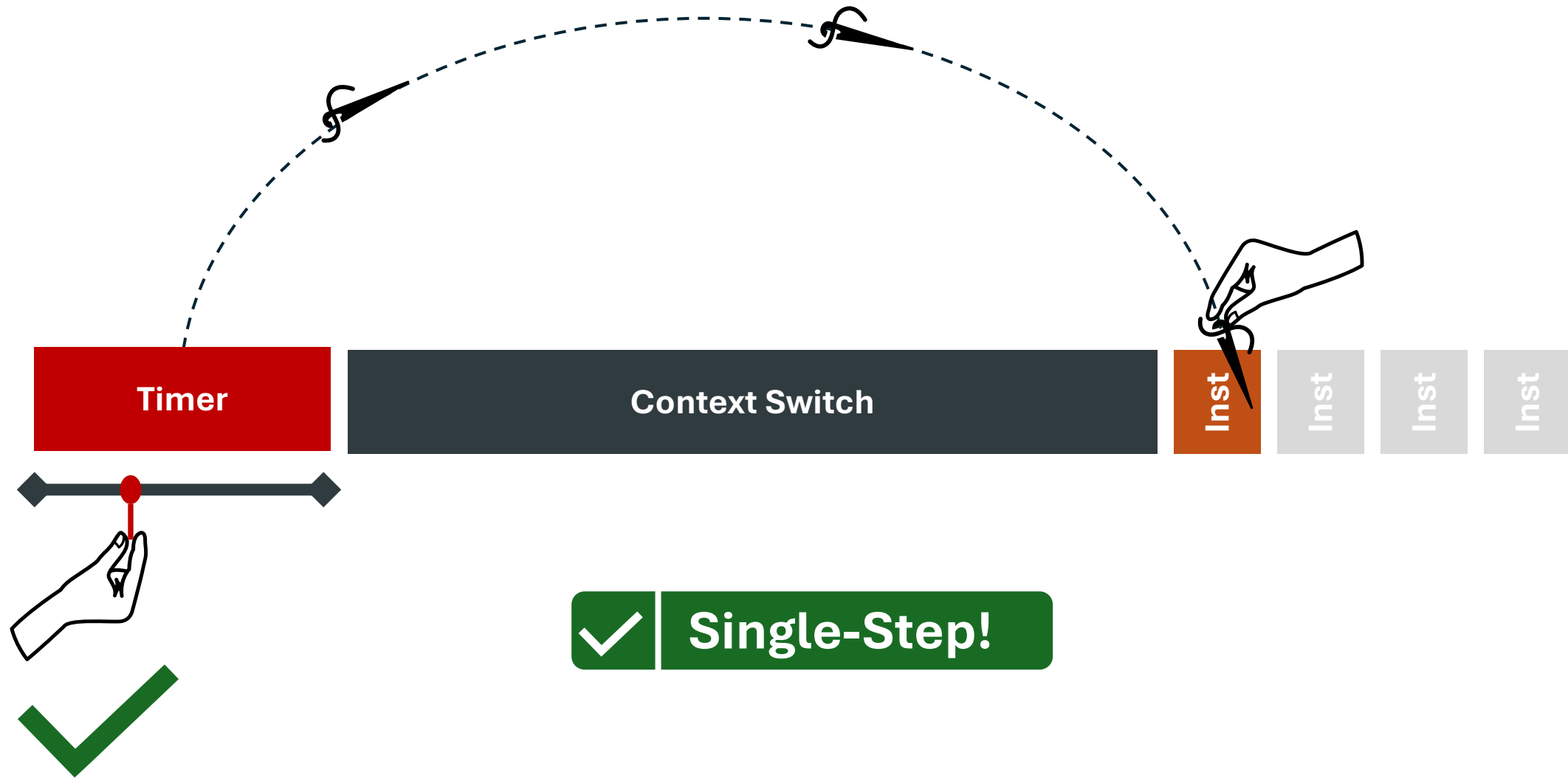


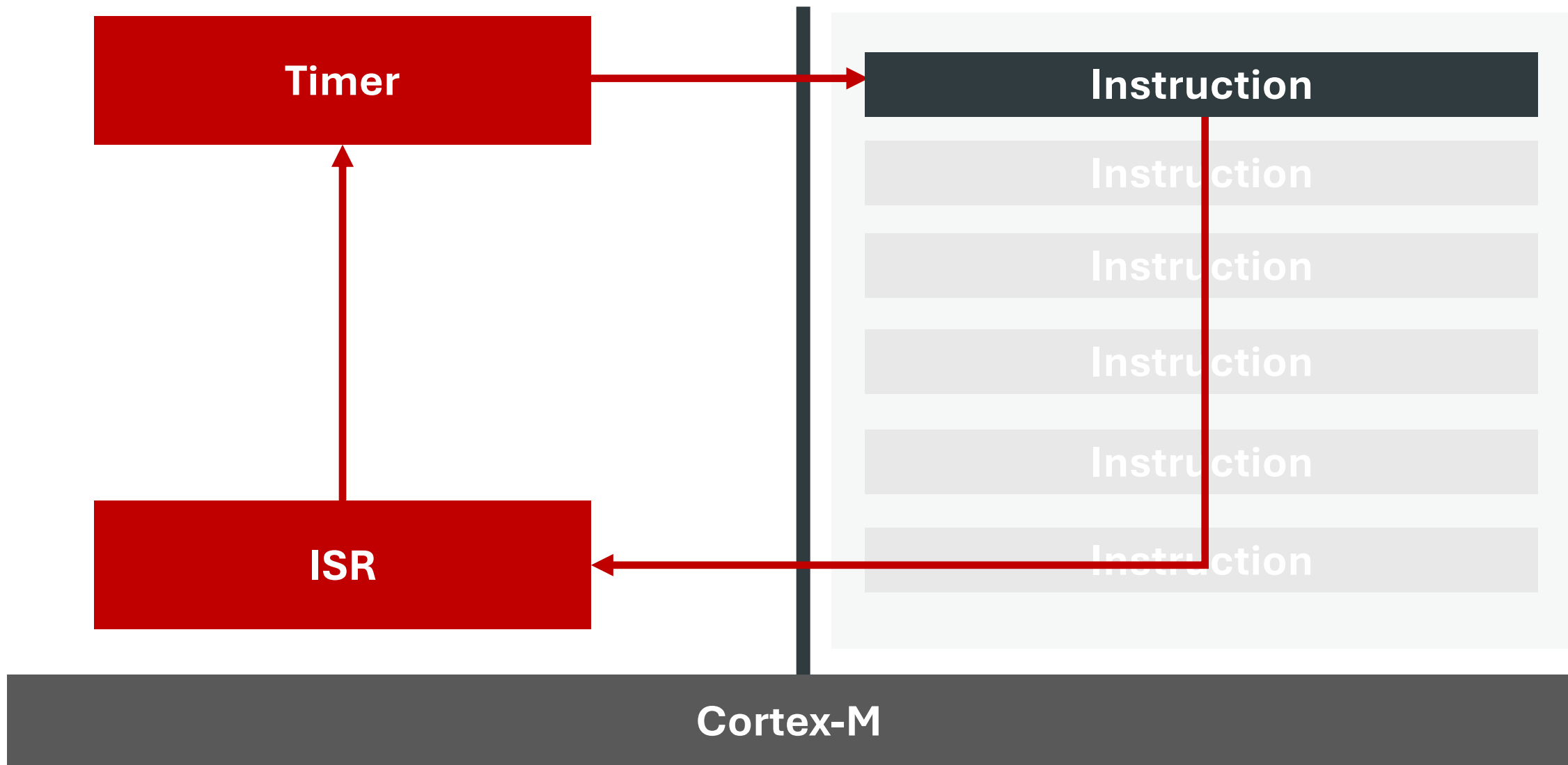


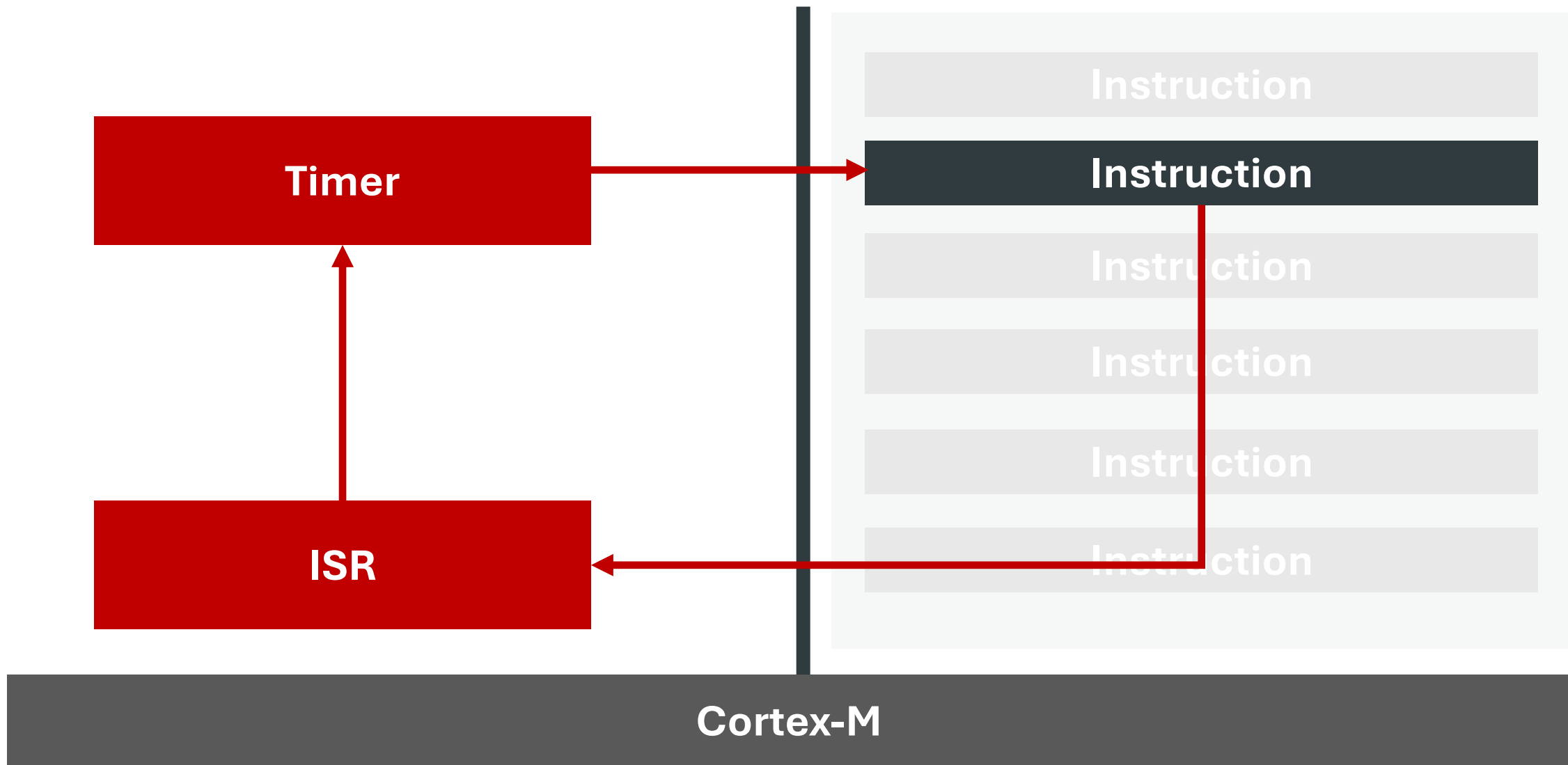


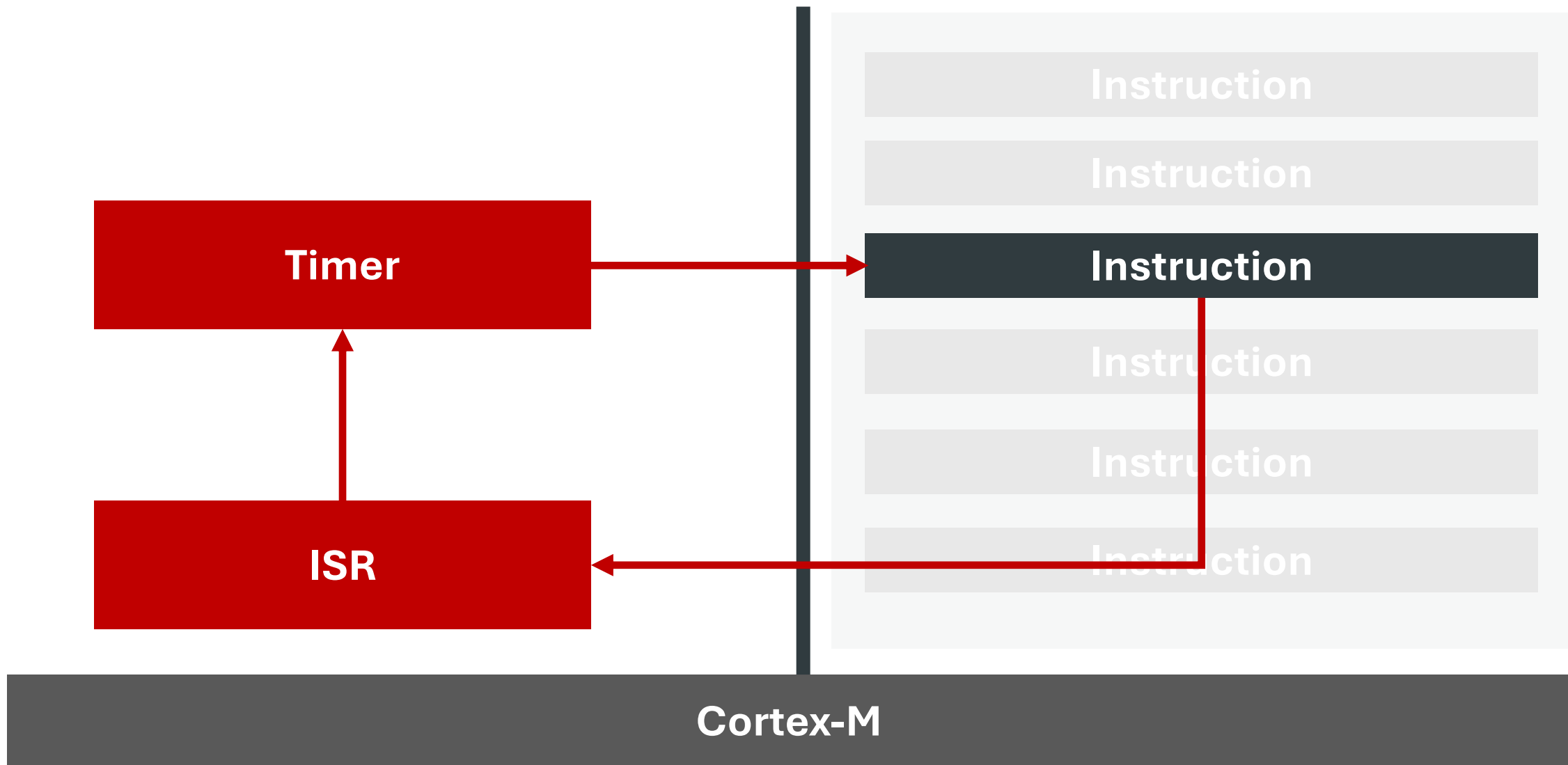


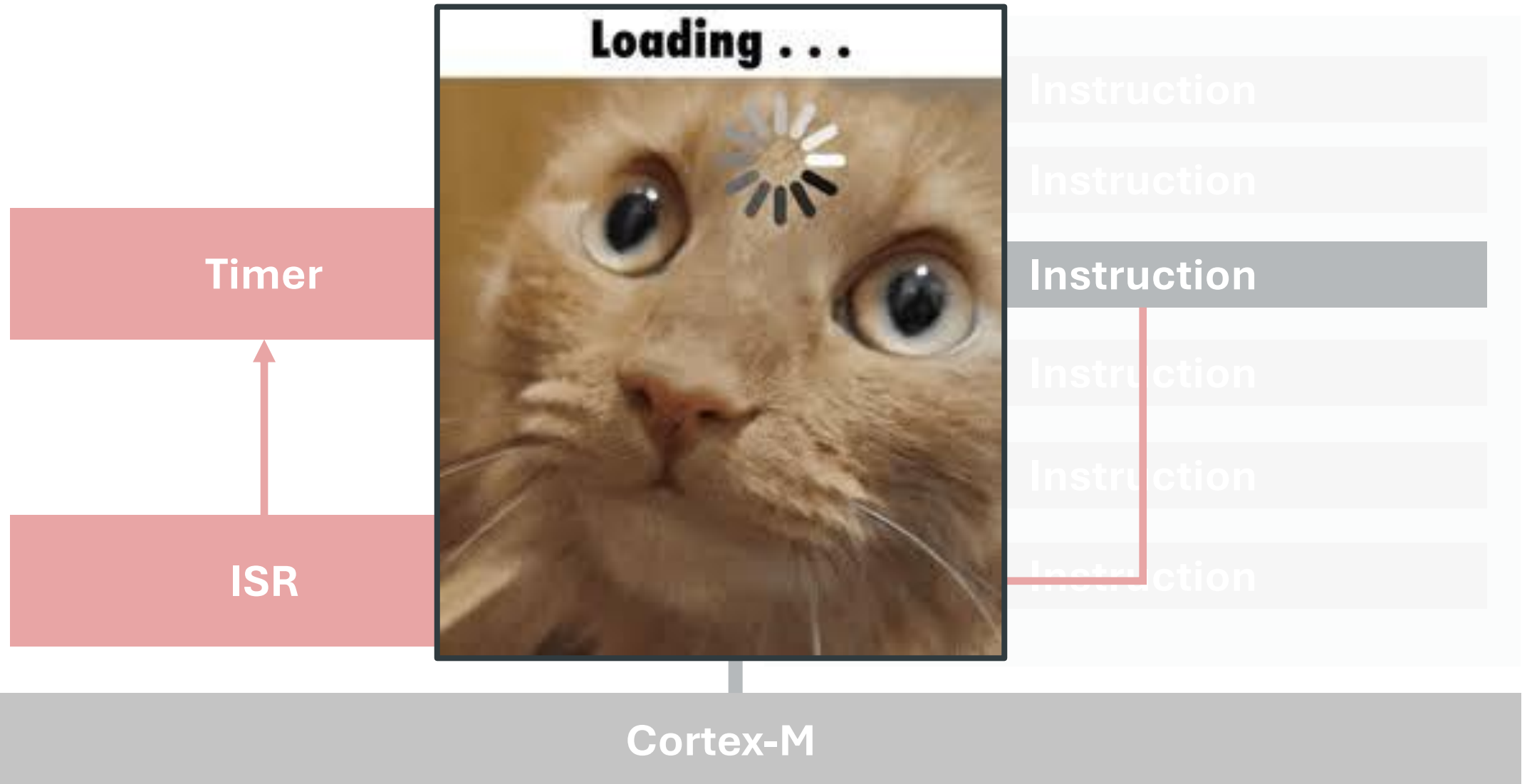










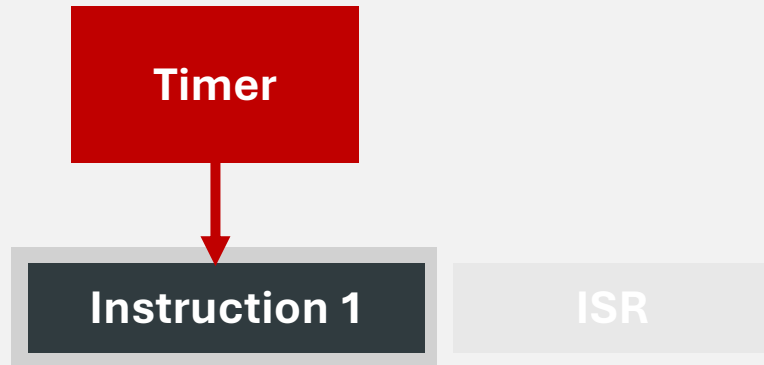


Single-Step Challenges On Cortex-M

Normal Instructions

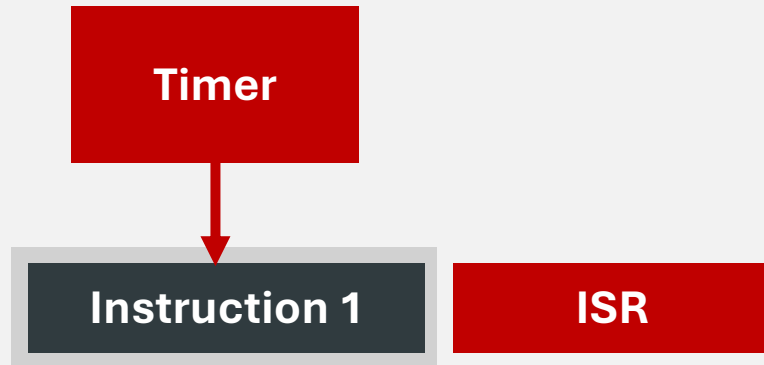
Interruptible Instructions

Normal Instructions



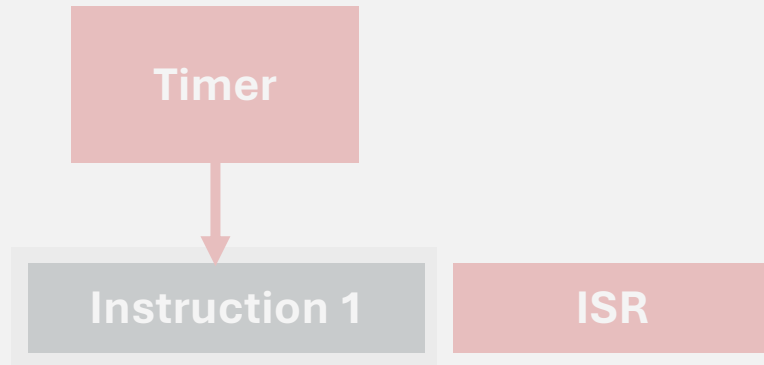
Interruptible Instructions

Normal Instructions

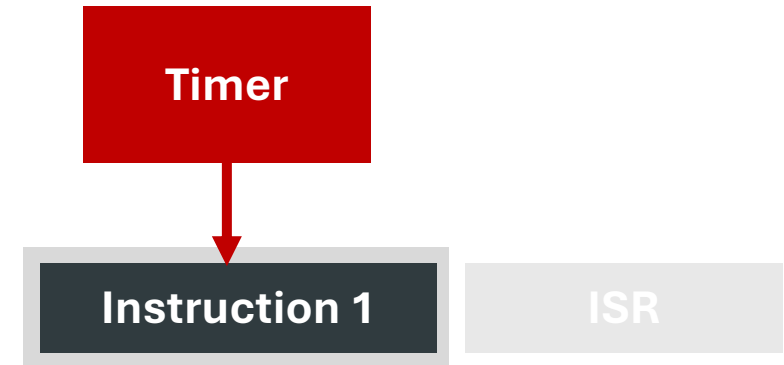


Interruptible Instructions

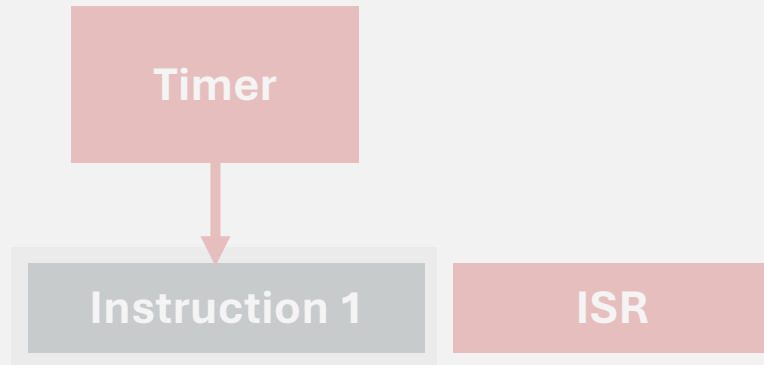
Normal Instructions



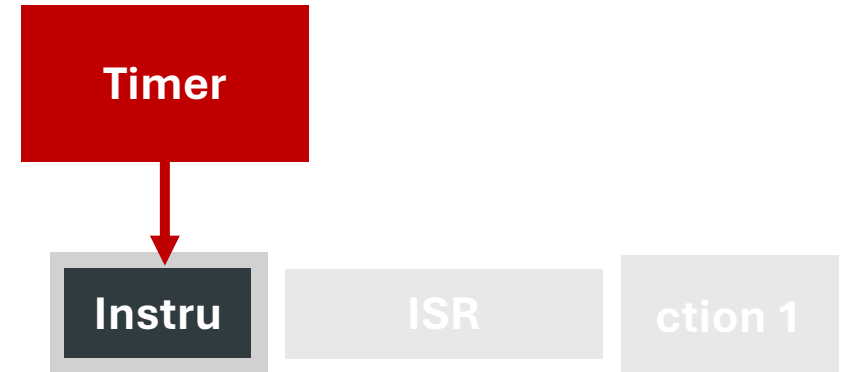
Interruptible Instructions



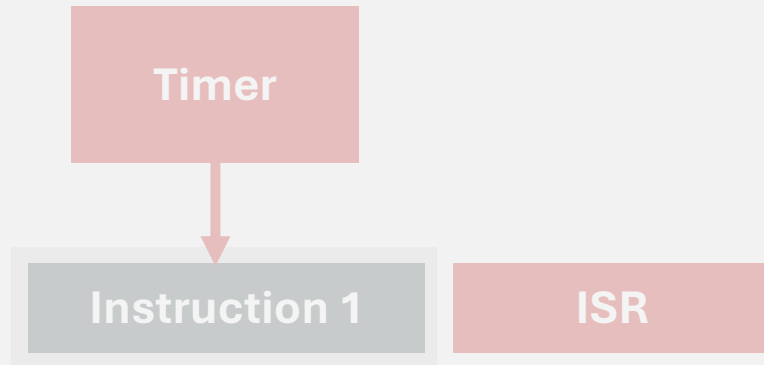
Normal Instructions



Interruptible Instructions



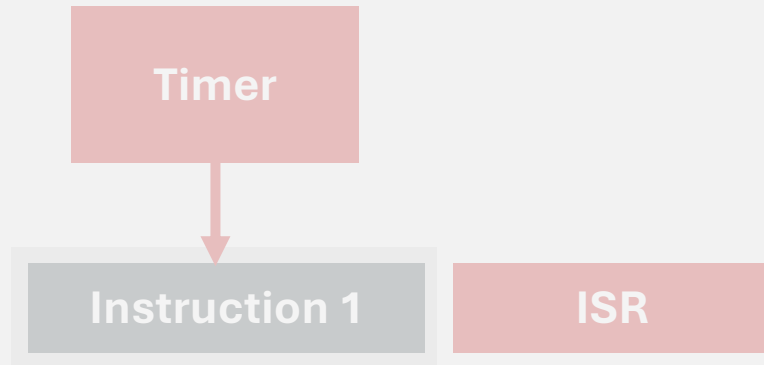
Normal Instructions



Interruptible Instructions



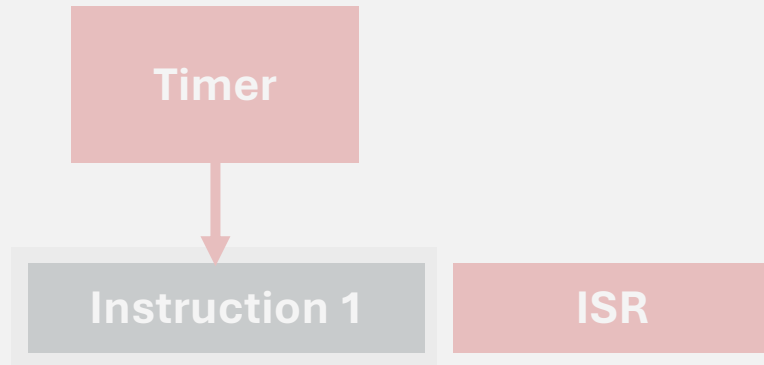
Normal Instructions



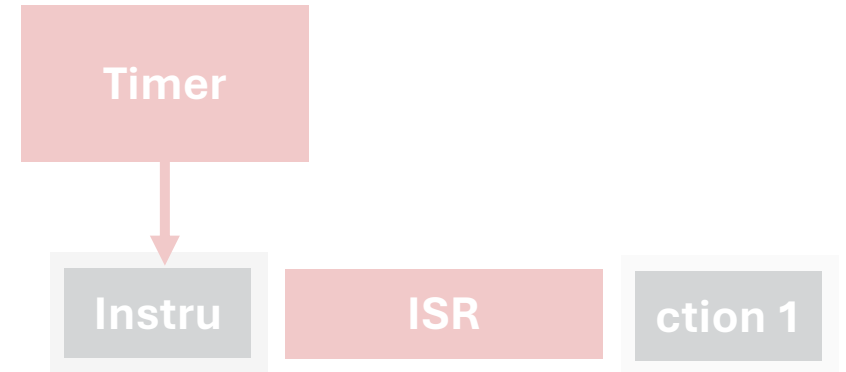
Interruptible Instructions



Normal Instructions



Interruptible Instructions



Heterogenous Interrupt Behavior!!

Cortex-M

Instruction Classes

Cortex-M Instruction Classes

Atomic

Resumable

Restartable

Cortex-M Instruction Classes

Atomic

Resumable

Restartable

Cortex-M Instruction Classes

Atomic

Resumable

Restartable

Cortex-M Instruction Classes

Atomic

Resumable

Restartable

Cortex-M Instruction Classes

Atomic



Resumable

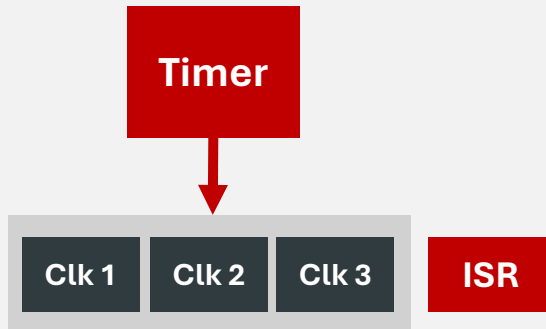
Restartable



Interrupts can
arrive anywhere.

Cortex-M Instruction Classes

Atomic



Resumable

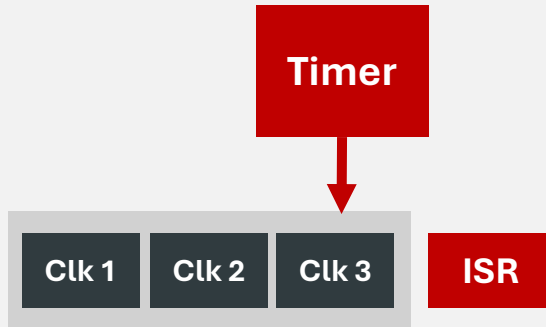
Restartable



Interrupts can
arrive anywhere.

Cortex-M Instruction Classes

Atomic



Resumable

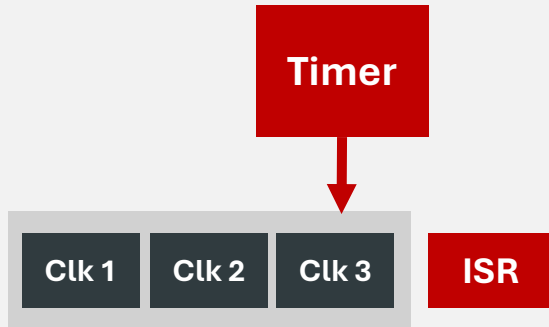
Restartable



Interrupts can
arrive anywhere.

Cortex-M Instruction Classes

Atomic



Resumable

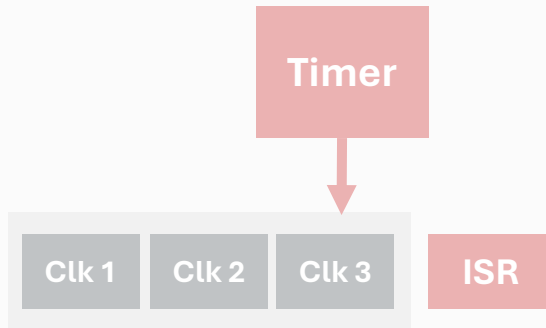
Restartable



Interrupts can
arrive **anywhere**.

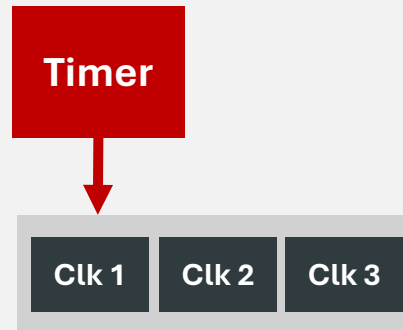
Cortex-M Instruction Classes

Atomic



🎯 Interrupts can arrive **anywhere**.

Resumable

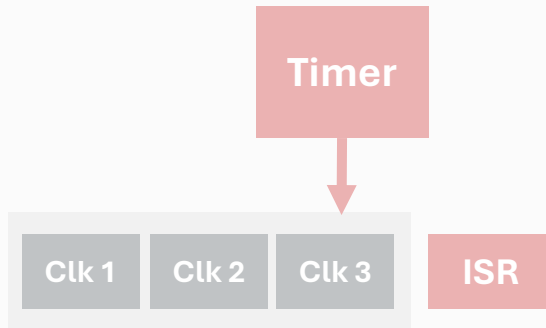


🎯 Interrupts require extra clock cycle.

Restartable

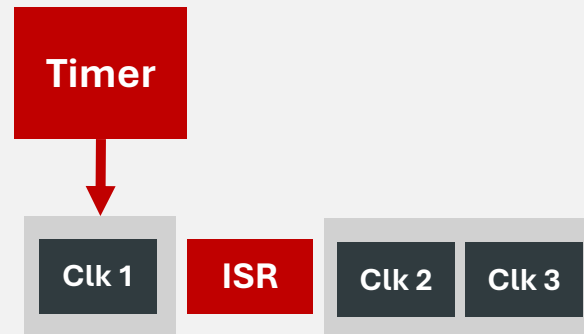
Cortex-M Instruction Classes

Atomic



🎯 Interrupts can arrive **anywhere**.

Resumable

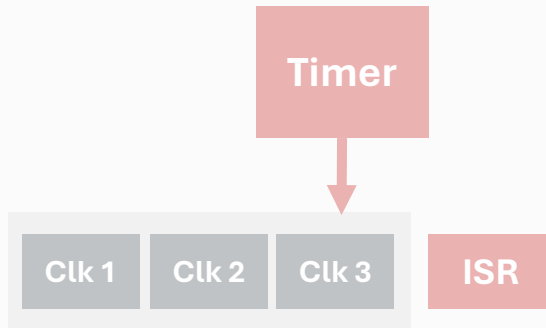


🎯 Interrupts require extra clock cycle.

Restartable

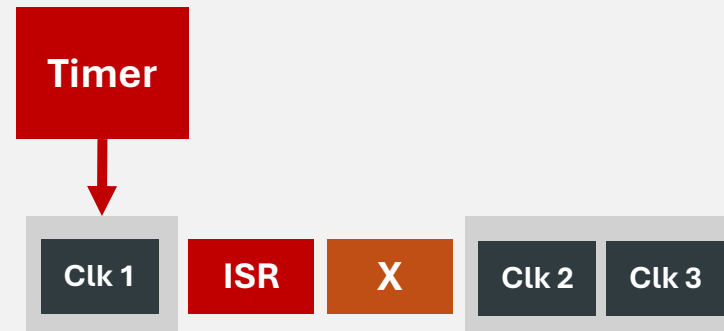
Cortex-M Instruction Classes

Atomic



🎯 Interrupts can arrive **anywhere**.

Resumable

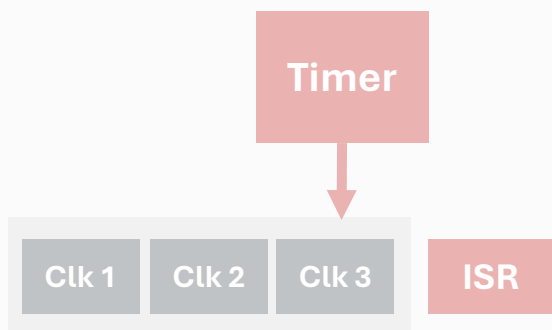


🎯 Interrupts require extra clock cycle.

Restartable

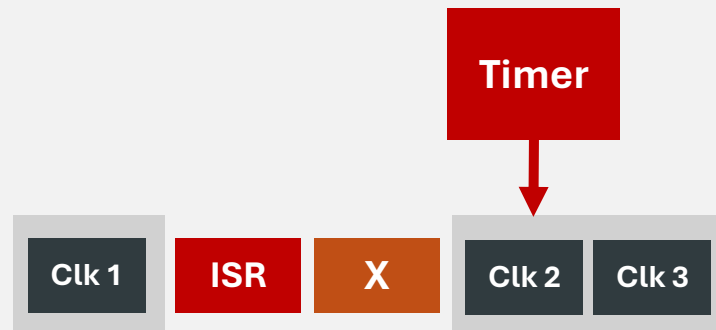
Cortex-M Instruction Classes

Atomic



Interrupts can arrive **anywhere**.

Resumable

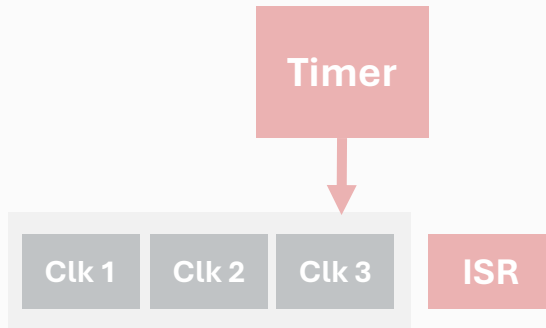


Interrupts require extra clock cycle.

Restartable

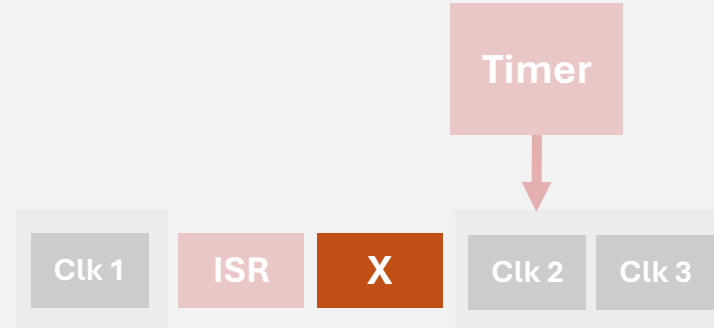
Cortex-M Instruction Classes

Atomic



🎯 Interrupts can arrive **anywhere**.

Resumable

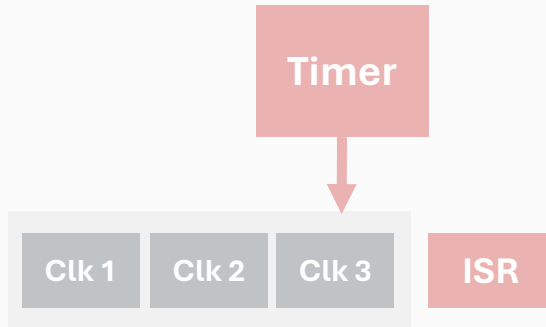


🎯 Interrupts require extra clock cycle.

Restartable

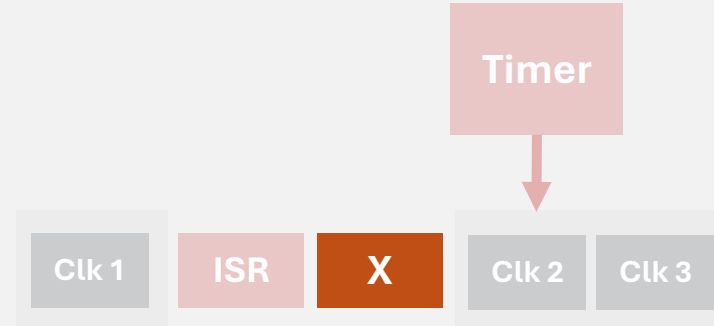
Cortex-M Instruction Classes

Atomic



Interrupts can arrive **anywhere**.

Resumable

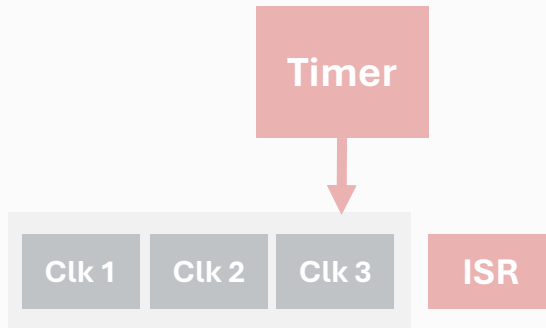


Interrupts require **extra clock cycle**.

Restartable

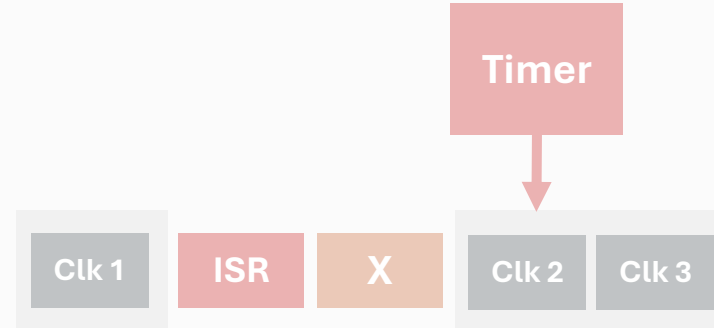
Cortex-M Instruction Classes

Atomic



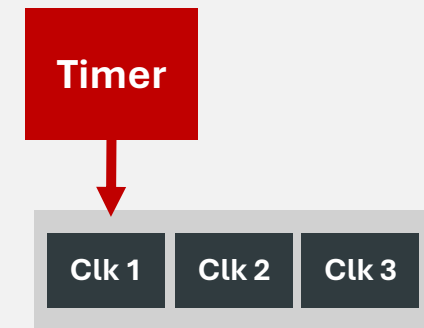
🎯 Interrupts can arrive **anywhere**.

Resumable



🎯 Interrupts require **extra clock cycle**.

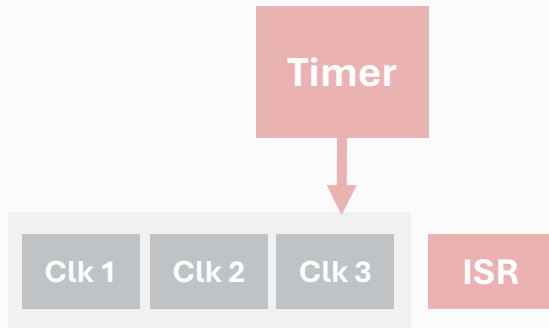
Restartable



🎯 Interrupt must arrive in the last clock cycle.

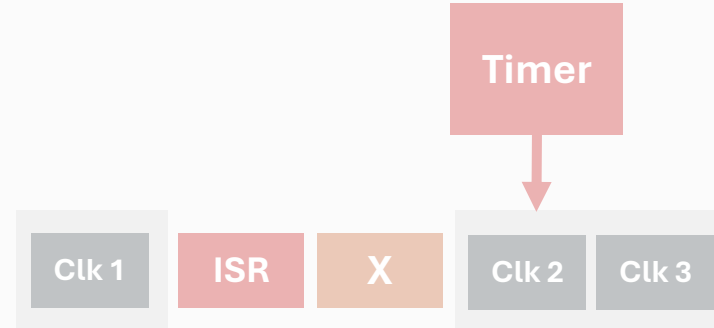
Cortex-M Instruction Classes

Atomic



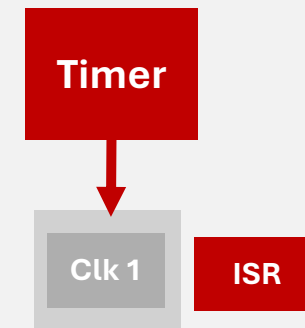
🎯 Interrupts can arrive **anywhere**.

Resumable



🎯 Interrupts require **extra clock cycle**.

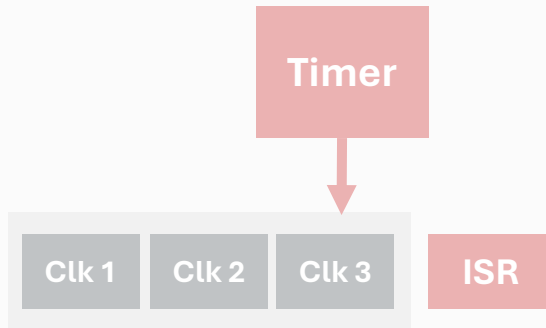
Restartable



🎯 Interrupt must arrive in the last clock cycle.

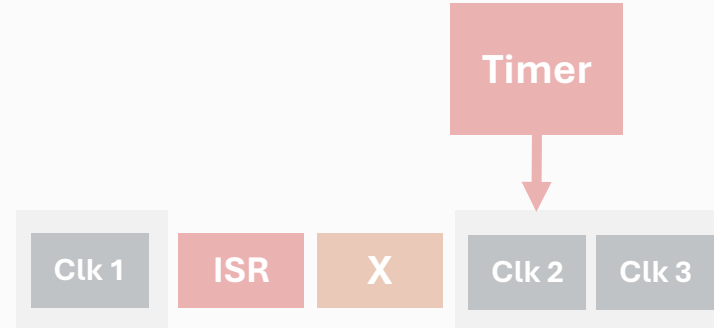
Cortex-M Instruction Classes

Atomic



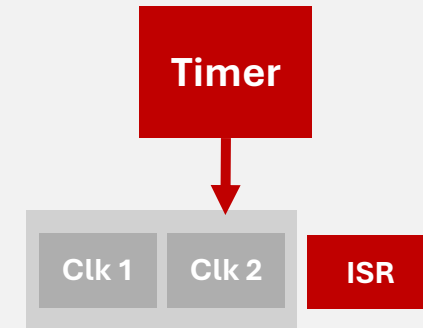
🎯 Interrupts can arrive **anywhere**.

Resumable



🎯 Interrupts require **extra clock cycle**.

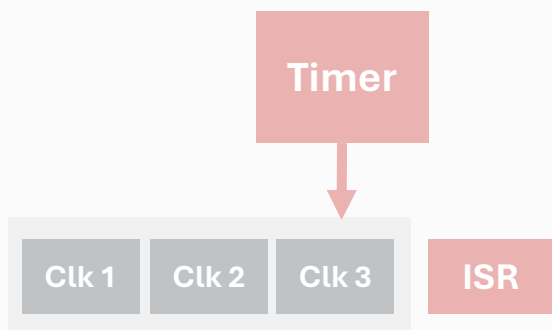
Restartable



🎯 Interrupt must arrive in the last clock cycle.

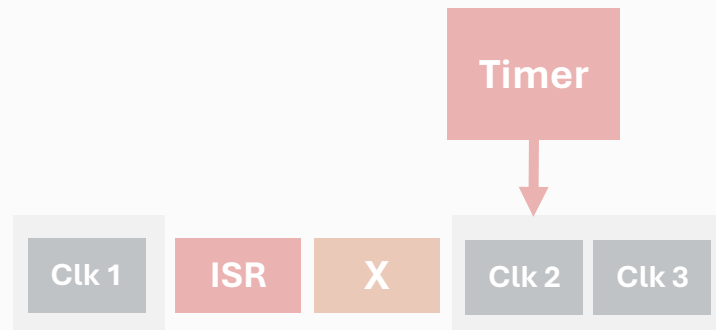
Cortex-M Instruction Classes

Atomic



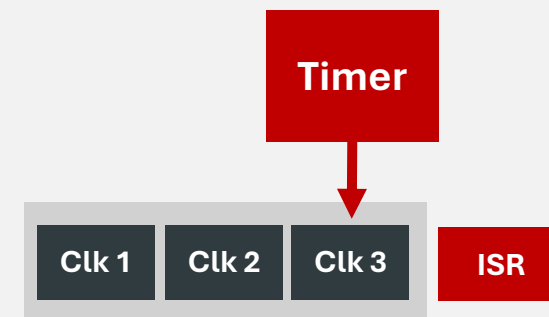
🎯 Interrupts can arrive **anywhere**.

Resumable



🎯 Interrupts require **extra clock cycle**.

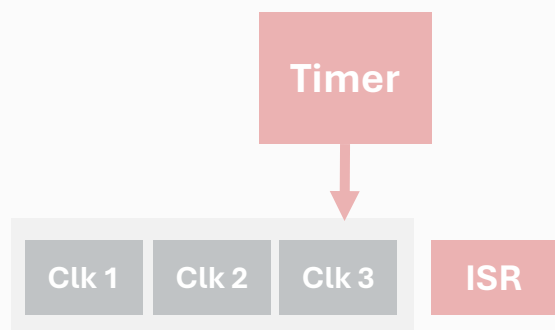
Restartable



🎯 Interrupt must arrive in the **last clock cycle**.

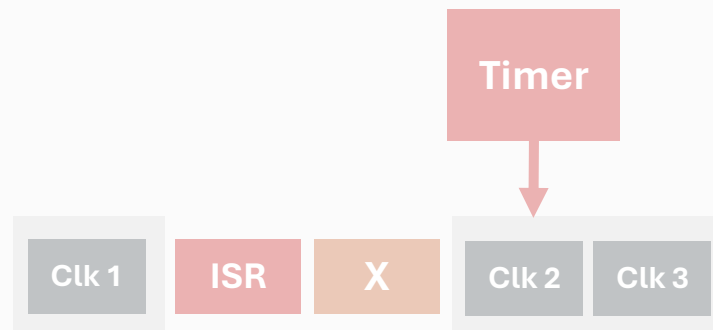
Cortex-M Instruction Classes

Atomic



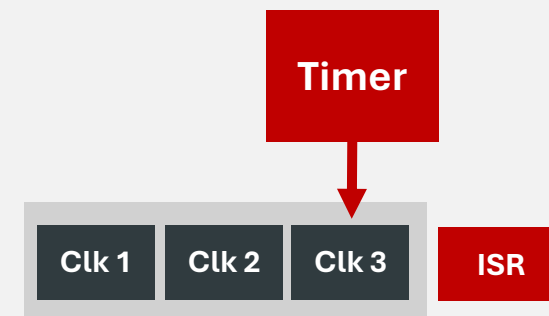
🎯 Interrupts can arrive **anywhere**.

Resumable



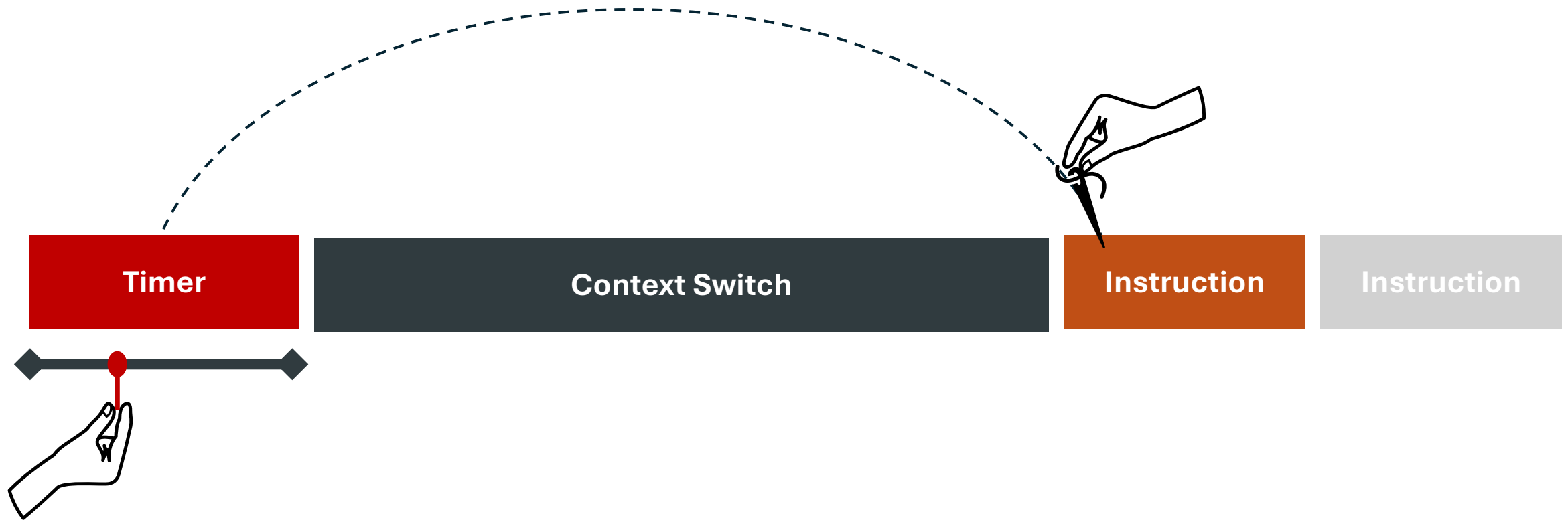
🎯 Interrupts require **extra clock cycle**.

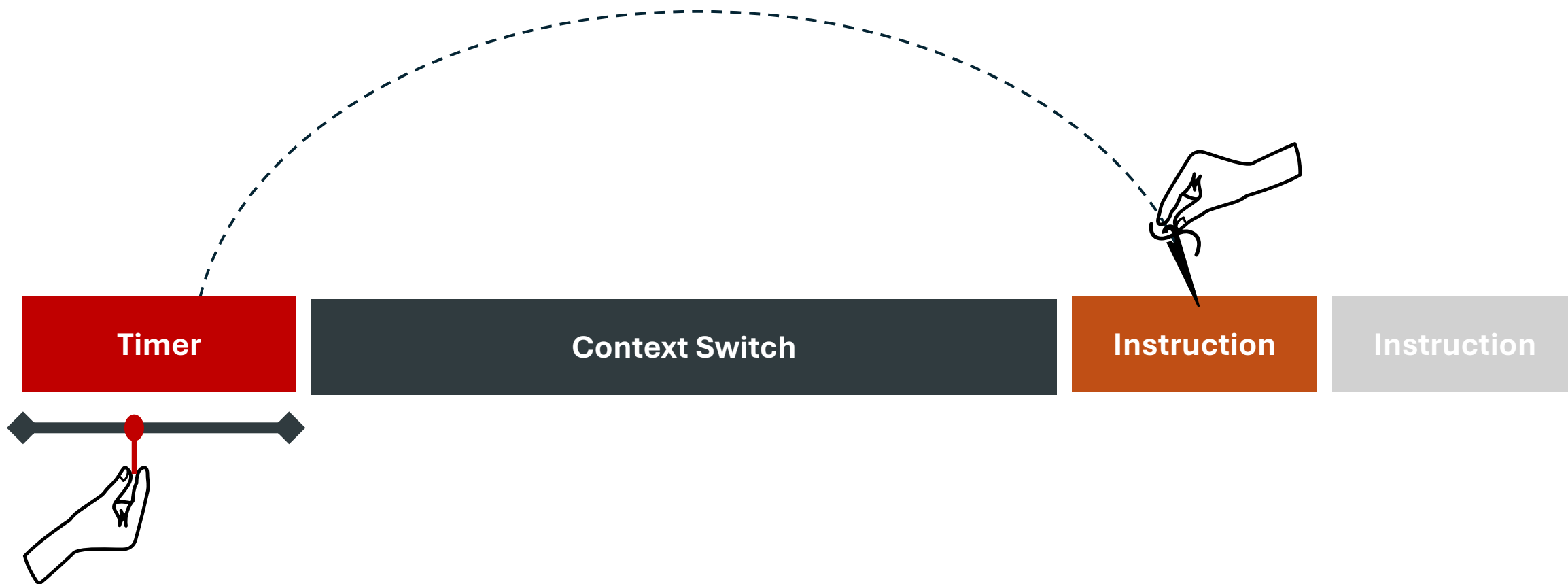
Restartable

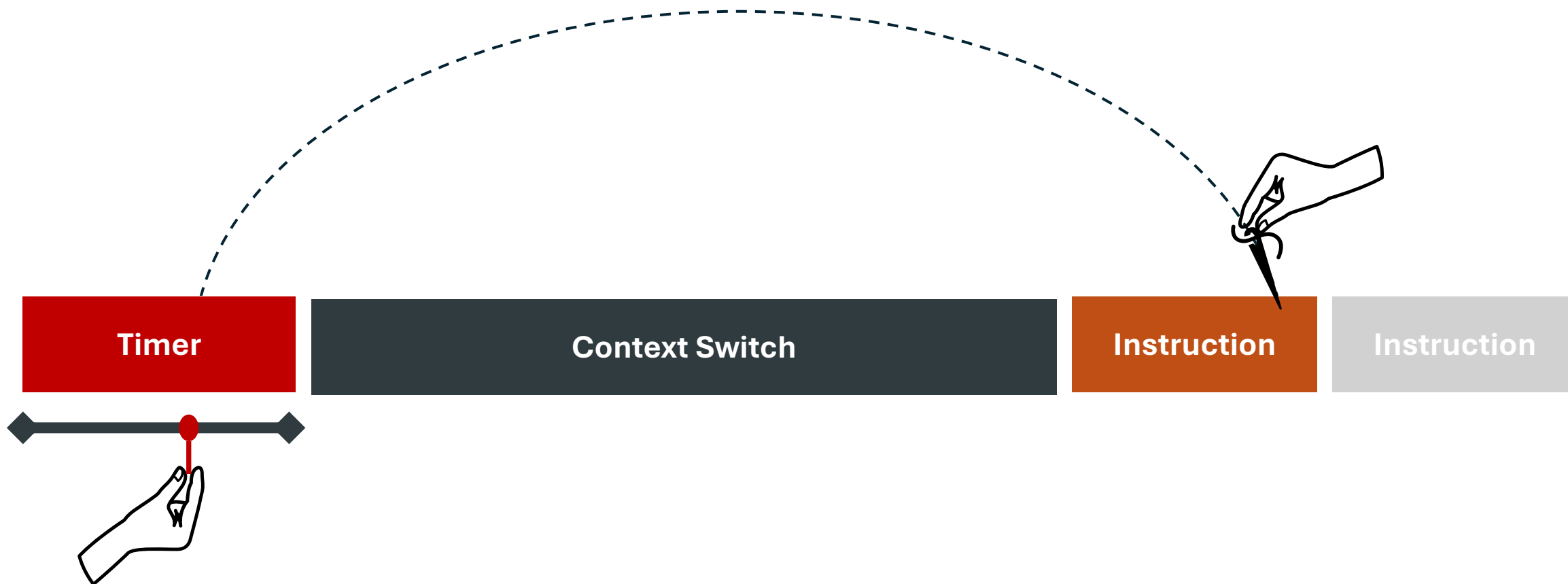


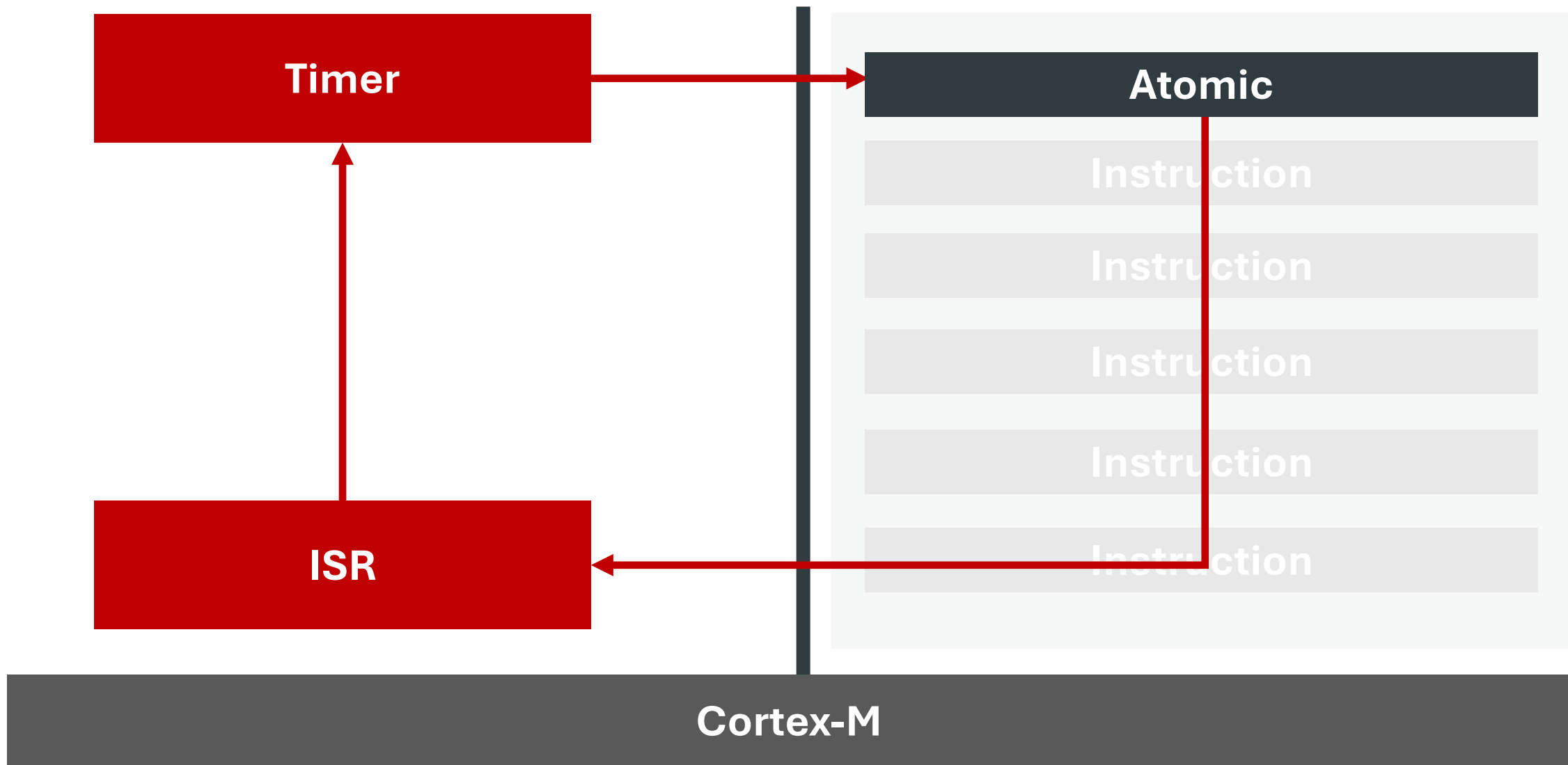
🎯 Interrupt must arrive in the **last clock cycle**.

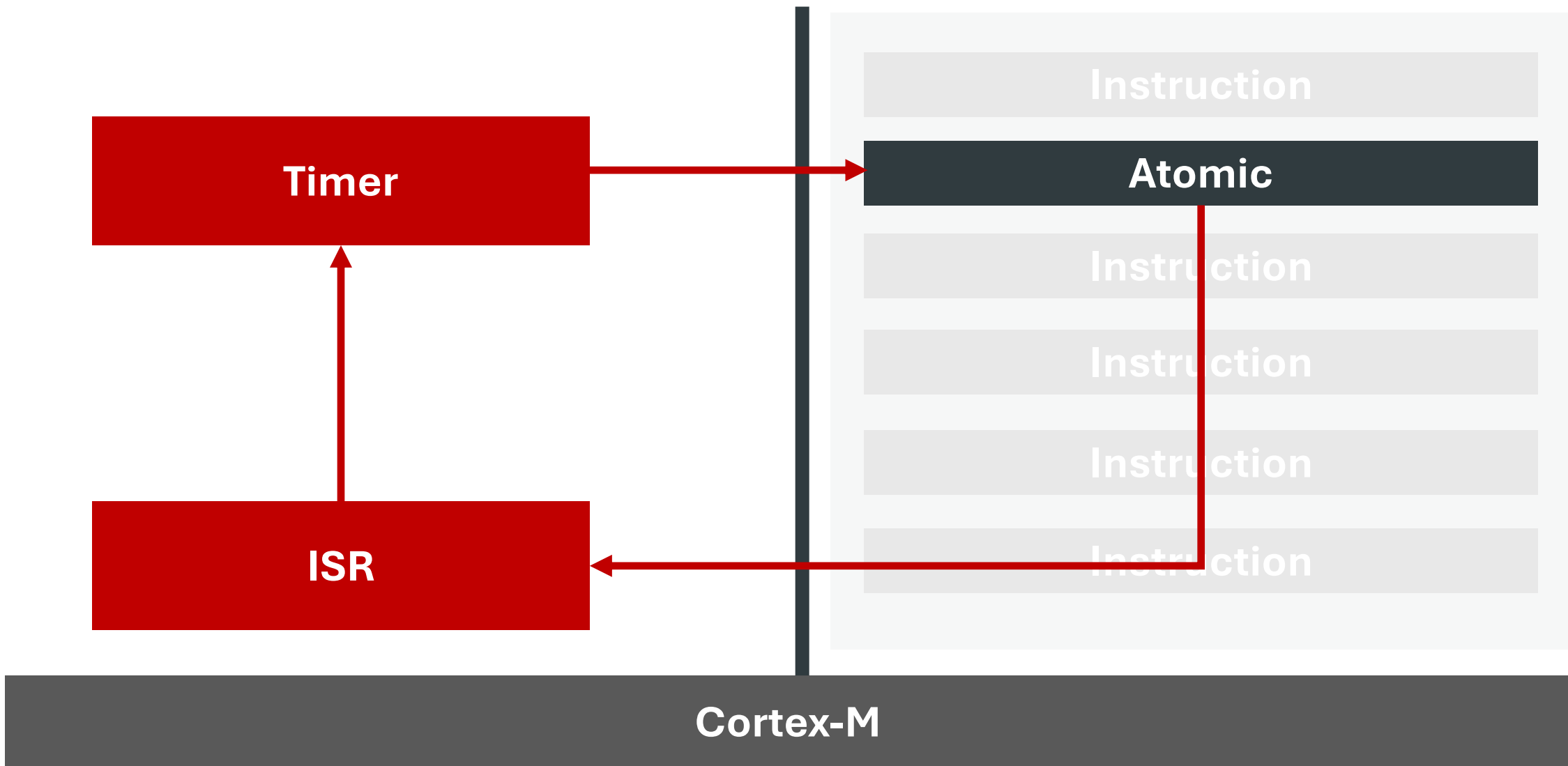
Dynamic Single-Step Algorithm

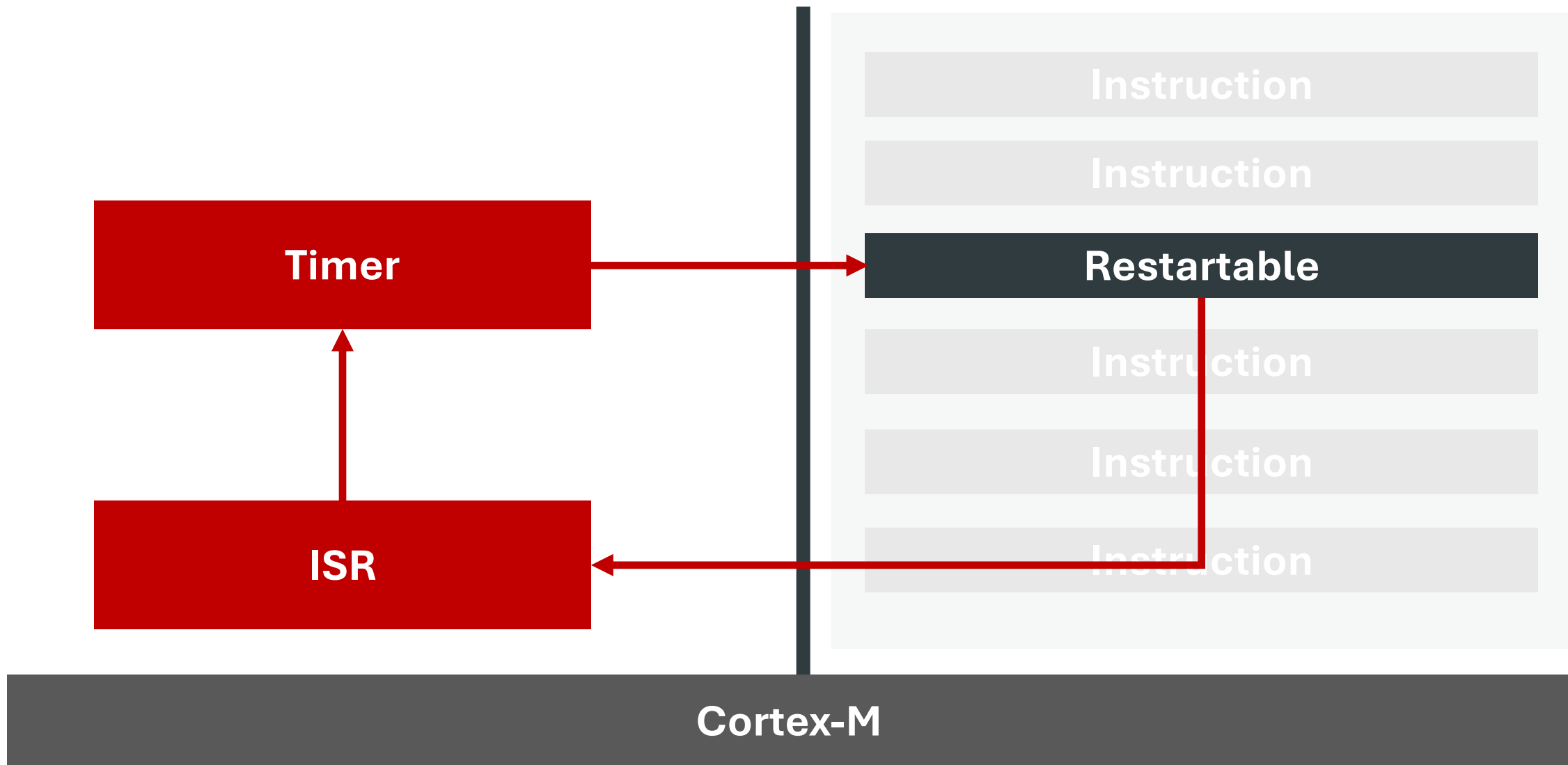


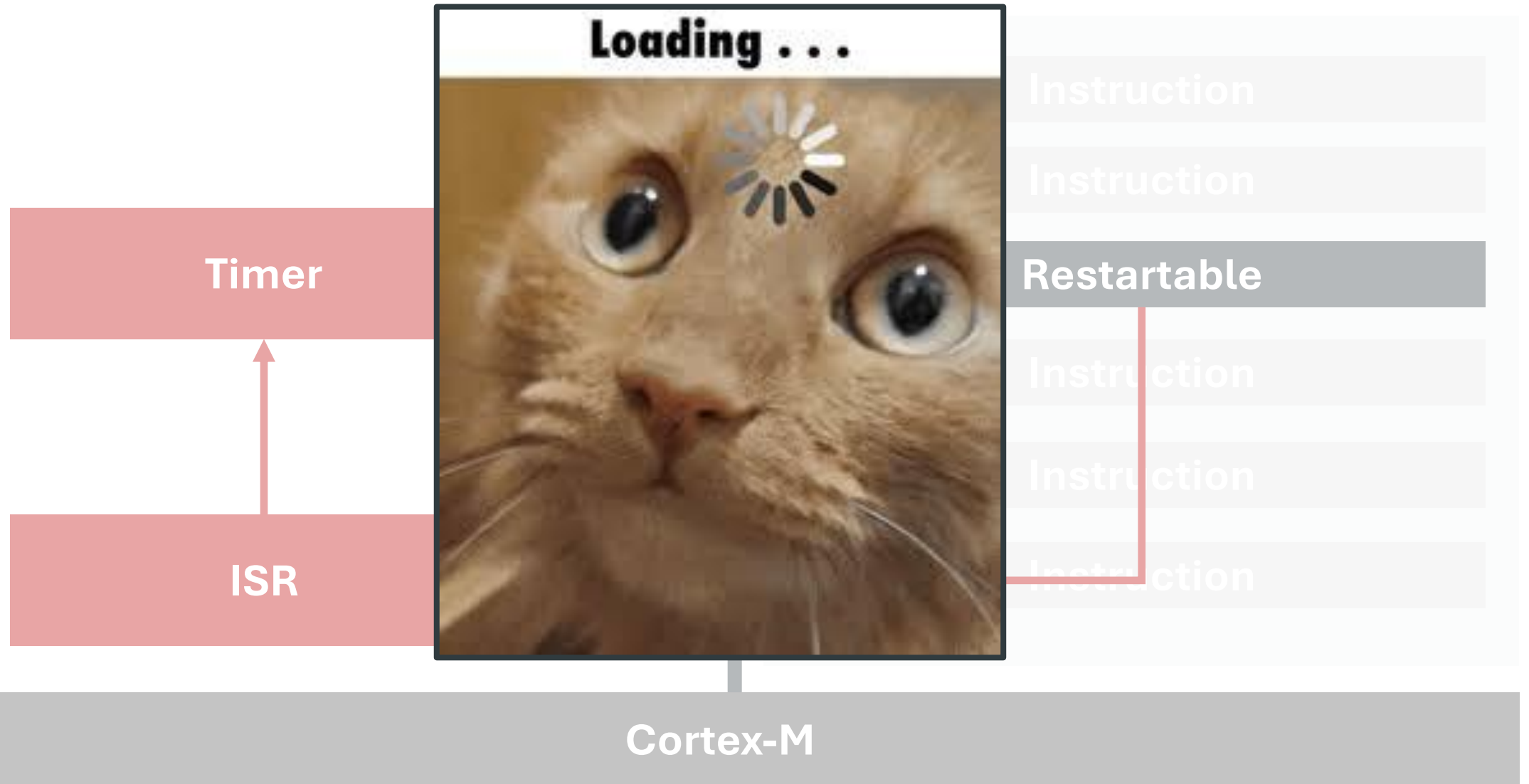


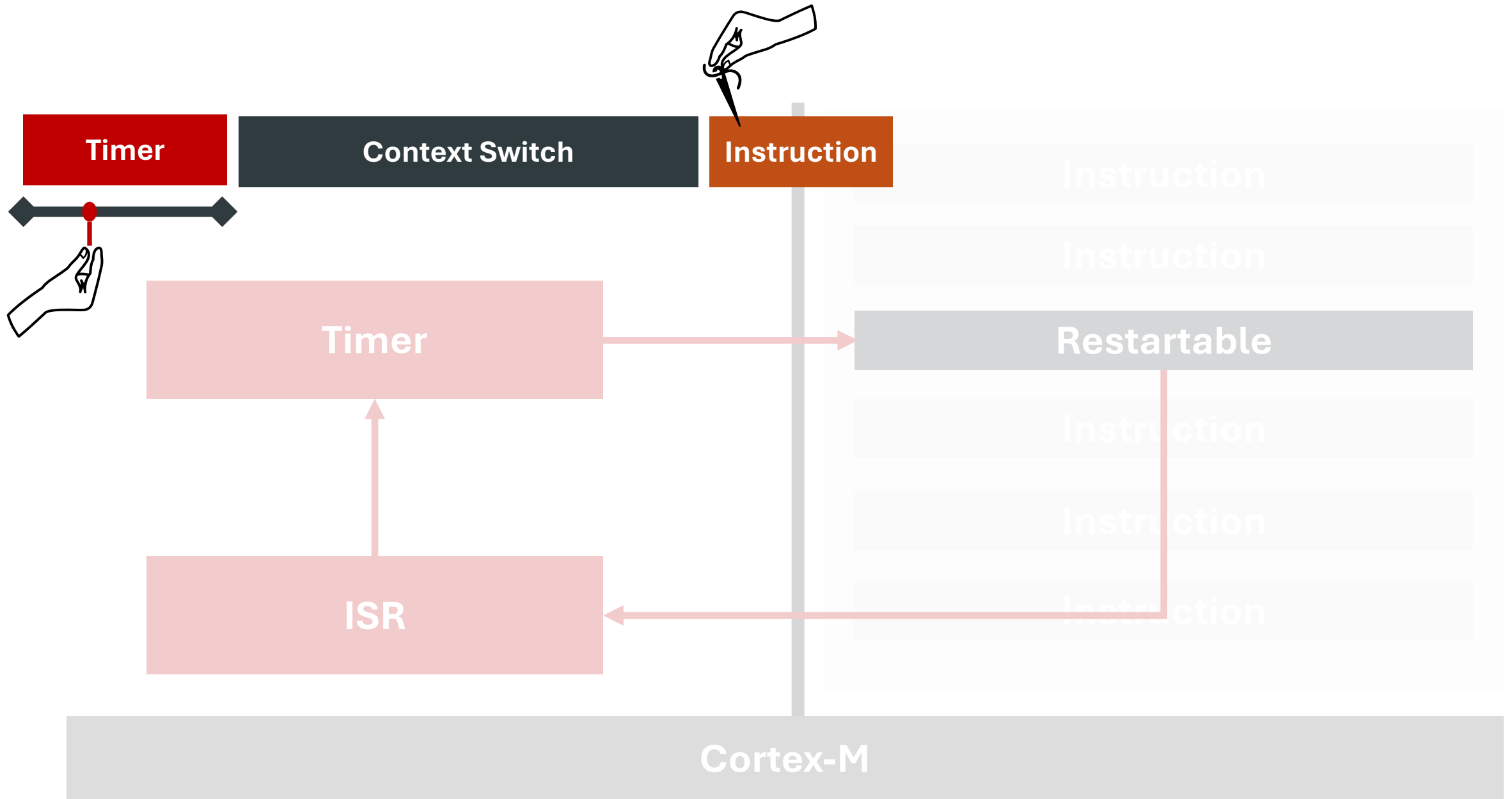


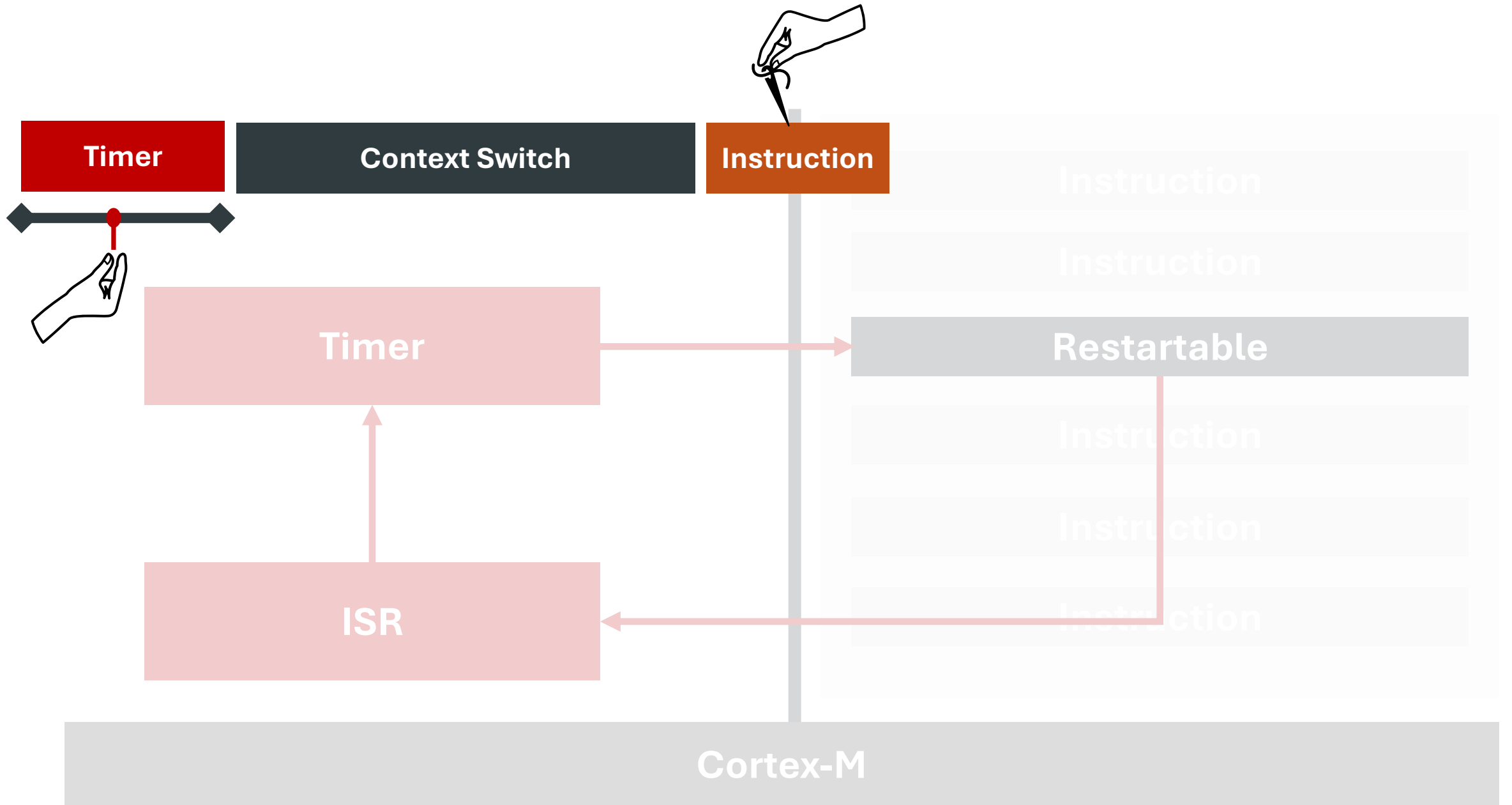


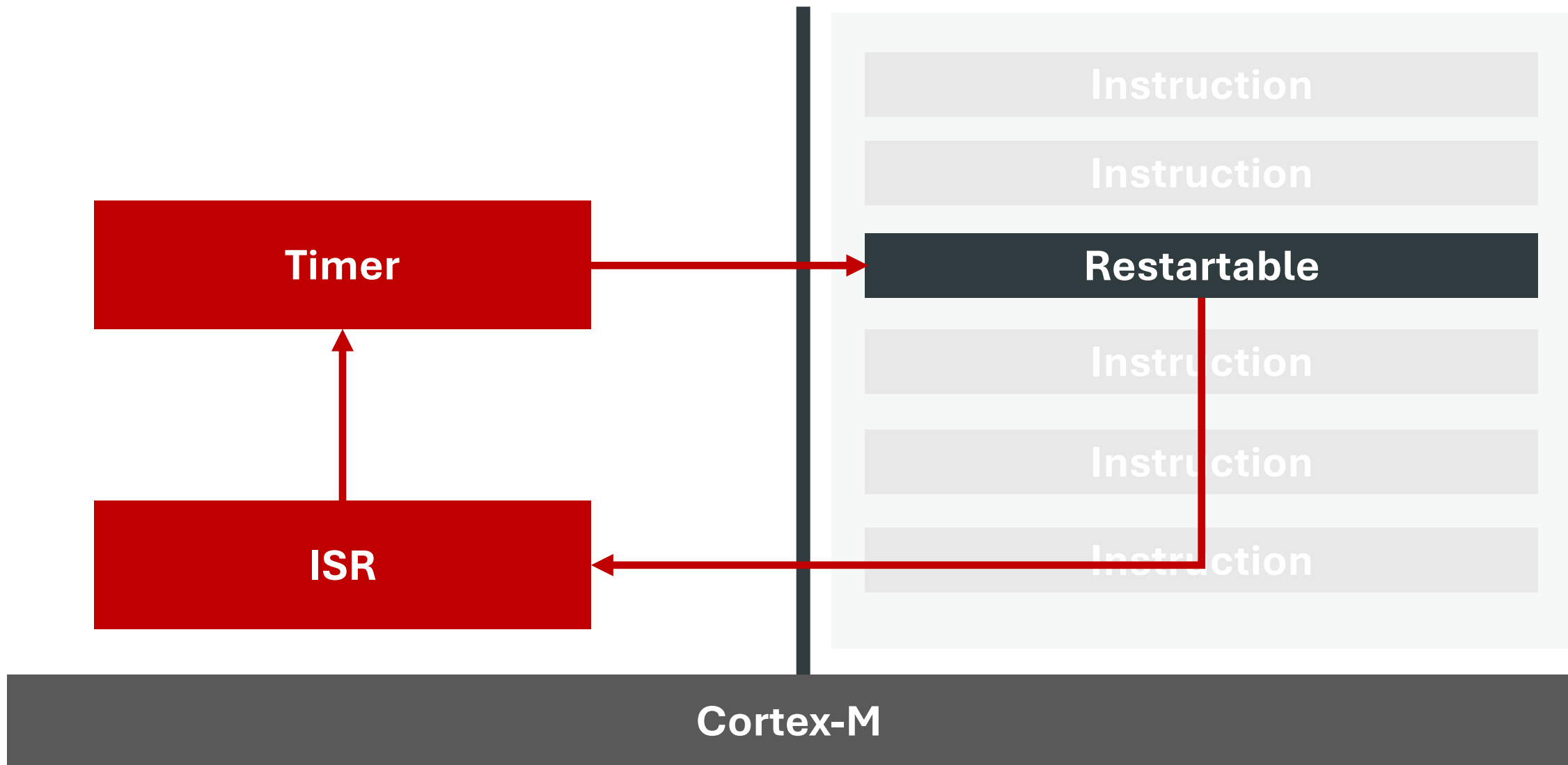


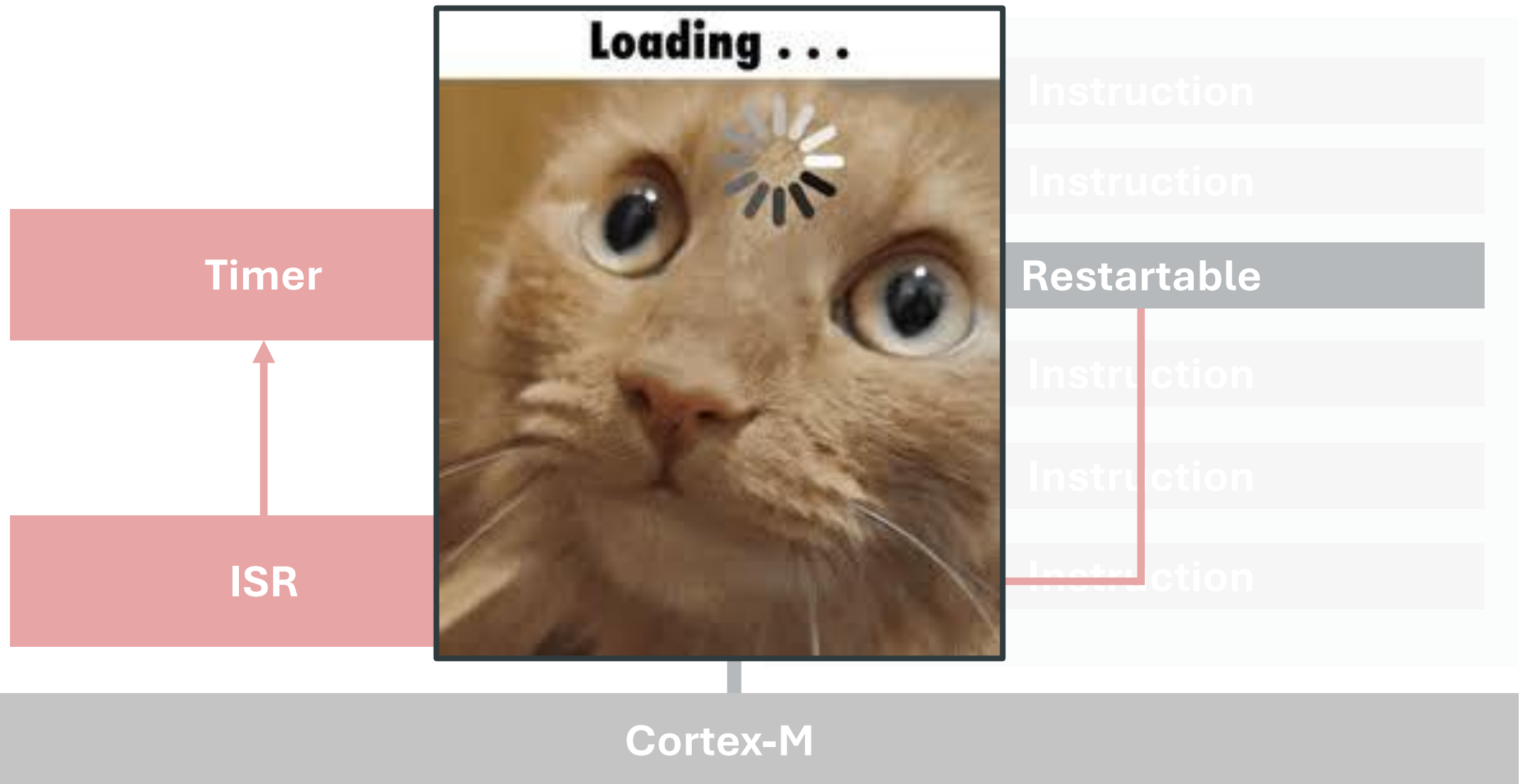


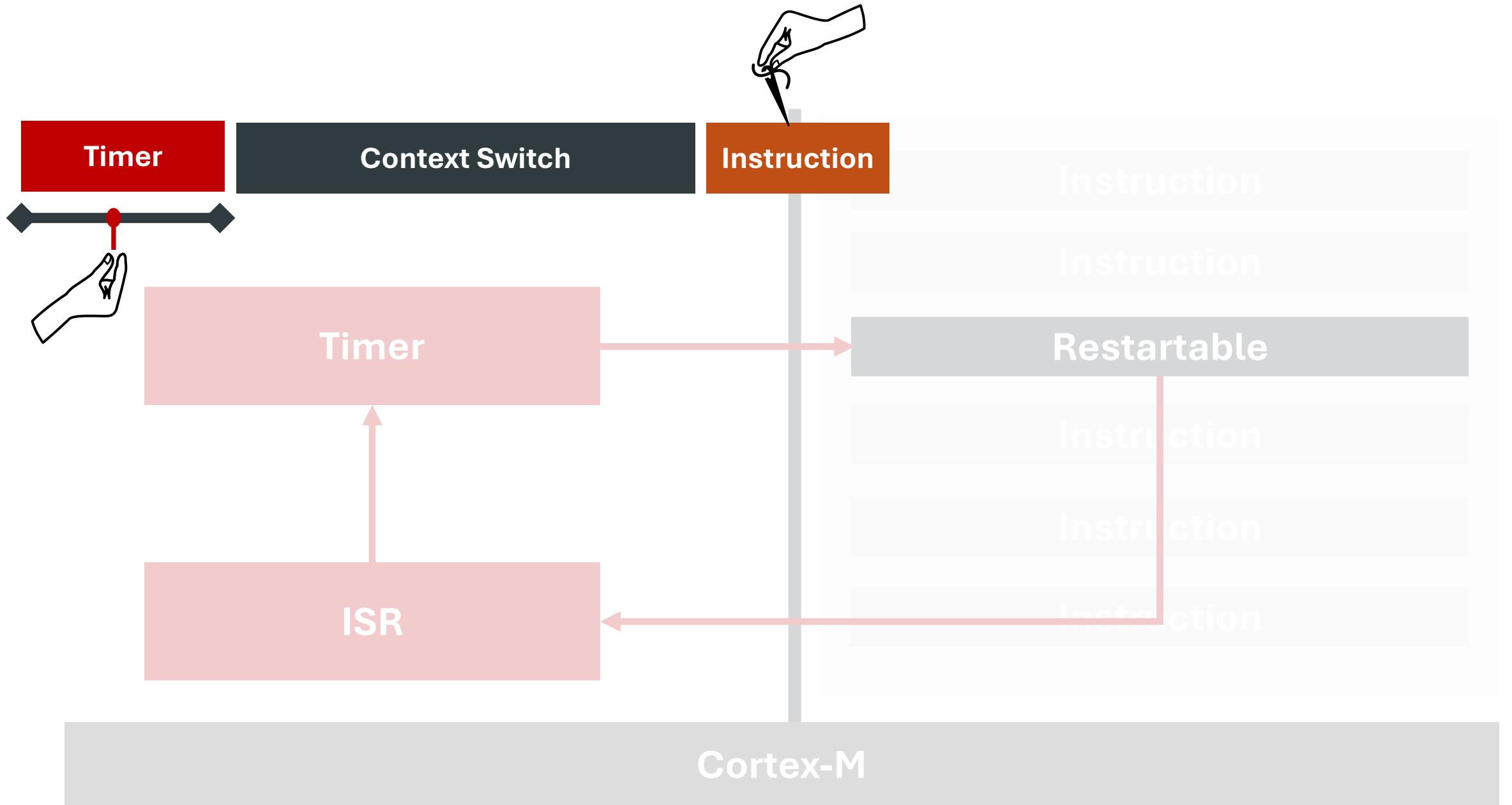


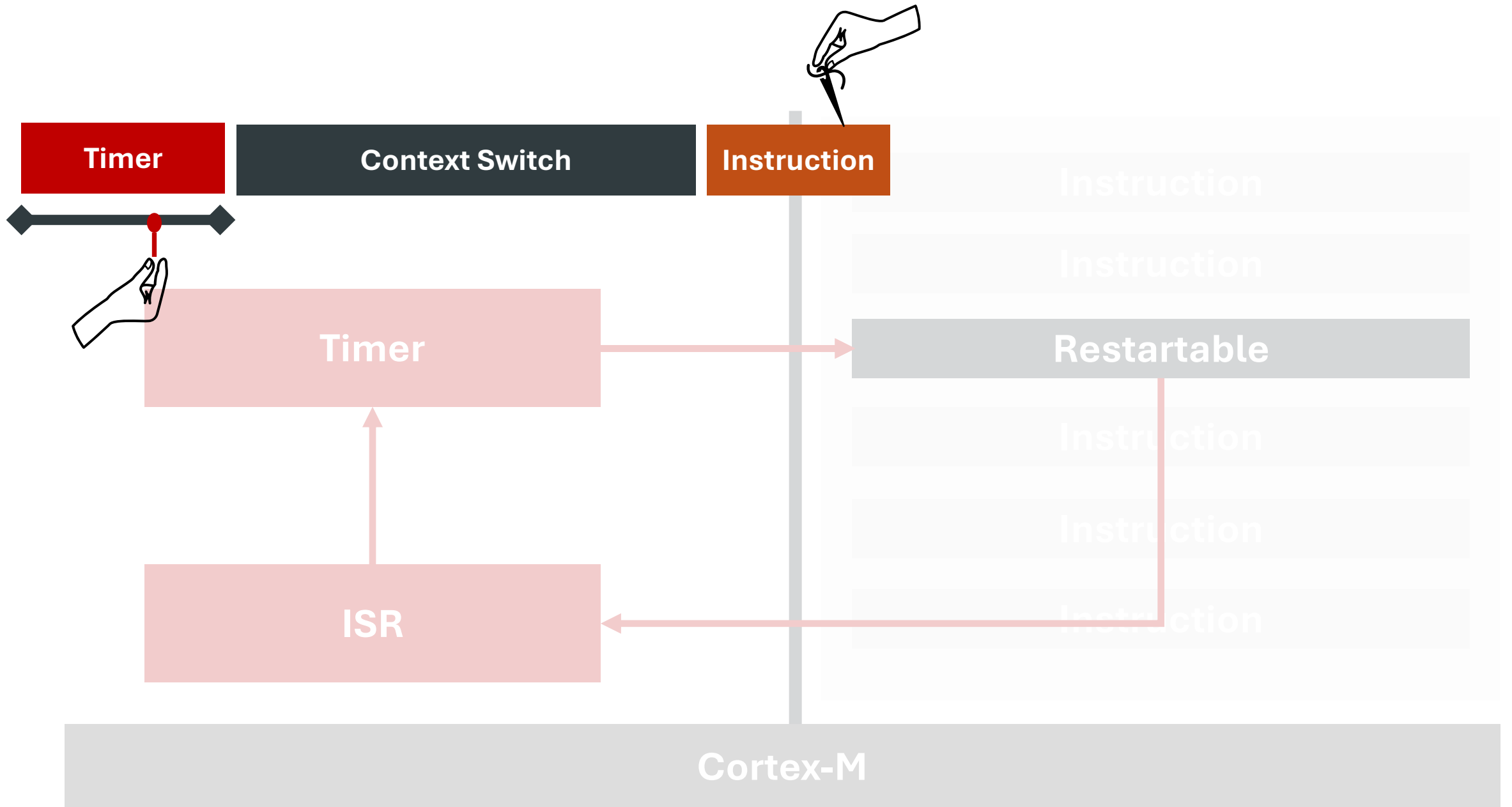


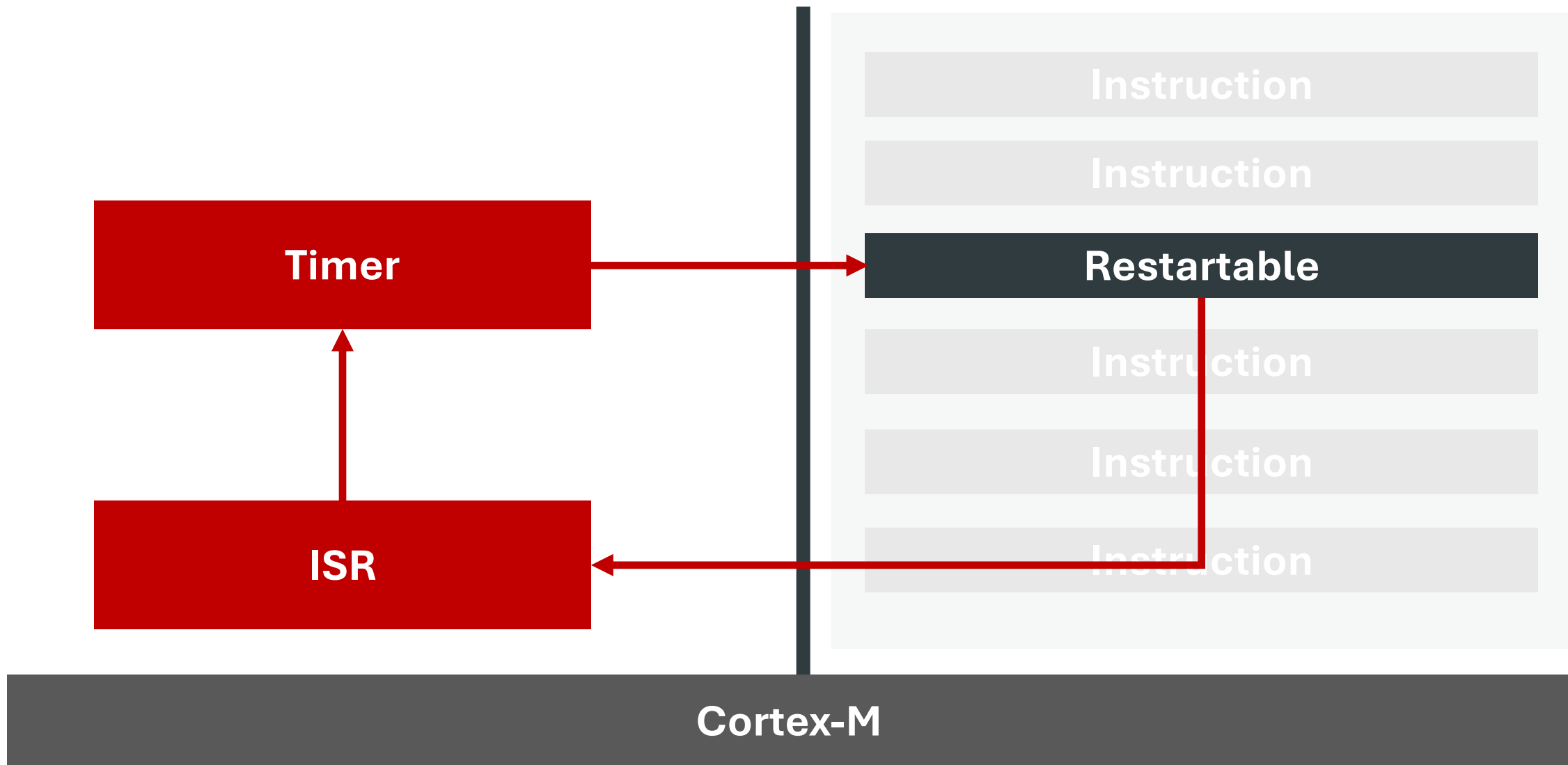


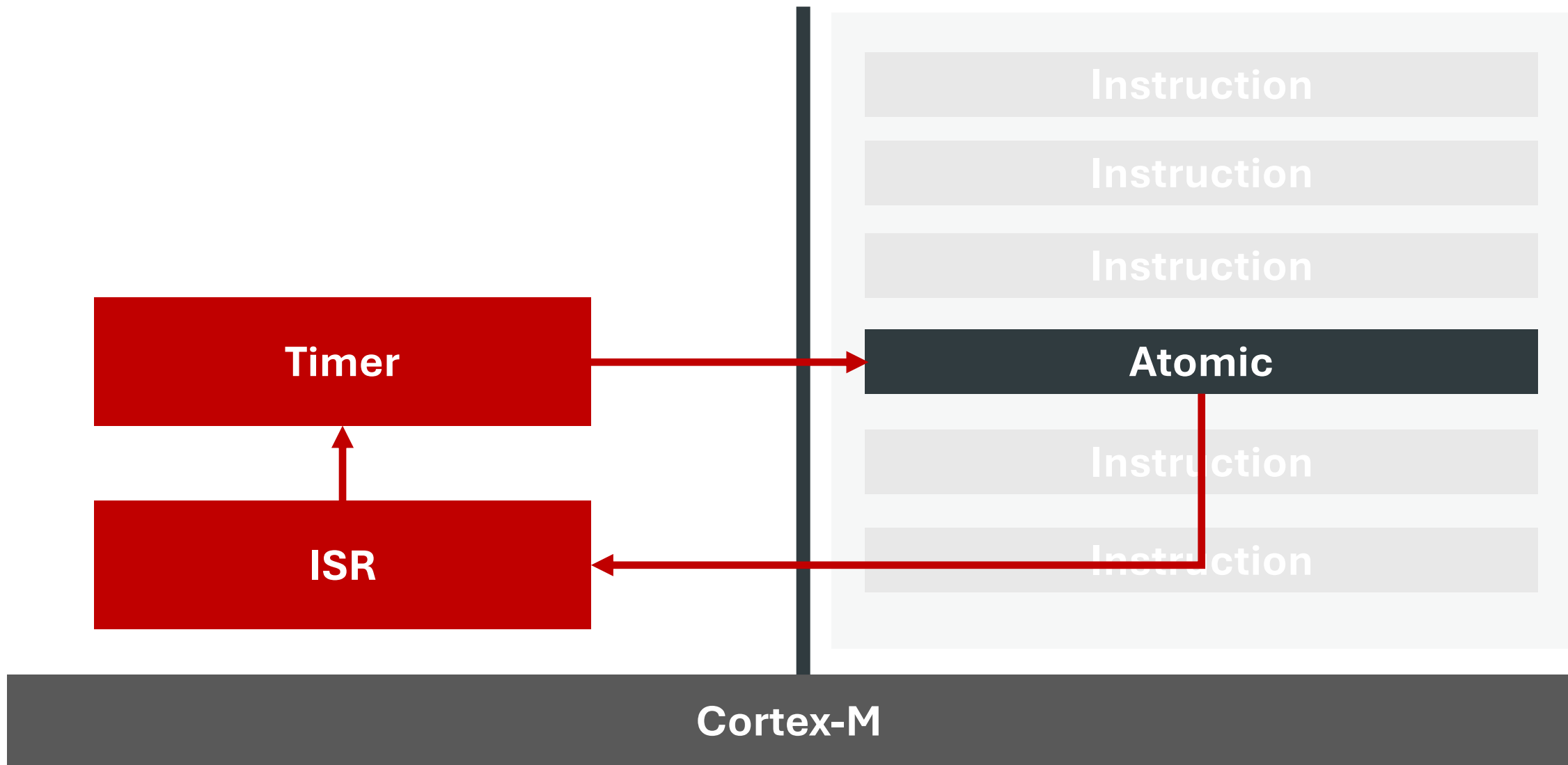


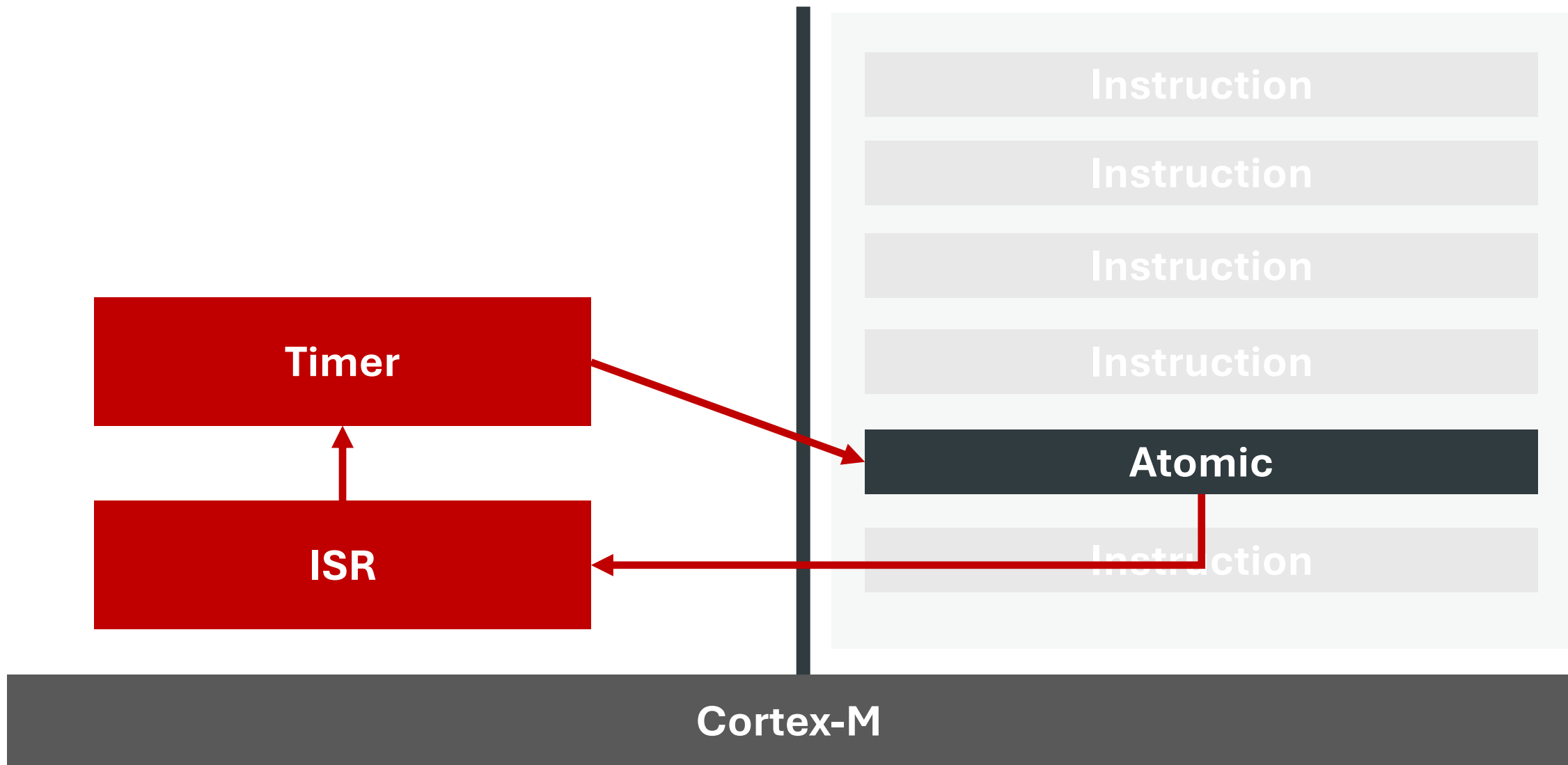


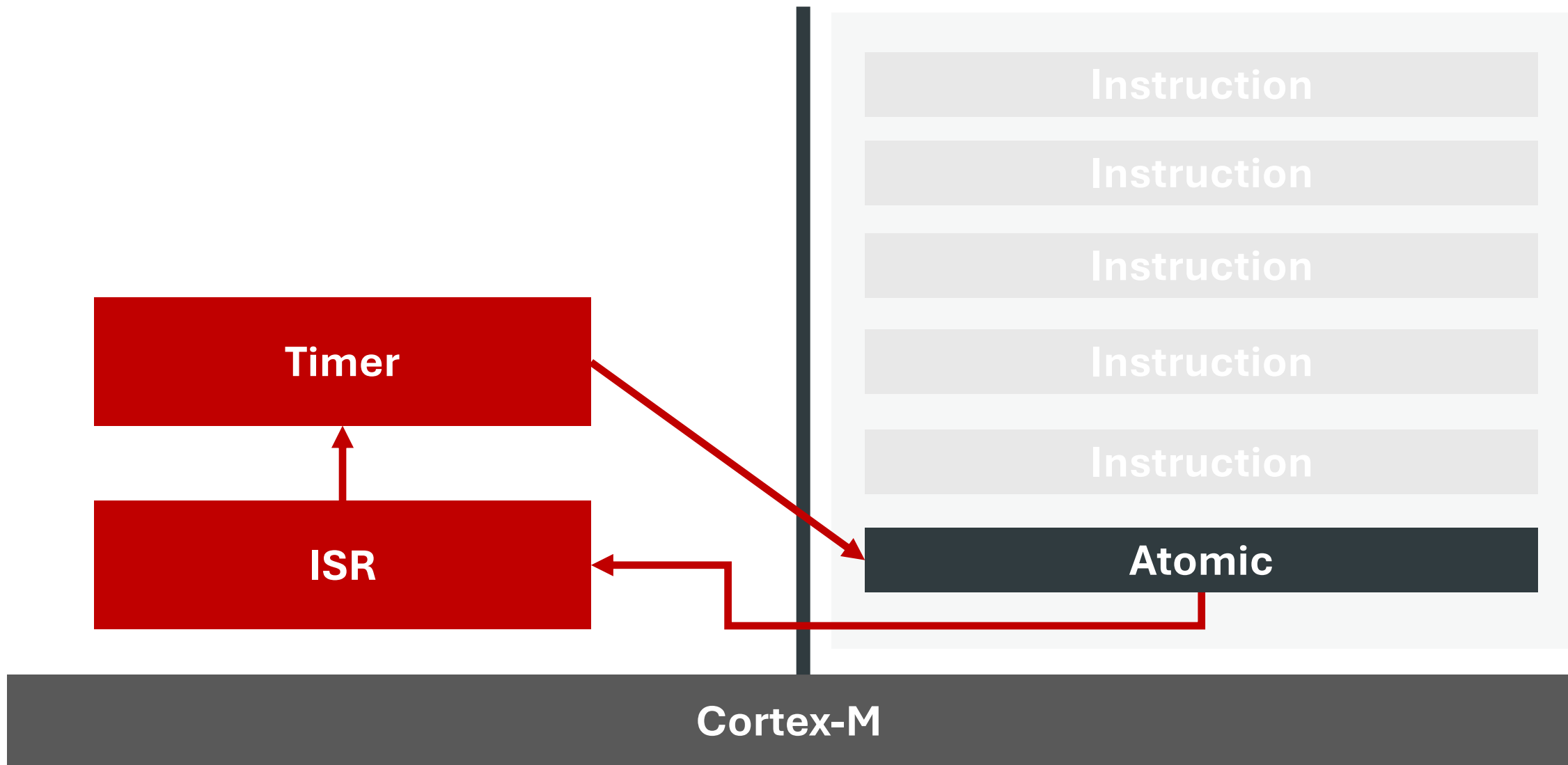












But....



**How can we detect
that we are making
progress?**

Progress Detection on Cortex-M

**Detect Zero-Steps
on APUs**

**Detect Zero-Steps
on Cortex-M**

Detect Zero-Steps on APUs

Zero-Step

Instruction 1

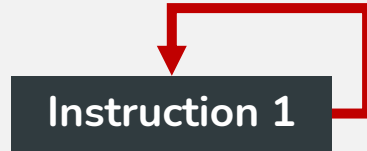
Single-Step

Instruction 1

**Detect Progression
on Cortex-M**

Detect Zero-Steps on APUs

Zero-Step



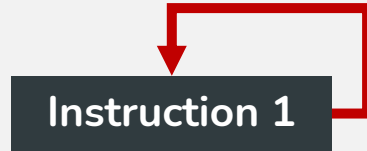
Single-Step



Detect Progression on Cortex-M

Detect Zero-Steps on APUs

Zero-Step



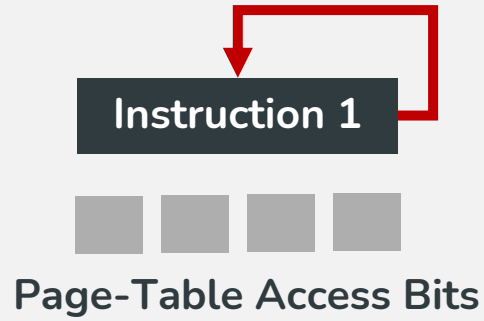
Single-Step



**Detect Progression
on Cortex-M**

Detect Zero-Steps on APUs

Zero-Step



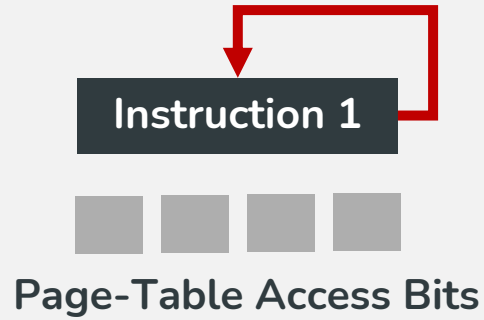
Single-Step



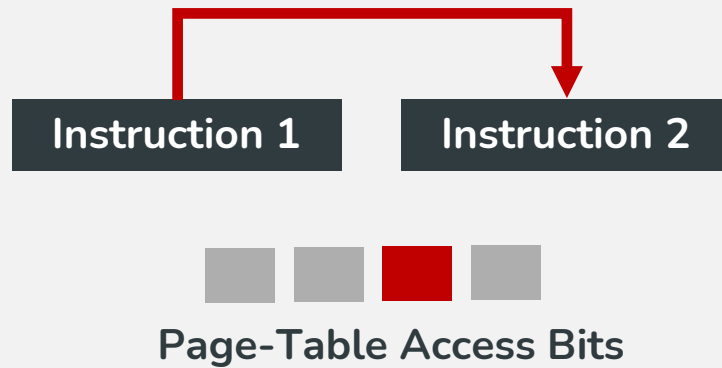
Detect Progression on Cortex-M

Detect Zero-Steps on APUs

Zero-Step



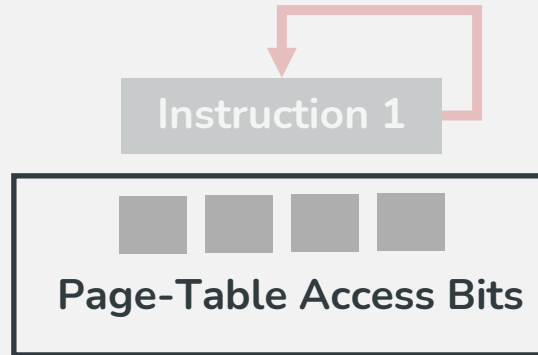
Single-Step



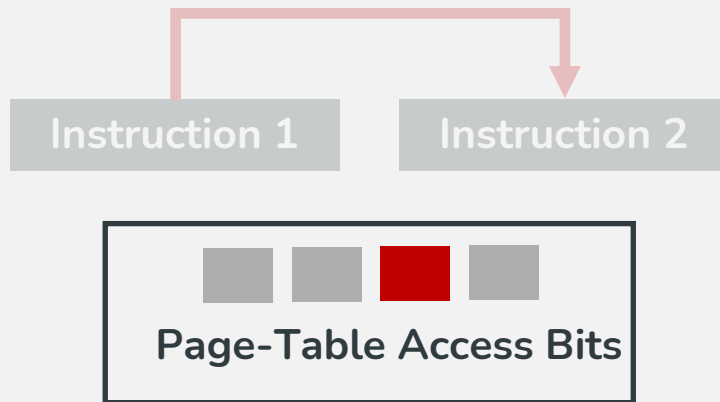
Detect Progression on Cortex-M

Detect Zero-Steps on APUs

Zero-Step



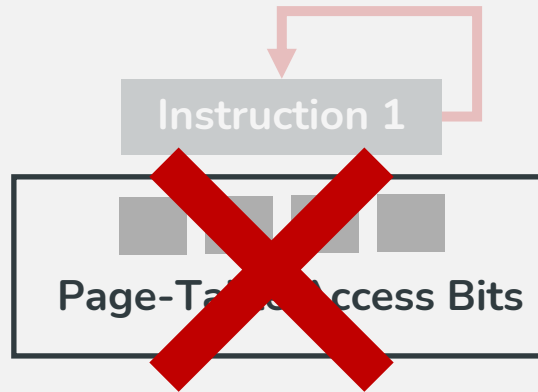
Single-Step



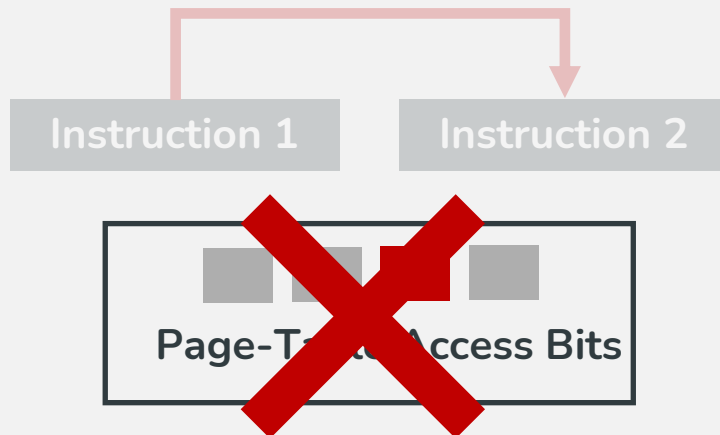
Detect Progression on Cortex-M

Detect Zero-Steps on APUs

Zero-Step



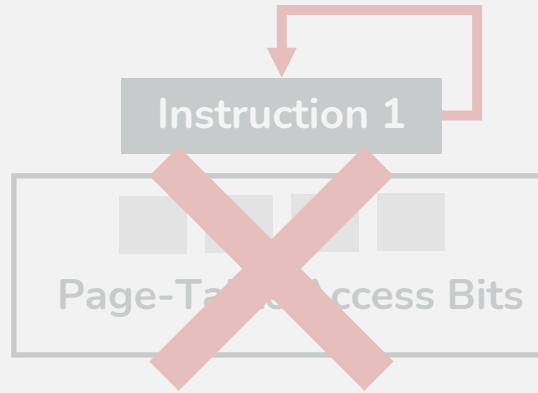
Single-Step



**Detect Progression
on Cortex-M**

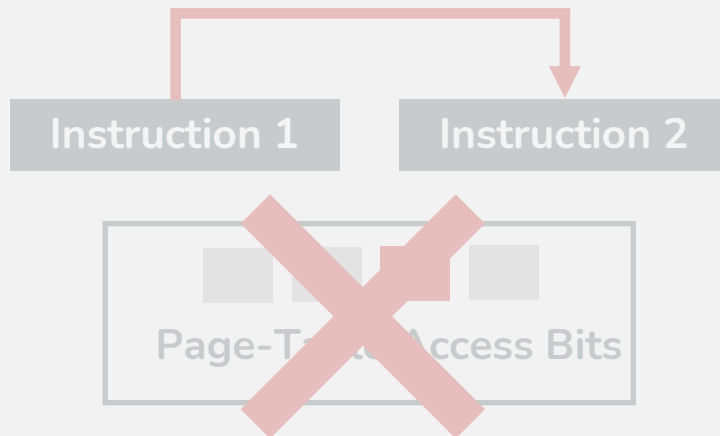
Detect Zero-Steps on APUs

Zero-Step



No Virtual Memory on Cortex-M!!

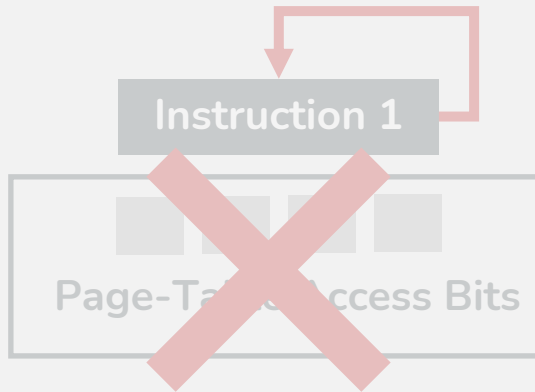
Single-Step



Detect Progression on Cortex-M

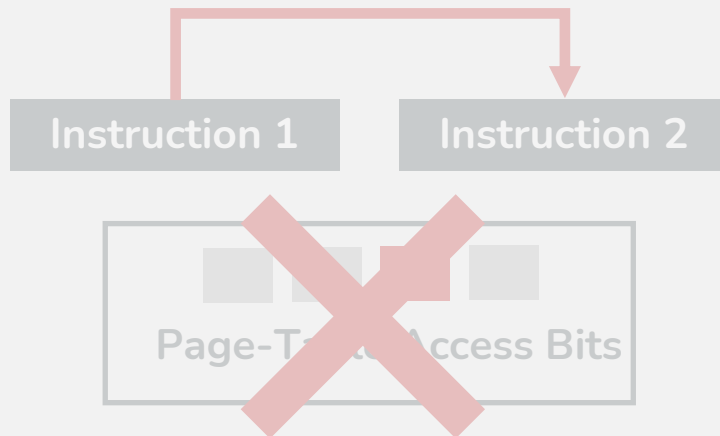
Detect Zero-Steps on APUs

Zero-Step



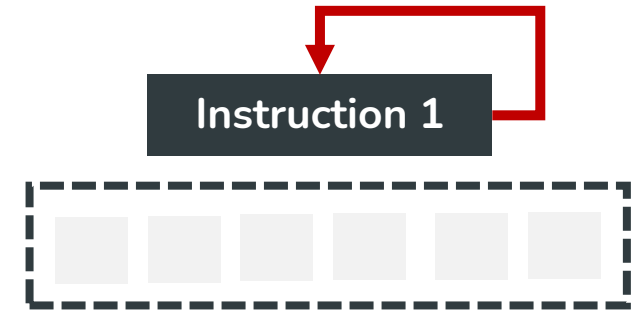
No Virtual Memory on Cortex-M!!

Single-Step



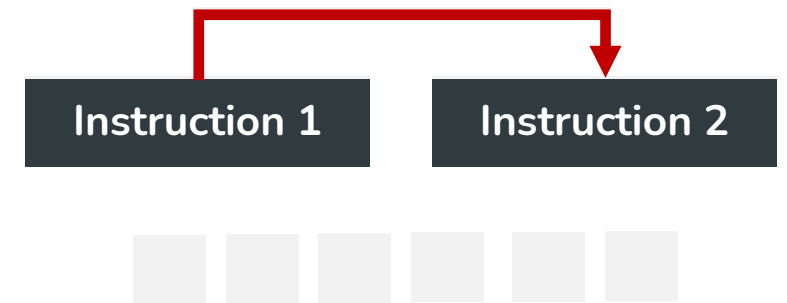
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

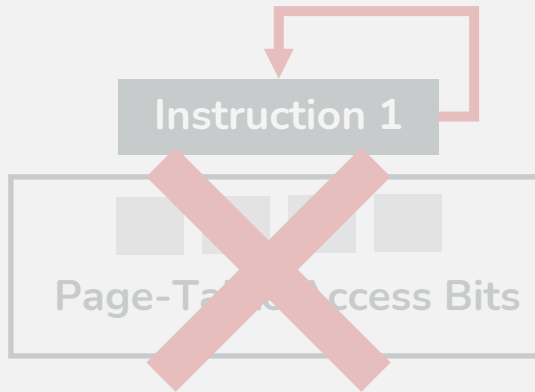
Single-Step



Latency Oracle

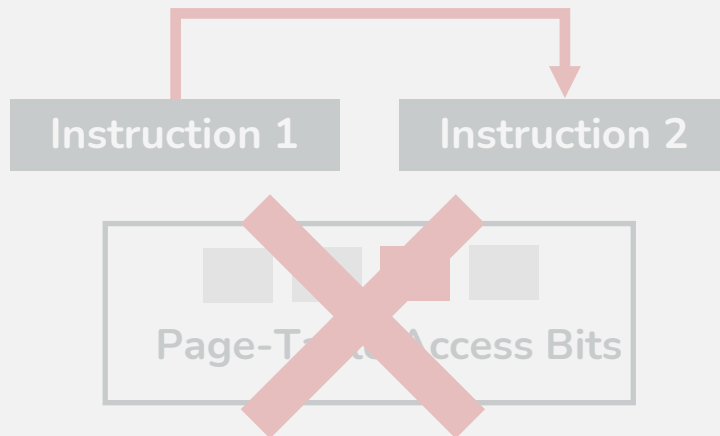
Detect Zero-Steps on APUs

Zero-Step



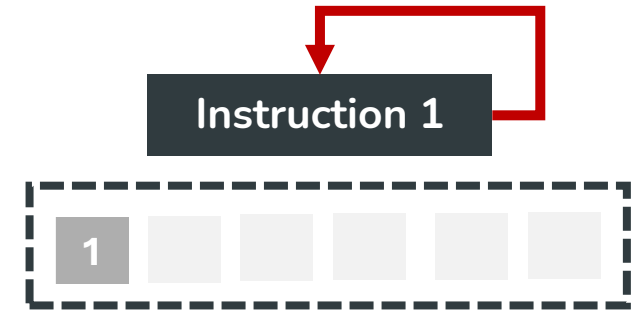
No Virtual Memory on Cortex-M!!

Single-Step



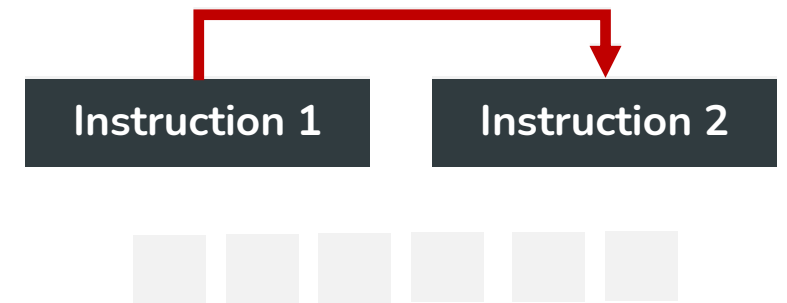
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

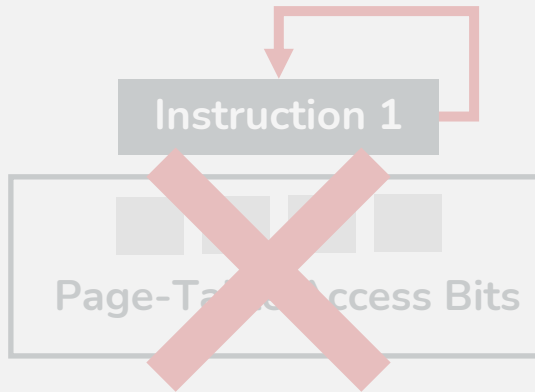
Single-Step



Latency Oracle

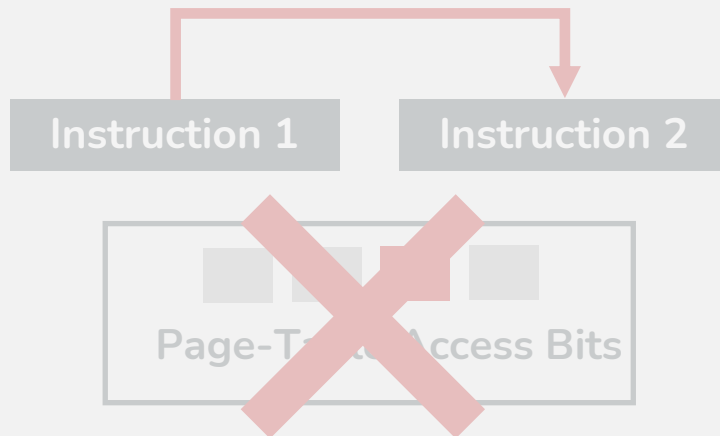
Detect Zero-Steps on APUs

Zero-Step



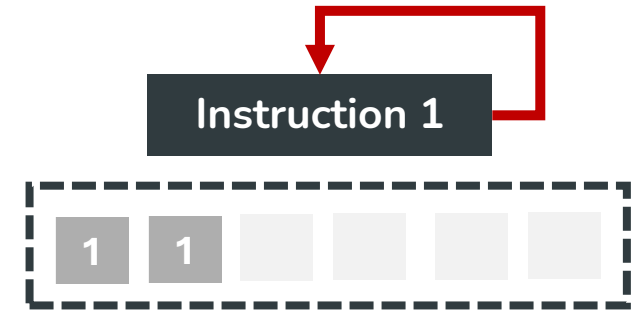
No Virtual Memory on Cortex-M!!

Single-Step



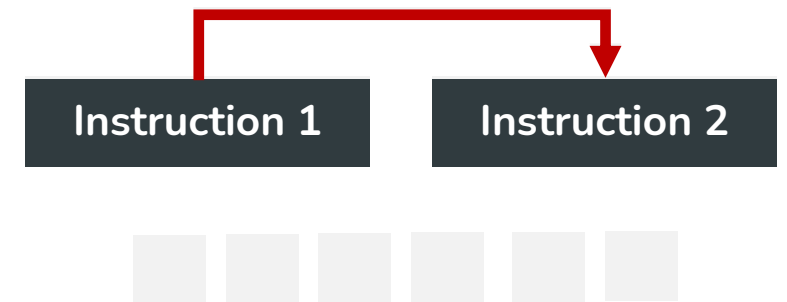
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

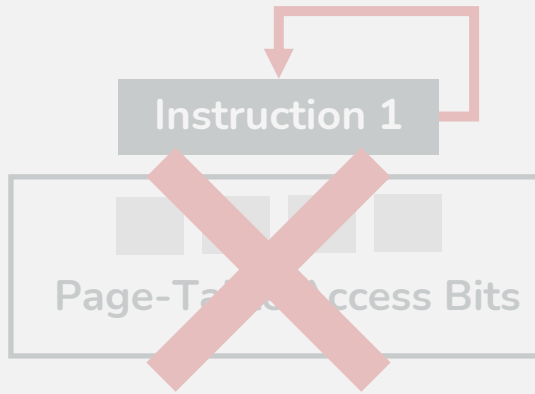
Single-Step



Latency Oracle

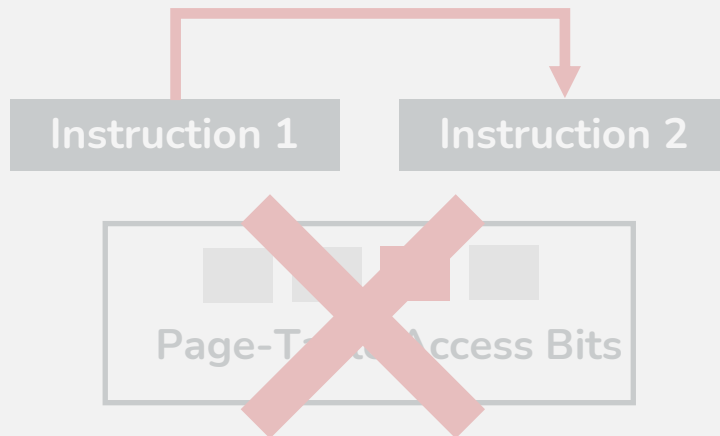
Detect Zero-Steps on APUs

Zero-Step



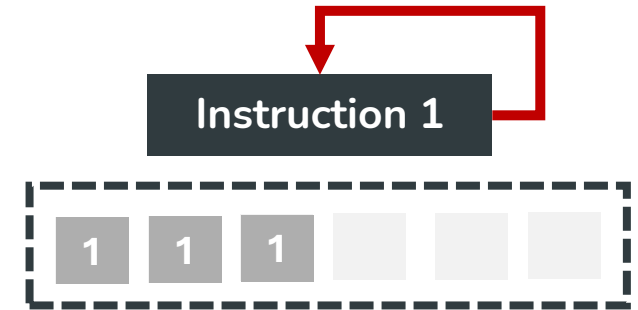
No Virtual Memory on Cortex-M!!

Single-Step



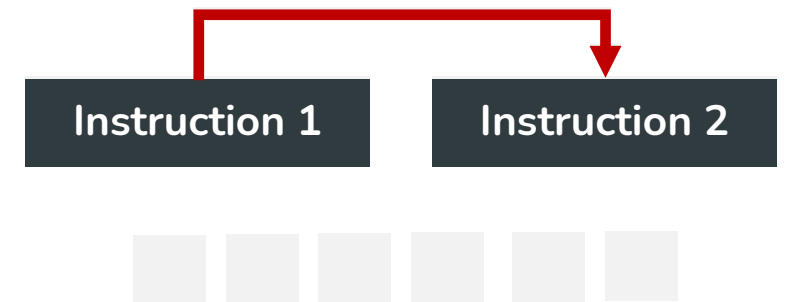
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

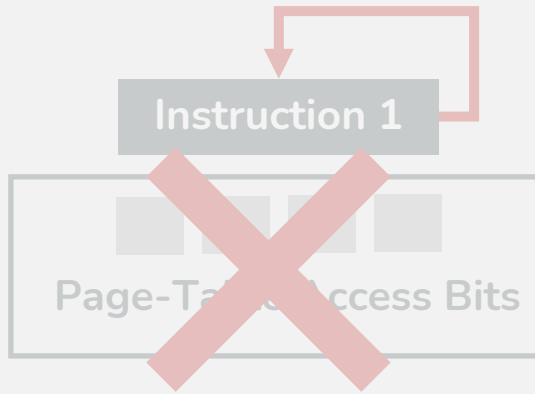
Single-Step



Latency Oracle

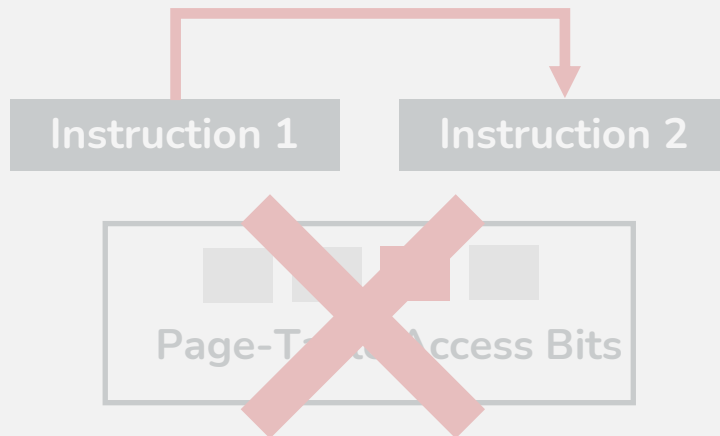
Detect Zero-Steps on APUs

Zero-Step



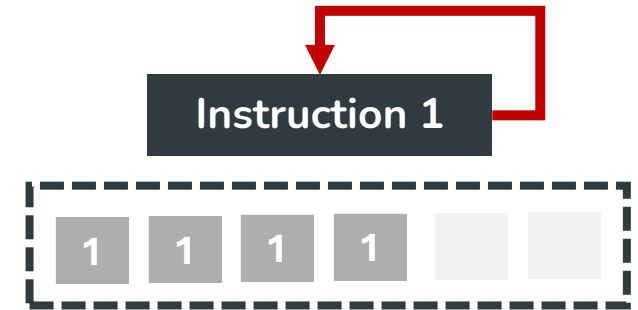
No Virtual Memory on Cortex-M!!

Single-Step



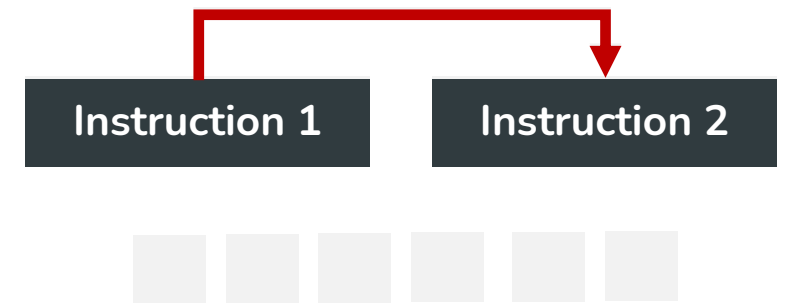
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

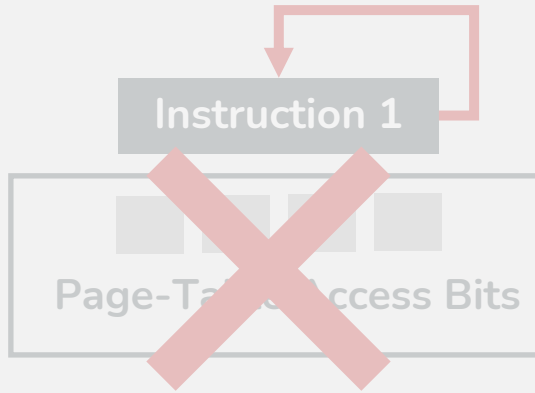
Single-Step



Latency Oracle

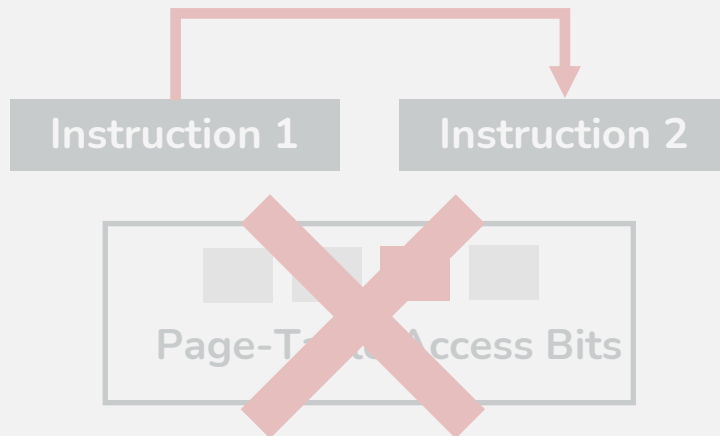
Detect Zero-Steps on APUs

Zero-Step



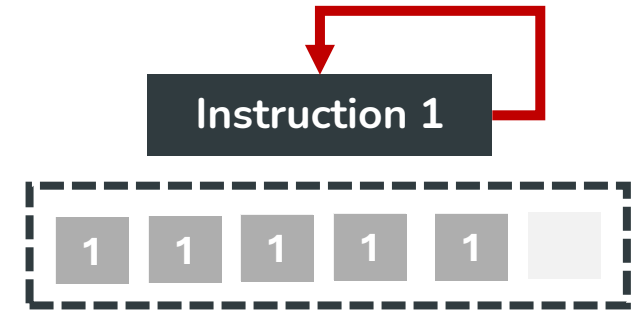
No Virtual Memory on Cortex-M!!

Single-Step



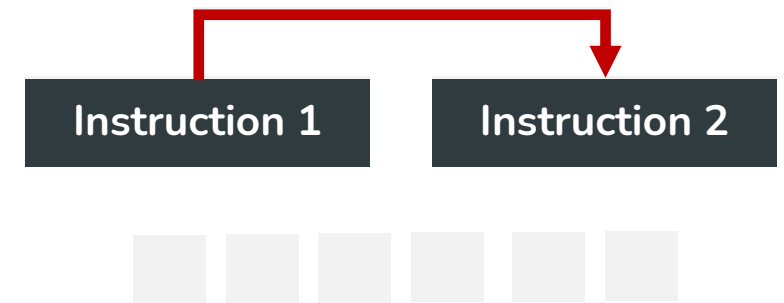
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

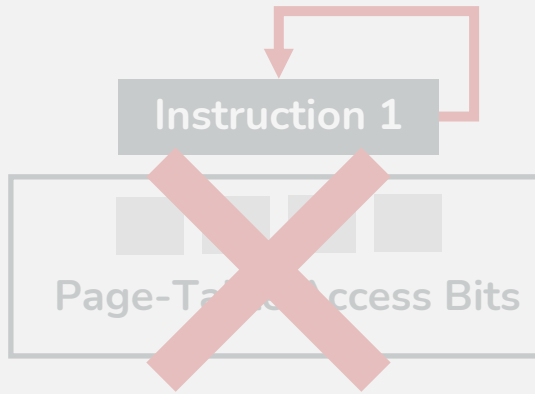
Single-Step



Latency Oracle

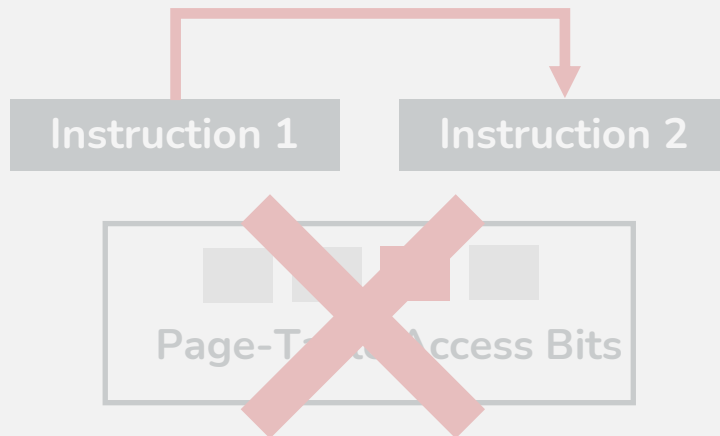
Detect Zero-Steps on APUs

Zero-Step



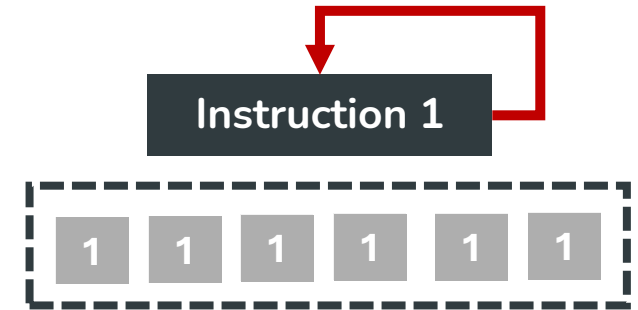
No Virtual Memory on Cortex-M!!

Single-Step



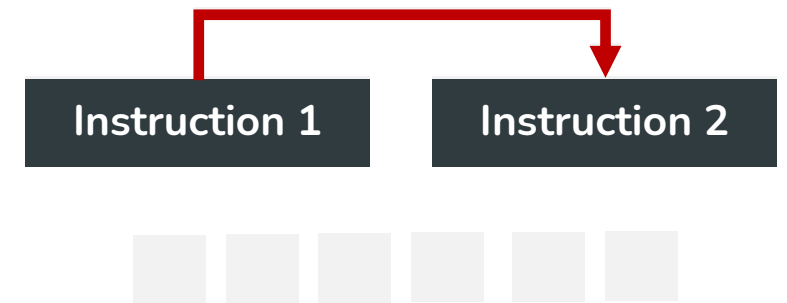
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

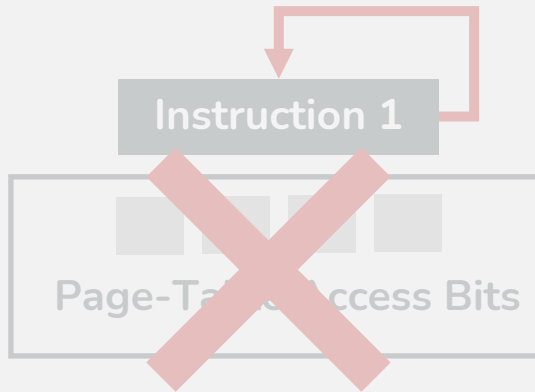
Single-Step



Latency Oracle

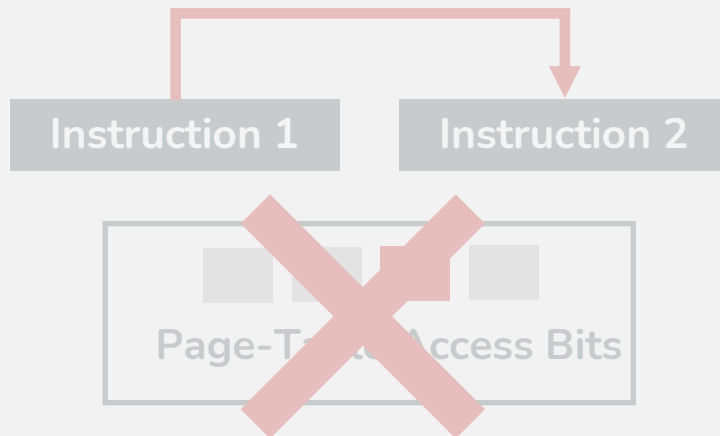
Detect Zero-Steps on APUs

Zero-Step



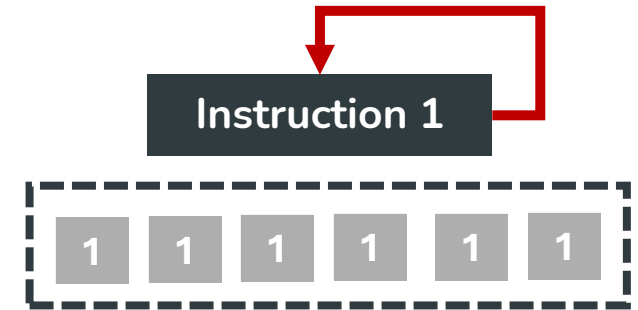
No Virtual Memory on Cortex-M!!

Single-Step



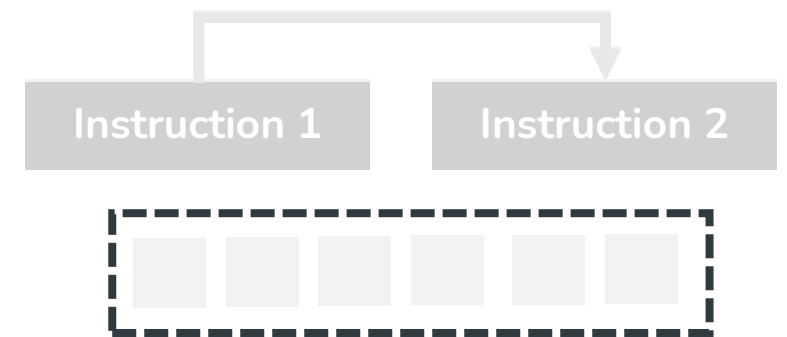
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

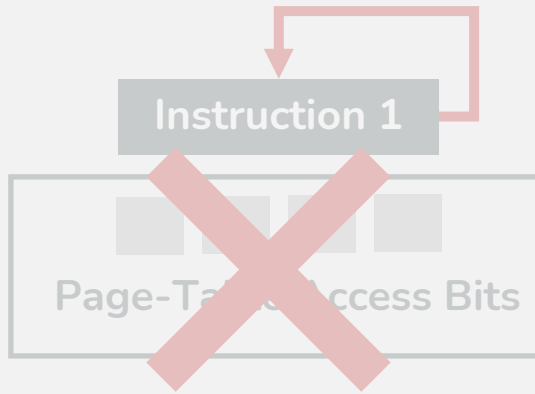
Single-Step



Latency Oracle

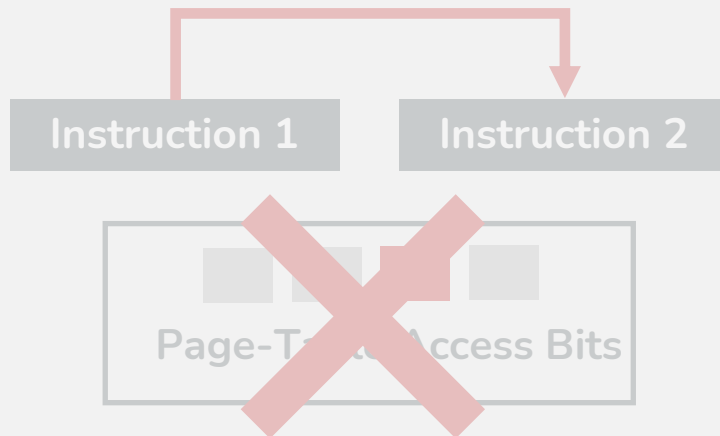
Detect Zero-Steps on APUs

Zero-Step



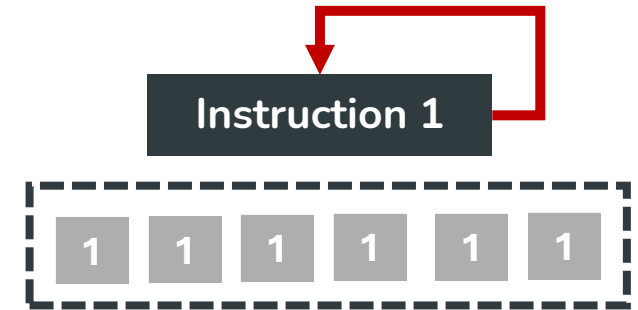
No Virtual Memory on Cortex-M!!

Single-Step



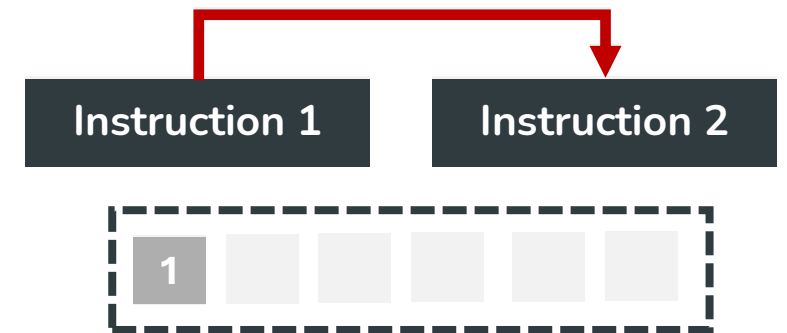
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

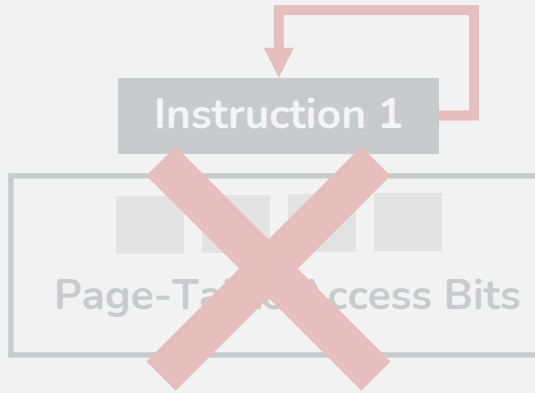
Single-Step



Latency Oracle

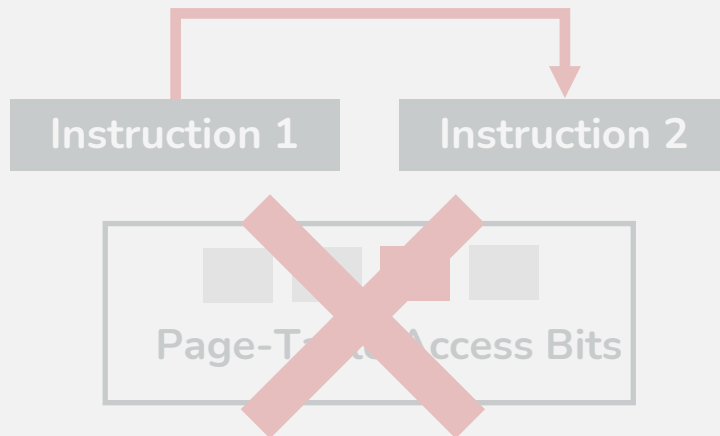
Detect Zero-Steps on APUs

Zero-Step



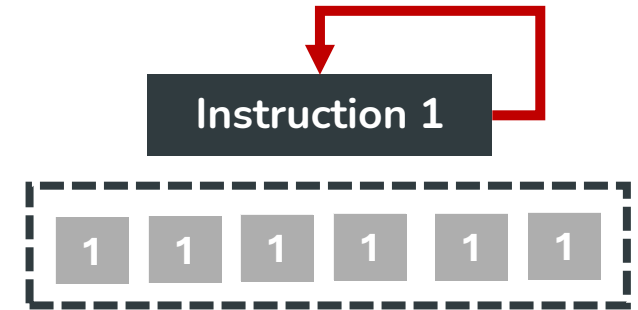
No Virtual Memory on Cortex-M!!

Single-Step



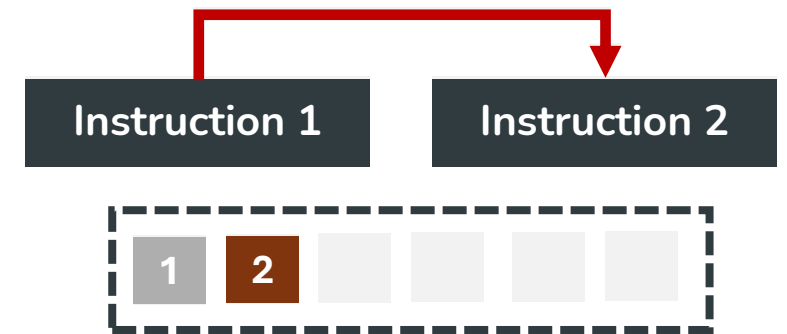
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

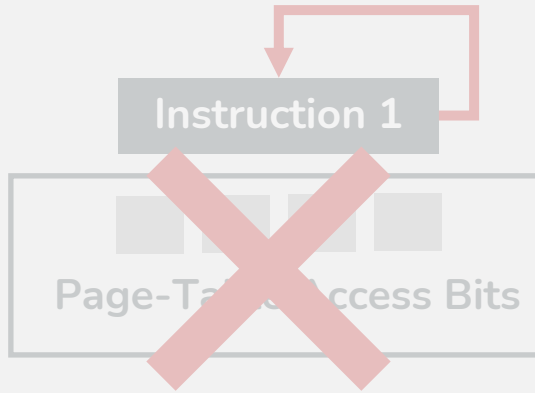
Single-Step



Latency Oracle

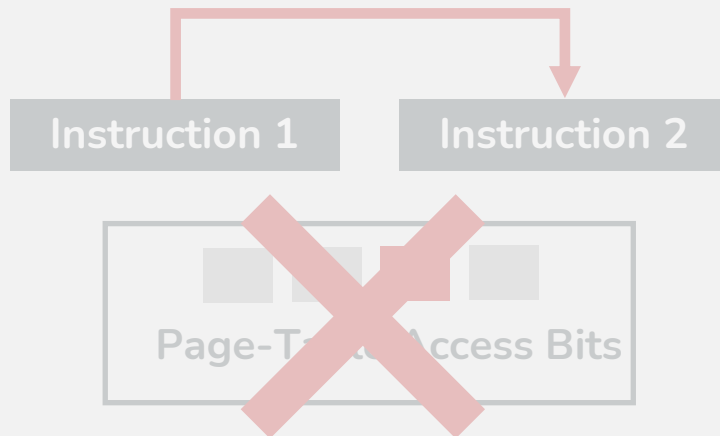
Detect Zero-Steps on APUs

Zero-Step



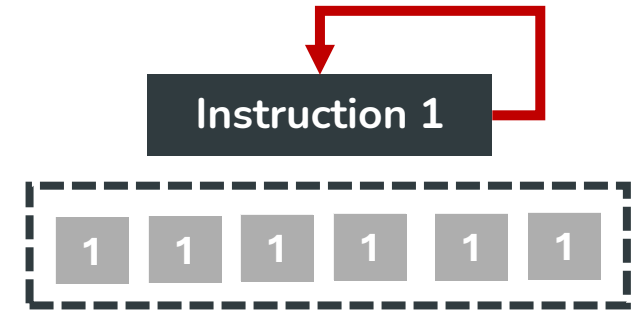
No Virtual Memory on Cortex-M!!

Single-Step



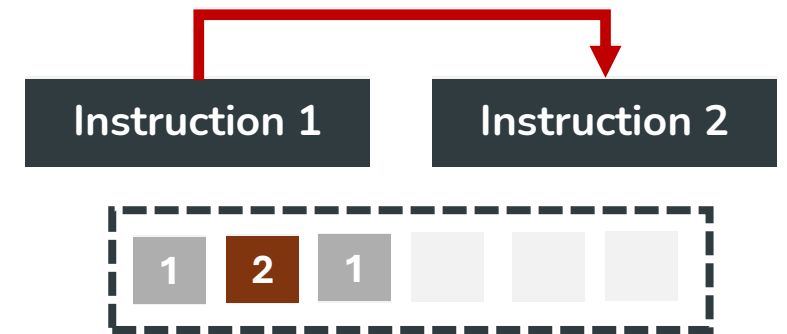
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

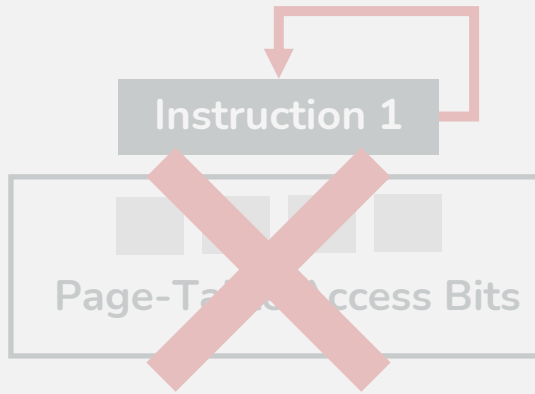
Single-Step



Latency Oracle

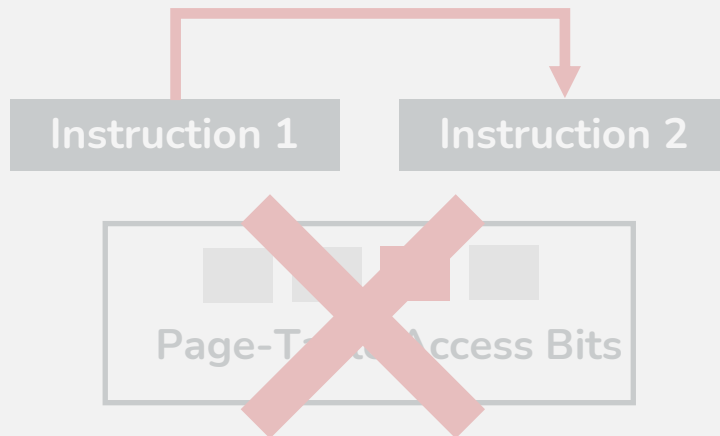
Detect Zero-Steps on APUs

Zero-Step



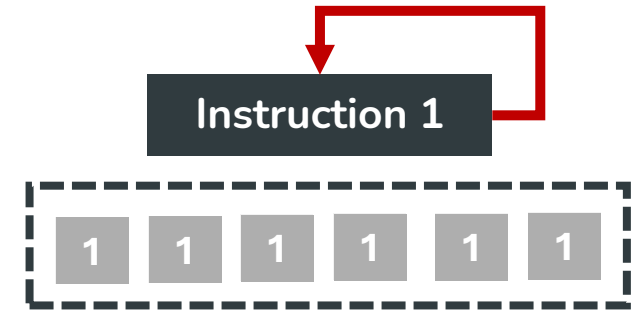
No Virtual Memory on Cortex-M!!

Single-Step



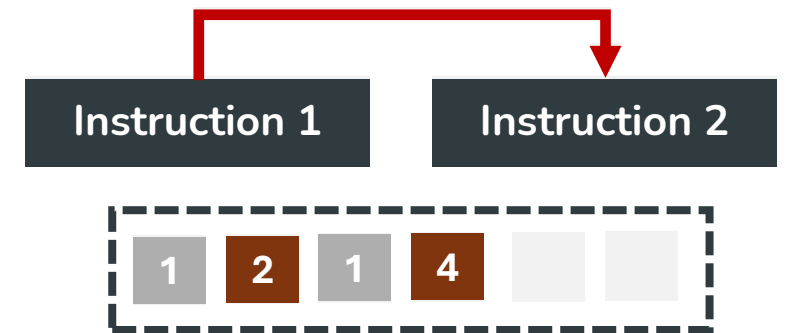
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

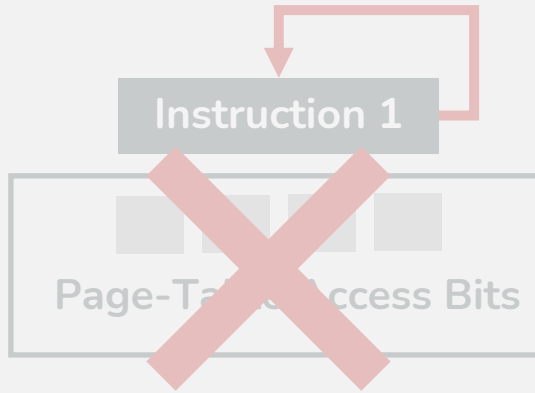
Single-Step



Latency Oracle

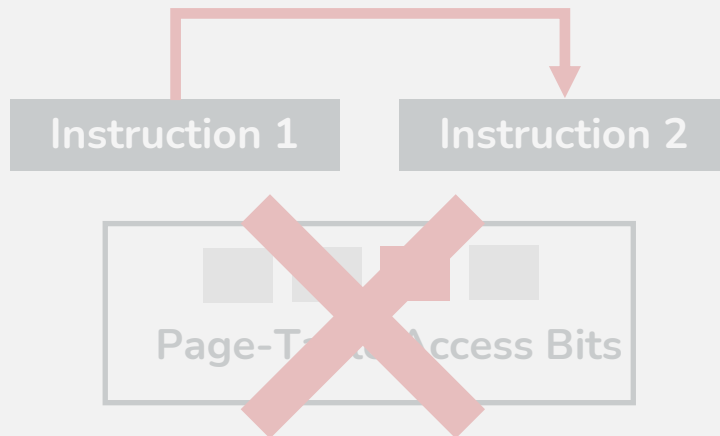
Detect Zero-Steps on APUs

Zero-Step



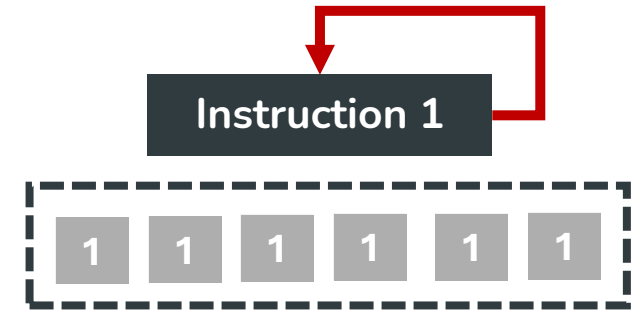
No Virtual Memory on Cortex-M!!

Single-Step



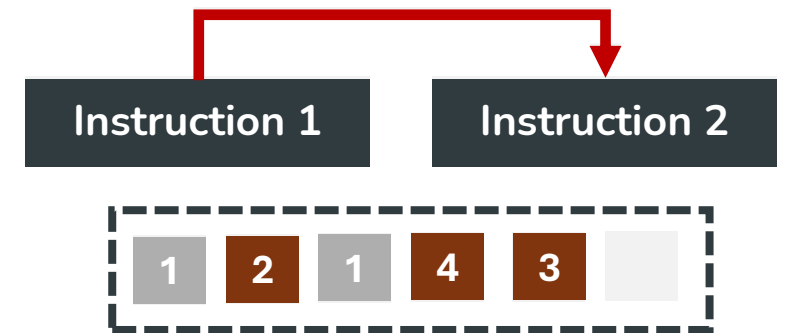
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

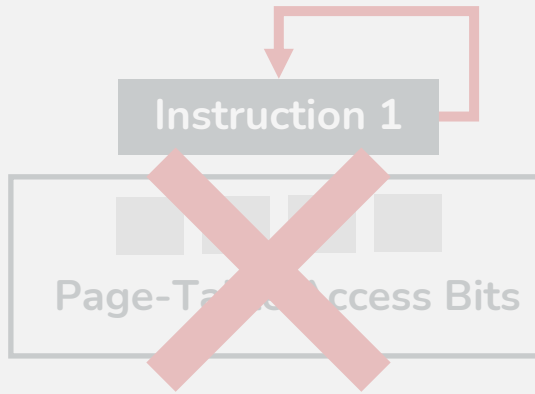
Single-Step



Latency Oracle

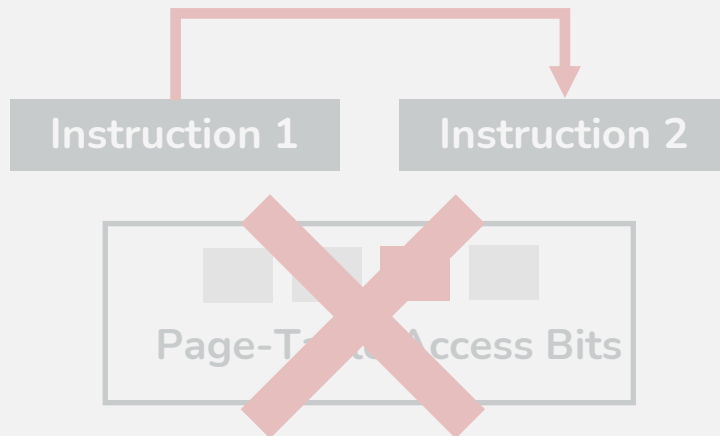
Detect Zero-Steps on APUs

Zero-Step



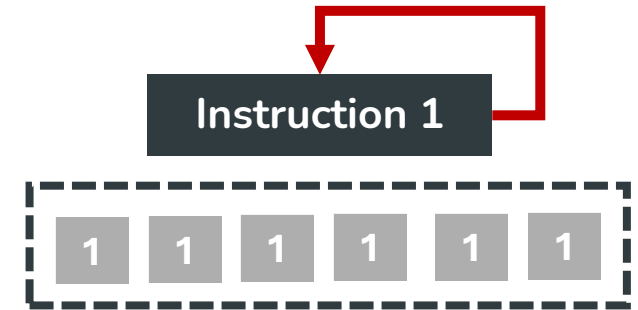
No Virtual Memory on Cortex-M!!

Single-Step



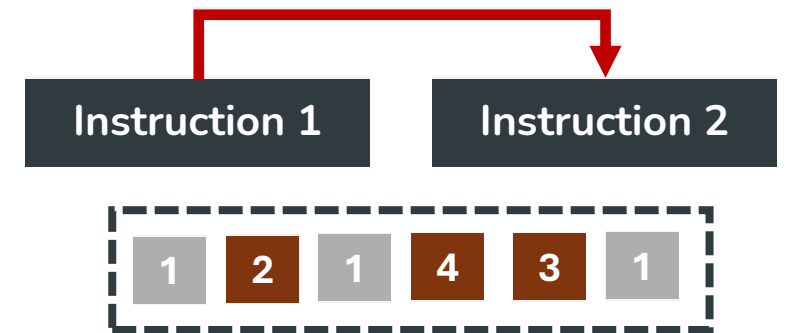
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

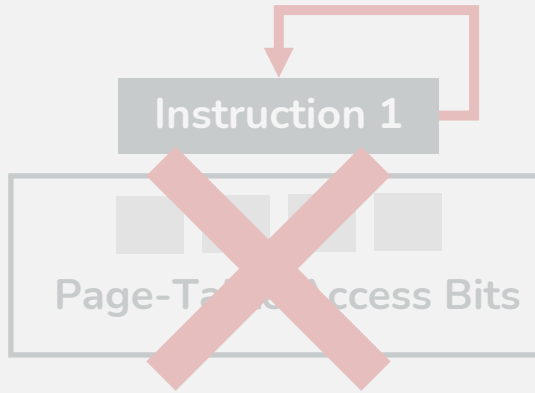
Single-Step



Latency Oracle

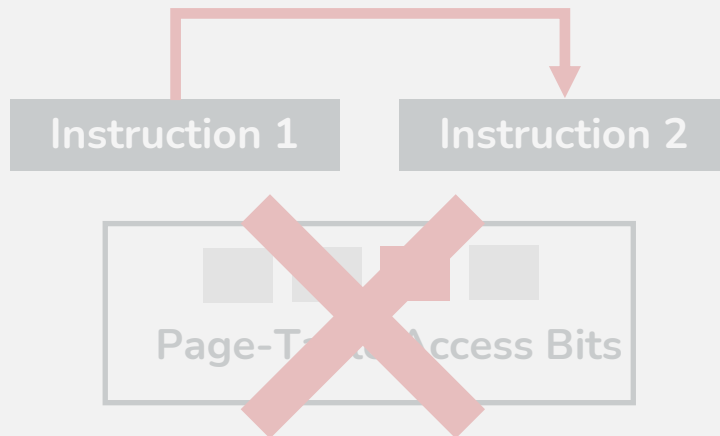
Detect Zero-Steps on APUs

Zero-Step



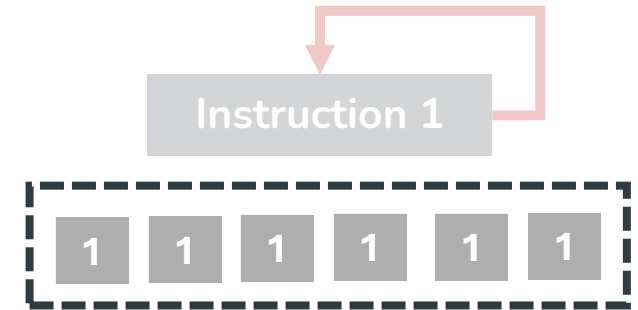
No Virtual Memory on Cortex-M!!

Single-Step



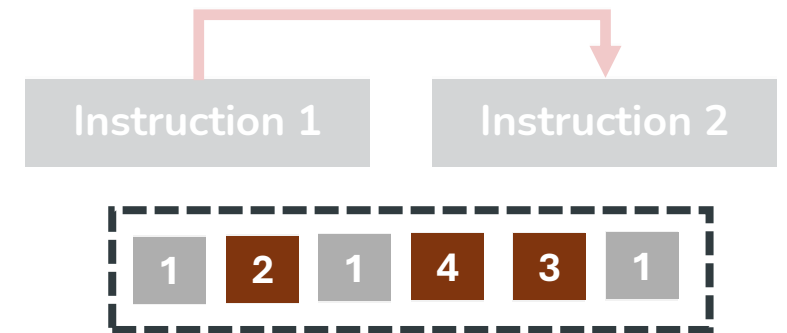
Detect Zero-Steps on Cortex-M

Zero-Step



Latency Oracle

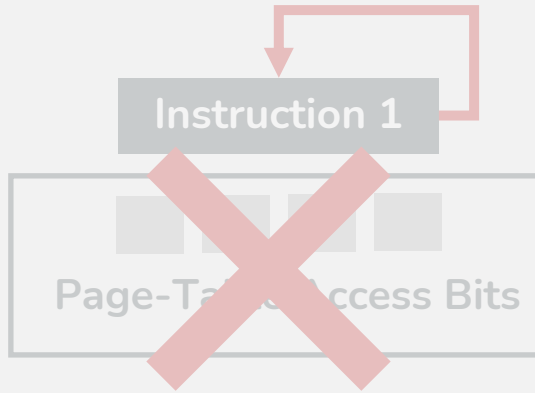
Single-Step



Latency Oracle

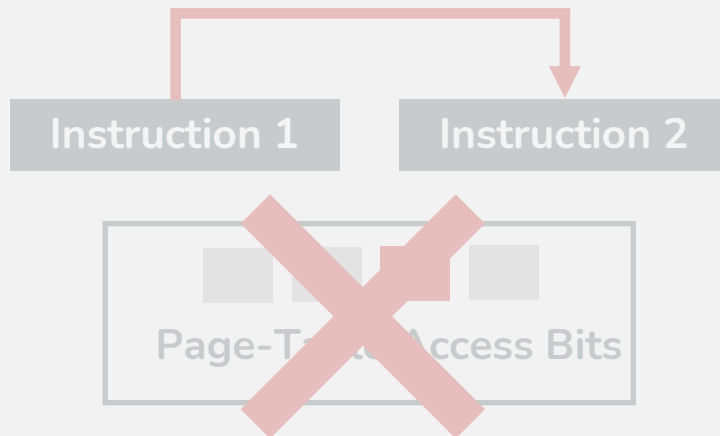
Detect Zero-Steps on APUs

Zero-Step



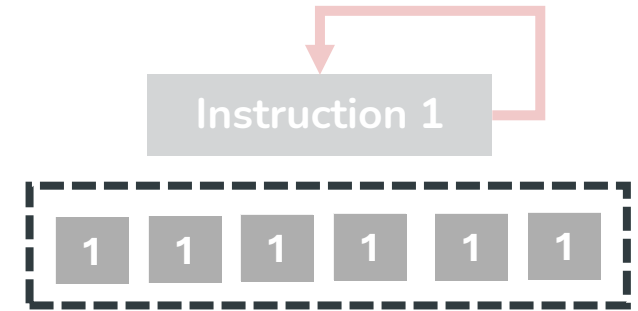
No Virtual Memory on Cortex-M!!

Single-Step



Detect Zero-Steps on Cortex-M

Zero-Step

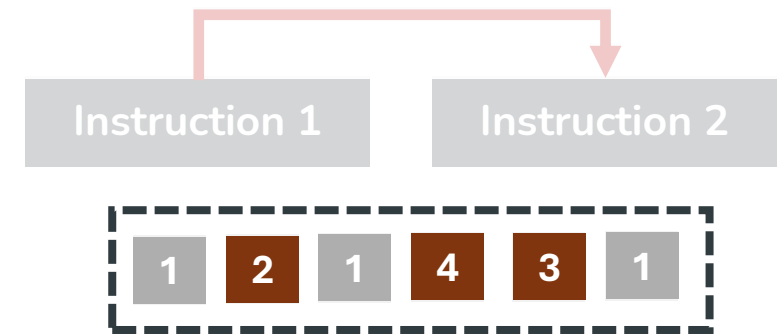


Latency Oracle



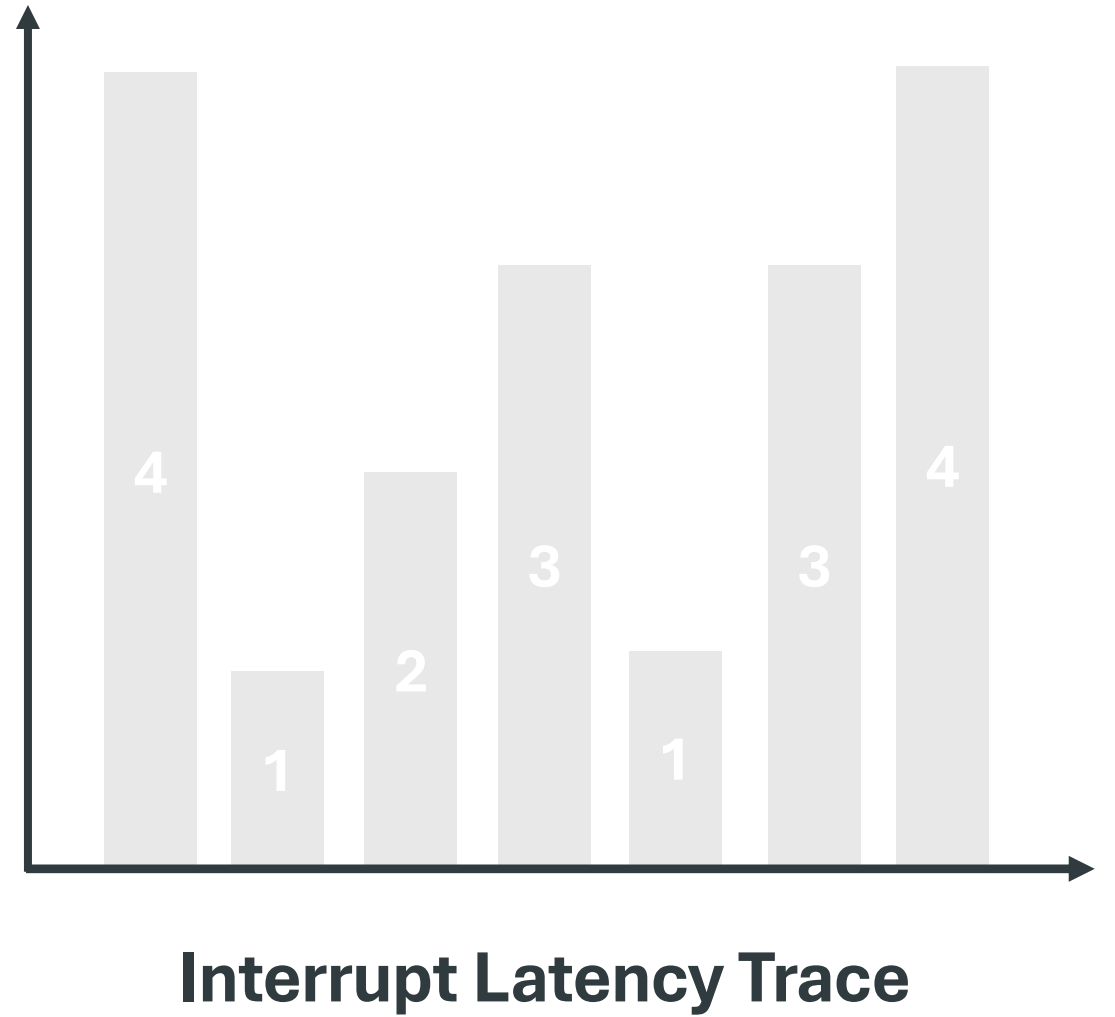
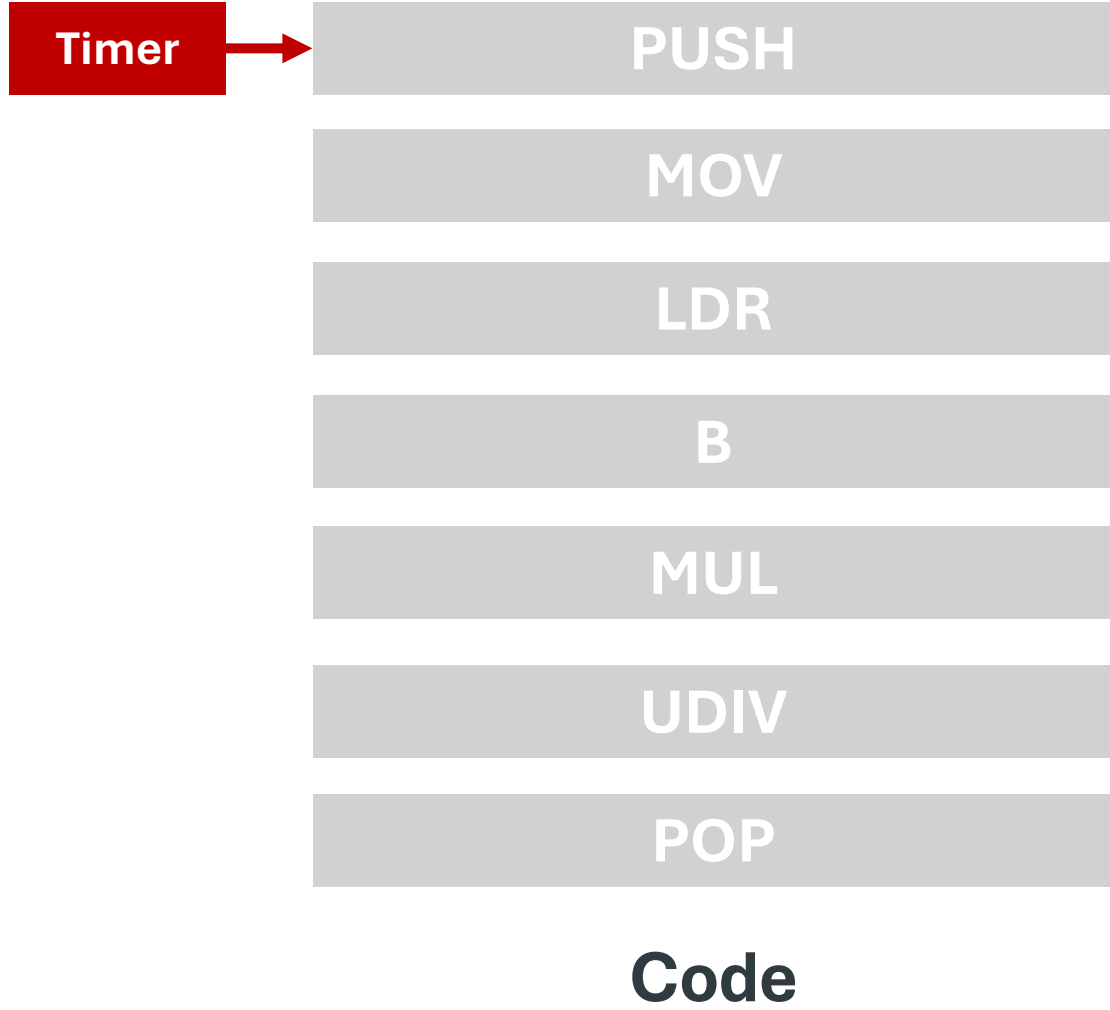
Detect Zero-Steps via Latency Oracle!

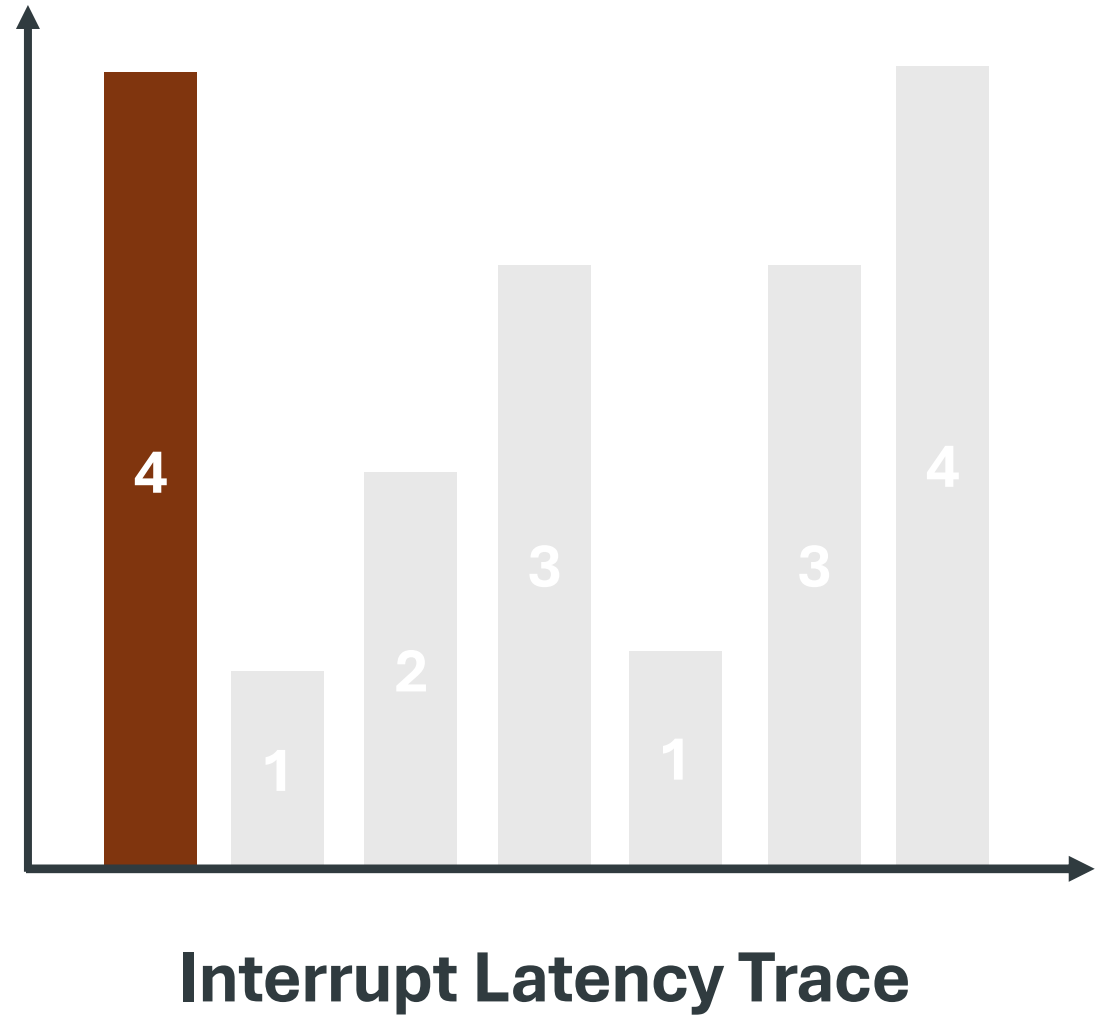
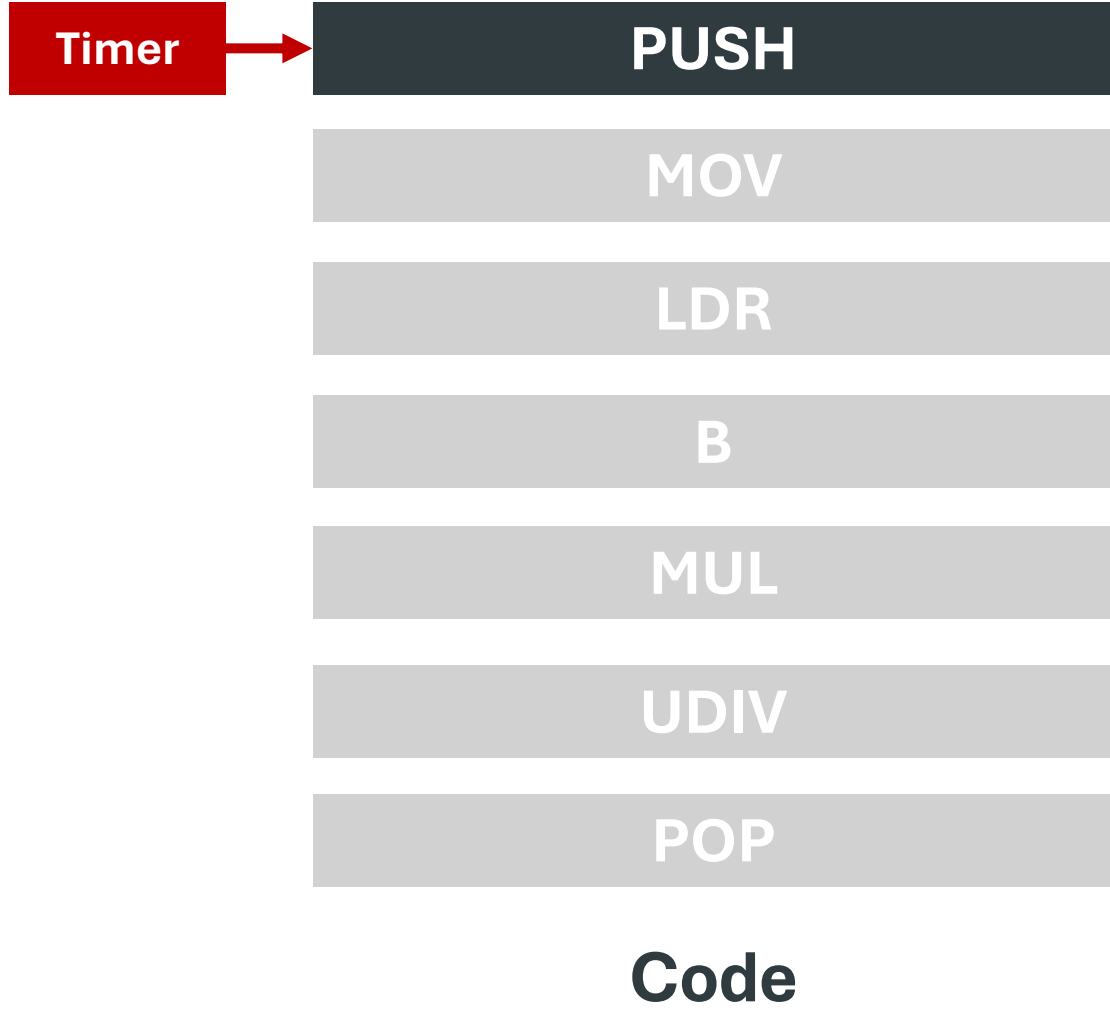
Single-Step

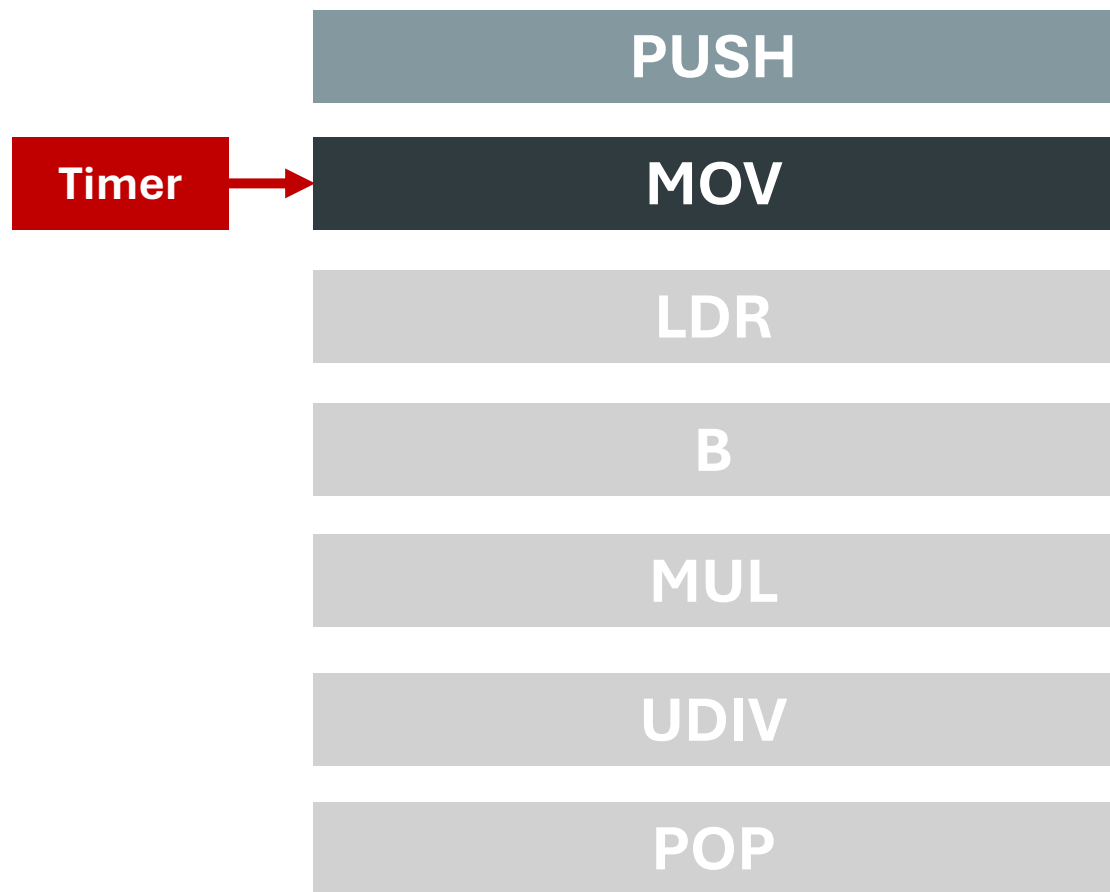


Latency Oracle

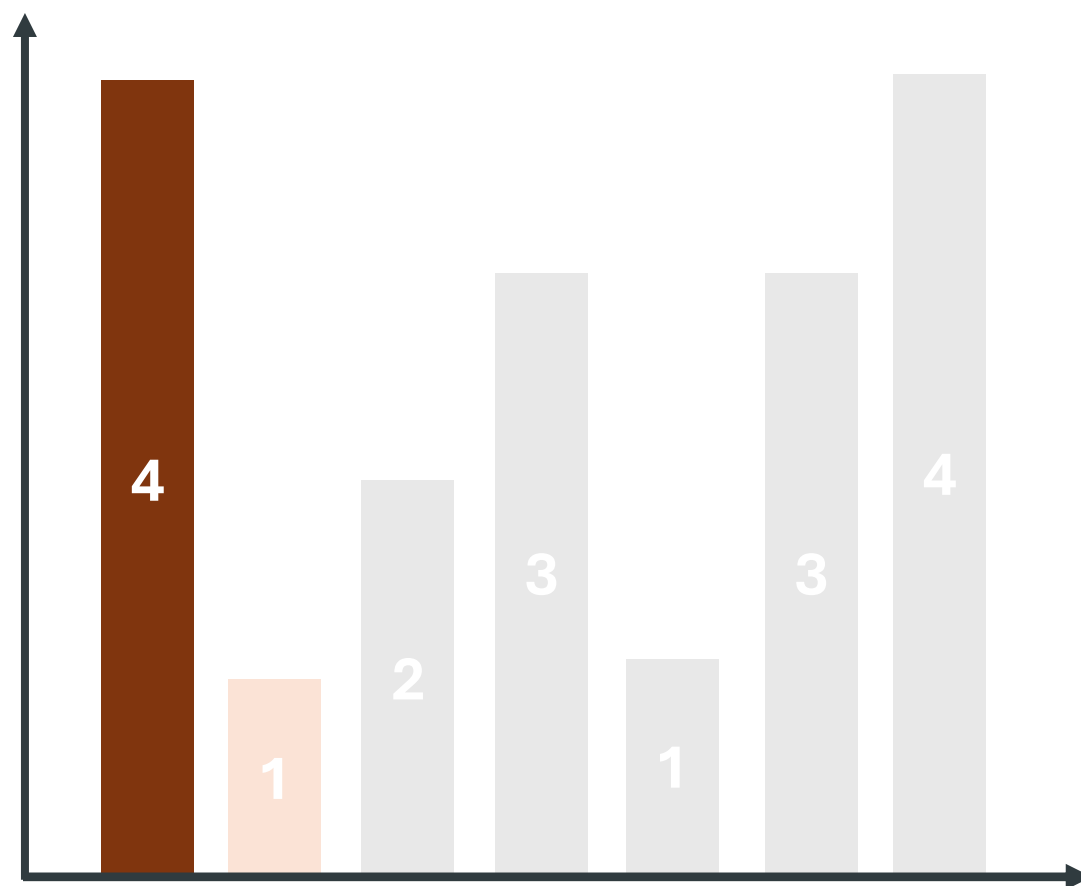
Voilà!
Single-Step on Cortex-M



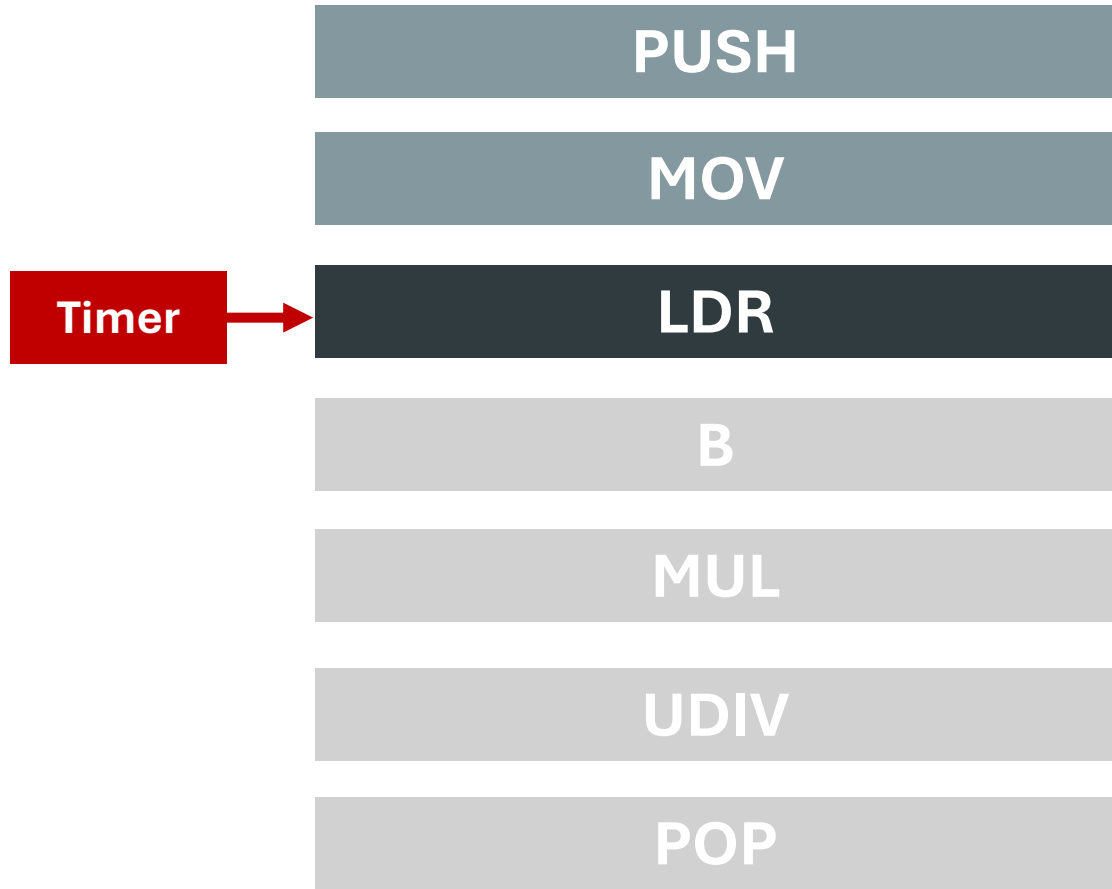




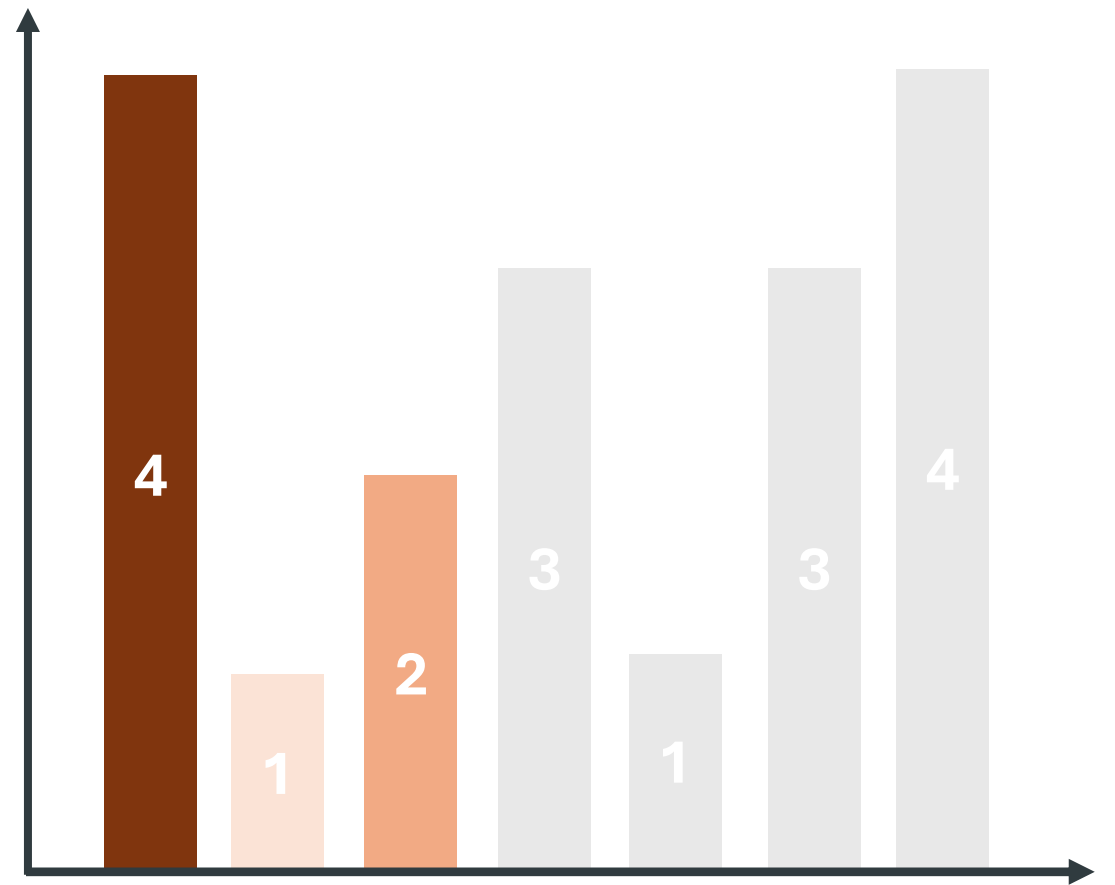
Code



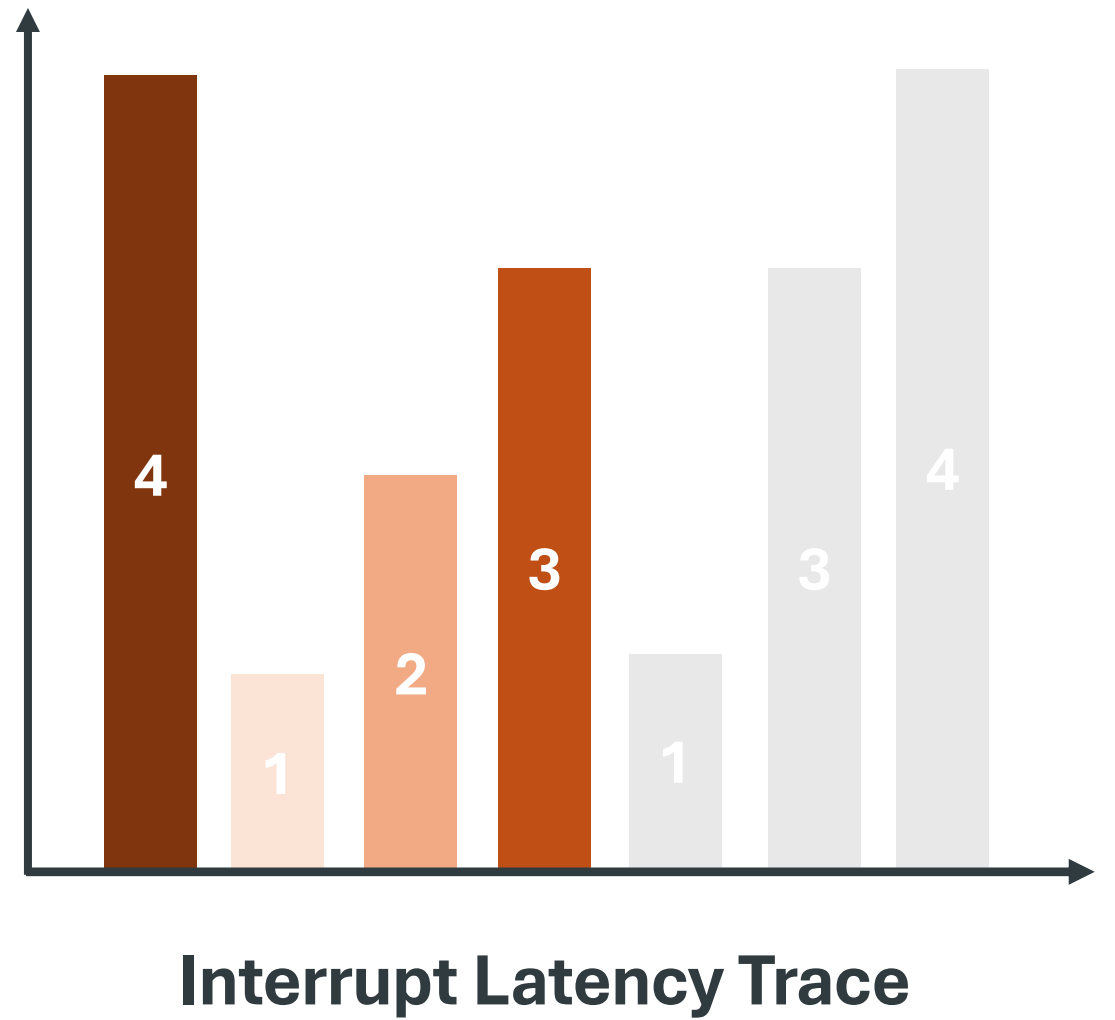
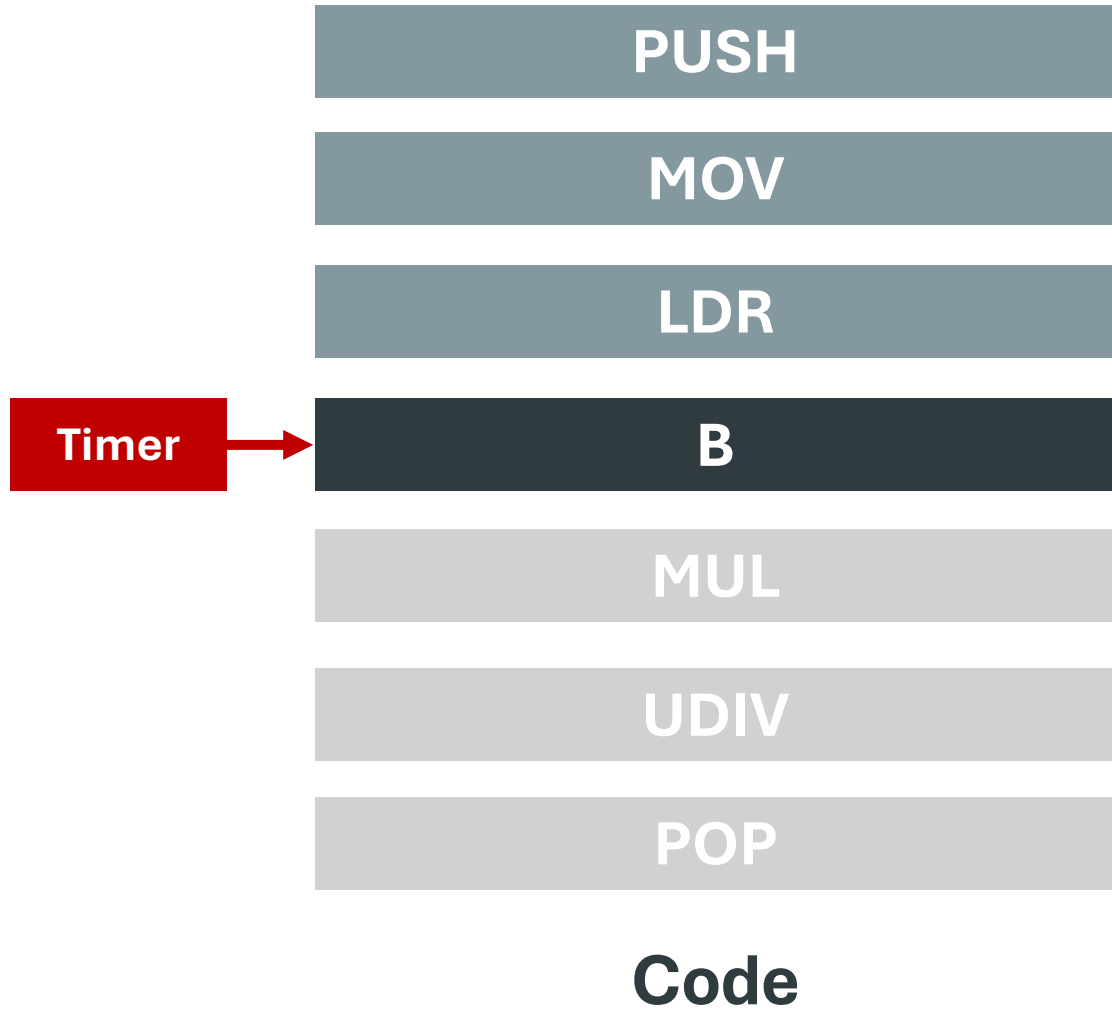
Interrupt Latency Trace

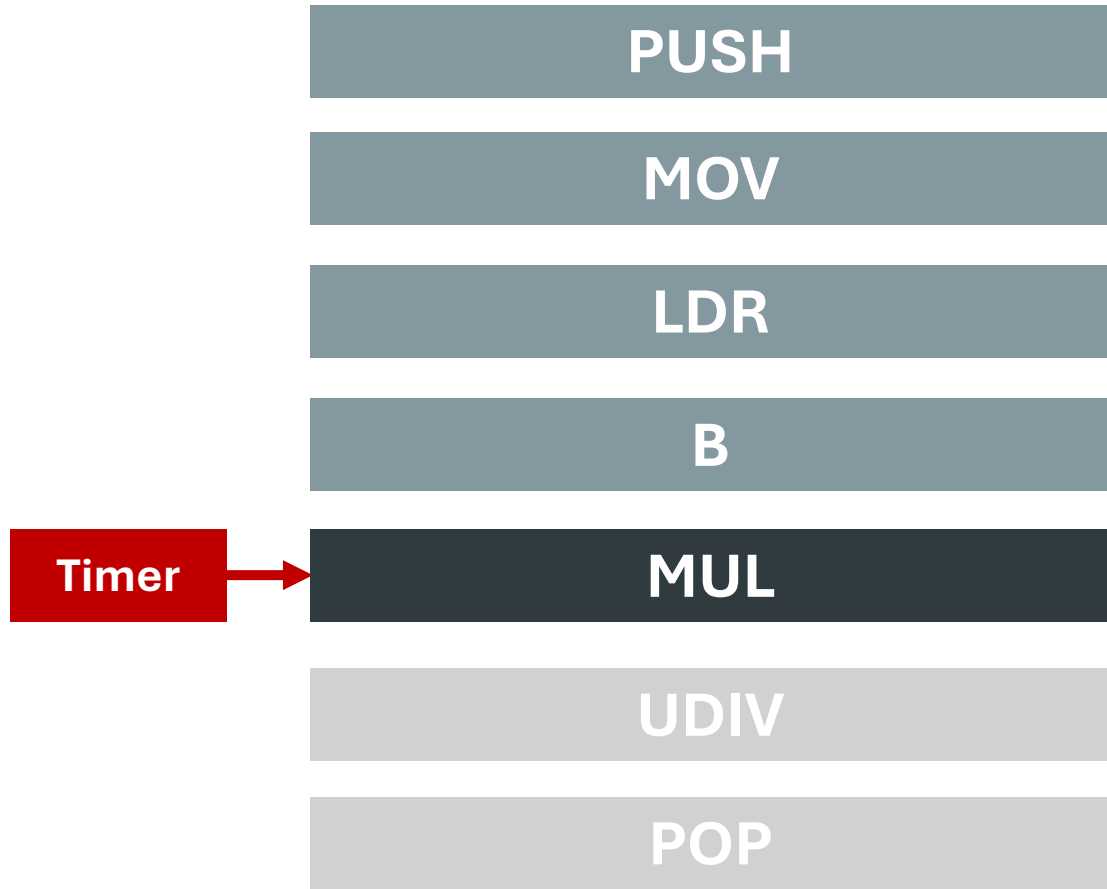


Code

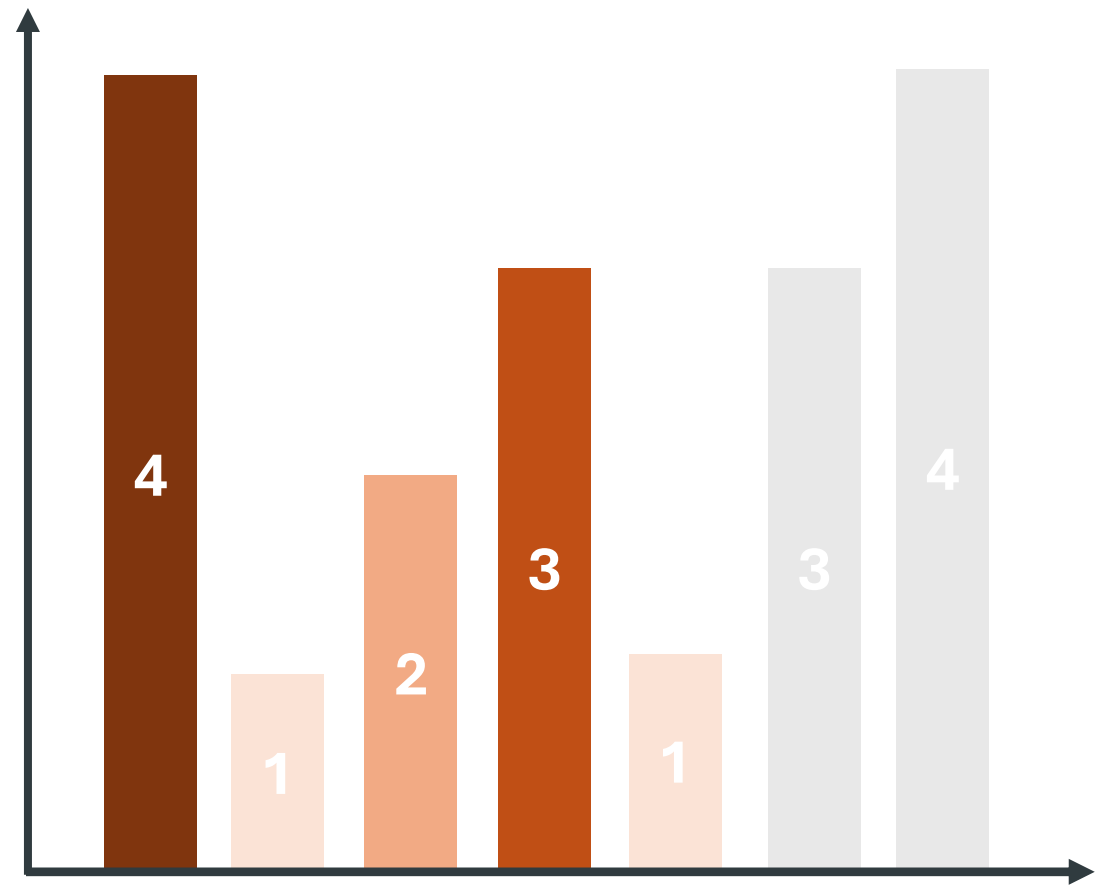


Interrupt Latency Trace

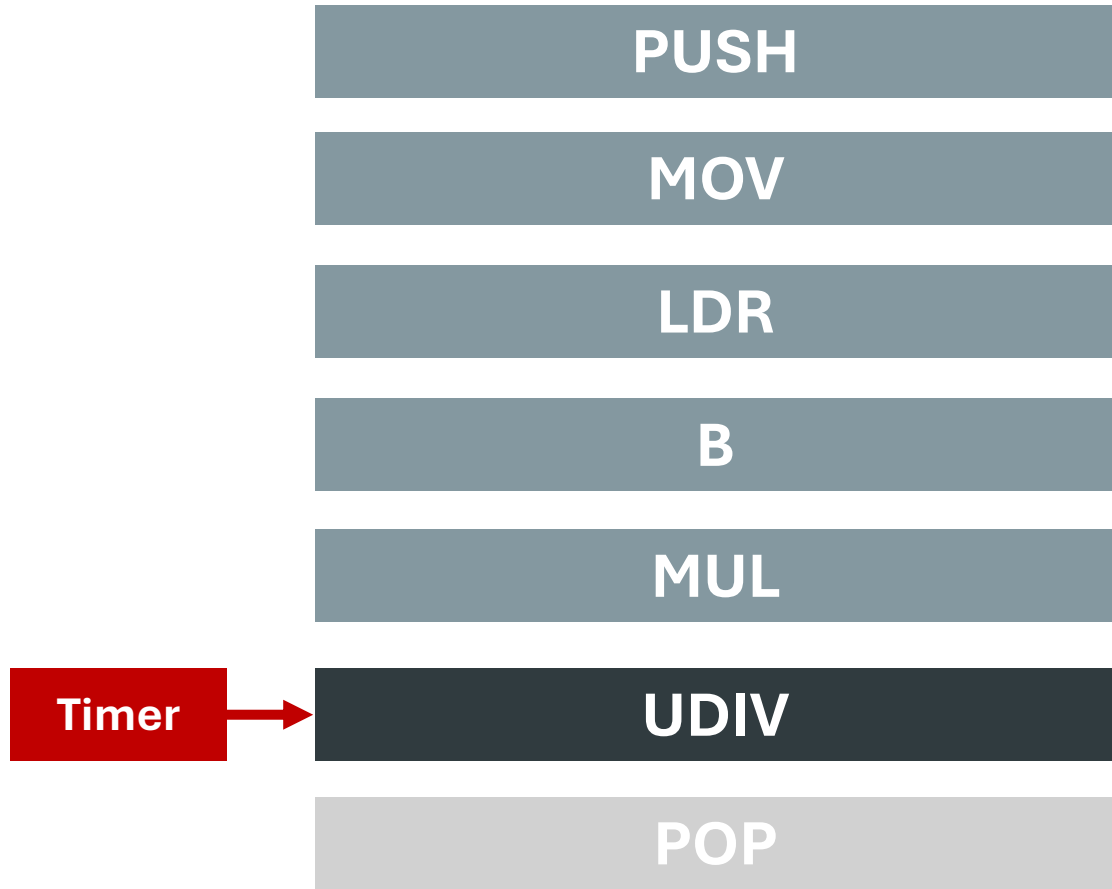




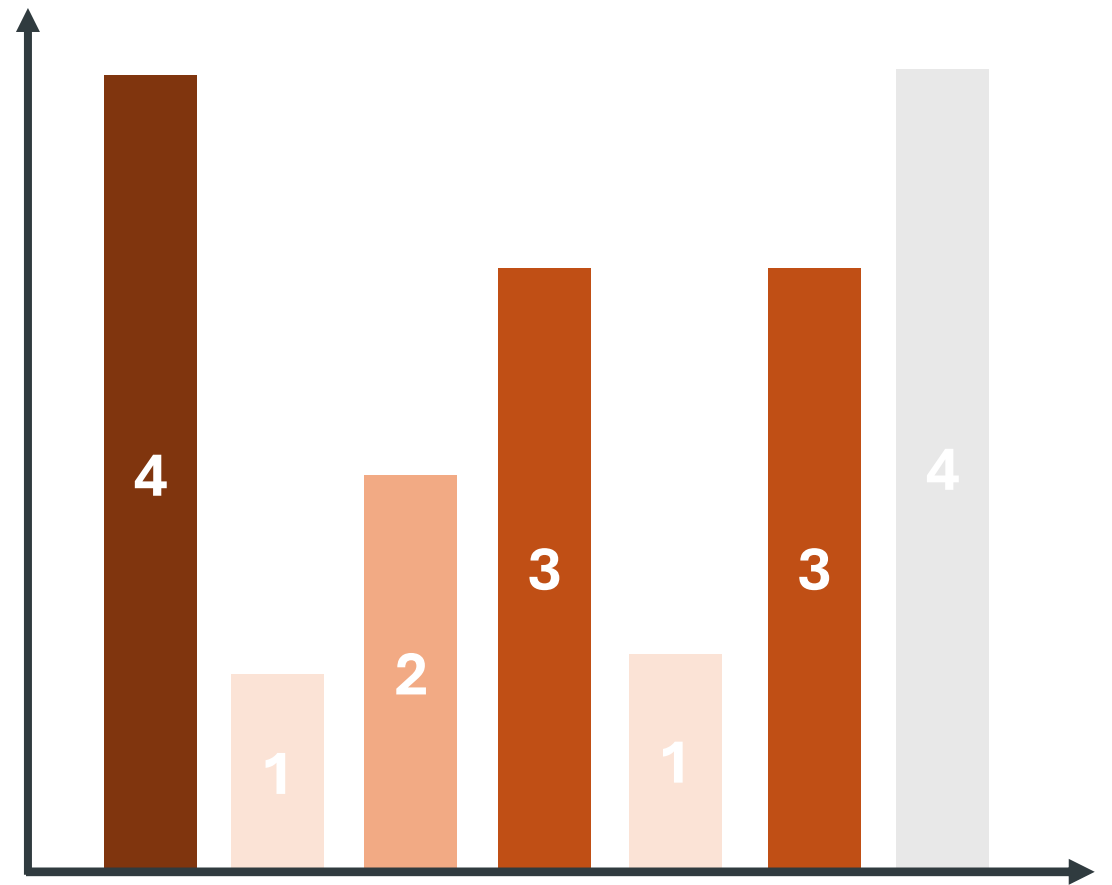
Code



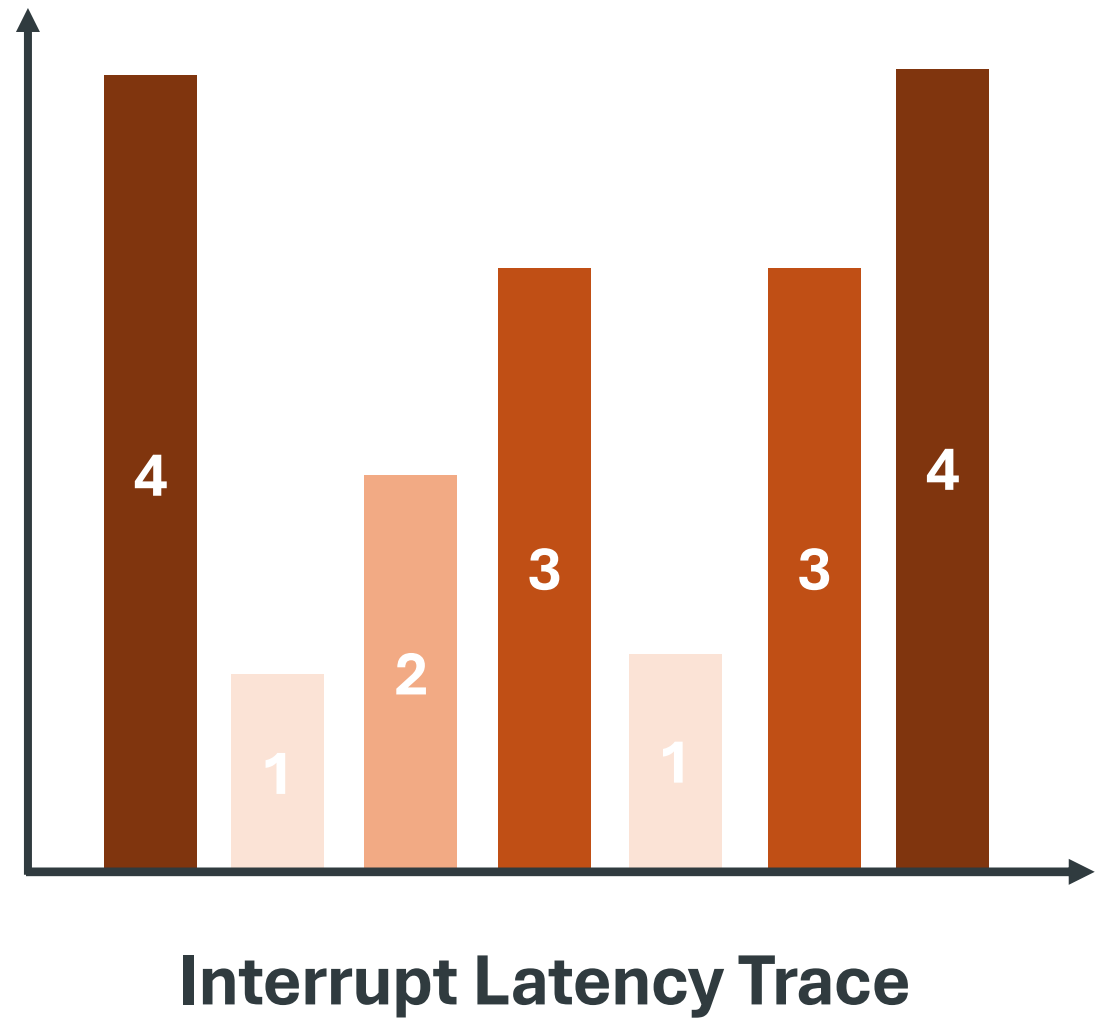
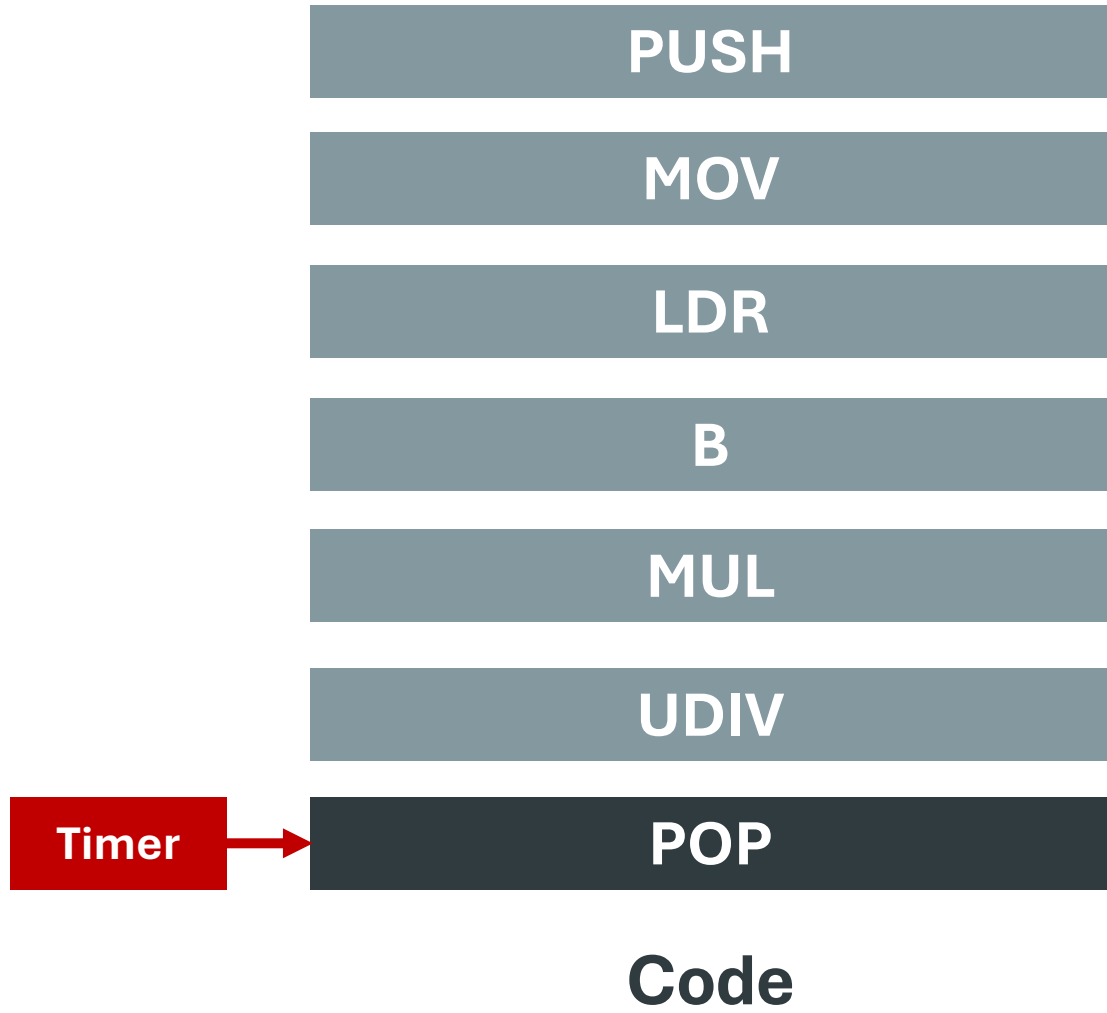
Interrupt Latency Trace



Code



Interrupt Latency Trace



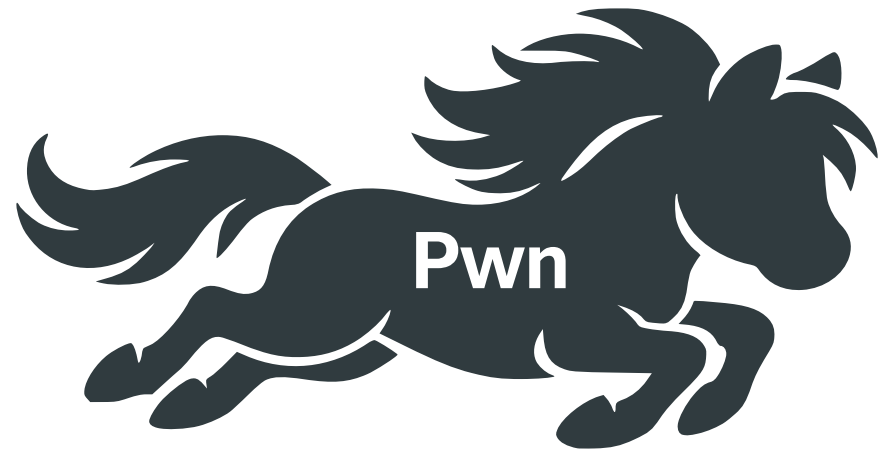


M-Step Framework



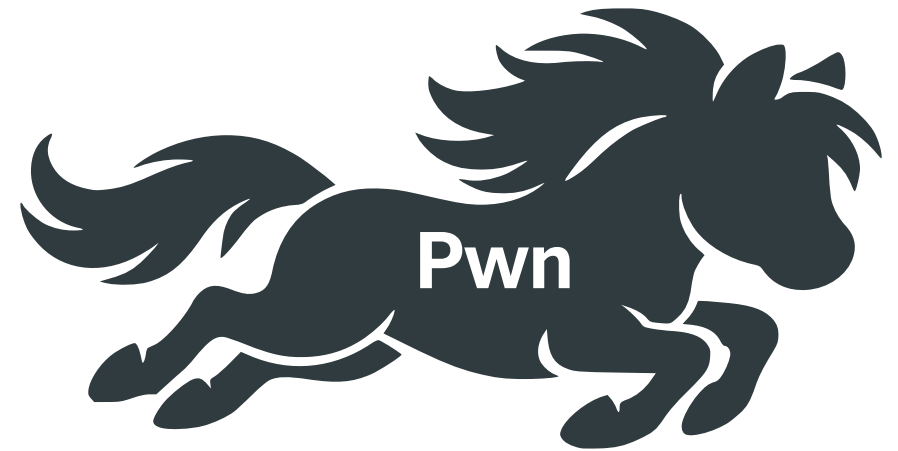
M-Step Framework





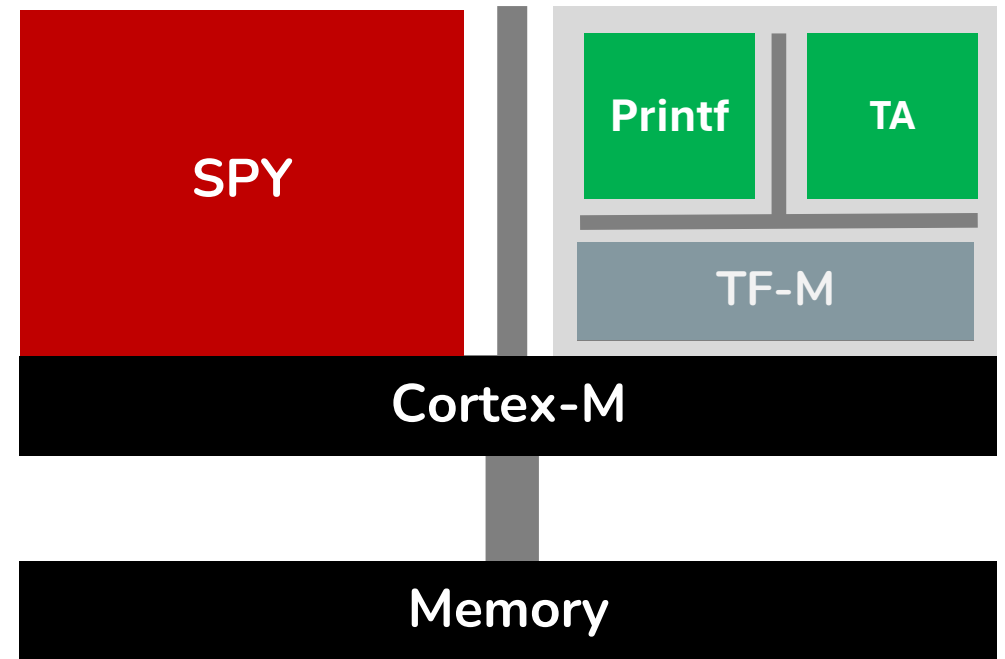


Develop the Exploit

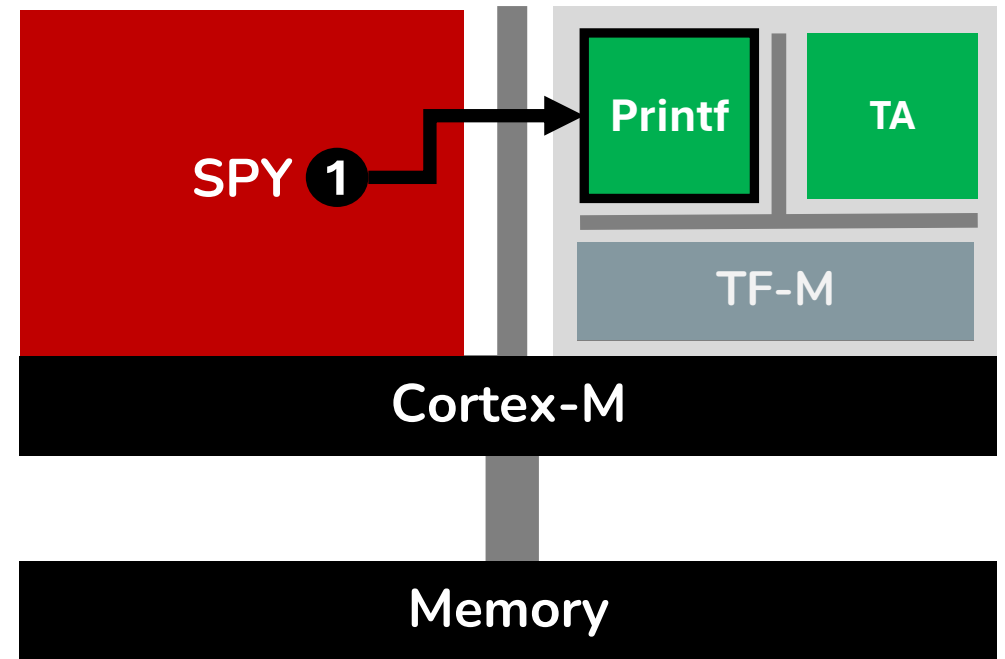


Recover Secrets

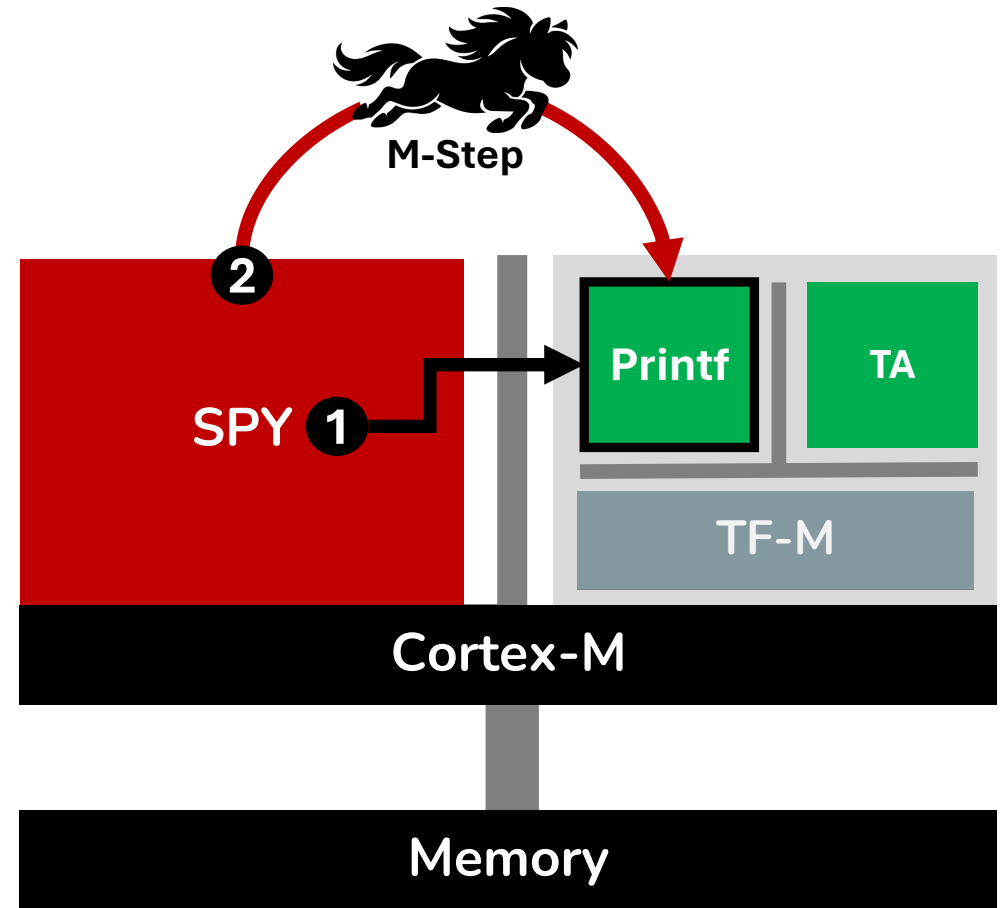
1st M-Step Use Case- Printf



1st M-Step Use Case- Printf

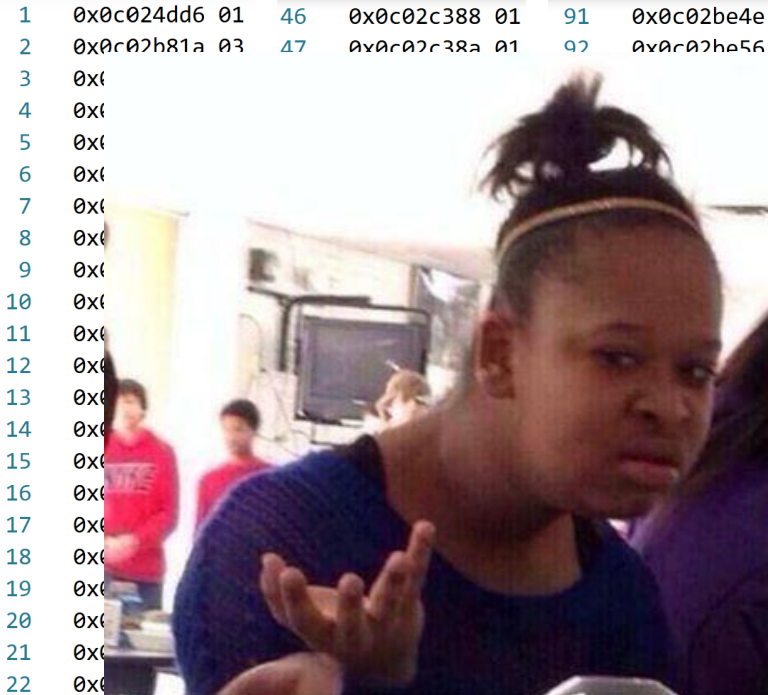


1st M-Step Use Case- Printf



Raw Trace

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2	0x0c02b81a 03	47	0x0c02c38a 01	92	0x0c02be56 01	137	0x0c01f1b4 02	182	0x0c029562 01	227	0x0c029542 01	272	0x0c029696 01	317	0x0c0296c6 02	362	0x0c0296cc 01
3	0x0c02b81e 01	48	0x0c02c38c 01	93	0x0c02be58 01	138	0x0c01f1b6 03	183	0x0c029542 01	228	0x0c02955e 01	273	0x0c029698 01	318	0x0c0296c8 01	363	0x0c029696 01
4	0x0c02b820 03	49	0x0c02c11e 03	94	0x0c02be5a 01	139	0x0c01f1ba 01	184	0x0c02955e 01	229	0x0c029560 02	274	0x0c0296c4 01	319	0x0c0296ca 02	364	0x0c029698 01
5	0x0c02b824 01	50	0x0c02c120 02	95	0x0c01f33c 01	140	0x0c01f1ce 02	185	0x0c029560 02	230	0x0c029562 01	275	0x0c0296c6 02	320	0x0c0296cc 01	365	0x0c0296c4 01
6	0x0c02b826 01	51	0x0c02c122 01	96	0x0c01f33e 02	141	0x0c01f1be 02	186	0x0c029562 01	231	0x0c029542 01	276	0x0c0296c8 01	321	0x0c029696 01	366	0x0c0296c6 02
7	0x0c02b828 01	52	0x0c02c12e 02	97	0x0c01f340 01	142	0x0c01f1c0 01	187	0x0c029542 01	232	0x0c029546 01	277	0x0c0296ca 02	322	0x0c029698 01	367	0x0c0296c8 01
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9	0x0c02b82c 01	54	0x0c02c134 02	99	0x0c01f0ee 03	144	0x0c01f1c4 01	189	0x0c029560 02	234	0x0c02954c 01	279	0x0c029696 01	324	0x0c0296c6 02	369	0x0c0296cc 01
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11	0x0c02be14 01	56	0x0c02c124 02	101	0x0c01f0f4 02	146	0x0c01f1c8 03	191	0x0c029542 01	236	0x0c01f186 01	281	0x0c0296c4 01	326	0x0c0296ca 02	371	0x0c029698 01
12	0x0c02be16 03	57	0x0c02c126 01	102	0x0c01f0f6 01	147	0x0c01f1cc 01	192	0x0c02955e 01	237	0x0c01f12e 01	282	0x0c0296c6 02	327	0x0c0296cc 01	372	0x0c02969c 01
13	0x0c02be1a 01	58	0x0c02c140 02	103	0x0c01f0f8 02	148	0x0c01f166 01	193	0x0c029560 02	238	0x0c01f130 04	283	0x0c0296c8 01	328	0x0c029696 01	373	0x0c0296a0 01
14	0x0c02b830 01	59	0x0c02c142 01	104	0x0c01f0fa 01	149	0x0c01f168 03	194	0x0c029562 01	239	0x0c02be5e 01	284	0x0c0296ca 02	329	0x0c029698 01	374	0x0c0296a2 01
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17	0x0c02b848 01	62	0x0c02c150 01	107	0x0c01f104 01	152	0x0c01f17a 01	197	0x0c029560 02	242	0x0c02be66 01	287	0x0c029698 01	332	0x0c0296c8 01	377	0x0c0296aa 02
18	0x0c02b84a 01	63	0x0c02c12a 01	108	0x0c01f108 03	153	0x0c01f17c 01	198	0x0c029562 01	243	0x0c02be7a 02	288	0x0c0296c4 01	333	0x0c0296ca 02	378	0x0c0296ae 04
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0x0c02952e 03	158	0x0c02952e 03	203	0x0c029542 01	248



0x0c02955e 01	225	0x0c029560 02	270
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Mathematical diagrams and formulas overlaid on a background image of a woman's face.

Formulas shown include:

- $V = \frac{1}{3} \pi r^2 h$ (Volume of a cone)
- $A = \pi r^2$ (Area of a circle)
- $C = 2\pi r$ (Circumference of a circle)
- $\sin x dx = -\cos x + C$
- $\int \frac{dx}{\cos^2 x} = \tan x + C$
- $\int \tan x dx = -\ln|\cos x| + C$
- $\int \frac{dx}{\sin x} = \ln \left| \frac{x}{2} \right| + C$
- $\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan \frac{x}{a} + C$
- $\int \frac{dx}{x} = \ln|x| + C$
- $\tan(\theta)$ graph
- $ax^2 + bx + c = 0$
- $a(x^2 + \frac{b}{a}x + \frac{c}{a}) = 0$
- $x^2 + 2\frac{b}{2a}x + (\frac{b}{2a})^2 - (\frac{b}{2a})^2 + \frac{c}{a} = 0$
- $(x + \frac{b}{2a})^2 - \frac{b^2 - 4ac}{4a^2} = 0$

0x0c02c2ec 01	296	0x0c0296c6 02	341	0x0c0296cc 01	386	0x0c02c326 01
0x0c02c2f0 02	297	0x0c0296c8 01	342	0x0c029696 01	387	0x0c02d102 03
0x0c02c2f2 02	298	0x0c0296ca 02	343	0x0c029698 01	388	0x0c02d104 01
0x0c02c2f4 01	299	0x0c0296cc 01	344	0x0c0296c4 01	389	0x0c02d106 01
0x0c02c2f6 01	300	0x0c029696 01	345	0x0c0296c6 02	390	0x0c02d108 01
0x0c02c2f8 01	301	0x0c029698 01	346	0x0c0296c8 01	391	0x0c02d10a 01
0x0c02c2fc 01	302	0x0c0296c4 01	347	0x0c0296ca 02	392	0x0c02d110 02
0x0c02c300 01	303	0x0c0296c6 02	348	0x0c0296cc 01	393	0x0c02d114 02
0x0c02968e 01	304	0x0c0296c8 01	349	0x0c029696 01	394	0x0c02d118 01
0x0c029690 03	305	0x0c0296ca 02	350	0x0c029698 01	395	0x0c02d11c 01
0x0c029692 01	306	0x0c0296cc 01	351	0x0c0296c4 01	396	0x0c02d120 01
0x0c0296b0 01	307	0x0c029696 01	352	0x0c0296c6 02	397	0x0c02d124 01
0x0c0296b4 01	308	0x0c029698 01	353	0x0c0296c8 01	398	0x0c02d128 01
0x0c029694 01	309	0x0c0296c4 01	354	0x0c0296ca 02	399	0x0c02d12a 02
0x0c029696 01	310	0x0c0296c6 02	355	0x0c0296cc 01	400	0x0c02d12e 01
0x0c029698 01	311	0x0c0296c8 01	356	0x0c029696 01	401	0x0c02d132 01
0x0c0296c4 01	312	0x0c0296ca 02	357	0x0c029698 01	402	0x0c02d134 01
0x0c0296c6 02	313	0x0c0296cc 01	358	0x0c0296c4 01	403	0x0c02d138 01
0x0c0296c8 01	314	0x0c029696 01	359	0x0c0296c6 02	404	0x0c02d10a 01
0x0c0296ca 02	315	0x0c029698 01	360	0x0c0296c8 01	405	0x0c02d10e 04

1	0x0c024dd6	01	46	0x0c02c388	01	91	0x0c02be4e	01	136	0x0c01f1b2	01	181	0x0c029560	02	226	0x0c029562	01	271	0x0c0296c0	01	316	0x0c0296c4	01	361	0x0c0296ca	01
2	0x0c02b81a	03	47	0x0c02c38a	01	92	0x0c02be56	01	137	0x0c01f1b4	02	182	0x0c029562	01	227	0x0c029542	01	272	0x0c029696	01	317	0x0c0296c6	02	362	0x0c0296cc	01
3	0x0c02b81e	01	48	0x0c02c38c	01	93	0x0c02be58	01	138	0x0c01f1b6	03	183	0x0c029542	01	228	0x0c02955e	01	273	0x0c029698	01	318	0x0c0296c8	01	363	0x0c029696	01
4	0x0c02b820	03	49	0x0c02c11e	03	94	0x0c02be5a	01	139	0x0c01f1ba	01	184	0x0c02955e	01	229	0x0c029560	02	274	0x0c0296c4	01	319	0x0c0296ca	02	364	0x0c029698	01
5	0x0c02b824	01	50	0x0c02c120	02	95	0x0c01f33c	01	140	0x0c01f1ce	02	185	0x0c029560	02	230	0x0c029562	01	275	0x0c0296c6	02	320	0x0c0296cc	01	365	0x0c0296c4	01
6	0x0c02b826	01	51	0x0c02c122	01	96	0x0c01f33e	02	141	0x0c01f1be	02	186	0x0c029562	01	231	0x0c029542	01	276	0x0c0296c8	01	321	0x0c029696	01	366	0x0c0296c6	02
7	0x0c02b828	01	52	0x0c02c12e	02	97	0x0c01f340	01	142	0x0c01f1c0	01	187	0x0c029542	01	232	0x0c029546	01	277	0x0c0296ca	02	322	0x0c029698	01	367	0x0c0296c8	01
8	0x0c02b82a	01	53	0x0c02c130	01	98	0x0c01f0ec	01	143	0x0c01f1c2	02	188	0x0c02955e	01	233	0x0c029548	01	278	0x0c0296cc	01	323	0x0c0296c4	01	368	0x0c0296ca	02
9	0x0c02b82c	01	54	0x0c02c134	02	99	0x0c01f0ee	03	144	0x0c01f1c4	01	189	0x0c029560	02	234	0x0c02954c	01	279	0x0c029696	01	324	0x0c0296c6	02	369	0x0c0296cc	01
10	0x0c02be12	01	55	0x0c02c138	01	100	0x0c01f0f2	02	145	0x0c01f1c6	02	190	0x0c029562	01	235	0x0c029550	04	280	0x0c029698	01	325	0x0c0296c8	01	370	0x0c029696	01
	0x0c02be14	01	56	0x0c02c124	02	101	0x0c01f0f4	02	146	0x0c01f1c8	03	191	0x0c029542	01	236	0x0c01f186	01	281	0x0c0296c4	01	326	0x0c0296ca	02	371	0x0c029698	01
12	0x0c02be16	03	57	0x0c02c126	01	102	0x0c01f0f6	01	147	0x0c01f1cc	01	192	0x0c02955e	01	237	0x0c01f12e	01	282	0x0c0296c6	02	327	0x0c0296cc	01	372	0x0c02969c	01
13	0x0c02be1a	01	58	0x0c02c12a	02								0x0c029560	02	238	0x0c01f130	04	283	0x0c0296c8	01	328	0x0c029696	01	373	0x0c0296a0	01
14	0x0c02b830	01	59	0x0c02c12e	02								0x0c029562	01	239	0x0c02be5e	01	284	0x0c0296ca	02	329	0x0c029698	01	374	0x0c0296a2	01
15	0x0c02b834	01	60	0x0c02c132	02								0x0c029542	01	240	0x0c02be60	01	285	0x0c0296cc	01	330	0x0c0296c4	01	375	0x0c0296a4	01
16	0x0c02b846	01	61	0x0c02c136	02								0x0c02955e	01	241	0x0c02be64	02	286	0x0c029696	01	331	0x0c0296c6	02	376	0x0c0296a6	01
17	0x0c02b848	01											0x0c029560	02	242	0x0c02be66	01	287	0x0c029698	01	332	0x0c0296c8	01	377	0x0c0296aa	01
18	0x0c02b84a	01											0x0c029562	01	243	0x0c02be7a	02	288	0x0c0296c4	01	333	0x0c0296ca	02	378	0x0c0296ae	04
19	0x0c02b84c	01											0x0c029542	01	244	0x0c02be7c	02	289	0x0c0296c6	02	334	0x0c0296cc	01	379	0x0c02c304	02
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22	0x0c02c40c	01											0x0c029562	01	247	0x0c02c2e2	01	292	0x0c0296cc	01	337	0x0c0296c4	01	382	0x0c02c31c	02
23	0x0c02c410	02											0x0c029542	01	248	0x0c02c2e4	01	293	0x0c029696	01	338	0x0c0296c6	02	383	0x0c02c320	01
24	0x0c02c412	01											0x0c02955e	01	249	0x0c02c2e6	02	294	0x0c029698	01	339	0x0c0296c8	01	384	0x0c02c322	02
25	0x0c02c414	01	70	0x0c02c410	01								0x0c029560	02	250	0x0c02c2e8	01	295	0x0c0296c4	01	340	0x0c0296ca	02	385	0x0c02c324	02
26	0x0c02c416	01	71	0x0c02c412	01								0x0c029562	01	251	0x0c02c2ec	01	296	0x0c0296c6	02	341	0x0c0296cc	01	386	0x0c02c326	01
27	0x0c02c41a	01	72	0x0c02c414	01								0x0c029542	01	252	0x0c02c2f0	02	297	0x0c0296c8	01	342	0x0c029696	01	387	0x0c02d102	03
28	0x0c02c41c	01	73	0x0c02c416	01								0x0c02955e	01	253	0x0c02c2f2	02	298	0x0c0296ca	02	343	0x0c029698	01	388	0x0c02d104	01
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30	0x0c02c422	02	75	0x0c02c2d6	02	120	0x0c01f148	01	165	0x0c02953e	01	210	0x0c029562	01	255	0x0c02c2f6	01	300	0x0c029696	01	345	0x0c0296c6	02	390	0x0c02d108	01
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35	0x0c02c3ce	01	80	0x0c02c340	01	125	0x0c01f18e	01	170	0x0c029562	01	215	0x0c029542	01	260	0x0c029690	03	305	0x0c0296ca	02	350	0x0c029698	01	395	0x0c02d11c	01
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37	0x0c02c36c	03	82	0x0c02c348	01	127	0x0c01f196	01	172	0x0c02955e	01	217	0x0c029560	02	262	0x0c0296b0	01	307	0x0c029696	01	352	0x0c0296c6	02	397	0x0c02d124	01
38	0x0c02c370	02	83	0x0c02c2dc	01	128	0x0c01f198	03	173	0x0c029560	02	218	0x0c029562	01	263	0x0c0296b4	01	308	0x0c029698	01	353	0x0c0296c8	01	398	0x0c02d128	01
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41	0x0c02c37a	01	86	0x0c02be3c	01	131	0x0c01f1a2	02	176	0x0c02955e	01	221	0x0c029560	02	266	0x0c029698	01	311	0x0c0296c8	01	356	0x0c029696	01	401	0x0c02d132	01
42	0x0c02c37e	01	87	0x0c02be3e	03	132	0x0c01f1a4	02	177	0x0c029560	02	222	0x0c029562	01	267	0x0c0296c4	01	312	0x0c0296ca	02	357	0x0c029698	01	402	0x0c02d134	01
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2	0x0c02b81a 03	47	0x0c02c38a 01	92	0x0c02be56 01	137	0x0c01f1b4 02	182	0x0c029562 01	227	0x0c029542 01	272	0x0c029696 01	317	0x0c0296c6 02	362	0x0c0296cc 01
3	0x0c02b81e 01	48	0x0c02c38c 01	93	0x0c02be58 01	138	0x0c01f1b6 03	183	0x0c029542 01	228	0x0c02955e 01	273	0x0c029698 01	318	0x0c0296c8 01	363	0x0c029696 01
4	0x0c02b820 03	49	0x0c02c11e 03	94	0x0c02be5a 01	139	0x0c01f1ba 01	184	0x0c02955e 01	229	0x0c029560 02	274	0x0c0296c4 01	319	0x0c0296ca 02	364	0x0c029698 01
5	0x0c02b824 01	50	0x0c02c120 02	95	0x0c01f33c 01	140	0x0c01f1ce 02	185	0x0c029560 02	230	0x0c029562 01	275	0x0c0296c6 02	320	0x0c0296cc 01	365	0x0c0296c4 01
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7	0x0c02b828 01	52	0x0c02c12e 02	97	0x0c01f340 01	142	0x0c01f1c0 01	187	0x0c029542 01	232	0x0c029546 01	277	0x0c0296ca 02	322	0x0c029698 01	367	0x0c0296c8 01
8	0x0c02b82a 01	53	0x0c02c130 01	98	0x0c01f0ec 01	143	0x0c01f1c2 02	188	0x0c02955e 01	233	0x0c029548 01	278	0x0c0296cc 01	323	0x0c0296c4 01	368	0x0c0296ca 02
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10	0x0c02be12 01	55	0x0c02c138 01	100	0x0c01f0f2 02	145	0x0c01f1c6 02	190	0x0c029562 01	235	0x0c029550 04	280	0x0c029698 01	325	0x0c0296c8 01	370	0x0c029696 01
	0x0c02be14 01	56	0x0c02c124 02	101	0x0c01f0f4 02	146	0x0c01f1c8 03	191	0x0c029542 01	236	0x0c01f186 01	281	0x0c0296c4 01	326	0x0c0296ca 02	371	0x0c029698 01
12	0x0c02be16 03	57	0x0c02c126 01	102	0x0c01f0f6 01	147	0x0c01f1cc 01	192	0x0c02955e 01	237	0x0c01f12e 01	282	0x0c0296c6 02	327	0x0c0296cc 01	372	0x0c02969c 01
13	0x0c02be1a 01	58	0x0c02c128 01						0x0c029560 02	238	0x0c01f130 04	283	0x0c0296c8 01	328	0x0c029696 01	373	0x0c0296a0 01
14	0x0c02b830 01	59	0x0c02c12a 01						0x0c029562 01	239	0x0c02be5e 01	284	0x0c0296ca 02	329	0x0c029698 01	374	0x0c0296a2 01
15	0x0c02b834 01	60	0x0c02c12c 01						0x0c029542 01	240	0x0c02be60 01	285	0x0c0296cc 01	330	0x0c0296c4 01	375	0x0c0296a4 01
16	0x0c02b846 01	61	0x0c02c12e 01						0x0c02955e 01	241	0x0c02be64 02	286	0x0c029696 01	331	0x0c0296c6 02	376	0x0c0296a6 01
17	0x0c02b848 01								0x0c029560 02	242	0x0c02be66 01	287	0x0c029698 01	332	0x0c0296c8 01	377	0x0c0296aa 01
18	0x0c02b84a 01								0x0c029562 01	243	0x0c02be7a 02	288	0x0c0296c4 01	333	0x0c0296ca 02	378	0x0c0296ae 04
19	0x0c02b84c 01								0x0c029542 01	244	0x0c02be7c 02	289	0x0c0296c6 02	334	0x0c0296cc 01	379	0x0c02c304 02
20	0x0c02c406 03								0x0c02955e 01	245	0x0c02be52 01	290	0x0c0296c8 01	335	0x0c029696 01	380	0x0c02c306 02
21	0x0c02c408 01								0x0c029560 02	246	0x0c02be4a 04	291	0x0c0296ca 02	336	0x0c029698 01	381	0x0c02c308 01
22	0x0c02c40c 01								0x0c029562 01	247	0x0c02c2e2 01	292	0x0c0296cc 01	337	0x0c0296c4 01	382	0x0c02c31c 02
23	0x0c02c410 02								0x0c029542 01	248	0x0c02c2e4 01	293	0x0c029696 01	338	0x0c0296c6 02	383	0x0c02c320 01
24	0x0c02c412 01								0x0c02955e 01	249	0x0c02c2e6 02	294	0x0c029698 01	339	0x0c0296c8 01	384	0x0c02c322 02
25	0x0c02c414 01	70	0x0c02c410 01						0x0c029560 02	250	0x0c02c2e8 01	295	0x0c0296c4 01	340	0x0c0296ca 02	385	0x0c02c324 02
26	0x0c02c416 01	71	0x0c02c412 01						0x0c029562 01	251	0x0c02c2ec 01	296	0x0c0296c6 02	341	0x0c0296cc 01	386	0x0c02c326 01
27	0x0c02c41a 01	72	0x0c02c414 01						0x0c029542 01	252	0x0c02c2f0 02	297	0x0c0296c8 01	342	0x0c029696 01	387	0x0c02d102 03
28	0x0c02c41c 01	73	0x0c02c416 01						0x0c02955e 01	253	0x0c02c2f2 02	298	0x0c0296ca 02	343	0x0c029698 01	388	0x0c02d104 01
29	0x0c02c41e 02	74	0x0c02c418 01						0x0c029560 02	254	0x0c02c2f4 01	299	0x0c0296cc 01	344	0x0c0296c4 01	389	0x0c02d106 01
30	0x0c02c422 02	75	0x0c02c2d6 02	120	0x0c01f148 01	165	0x0c02953e 01	210	0x0c029562 01	255	0x0c02c2f6 01	300	0x0c029696 01	345	0x0c0296c6 02	390	0x0c02d108 01
31	0x0c02c426 01	76	0x0c02c2d8 01	121	0x0c01f14a 01	166	0x0c029540 01	211	0x0c029542 01	256	0x0c02c2f8 01	301	0x0c029698 01	346	0x0c0296c8 01	391	0x0c02d10a 01
32	0x0c02c428 01	77	0x0c02c334 02	122	0x0c01f14c 01	167	0x0c029542 01	212	0x0c02955e 01	257	0x0c02c2fc 01	302	0x0c0296c4 01	347	0x0c0296ca 02	392	0x0c02d110 02
33	0x0c02c42a 02	78	0x0c02c338 01	123	0x0c01f150 03	168	0x0c02955e 01	213	0x0c029560 02	258	0x0c02c300 01	303	0x0c0296c6 02	348	0x0c0296cc 01	393	0x0c02d114 02
34	0x0c02c42c 01	79	0x0c02c33c 02	124	0x0c01f154 01	169	0x0c029560 02	214	0x0c029562 01	259	0x0c02968e 01	304	0x0c0296c8 01	349	0x0c029696 01	394	0x0c02d118 01
35	0x0c02c3ce 01	80	0x0c02c340 01	125	0x0c01f18e 01	170	0x0c029562 01	215	0x0c029542 01	260	0x0c029690 03	305	0x0c0296ca 02	350	0x0c029698 01	395	0x0c02d11c 01
36	0x0c02c3d2 01	81	0x0c02c346 02	126	0x0c01f192 01	171	0x0c029542 01	216	0x0c02955e 01	261	0x0c029692 01	306	0x0c0296cc 01	351	0x0c0296c4 01	396	0x0c02d120 01
37	0x0c02c36c 03	82	0x0c02c348 01	127	0x0c01f196 01	172	0x0c02955e 01	217	0x0c029560 02	262	0x0c0296b0 01	307	0x0c029696 01	352	0x0c0296c6 02	397	0x0c02d124 01
38	0x0c02c370 02	83	0x0c02c2dc 01	128	0x0c01f198 03	173	0x0c029560 02	218	0x0c029562 01	263	0x0c0296b4 01	308	0x0c029698 01	353	0x0c0296c8 01	398	0x0c02d128 01
39	0x0c02c374 02	84	0x0c02c2de 01	129	0x0c01f19c 02	174	0x0c029562 01	219	0x0c029542 01	264	0x0c029694 01	309	0x0c0296c4 01	354	0x0c0296ca 02	399	0x0c02d12a 02
40	0x0c02c378 01	85	0x0c02be38 01	130	0x0c01f19e 01	175	0x0c029542 01	220	0x0c02955e 01	265	0x0c029696 01	310	0x0c0296c6 02	355	0x0c0296cc 01	400	0x0c02d12e 01
41	0x0c02c37a 01	86	0x0c02be3c 01	131	0x0c01f1a2 02	176	0x0c02955e 01	221	0x0c029560 02	266	0x0c029698 01	311	0x0c0296c8 01	356	0x0c029696 01	401	0x0c02d132 01
42	0x0c02c37e 01	87	0x0c02be3e 03	132	0x0c01f1a4 02	177	0x0c029560 02	222	0x0c029562 01	267	0x0c0296c4 01	312	0x0c0296ca 02	357	0x0c029698 01	402	0x0c02d134 01
43	0x0c02c380 01	88	0x0c02be40 01	133	0x0c01f1a6 02	178	0x0c029562 01	223	0x0c029542 01	268	0x0c0296c6 02	313	0x0c0296cc 01	358	0x0c0296c4 01	403	0x0c02d138 01
44	0x0c02c382 01	89	0x0c02be42 01	134	0x0c01f1aa 01	179	0x0c029542 01	224	0x0c02955e 01	269	0x0c0296c8 01	314	0x0c029696 01	359	0x0c0296c6 02	404	0x0c02d10a 01
45	0x0c02c384 01	90	0x0c02be4c 02	135	0x0c01f1ae 02	180	0x0c02955e 01	225	0x0c029560 02	270	0x0c0296ca 02	315	0x0c029698 01	360	0x0c0296c8 01	405	0x0c02d10e 04

1	0x0c024dd0	01	46	0x0c02c388	01	91	0x0c02be4e	01	136	0x0c01f1b2	01	181	0x0c029560	02	226	0x0c029562	01	271	0x0c0296c0	01	316	0x0c0296c4	01	361	0x0c0296ca	01												
2	0x0c02b81a	03	47	0x0c02c38a	01	92	0x0c02be56	01	137	0x0c01f1b4	02	182	0x0c029562	01	227	0x0c029542	01	272	0x0c029696	01	317	0x0c0296c6	02	362	0x0c0296cc	01												
3	0x0c02b81e	01	48	0x0c02c38c	01	93	0x0c02be58	01	138	0x0c01f1b6	03	183	0x0c029542	01	228	0x0c02955e	01	273	0x0c029698	01	318	0x0c0296c8	01	363	0x0c029696	01												
4	0x0c02b820	03	49	0x0c02c11e	03	94	0x0c02be5a	01	139	0x0c01f1ba	01	184	0x0c02955e	01	229	0x0c029560	02	274	0x0c0296c4	01	319	0x0c0296ca	02	364	0x0c029698	01												
5	0x0c02b824	01	50	0x0c02c120	02	95	0x0c01f33c	01	140	0x0c01f1ce	02	185	0x0c029560	02	230	0x0c029562	01	275	0x0c0296c6	02	320	0x0c0296cc	01	365	0x0c0296c4	01												
6	0x0c02b826	01	51	0x0c02c122	01	96	0x0c01f33e	02	141	0x0c01f1be	02	186	0x0c029562	01	231	0x0c029542	01	276	0x0c0296c8	01	321	0x0c029696	01	366	0x0c0296c6	02												
7	0x0c02b828	01	52	0x0c02c12e	02	97	0x0c01f340	01	142	0x0c01f1c0	01	187	0x0c029542	01	232	0x0c029546	01	277	0x0c0296ca	02	322	0x0c029698	01	367	0x0c0296c8	01												
8	0x0c02b82a	01	53	0x0c02c130	01	98	0x0c01f0ec	01	143	0x0c01f1c2	02	188	0x0c02955e	01	233	0x0c029548	01	278	0x0c0296cc	01	323	0x0c0296c4	01	368	0x0c0296ca	02												
9	0x0c02b82c	01	54	0x0c02c134	02	99	0x0c01f0ee	03	144	0x0c01f1c4	01	189	0x0c029560	02	234	0x0c02954c	01	279	0x0c029696	01	324	0x0c0296c6	02	369	0x0c0296cc	01												
10	0x0c02be12	01	55	0x0c02c138	01	100	0x0c01f0f2	02	145	0x0c01f1c6	02	190	0x0c029562	01	235	0x0c029550	04	280	0x0c029698	01	325	0x0c0296c8	01	370	0x0c029696	01												
	0x0c02be14	01	56	0x0c02c124	02	101	0x0c01f0f4	02	146	0x0c01f1c8	03	191	0x0c029542	01	236	0x0c01f186	01	281	0x0c0296c4	01	326	0x0c0296ca	02	371	0x0c029698	01												
12	0x0c02be16	03	57	0x0c02c126	01	102	0x0c01f0f6	01	147	0x0c01f18a	01																											
13	0x0c02be1a	01	58	0x0c02c128	01																																	
14	0x0c02b830	01	59	0x0c02c12a	01																																	
15	0x0c02b834	01	60	0x0c02c12c	01																																	
16	0x0c02b846	01	61	0x0c02c12e	01																																	
17	0x0c02b848	01																																				
18	0x0c02b84a	01																																				
19	0x0c02b84c	01																																				
20	0x0c02c406	03																																				
21	0x0c02c408	01																																				
22	0x0c02c40c	01																																				
23	0x0c02c410	02																																				
24	0x0c02c412	01																																				
25	0x0c02c414	01	70	0x0c02c416	01	71	0x0c02c418	01	72	0x0c02c41a	01	73	0x0c02c41c	01	74	0x0c02c41e	02	75	0x0c02c420	02	120	0x0c01f148	01	165	0x0c02953e	01	210											
26	0x0c02c416	01	71	0x0c02c418	01	72	0x0c02c41a	01	73	0x0c02c41c	01	74	0x0c02c41e	02	75	0x0c02c420	02	120	0x0c01f148	01	165	0x0c02953e	01	210	0x0c029562	01	255	0x0c02c2f6	01	300	0x0c029696	01	345	0x0c0296c6	02	390	0x0c02d108	01
27	0x0c02c41a	01	72	0x0c02c41c	01	73	0x0c02c41e	02	74	0x0c02c420	02	75	0x0c02c422	02	76	0x0c02c424	01	121	0x0c01f14a	01	166	0x0c029540	01	211	0x0c029542	01	256	0x0c02c2f8	01	301	0x0c029698	01	346	0x0c0296c8	01	391	0x0c02d10a	01
28	0x0c02c41c	01	73	0x0c02c41e	02	74	0x0c02c420	02	75	0x0c02c422	02	76	0x0c02c424	01	77	0x0c02c426	01	122	0x0c01f14c	01	167	0x0c029542	01	212	0x0c02955e	01	257	0x0c02c2fc	01	302	0x0c0296c4	01	347	0x0c0296ca	02	392	0x0c02d110	02
29	0x0c02c41e	02	74	0x0c02c420	02	75	0x0c02c422	02	76	0x0c02c424	01	77	0x0c02c426	01	78	0x0c02c428	01	123	0x0c01f150	03	168	0x0c02955e	01	213	0x0c029560	02	258	0x0c02c300	01	303	0x0c0296c6	02	348	0x0c0296cc	01	393	0x0c02d114	02
30	0x0c02c422	02	75	0x0c02c424	01	76	0x0c02c426	01	77	0x0c02c428	01	78	0x0c02c42a	02	79	0x0c02c42c	01	124	0x0c01f154	01	169	0x0c029560	02	214	0x0c029562	01	259	0x0c02968e	01	304	0x0c0296c8	01	349	0x0c029696	01	394	0x0c02d118	01
31	0x0c02c426	01	76	0x0c02c428	01	77	0x0c02c42a	02	78	0x0c02c42c	01	79	0x0c02c42e	01	80	0x0c02c430	01	125	0x0c01f158	01	170	0x0c029562	01	215	0x0c029542	01	260	0x0c029690	03	305	0x0c0296ca	02	350	0x0c029698	01	395	0x0c02d11c	01
32	0x0c02c428	01	77	0x0c02c42a	02	78	0x0c02c42c	01	79	0x0c02c42e	01	80	0x0c02c430	01	81	0x0c02c432	01	126	0x0c01f15c	01	171	0x0c029542	01	216	0x0c02955e	01	261	0x0c029692	01	306	0x0c0296cc	01	351	0x0c0296c4	01	396	0x0c02d120	01
33	0x0c02c42a	02	78	0x0c02c42c	01	79	0x0c02c42e	01	80	0x0c02c430	01	81	0x0c02c432	01	82	0x0c02c434	01	127	0x0c01f160	01	172	0x0c02955e	01	217	0x0c029560	02	262	0x0c0296b0	01	307	0x0c029696	01	352	0x0c0296c6	02	397	0x0c02d124	01
34	0x0c02c42c	01	79	0x0c02c42e	01	80	0x0c02c430	01	81	0x0c02c432	01	82	0x0c02c434	01	83	0x0c02c436	01	128	0x0c01f164	01	173	0x0c029560	02	218	0x0c029562	01	263	0x0c0296b4	01	308	0x0c029698	01	353	0x0c0296c8	01	398	0x0c02d128	01
35	0x0c02c3ce	01	80	0x0c02c438	01	81	0x0c02c43a	01	82	0x0c02c43c	01	83	0x0c02c43e	01	84	0x0c02c440	01	129	0x0c01f168	01	174	0x0c029562	01	219	0x0c029542	01	264	0x0c02969a	01	309	0x0c0296c4	01	354	0x0c0296ca	02	399	0x0c02d12a	02
36	0x0c02c3d2	01	81</																																			

Program Counter of the Instruction

0x0c02c406 03

1	0x0c024dd6	01	46	0x0c02c388	01	91	0x0c02be4e	01	136	0x0c01f1b2	01	181	0x0c029560	02	226	0x0c029562	01	271	0x0c0296cc	01	316	0x0c0296c4	01	361	0x0c0296ca	01
2	0x0c02b81a	03	47	0x0c02c38a	01	92	0x0c02be56	01	137	0x0c01f1b4	02	182	0x0c029562	01	227	0x0c029542	01	272	0x0c029696	01	317	0x0c0296c6	02	362	0x0c0296cc	01
3	0x0c02b81e	01	48	0x0c02c38c	01	93	0x0c02be58	01	138	0x0c01f1b6	03	183	0x0c029542	01	228	0x0c02955e	01	273	0x0c029698	01	318	0x0c0296c8	01	363	0x0c029696	01
4	0x0c02b820	03	49	0x0c02c11e	03	94	0x0c02be5a	01	139	0x0c01f1ba	01	184	0x0c02955e	01	229	0x0c029560	02	274	0x0c0296c4	01	319	0x0c0296ca	02	364	0x0c029698	01
5	0x0c02b824	01	50	0x0c02c120	02	95	0x0c01f33c	01	140	0x0c01f1ce	02	185	0x0c029560	02	230	0x0c029562	01	275	0x0c0296c6	02	320	0x0c0296cc	01	365	0x0c0296c4	01
6	0x0c02b826	01	51	0x0c02c122	01	96	0x0c01f33e	02	141	0x0c01f1be	02	186	0x0c029562	01	231	0x0c029542	01	276	0x0c0296c8	01	321	0x0c029696	01	366	0x0c0296c6	02
7	0x0c02b828	01	52	0x0c02c12e	02	97	0x0c01f340	01	142	0x0c01f1c0	01	187	0x0c029542	01	232	0x0c029546	01	277	0x0c0296ca	02	322	0x0c029698	01	367	0x0c0296c8	01
8	0x0c02b82a	01	53	0x0c02c130	01	98	0x0c01f0ec	01	143	0x0c01f1c2	02	188	0x0c02955e	01	233	0x0c029548	01	278	0x0c0296cc	01	323	0x0c0296c4	01	368	0x0c0296ca	02
9	0x0c02b82c	01	54	0x0c02c134	02	99	0x0c01f0ee	03	144	0x0c01f1c4	01	189	0x0c029560	02	234	0x0c02954c	01	279	0x0c029696	01	324	0x0c0296c6	02	369	0x0c0296cc	01
10	0x0c02be12	01	55	0x0c02c138	01	100	0x0c01f0f2	02	145	0x0c01f1c6	02	190	0x0c029562	01	235	0x0c029550	04	280	0x0c029698	01	325	0x0c0296c8	01	370	0x0c029696	01
11	0x0c02be14	01	56	0x0c02c124	02	101	0x0c01f0f4	02	146	0x0c01f1c8	03	191	0x0c029542	01	236	0x0c01f186	01	281	0x0c0296c4	01	326	0x0c0296ca	02	371	0x0c029698	01
12	0x0c02be16	03	57	0x0c02c126	01	102	0x0c01f0f6	01	147	0x0c01f188	01	192	0x0c029542	01	237	0x0c029544	01	282	0x0c0296cc	01	327	0x0c0296c6	02	372	0x0c02969c	01
13	0x0c02be1a	01	58	0x0c02c128	01	103	0x0c01f0f8	01	148	0x0c01f18a	01	193	0x0c029542	01	238	0x0c029546	01	283	0x0c0296cc	01	328	0x0c0296c8	01	373	0x0c0296a0	01
14	0x0c02b830	01	59	0x0c02c12a	01	104	0x0c01f0fa	01	149	0x0c01f18c	01	194	0x0c029542	01	239	0x0c029548	01	284	0x0c0296cc	01	329	0x0c0296c4	01	374	0x0c0296a2	01
15	0x0c02b834	01	60	0x0c02c12c	01	105	0x0c01f0fc	01	150	0x0c01f18e	01	195	0x0c029542	01	240	0x0c02be60	01	285	0x0c0296cc	01	330	0x0c0296c4	01	375	0x0c0296a4	01
16	0x0c02b846	01	61	0x0c02c12e	01	106	0x0c01f0fe	01	151	0x0c01f190	01	196	0x0c02955e	01	241	0x0c02be64	02	286	0x0c029696	01	331	0x0c0296c6	02	376	0x0c0296a6	01
17	0x0c02b848	01	62	0x0c02c130	01	107	0x0c01f100	01	152	0x0c01f192	01	197	0x0c029560	02	242	0x0c02be66	01	287	0x0c029698	01	332	0x0c0296c8	01	377	0x0c0296aa	01
18	0x0c02b84a	01	63	0x0c02c132	01	108	0x0c01f102	01	153	0x0c01f194	01	198	0x0c029562	01	243	0x0c02be7a	02	288	0x0c0296c4	01	333	0x0c0296ca	02	378	0x0c0296ae	04
19	0x0c02b84c	01	64	0x0c02c134	01	109	0x0c01f104	01	154	0x0c01f196	01	199	0x0c029542	01	244	0x0c02be7c	02	289	0x0c0296c6	02	334	0x0c0296cc	01	379	0x0c02c304	02
20	0x0c02c406	03	65	0x0c02c136	01	110	0x0c01f106	01	155	0x0c01f198	01	200	0x0c02955e	01	245	0x0c02be52	01	290	0x0c0296c8	01	335	0x0c029696	01	380	0x0c02c306	02
21	0x0c02c408	01	66	0x0c02c138	01	111	0x0c01f108	01	156	0x0c01f19a	01	201	0x0c029560	02	246	0x0c02be4a	04	291	0x0c0296ca	02	336	0x0c029698	01	381	0x0c02c308	01
22	0x0c02c40c	01	67	0x0c02c13a	01	112	0x0c01f10a	01	157	0x0c01f19c	01	202	0x0c029562	01	247	0x0c02c2e2	01	292	0x0c0296cc	01	337	0x0c0296c4	01	382	0x0c02c31c	02
23	0x0c02c410	02	68	0x0c02c13c	01	113	0x0c01f10c	01	158	0x0c01f19e	01	203	0x0c029542	01	248	0x0c02c2e4	01	293	0x0c0296cc	01	338	0x0c0296c6	02	383	0x0c02c320	01
24	0x0c02c412	01	69	0x0c02c13e	01	114	0x0c01f10e	01	159	0x0c01f1a0	01	204	0x0c029542	01	249	0x0c02c2e6	01	294	0x0c0296cc	01	339	0x0c0296c8	01	384	0x0c02c322	02
25	0x0c02c414	01	70	0x0c02c140	01	115	0x0c01f110	01	160	0x0c01f1a2	01	205	0x0c029542	01	250	0x0c02c2e8	01	295	0x0c0296cc	01	340	0x0c0296ca	02	385	0x0c02c324	02
26	0x0c02c416	01	71	0x0c02c142	01	116	0x0c01f112	01	161	0x0c01f1a4	01	206	0x0c029562	01	251	0x0c02c2ec	01	296	0x0c0296c6	02	341	0x0c0296cc	01	386	0x0c02c326	01
27	0x0c02c41a	01	72	0x0c02c144	01	117	0x0c01f114	01	162	0x0c01f1a6	01	207	0x0c029542	01	252	0x0c02c2f0	02	297	0x0c0296c8	01	342	0x0c029696	01	387	0x0c02d102	03
28	0x0c02c41c	01	73	0x0c02c146	01	118	0x0c01f116	01	163	0x0c01f1a8	01	208	0x0c02955e	01	253	0x0c02c2f2	02	298	0x0c0296ca	02	343	0x0c029698	01	388	0x0c02d104	01
29	0x0c02c41e	02	74	0x0c02c148	01	119	0x0c01f118	01	164	0x0c01f1aa	01	209	0x0c029560	02	254	0x0c02c2f4	01	299	0x0c0296cc	01	344	0x0c0296c4	01	389	0x0c02d106	01
30	0x0c02c422	02	75	0x0c02c2d6	02	120	0x0c01f148	01	165	0x0c02953e	01	210	0x0c029562	01	255	0x0c02c2f6	01	300	0x0c029696	01	345	0x0c0296c6	02	390	0x0c02d108	01
31	0x0c02c426	01	76	0x0c02c2d8	01	121	0x0c01f14a	01	166	0x0c029540	01	211	0x0c029542	01	256	0x0c02c2f8	01	301	0x0c029698	01	346	0x0c0296c8	01	391	0x0c02d10a	01
32	0x0c02c428	01	77	0x0c02c334	02	122	0x0c01f14c	01	167	0x0c029542	01	212	0x0c02955e	01	257	0x0c02c2fc	01	302	0x0c0296c4	01	347	0x0c0296ca	02	392	0x0c02d110	02
33	0x0c02c42a	02	78	0x0c02c338	01	123	0x0c01f150	03	168	0x0c02955e	01	213	0x0c029560	02	258	0x0c02c300	01	303	0x0c0296c6	02	348	0x0c0296cc	01	393	0x0c02d114	02
34	0x0c02c42c	01	79	0x0c02c33c	02	124	0x0c01f154	01	169	0x0c029560	02	214	0x0c029562	01	259	0x0c02968e	01	304	0x0c0296c8	01	349	0x0c029696	01	394	0x0c02d118	01
35	0x0c02c3ce	01	80	0x0c02c340	01	125	0x0c01f18e	01	170	0x0c029562	01	215	0x0c029542	01	260	0x0c029690	03	305	0x0c0296ca	02	350	0x0c029698	01	395	0x0c02d11c	01
36	0x0c02c3d2	01	81	0x0c02c346	02	126	0x0c01f192	01	171	0x0c029542	01	216	0x0c02955e	01	261	0x0c029692	01	306	0x0c0296cc	01	351	0x0c0296c4	01	396	0x0c02d120	01
37	0x0c02c36c	03	82	0x0c02c348	01	127	0x0c01f196	01	172	0x0c02955e	01	217	0x0c029560	02	262	0x0c0296b0	01	307	0x0c029696	01	352	0x0c0296c6	02	397	0x0c02d124	01
38	0x0c02c370	02	83	0x0c02c2dc	01	128	0x0c01f198	03	173	0x0c029560	02	218	0x0c029562	01	263	0x0c0296b4	01	308	0x0c029698	01	353	0x0c0296c8	01	398	0x0c02d128	01
39	0x0c02c374	02	84	0x0c02c2de	01	129	0x0c01f19c	02	174	0x0c029562	01	219	0x0c029542	01	264	0x0c029694	01	309	0x0c0296c4	01	354	0x0c0296ca	02	399	0x0c02d12a	02
40	0x0c02c378	01	85	0x0c02be38	01	130	0x0c01f19e	01	175	0x0c029542	01	220	0x0c02955e	01	265	0x0c029696	01	310	0x0c0296c6	02	355	0x0c0296cc	01	400	0x0c02d12e	01
41	0x0c02c37a	01	86	0x0c02be3c	01	131	0x0c01f1a2	02	176	0x0c02955e	01	221	0x0c029560	02	266	0x0c029698	01	311	0x0c0296c8	01	356	0x0c029696	01	401	0x0c02d132	01
42	0x0c02c37e	01	87	0x0c02be3e	03	132	0x0c01f1a4	02	177	0x0c029560	02	222	0x0c029562	01	267	0x0c0296c4	01	312	0x0c0296ca	02	357	0x0c029698	01	402	0x0c02d134	01
43	0x0c02c380	01	88	0x0c02be40	01	133	0x0c01f1a6	02	178	0x0c029562	01	223	0x0c029542	01	268	0x0c0296c6	02	313	0x0c0296cc	01	358	0x0c0296c4	01	403	0x0c02d138	01
44	0x0c02c382	01	89	0x0c02be42	01	134	0x0c01f1aa	01	179	0x0c029542	01	224	0x0c02955e	01	269	0x0c0296c8	01	314	0x0c029696	01	359	0x0c0296c6	02	404	0x0c02d10a	01
45	0x0c02c384	01	90	0x0c02be4c	02	135	0x0c01f1ae	02	180	0x0c02955e	01	225	0x0c029560	02	270	0x0c0296ca	02	315	0x0c029698	01	360	0x0c0296c8	01	405	0x0c02d10e	04

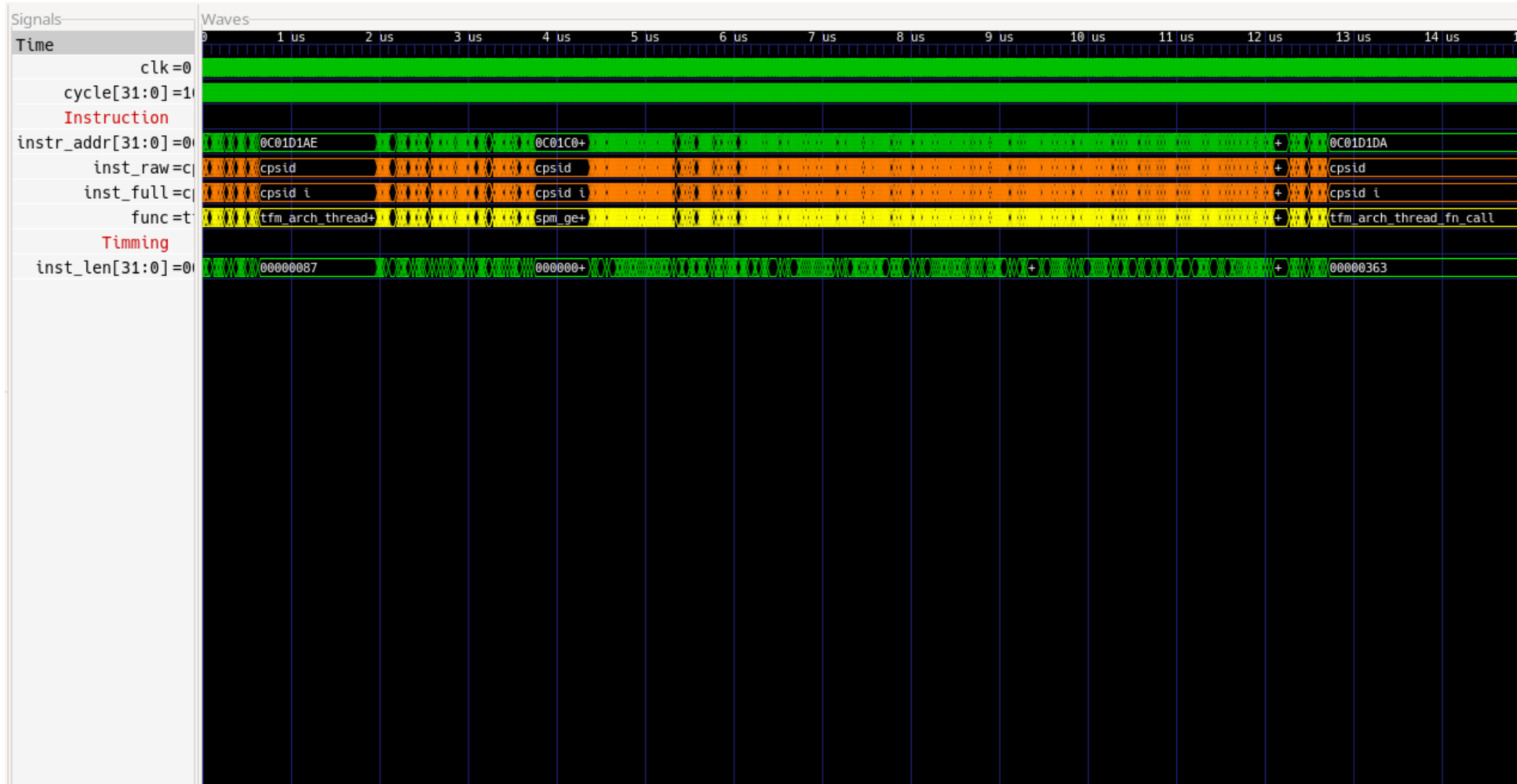
Program Counter of the Instruction

Interrupt Latency

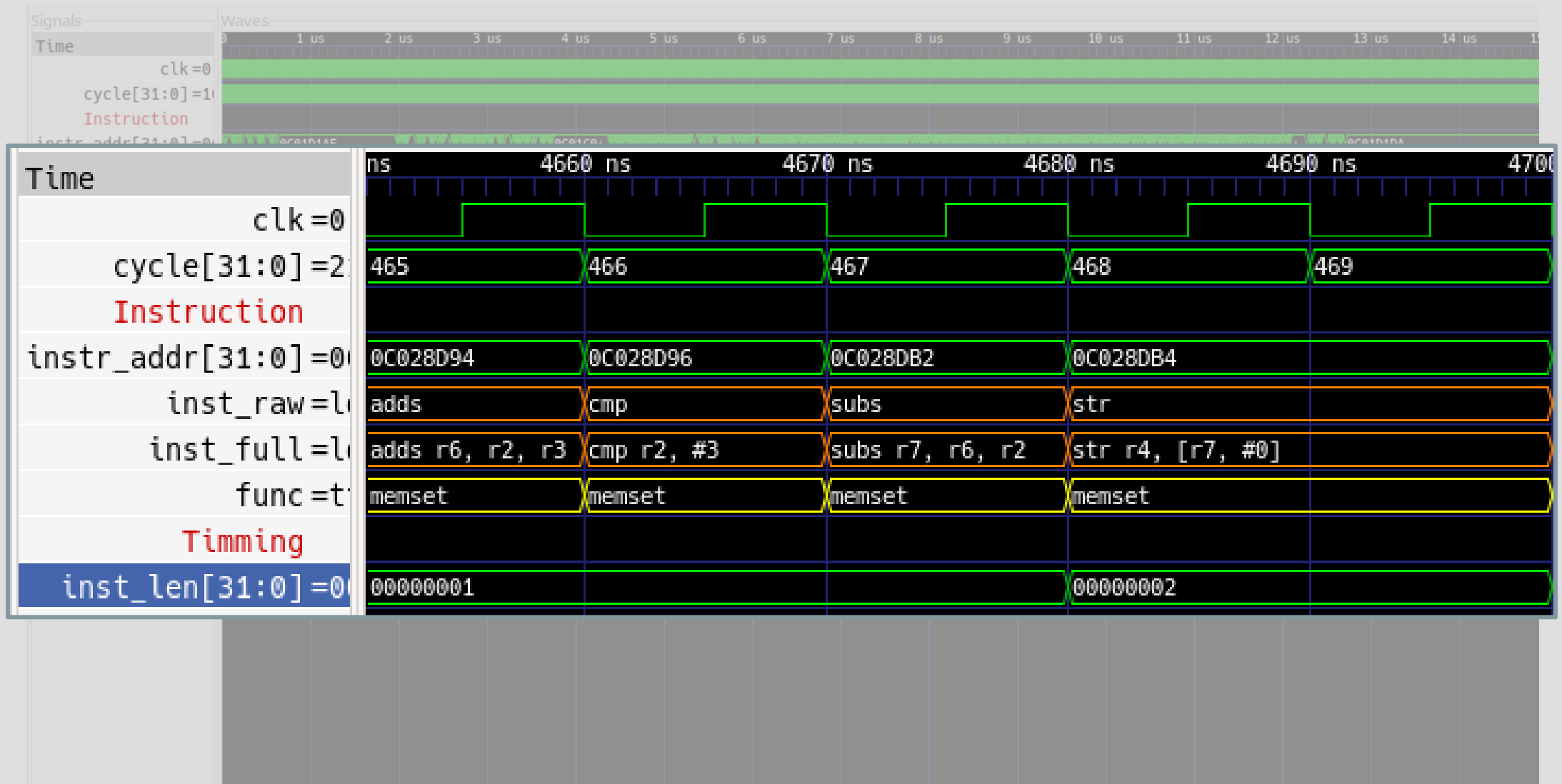


Mstp-Visualizer

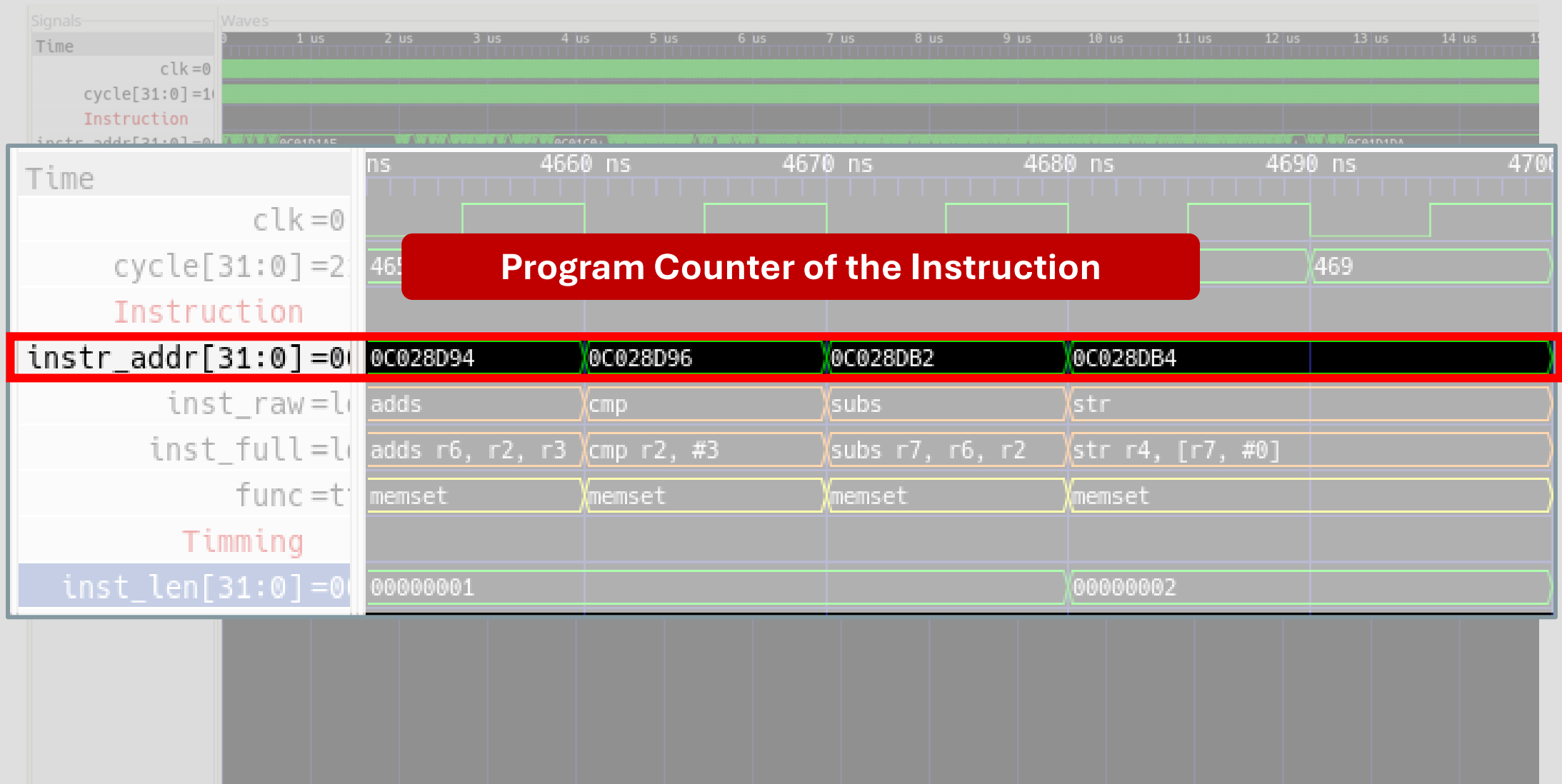
M-Step Visualizer (GTKwave)



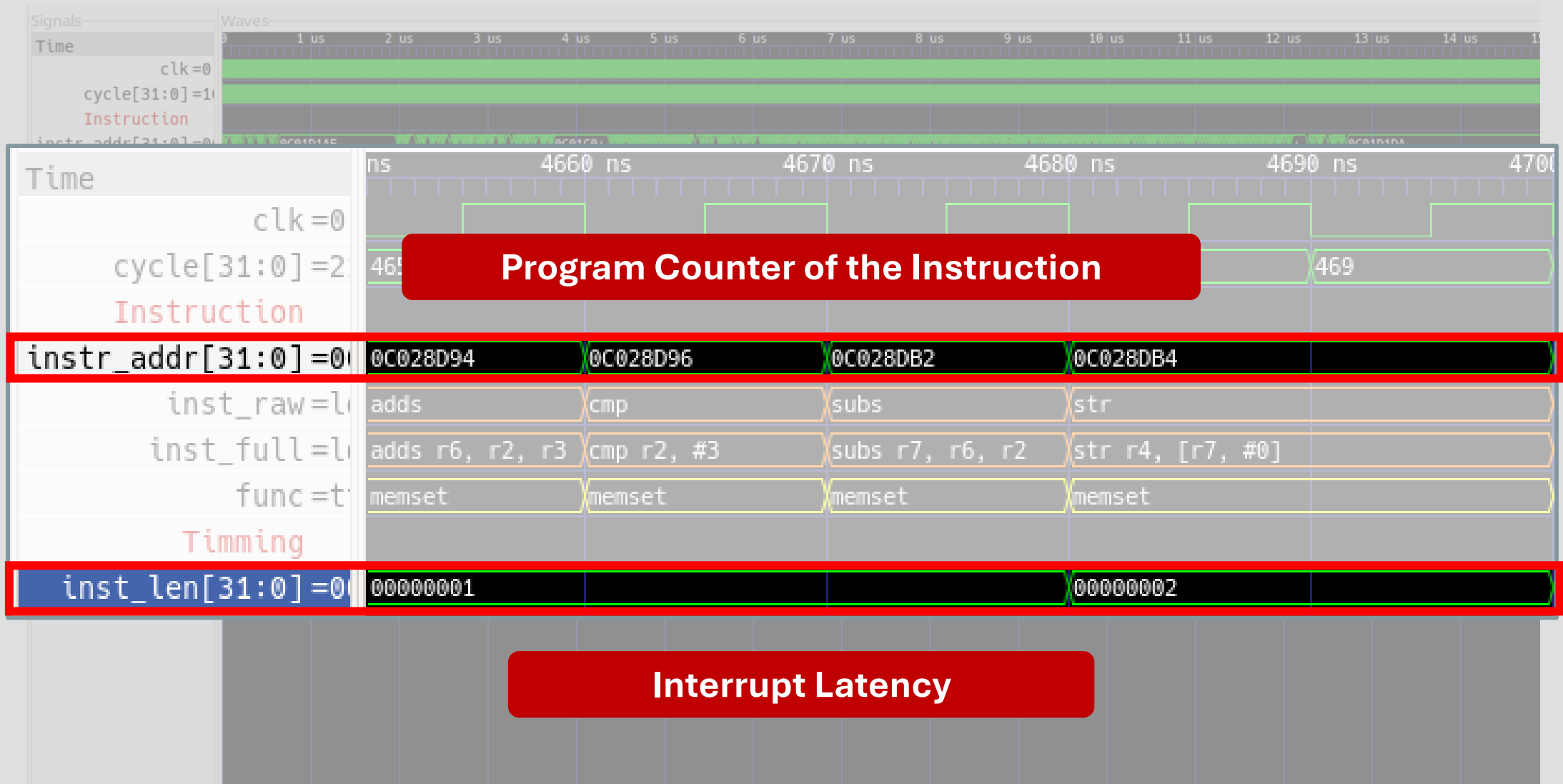
M-Step Visualizer (GTKwave)



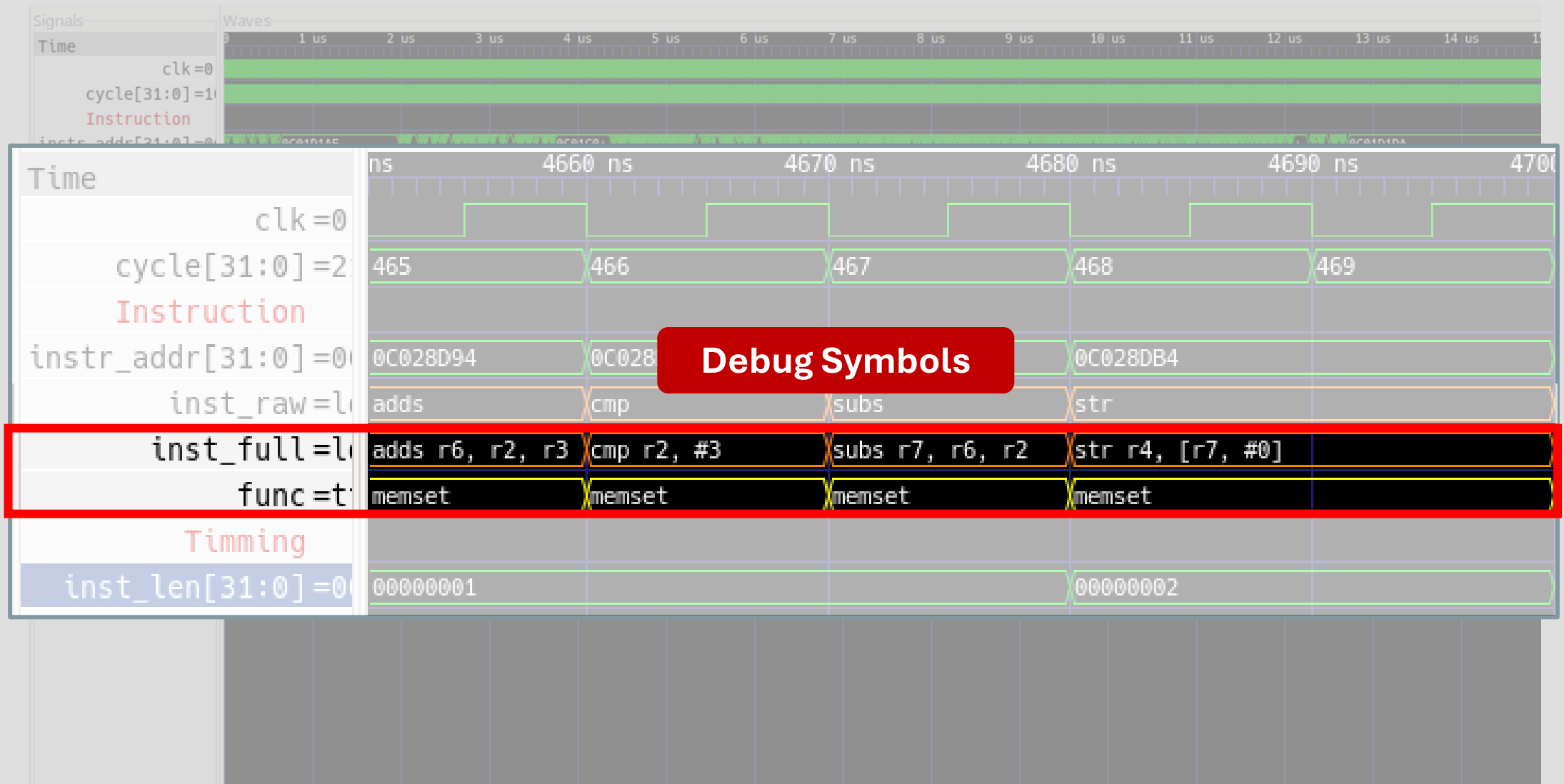
M-Step Visualizer (GTKwave)



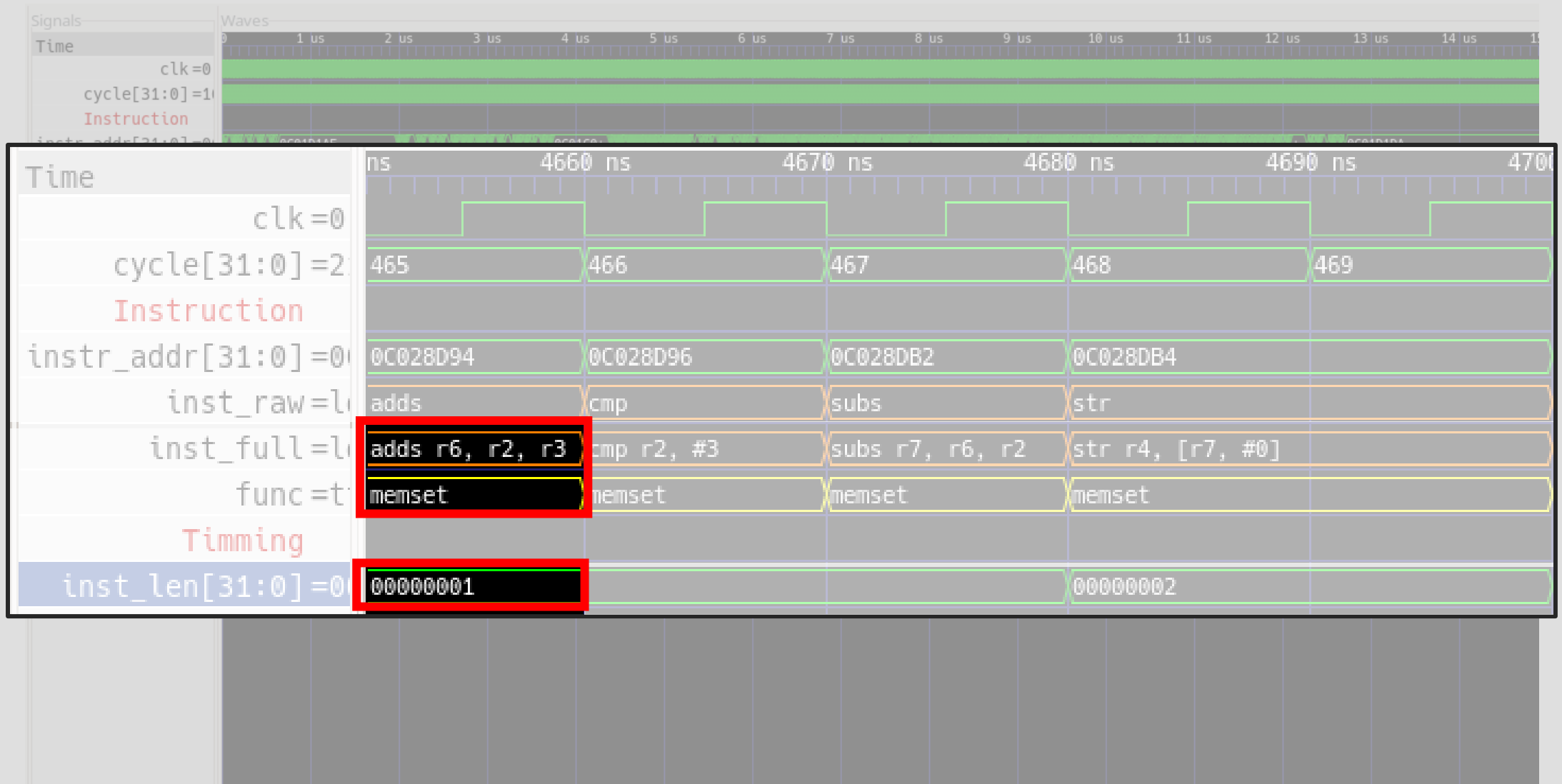
M-Step Visualizer (GTKwave)



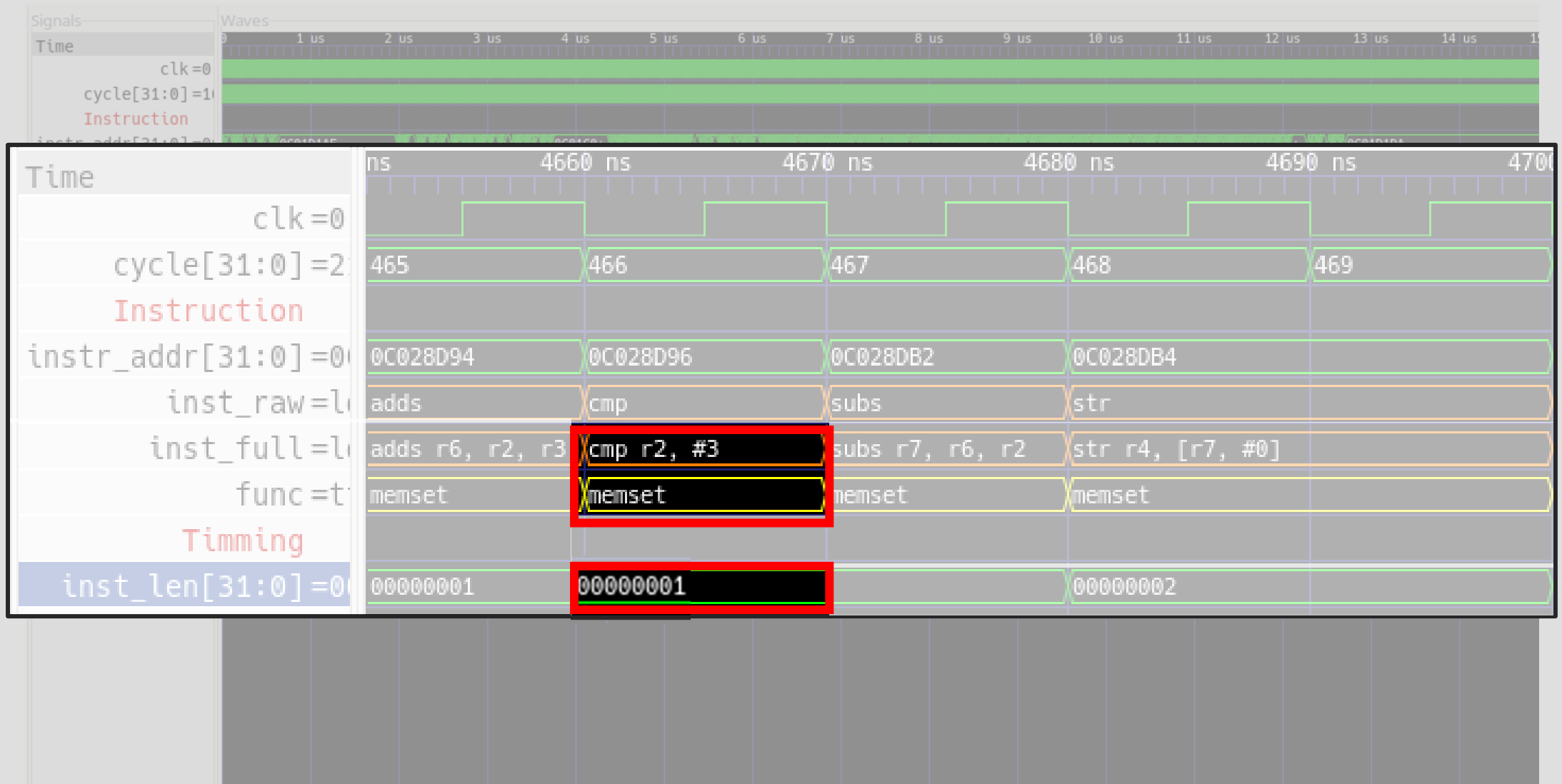
M-Step Visualizer (GTKwave)



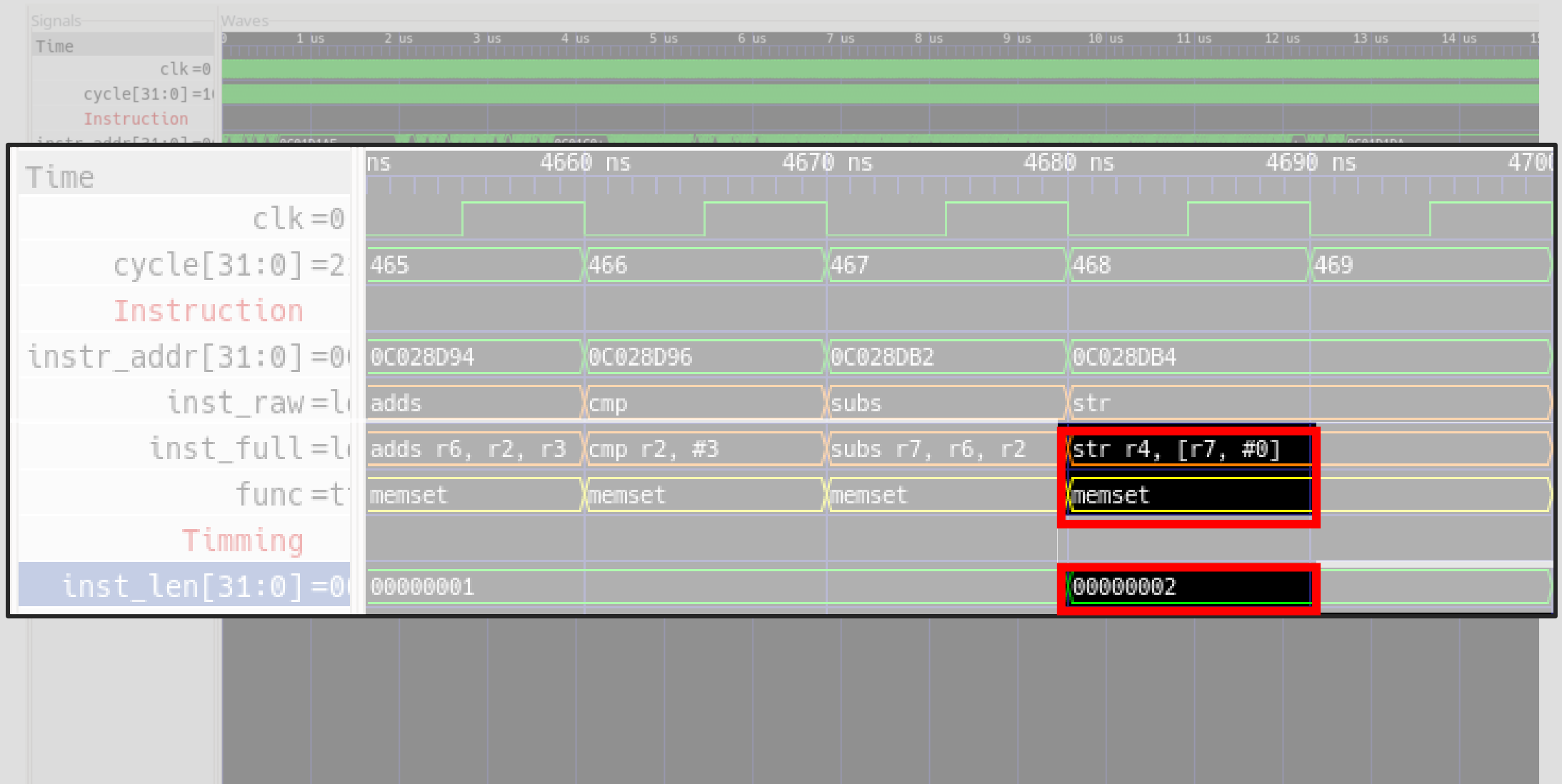
M-Step Visualizer (GTKwave)

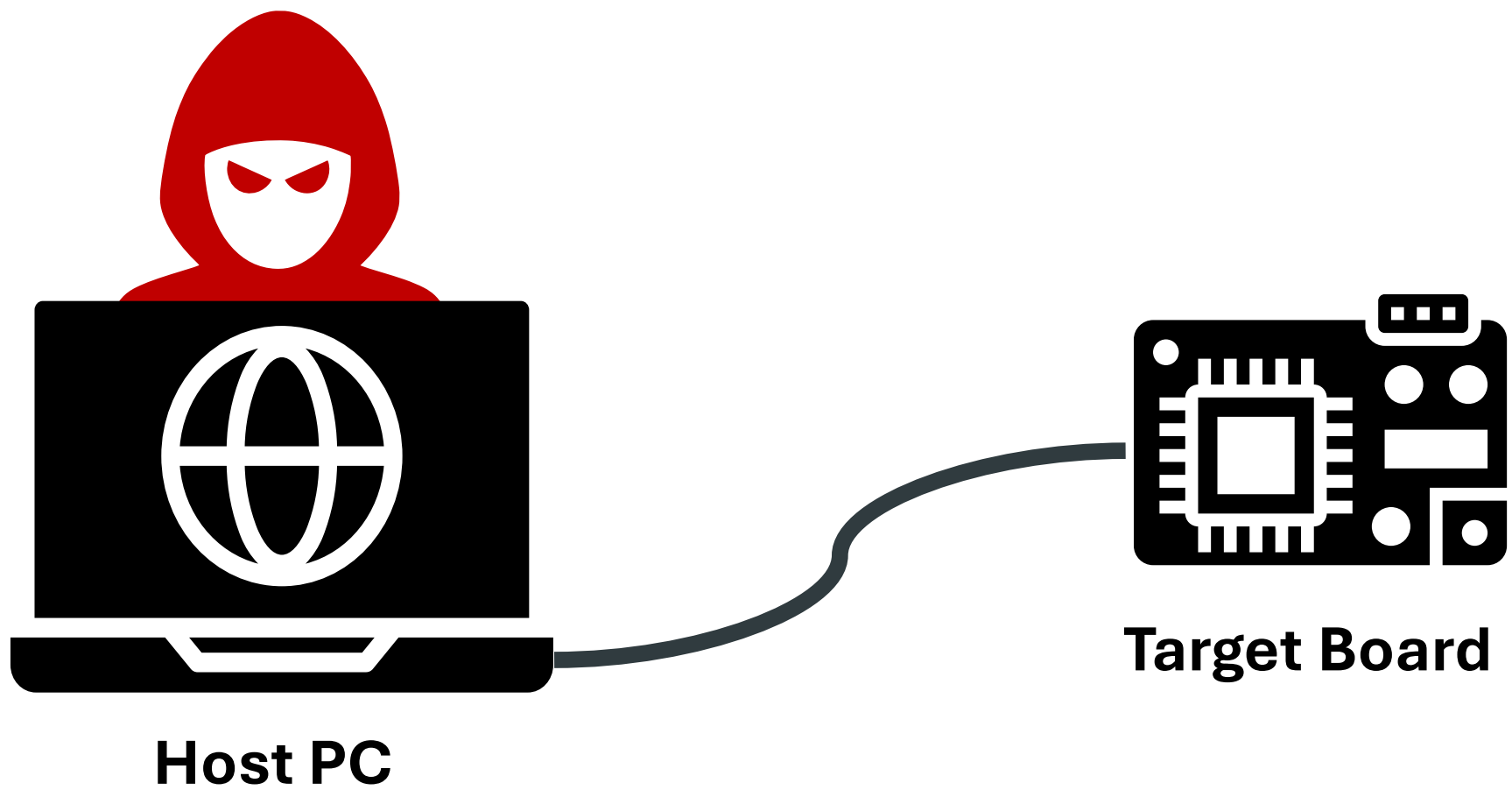


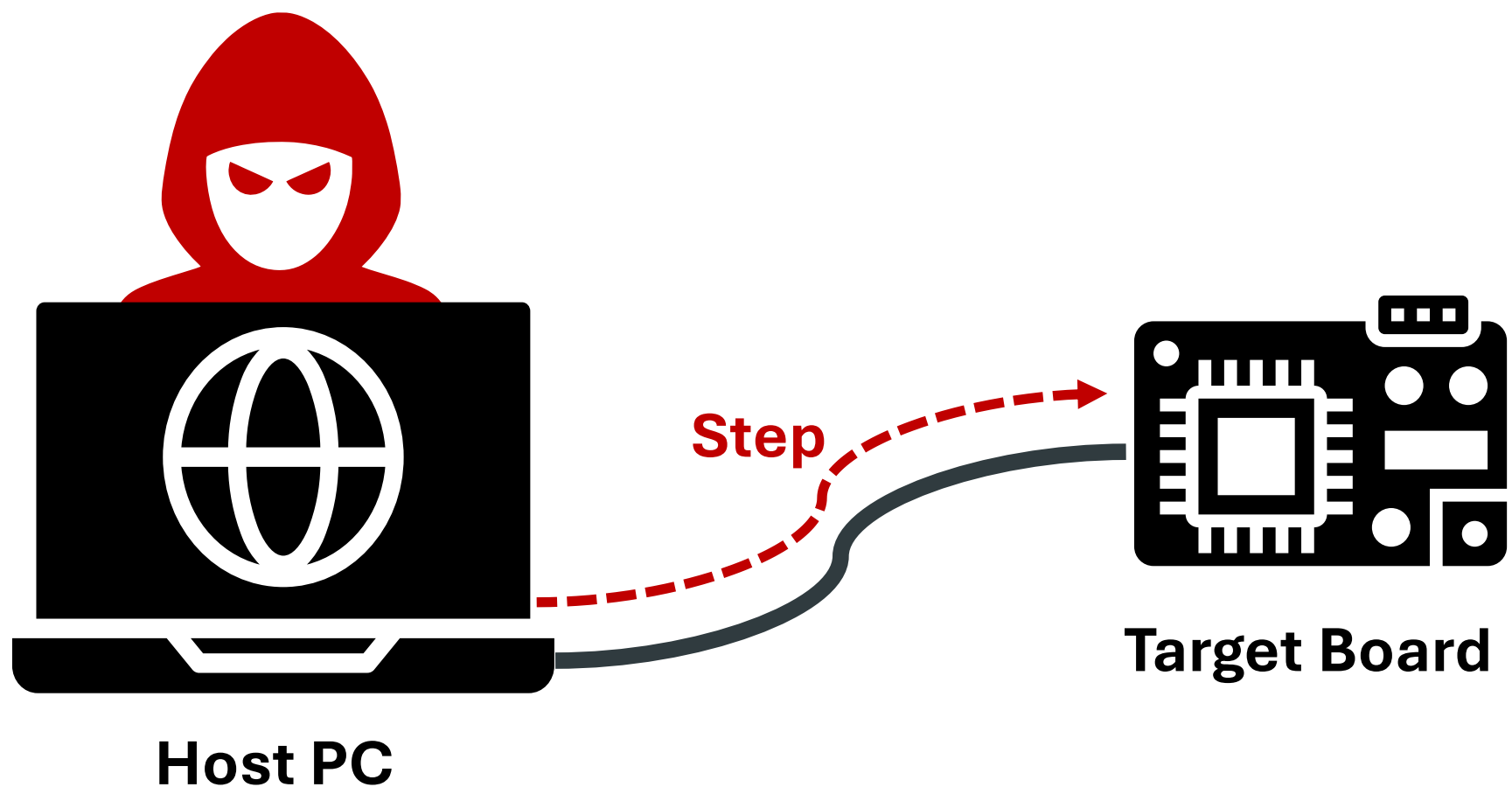
M-Step Visualizer (GTKwave)

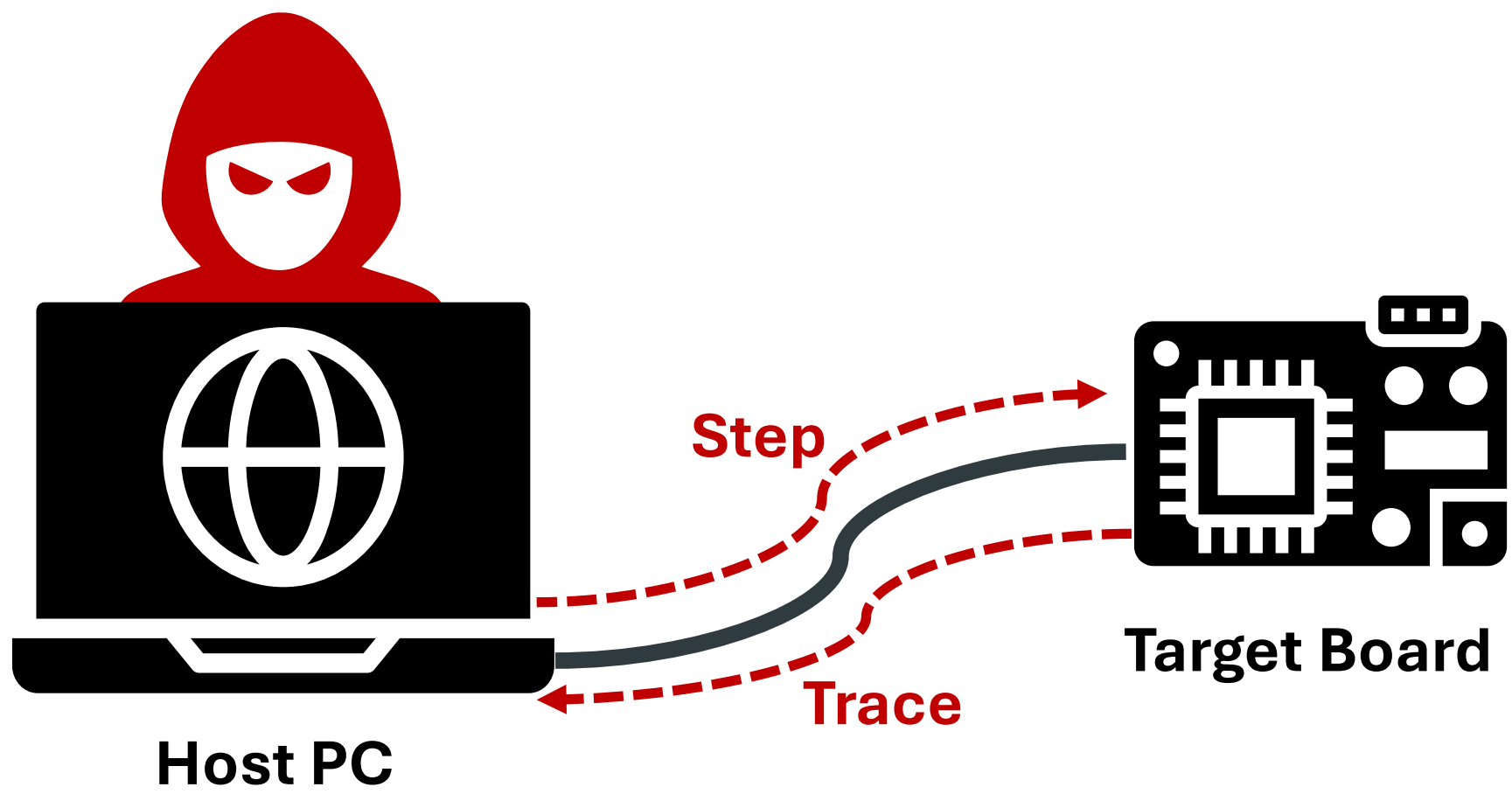


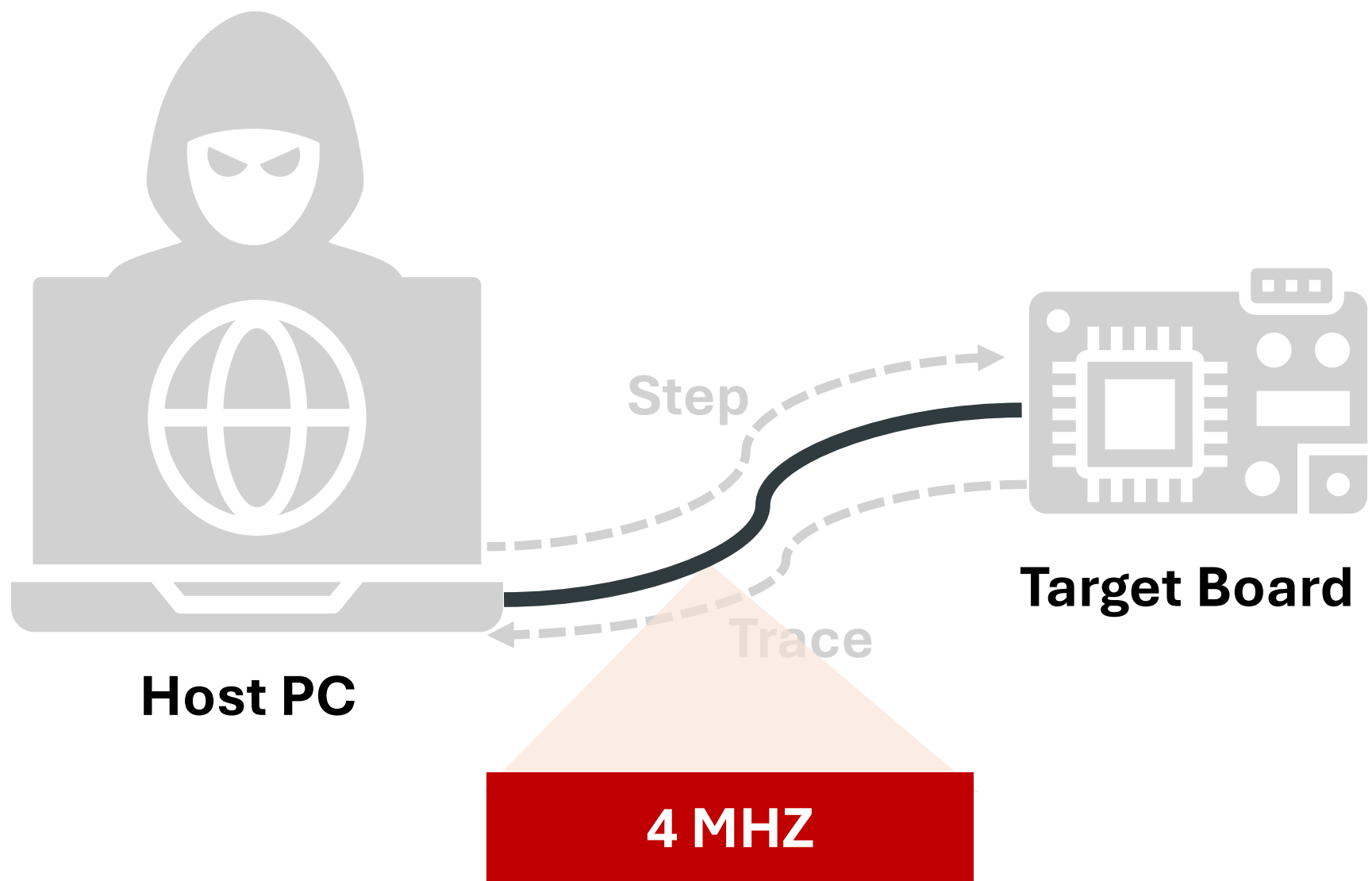
M-Step Visualizer (GTKwave)

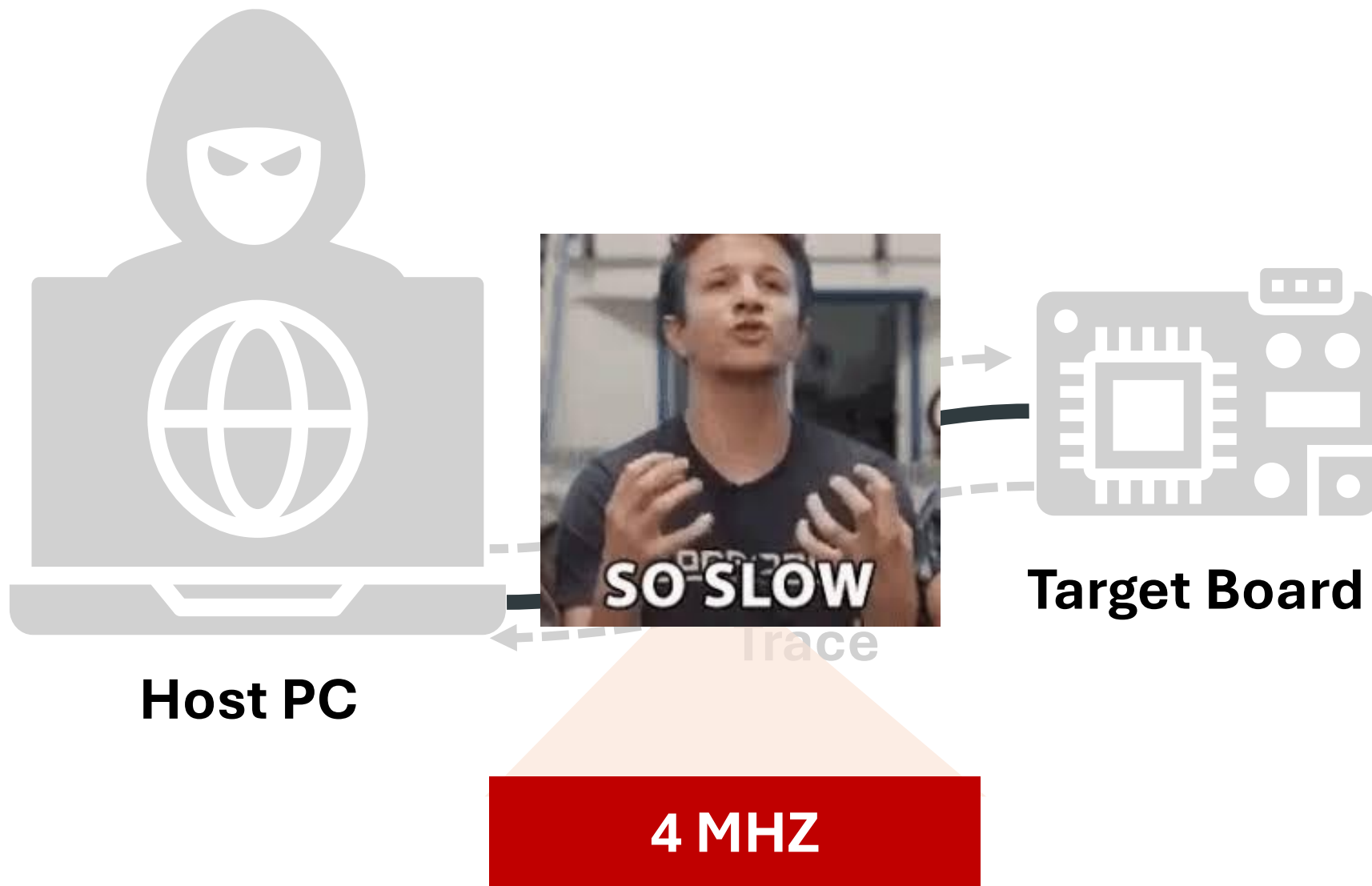






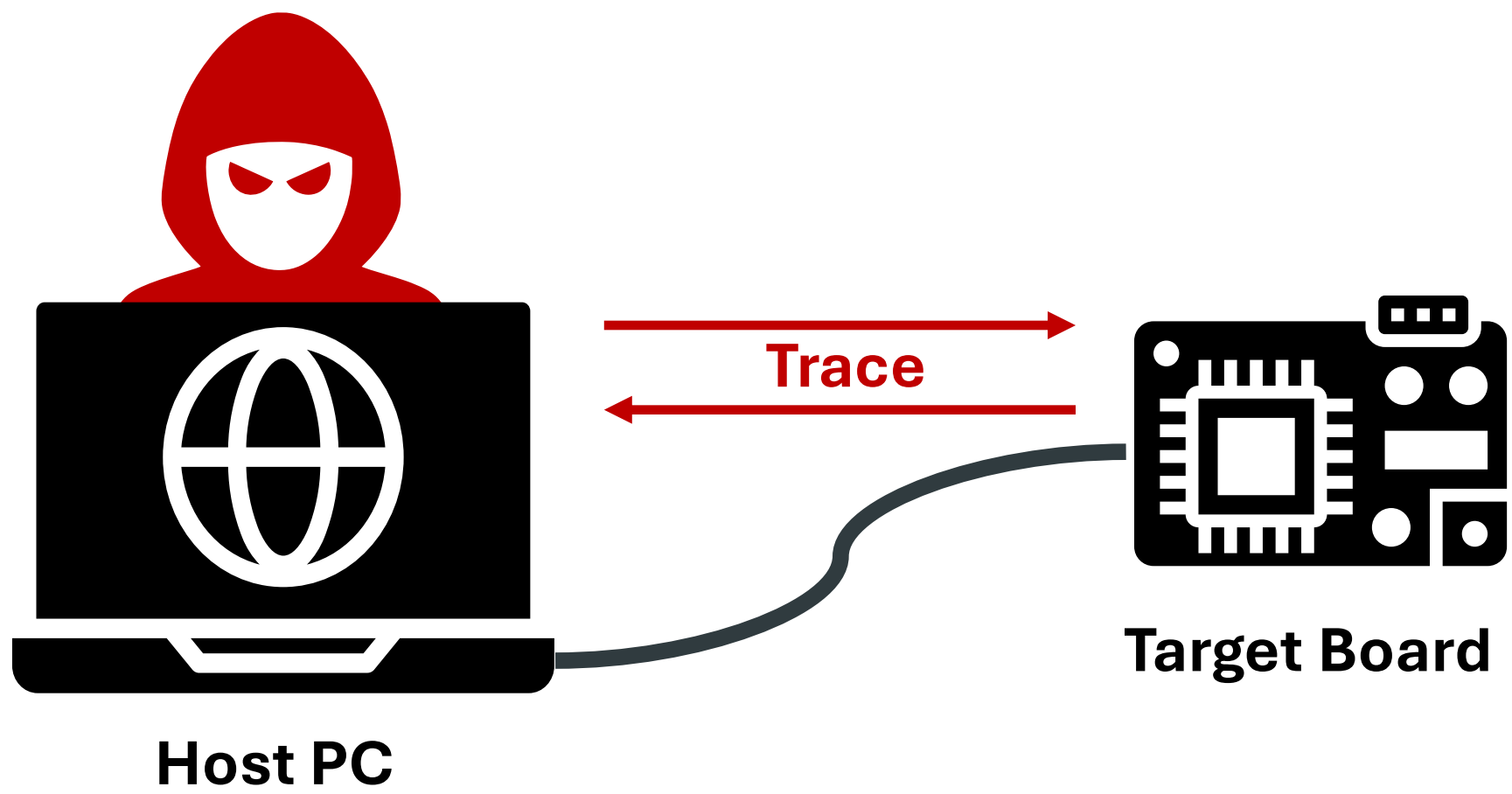


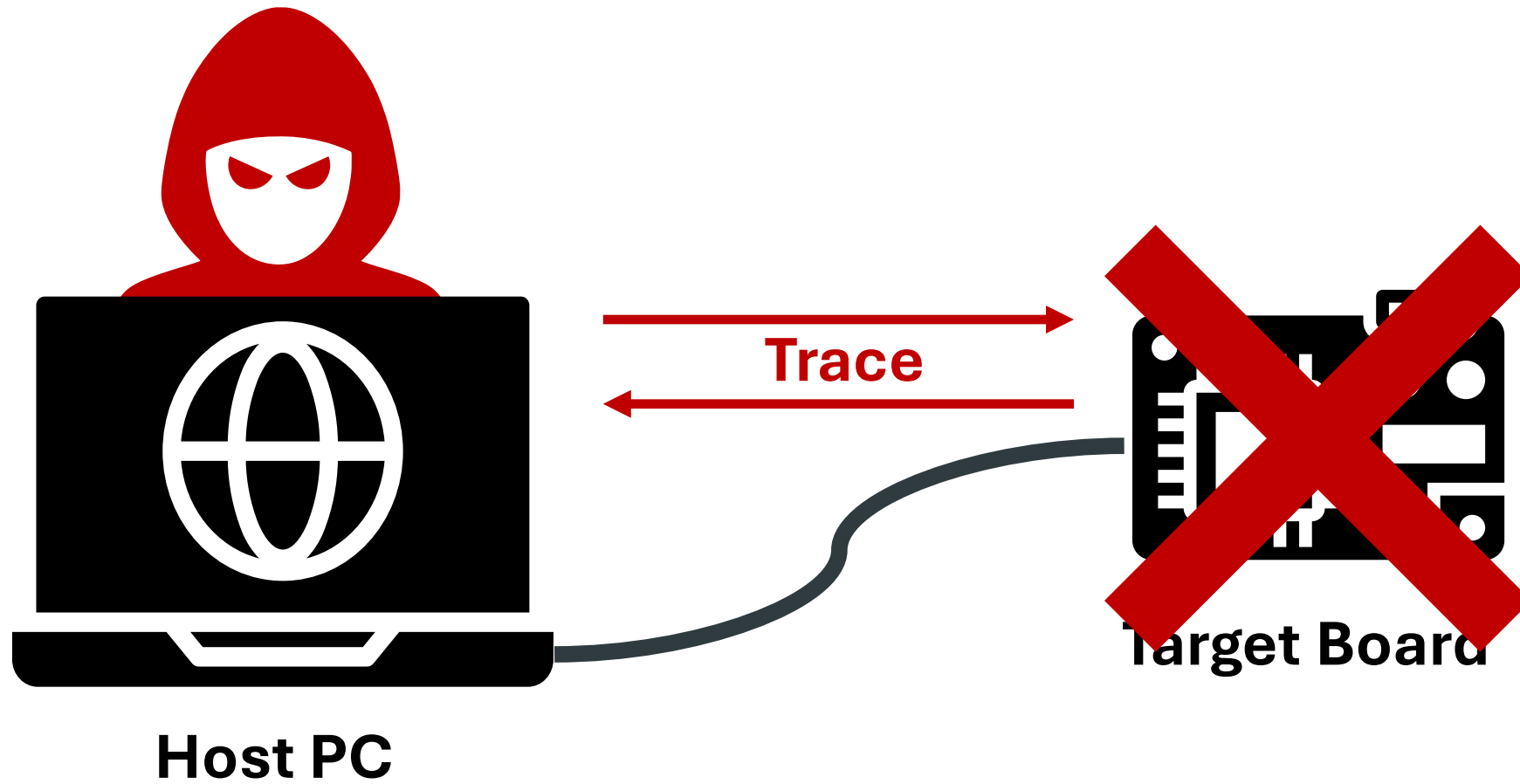


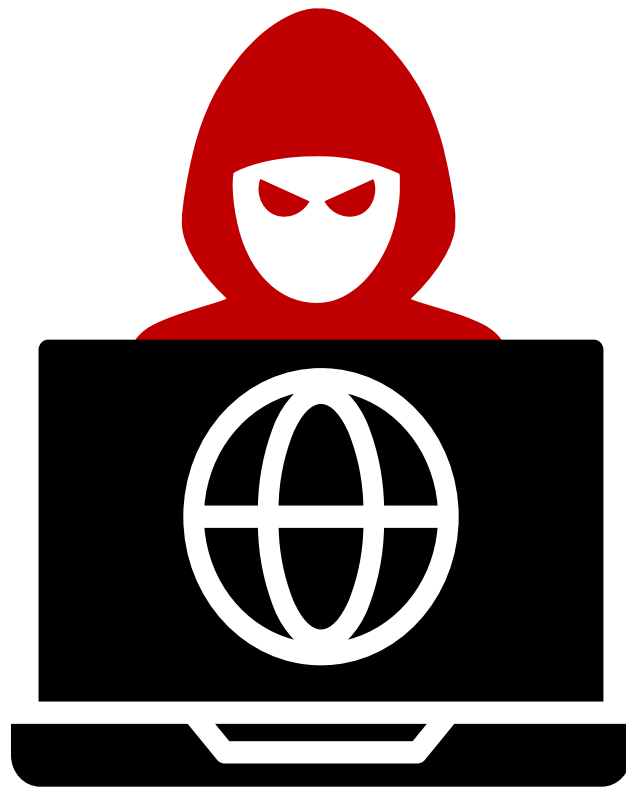




Mstp-Debug & Mstp-Emulator







Host PC

[No Source Available]

[No Assembly Available]

extended-r Thread 1 In: Reset_Handler

L893 PC: 0xc018e70

(gdb) █

[No Source Available]

[No Assembly Available]

extended-r Thread 1 In: Reset_Handler

L893 PC: 0xc018e70

(gdb) b *0x0c028868

Breakpoint 1 at 0xc028868: file /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c, line 6.

(gdb) █

```
/home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c
```

```
1 #include "victims.h"
2
3 #define STRING "M-Step-S\r\n"
4
5 void victim_printf_inst(){
6     printf(STRING);
7 }
8
9 void victim_strlen_inst(){
10     strlen(STRING);
11 }
12
```

```
B+>0xc028868 <victim_printf_inst>    ldr    r0, [pc, #4]    @ (0xc028870 <victim_printf_inst+8>)
0xc02886a <victim_printf_inst+2>      b.w    0xc028e8c <printf>
0xc02886e <victim_printf_inst+6>      nop
0xc028870 <victim_printf_inst+8>      asrs    r1, r6, #29
0xc028872 <victim_printf_inst+10>     lsrs    r3, r0, #16
0xc028874 <victim_strlen_inst>       ldr    r0, [pc, #4]    @ (0xc02887c <victim_strlen_inst+8>)
0xc028876 <victim_strlen_inst+2>      b.w    0xc02fbce <strlen>
0xc02887a <victim_strlen_inst+6>      nop
0xc02887c <victim_strlen_inst+8>      asrs    r1, r6, #29
0xc02887e <victim_strlen_inst+10>     lsrs    r3, r0, #16
0xc028880 <QCB0REncode_EncodeHead>      stmdb   sp!, {r0, r1, r2, r4, r5, r6, r7, r8, r9, lr}
0xc028884 <QCB0REncode_EncodeHead+4>    add     r4, sp, #8
0xc028886 <QCB0REncode_EncodeHead+6>    stmdb   r4, {r1, r2}
```

```
extended-r Thread 1 In: victim_printf_inst
```

```
L6      PC: 0xc028868
```

```
(gdb) b *0xc028868
```

```
Breakpoint 1 at 0xc028868: file /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c, line 6.
```

```
(gdb) c
```

```
Continuing.
```

```
Breakpoint 1, victim_printf_inst () at /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c:6
```

```
[Trace Info] PC=0xc028868 | 7 clk
```

```
(gdb) █
```

```
/home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c
```

```
1 #include "victims.h"
2
3 #define STRING "M-Step-S\r\n"
4
5 void victim_printf_inst(){
6     printf(STRING);
7 }
8
9 void victim_strlen_inst(){
10     strlen(STRING);
11 }
12
```

```
B+ 0xc028868 <victim_printf_inst>      ldr    r0, [pc, #4]      @ (0xc028870 <victim_printf_inst+8>)
>0xc02886a <victim_printf_inst+2>      b.w    0xc028e8c <printf>
0xc02886e <victim_printf_inst+6>      nop
0xc028870 <victim_printf_inst+8>      asrs   r1, r6, #29
0xc028872 <victim_printf_inst+10>     lsrs   r3, r0, #16
0xc028874 <victim_strlen_inst>        ldr    r0, [pc, #4]      @ (0xc02887c <victim_strlen_inst+8>)
0xc028876 <victim_strlen_inst+2>      b.w    0xc02fbce <strlen>
0xc02887a <victim_strlen_inst+6>      nop
0xc02887c <victim_strlen_inst+8>      asrs   r1, r6, #29
0xc02887e <victim_strlen_inst+10>     lsrs   r3, r0, #16
0xc028880 <QCB0REncode_EncodeHead>      stmdb  sp!, {r0, r1, r2, r4, r5, r6, r7, r8, r9, lr}
0xc028884 <QCB0REncode_EncodeHead+4>    add    r4, sp, #8
0xc028886 <QCB0REncode_EncodeHead+6>    stmdb  r4, {r1, r2}
```

```
extended-r Thread 1 In: victim_printf_inst
```

```
L6      PC: 0xc02886a
```

```
(gdb) b *0xc028868
```

```
Breakpoint 1 at 0xc028868: file /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c, line 6.
```

```
(gdb) c
```

```
Continuing.
```

```
Breakpoint 1, victim_printf_inst () at /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c:6
```

```
[Trace Info] PC=0x0C028868 | 7 clk
```

```
(gdb) ni
```

```
[Trace Info] PC=0x0C02886A | 1 clk
```

```
(gdb) 
```

```
[M-Step-De0: bash*Z 1:fish- 2:fish
```

```
"/1-gdb-py.sh -s exp " 12:50 18-jun-25
```

```
/home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c
```

```
1 #include "victims.h"
2
3 #define STRING "M-Step-S\r\n"
4
5 void victim_printf_inst(){
6     printf(STRING);
7 }
8
9 void victim_strlen_inst(){
10     strlen(STRING);
11 }
12
```

```
B+ 0xc028868 <victim_printf_inst>    ldr    r0, [pc, #4]    @ (0xc028870 <victim_printf_inst+8>)
>0xc02886a <victim_printf_inst+2>    b.w    0xc028e8c <printf>
0xc02886e <victim_printf_inst+6>    nop
0xc028870 <victim_printf_inst+8>    asrs    r1, r6, #29
0xc028872 <victim_printf_inst+10>    lsrs    r3, r0, #16
0xc028874 <victim_strlen_inst>       ldr    r0, [pc, #4]    @ (0xc02887c <victim_strlen_inst+8>)
0xc028876 <victim_strlen_inst+2>    b.w    0xc02fbce <strlen>
0xc02887a <victim_strlen_inst+6>    nop
0xc02887c <victim_strlen_inst+8>    asrs    r1, r6, #29
0xc02887e <victim_strlen_inst+10>    lsrs    r3, r0, #16
0xc028880 <QCB0REncode_EncodeHead>    stmdb    sp!, {r0, r1, r2, r4, r5, r6, r7, r8, r9, lr}
0xc028884 <QCB0REncode_EncodeHead+4>    add     r4, sp, #8
0xc028886 <QCB0REncode_EncodeHead+6>    stmdb    r4, {r1, r2}
```

```
extended-r Thread 1 In: victim_printf_inst
```

```
L6    PC: 0xc02886a
```

```
(gdb) b *0xc028868
```

```
Breakpoint 1 at 0xc028868: file /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c, line 6.
```

```
(gdb) c
```

```
Continuing.
```

```
Breakpoint 1, victim_printf_inst () at /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c:6
```

```
[Trace Info] PC=0x0C028868 | 7 clk
```

```
(gdb) ni
```

```
[Trace Info] PC=0x0C02886A | 1 clk
```

```
(gdb) 
```

```
[M-Step-De0: bash*Z 1:fish- 2:fish
```

```
"/1-gdb-py.sh -s exp " 12:50 18-jun-25
```

```
/home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c
```

```
165 int printf(const char *fmt, ...)
166 {
167     int count = 0;
168     va_list ap;
169
> 170     va_start(ap, fmt);
171     count = vprintf(fmt, ap);
172     va_end(ap);
173
174     return count;
175 }
```

```
>0xc028e8c <printf>          push    {r0, r1, r2, r3}
0xc028e8e <printf+2>         push    {r0, r1, r2, lr}
0xc028e90 <printf+4>         add     r1, sp, #16
0xc028e92 <printf+6>         ldr.w   r0, [r1], #4
0xc028e96 <printf+10>        str     r1, [sp, #4]
0xc028e98 <printf+12>        bl      0xc01e848 <vprintf>
0xc028e9c <printf+16>        add     sp, #12
0xc028e9e <printf+18>        ldr.w   lr, [sp], #4
0xc028ea2 <printf+22>        add     sp, #16
0xc028ea4 <printf+24>        bx      lr
0xc028ea6 <tfm_output_unpriv_string> svc     2
0xc028ea8 <tfm_output_unpriv_string+2> bx      lr
0xc028eaa <tfm_hal_output_sp_log> b.w     0xc028ea6 <tfm_output_unpriv_string>
```

extended-r Thread 1 In: printf

L170 PC: 0xc028e8c

(gdb) b *0xc028868

Breakpoint 1 at 0xc028868: file /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c, line 6.

(gdb) c

Continuing.

Breakpoint 1, victim_printf_inst () at /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c:6

[Trace Info] PC=0x0C028868 | 7 clk

(gdb) ni

[Trace Info] PC=0x0C02886A | 1 clk

printf (fmt=0x0) at /home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c:170

[Trace Info] PC=0x0C028E8C | 3 clk

(gdb)

[M-Step-De0: bash*Z 1: fish- 2: fish

"/1-gdb-py.sh -s exp " 12:50 18-jun-25


```
/home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c
```

```
165 int printf(const char *fmt, ...)
166 {
167     int count = 0;
168     va_list ap;
169
> 170     va_start(ap, fmt);
171     count = vprintf(fmt, ap);
172     va_end(ap);
173
174     return count;
175 }
```

```
0xc028e8c <printf>          push    {r0, r1, r2, r3}
>0xc028e8e <printf+2>       push    {r0, r1, r2, lr}
0xc028e90 <printf+4>        add     r1, sp, #16
0xc028e92 <printf+6>        ldr.w   r0, [r1], #4
0xc028e96 <printf+10>       str     r1, [sp, #4]
0xc028e98 <printf+12>       bl      0xc01e848 <vprintf>
0xc028e9c <printf+16>       add     sp, #12
0xc028e9e <printf+18>       ldr.w   lr, [sp], #4
0xc028ea2 <printf+22>       add     sp, #16
0xc028ea4 <printf+24>       bx      lr
0xc028ea6 <tfm_output_unpriv_string> svc     2
0xc028ea8 <tfm_output_unpriv_string+2> bx      lr
0xc028eaa <tfm_hal_output_sp_log> b.w     0xc028ea6 <tfm_output_unpriv_string>
```

```
extended-r Thread 1 In: printf
```

```
L170 PC: 0xc028e8e
```

```
(gdb) b *0x0c028868
```

```
Breakpoint 1 at 0xc028868: file /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c, line 6.
```

```
(gdb) c
```

```
Continuing.
```

```
Breakpoint 1, victim_printf_inst () at /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c:6
```

```
[Trace Info] PC=0x0C028868 | 7 clk
```

```
(gdb) ni
```

```
[Trace Info] PC=0x0C02886A | 1 clk
```

```
printf (fmt=0x0) at /home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c:170
```

```
[Trace Info] PC=0x0C028E8C | 3 clk
```

```
[Trace Info] PC=0x0C028E8E | 3 clk
```

```
(gdb) 
```

```
[M-Step-De0: bash*Z 1: fish- 2: fish
```

```
"/1-gdb-py.sh -s exp " 12:50 18-jun-25
```

```
/home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c
```

```
165 int printf(const char *fmt, ...)
> 166 {
167     int count = 0;
168     va_list ap;
169
170     va_start(ap, fmt);
171     count = vprintf(fmt, ap);
172     va_end(ap);
173
174     return count;
175 }
```

```
0xc028e8c <printf>          push    {r0, r1, r2, r3}
0xc028e8e <printf+2>        push    {r0, r1, r2, lr}
>0xc028e90 <printf+4>       add     r1, sp, #16
0xc028e92 <printf+6>       ldr.w   r0, [r1], #4
0xc028e96 <printf+10>      str     r1, [sp, #4]
0xc028e98 <printf+12>      bl      0xc01e848 <vprintf>
0xc028e9c <printf+16>      add     sp, #12
0xc028e9e <printf+18>      ldr.w   lr, [sp], #4
0xc028ea2 <printf+22>      add     sp, #16
0xc028ea4 <printf+24>      bx      lr
0xc028ea6 <tfm_output_unpriv_string> svc     2
0xc028ea8 <tfm_output_unpriv_string+2> bx      lr
0xc028eaa <tfm_hal_output_sp_log> b.w     0xc028ea6 <tfm_output_unpriv_string>
```

extended-r Thread 1 In: printf

L166 PC: 0xc028e90

(gdb) b *0x0c028868

Breakpoint 1 at 0xc028868: file /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c, line 6.

(gdb) c

Continuing.

Breakpoint 1, victim_printf_inst () at /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c:6

[Trace Info] PC=0x0C028868 | 7 clk

(gdb) ni

[Trace Info] PC=0x0C02886A | 1 clk

printf (fmt=0x0) at /home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c:170

[Trace Info] PC=0x0C028E8C | 3 clk

[Trace Info] PC=0x0C028E8E | 3 clk

[Trace Info] PC=0x0C028E90 | 1 clk

(gdb)

[M-Step-De0: bash*Z 1: fish- 2: fish

"/1-gdb-py.sh -s exp " 12:51 18-jun-25

```
165 int printf(const char *fmt, ...)
> 166 {
167     int count = 0;
168     va_list ap;
169
170     va_start(ap, fmt);
171     count = vprintf(fmt, ap);
172     va_end(ap);
173
174     return count;
175 }
```

```
0xc028e8c <printf>          push    {r0, r1, r2, r3}
0xc028e8e <printf+2>        push    {r0, r1, r2, lr}
0xc028e90 <printf+4>        add     r1, sp, #16
>0xc028e92 <printf+6>      ldr.w   r0, [r1], #4
0xc028e96 <printf+10>       str     r1, [sp, #4]
0xc028e98 <printf+12>       bl      0xc01e848 <vprintf>
0xc028e9c <printf+16>       add     sp, #12
0xc028e9e <printf+18>       ldr.w   lr, [sp], #4
0xc028ea2 <printf+22>       add     sp, #16
0xc028ea4 <printf+24>       bx      lr
0xc028ea6 <tfm_output_unpriv_string> svc     2
0xc028ea8 <tfm_output_unpriv_string+2> bx      lr
0xc028eaa <tfm_hal_output_sp_log> b.w    0xc028ea6 <tfm_output_unpriv_string>
```

L166 PC: 0xc028e92

```
Breakpoint 1 at 0xc028868: file /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c, line 6.
```

Continuing.

```
Breakpoint 1: victim printf inst () at /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c:6
```

```
(qdb) ni
```

```
printf (fmt=0x0) at /home/cris/Documents/0_Projects/io-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c:170
```

[Trace Info] PC=0x0C028E8E | 3 clk

[Trace Info] PC=0x0C028E92 | 2 clk

Environ Biol Fish (2015) 98:1031–1041

```
"/1-gdb-py.sh -s exp " 12:51 18-jun-25
```

/home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c

```
165 int printf(const char *fmt, ...)
166 {
167     int count = 0;
168     va_list ap;
169
> 170     va_start(ap, fmt);
171     count = vprintf(fmt, ap);
172     va_end(ap);
173
174     return count;
175 }
```

```
0xc028e8c <printf>          push    {r0, r1, r2, r3}
0xc028e8e <printf+2>        push    {r0, r1, r2, lr}
0xc028e90 <printf+4>        add     r1, sp, #16
0xc028e92 <printf+6>        ldr.w   r0, [r1], #4
>0xc028e96 <printf+10>      str     r1, [sp, #4]
0xc028e98 <printf+12>        bl      0xc01e848 <vprintf>
0xc028e9c <printf+16>        add     sp, #12
0xc028e9e <printf+18>        ldr.w   lr, [sp], #4
0xc028ea2 <printf+22>        add     sp, #16
0xc028ea4 <printf+24>        bx      lr
0xc028ea6 <tfm_output_unpriv_string> svc     2
0xc028ea8 <tfm_output_unpriv_string+2> bx      lr
0xc028eaa <tfm_hal_output_sp_log> b.w     0xc028ea6 <tfm_output_unpriv_string>
```

extended-r Thread 1 In: printf

L170 PC: 0xc028e96

Breakpoint 1 at 0xc028868: file /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c, line 6.

(gdb) c

Continuing.

Breakpoint 1, victims_printf_init () at /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c:6

[Trace Info] PC=0x0C028868 | 7 clk

(gdb) ni

[Trace Info] PC=0x0C02886A | 1 clk

printf (fmt=0x0) at /home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c:170

[Trace Info] PC=0x0C028E8C | 3 clk

[Trace Info] PC=0x0C028E8E | 3 clk

[Trace Info] PC=0x0C028E90 | 1 clk

[Trace Info] PC=0x0C028E92 | 2 clk

[Trace Info] PC=0x0C028E96 | 2 clk

(gdb) □

In step 000.bash: 1:15h 2:15h

"/1-gdb-py.sh -s exp " 12:51 18-jun-25

/home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c

```
165 int printf(const char *fmt, ...)
166 {
167     int count = 0;
168     va_list ap;
169
170     va_start(ap, fmt);
> 171     count = vprintf(fmt, ap);
172     va_end(ap);
173
174     return count;
175 }
```

```
0xc028e8c <printf>          push    {r0, r1, r2, r3}
0xc028e8e <printf+2>        push    {r0, r1, r2, lr}
0xc028e90 <printf+4>        add     r1, sp, #16
0xc028e92 <printf+6>        ldr.w   r0, [r1], #4
0xc028e96 <printf+10>       str     r1, [sp, #4]
>0xc028e98 <printf+12>      bl      0xc01e848 <vprintf>
0xc028e9c <printf+16>       add     sp, #12
0xc028e9e <printf+18>       ldr.w   lr, [sp], #4
0xc028ea2 <printf+22>       add     sp, #16
0xc028ea4 <printf+24>       bx      lr
0xc028ea6 <tfm_output_unpriv_string> svc     2
0xc028ea8 <tfm_output_unpriv_string+2> bx      lr
0xc028eaa <tfm_hal_output_sp_log> b.w     0xc028ea6 <tfm_output_unpriv_string>
```

extended-r Thread 1 In: printf

L171 PC: 0xc028e98

(gdb) c

Continuing.

Breakpoint 1, victim_printf_inst () at /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c:6

[Trace Info] PC=0x0C028868 | 7 clk

(gdb) ni

[Trace Info] PC=0x0C02886A | 1 clk

printf (fmt=0x0) at /home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c:170

[Trace Info] PC=0x0C028E8C | 3 clk

[Trace Info] PC=0x0C028E8E | 3 clk

[Trace Info] PC=0x0C028E90 | 1 clk

[Trace Info] PC=0x0C028E92 | 2 clk

[Trace Info] PC=0x0C028E96 | 2 clk

[Trace Info] PC=0x0C028E98 | 1 clk

(gdb) □

In step 0000000000000000 1:15h 2:15h

"/1-gdb-py.sh -s exp " 12:52 18-jun-25

```

165 int printf(const char *fmt, ...)
166 {
167     int count = 0;
168     va_list ap;
169
170     va_start(ap, fmt);
171     count = vprintf(fmt, ap);
172     va_end(ap);
173
> 174     return count;
175 }

```

```

0xc028e8c <printf>          push    {r0, r1, r2, r3}
0xc028e8e <printf+2>        push    {r0, r1, r2, lr}
0xc028e90 <printf+4>        add     r1, sp, #16
0xc028e92 <printf+6>        ldr.w   r0, [r1], #4
0xc028e96 <printf+10>       str     r1, [sp, #4]
0xc028e98 <printf+12>       bl      0xc01e848 <vprintf>
>0xc028e9c <printf+16>      add     sp, #12
0xc028e9e <printf+18>       ldr.w   lr, [sp], #4
0xc028ea2 <printf+22>       add     sp, #16
0xc028ea4 <printf+24>       bx      lr
0xc028ea6 <tfm_output_unpriv_string> svc     2
0xc028ea8 <tfm_output_unpriv_string+2> bx      lr
0xc028eaa <tfm_hal_output_sp_log> b.w     0xc028ea6 <tfm_output_unpriv_string>

```

extended-r Thread 1 In: printf

L174 PC: 0xc028e9c

Continuing.

Breakpoint 1: victim_printf_inst () at /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c:6

[Trace Info] PC=0x0C028868 | 7 clk
(gdb) ni

[Trace Info] PC=0x0C02886A | 1 clk

printf (fmt=0x0) at /home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c:170

[Trace Info] PC=0x0C028E8C | 3 clk

[Trace Info] PC=0x0C028E8E | 3 clk

[Trace Info] PC=0x0C028E90 | 1 clk

[Trace Info] PC=0x0C028E92 | 2 clk

[Trace Info] PC=0x0C028E96 | 2 clk

[Trace Info] PC=0x0C028E98 | 1 clk

[Trace Info] PC=0x0C028E9C | 1 clk

(gdb) □

```

165 int printf(const char *fmt, ...)
166 {
167     int count = 0;
168     va_list ap;
169
170     va_start(ap, fmt);
171     count = vprintf(fmt, ap);
172     va_end(ap);
173
> 174     return count;
175 }

```

```

0xc028e8c <printf>          push    {r0, r1, r2, r3}
0xc028e8e <printf+2>        push    {r0, r1, r2, lr}
0xc028e90 <printf+4>        add     r1, sp, #16
0xc028e92 <printf+6>        ldr.w   r0, [r1], #4
0xc028e96 <printf+10>       str     r1, [sp, #4]
0xc028e98 <printf+12>       bl      0xc01e848 <vprintf>
0xc028e9c <printf+16>       add     sp, #12
>0xc028e9e <printf+18>      ldr.w   lr, [sp], #4
0xc028ea2 <printf+22>       add     sp, #16
0xc028ea4 <printf+24>       bx      lr
0xc028ea6 <tfm_output_unpriv_string> svc    2
0xc028ea8 <tfm_output_unpriv_string+2> bx     lr
0xc028eaa <tfm_hal_output_sp_log> b.w    0xc028ea6 <tfm_output_unpriv_string>

```

extended-r Thread 1 In: printf

L174 PC: 0xc028e9e

Breakpoint 1: victim_printf_inst () at /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c:6

```

[Trace Info] PC=0x0C028868 | 7 clk
(gdb) ni
[Trace Info] PC=0x0C02886A | 1 clk
printf (fmt=0x0) at /home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c:170
[Trace Info] PC=0x0C028E8C | 3 clk
[Trace Info] PC=0x0C028E8E | 3 clk
[Trace Info] PC=0x0C028E90 | 1 clk
[Trace Info] PC=0x0C028E92 | 2 clk
[Trace Info] PC=0x0C028E96 | 2 clk
[Trace Info] PC=0x0C028E98 | 1 clk
[Trace Info] PC=0x0C028E9C | 1 clk
[Trace Info] PC=0x0C028E9E | 2 clk
(gdb)

```

```

165 int printf(const char *fmt, ...)
166 {
167     int count = 0;
168     va_list ap;
169
170     va_start(ap, fmt);
171     count = vprintf(fmt, ap);
172     va_end(ap);
173
> 174     return count;
175 }

```

```

0xc028e8c <printf>          push    {r0, r1, r2, r3}
0xc028e8e <printf+2>        push    {r0, r1, r2, lr}
0xc028e90 <printf+4>        add     r1, sp, #16
0xc028e92 <printf+6>        ldr.w   r0, [r1], #4
0xc028e96 <printf+10>       str     r1, [sp, #4]
0xc028e98 <printf+12>       bl      0xc01e848 <vprintf>
0xc028e9c <printf+16>       add     sp, #12
0xc028e9e <printf+18>       ldr.w   lr, [sp], #4
>0xc028ea2 <printf+22>      add     sp, #16
0xc028ea4 <printf+24>       bx      lr
0xc028ea6 <tfm_output_unpriv_string> svc     2
0xc028ea8 <tfm_output_unpriv_string+2> bx      lr
0xc028eaa <tfm_hal_output_sp_log> b.w     0xc028ea6 <tfm_output_unpriv_string>

```

extended-r Thread 1 In: printf

L174 PC: 0xc028ea2

Breakpoint 1, victim_printf_inst () at /home/cris/Documents/0_Projects/jo-collab/m-step/mstp-victims/s/src/victims.c:6

```

[Trace Info] PC=0x0C028868 | 7 clk
(gdb) ni
[Trace Info] PC=0x0C02886A | 1 clk
printf (fmt=0x0) at /home/cris/Documents/0_Projects/jo-collab/tf-m/trusted-firmware-m/secure_fw/partitions/lib/runtime/tfm_sp_log_raw.c:170
[Trace Info] PC=0x0C028E8C | 3 clk
[Trace Info] PC=0x0C028E8E | 3 clk
[Trace Info] PC=0x0C028E90 | 1 clk
[Trace Info] PC=0x0C028E92 | 2 clk
[Trace Info] PC=0x0C028E96 | 2 clk
[Trace Info] PC=0x0C028E98 | 1 clk
[Trace Info] PC=0x0C028E9C | 1 clk
[Trace Info] PC=0x0C028E9E | 2 clk
[Trace Info] PC=0x0C028EA2 | 1 clk
(gdb)

```


M-Step Side-Channel Analysis



Mstp-Nemesis



Mstp-BUSted



Mstp-Cache

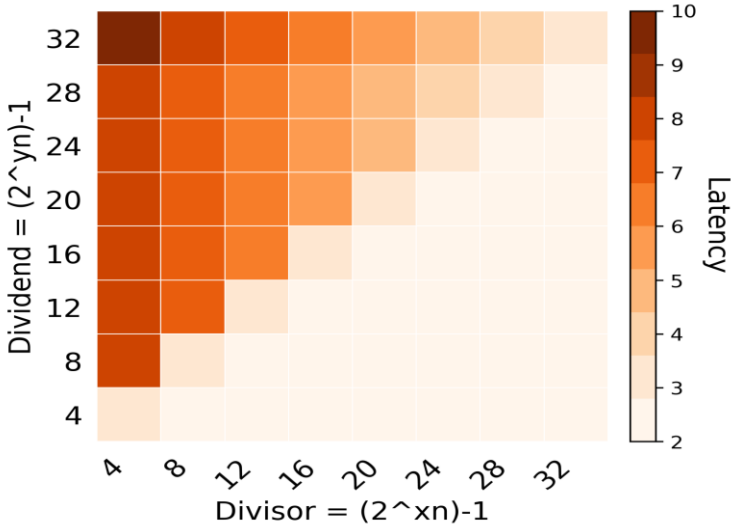
Mstp-Nemesis

Mstp-Nemesis

Mstp-Cache

Mstp-BUSted

Data-Dependent Divisions

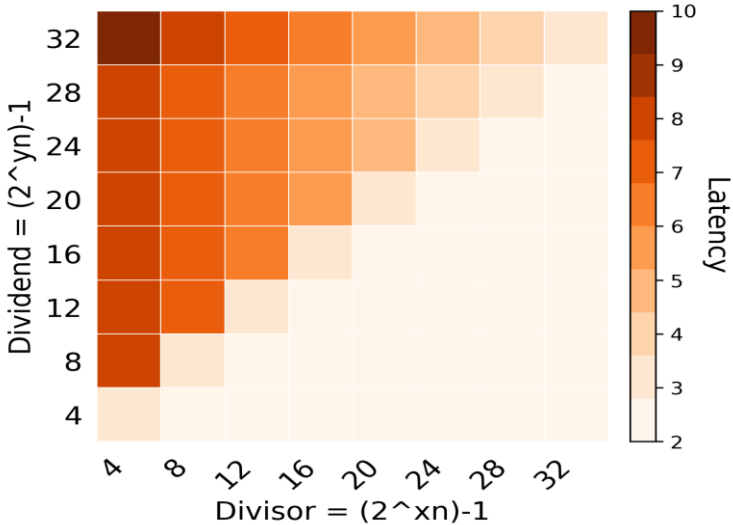


Mstp-Nemesis

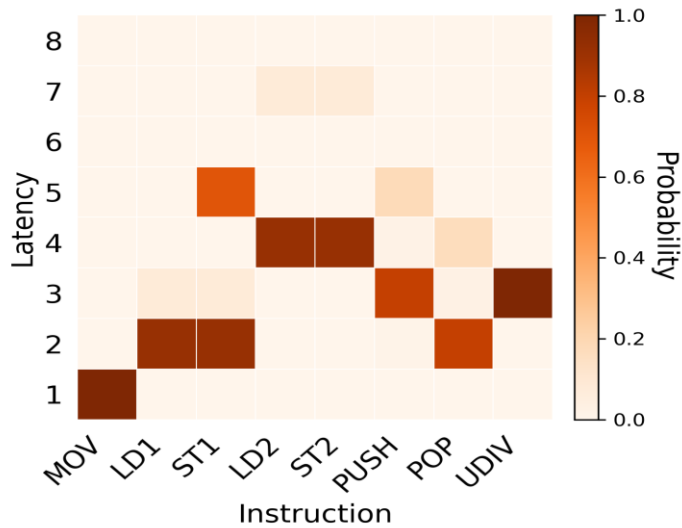
Mstp-Cache

Mstp-BUSted

Data-Dependent Divisions



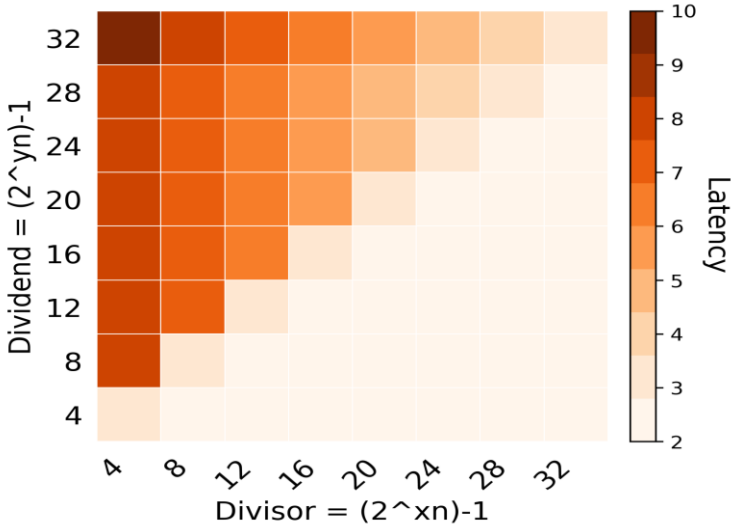
Differentiating Instructions



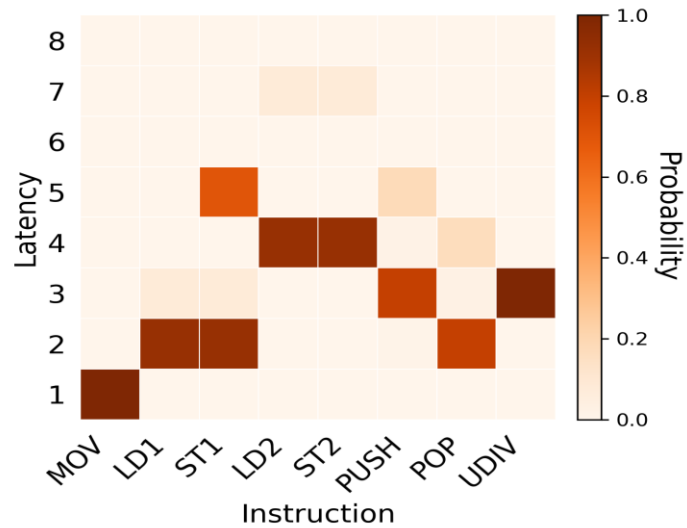
Mstp-Cache

Mstp-BUSted

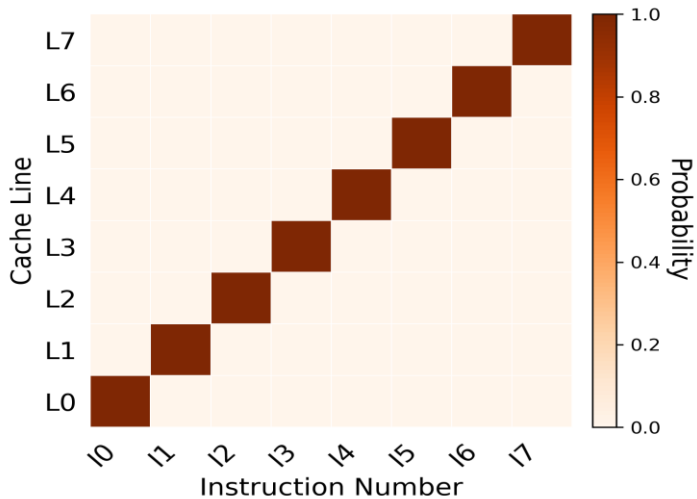
Data-Dependent Divisions



Differentiating Instructions

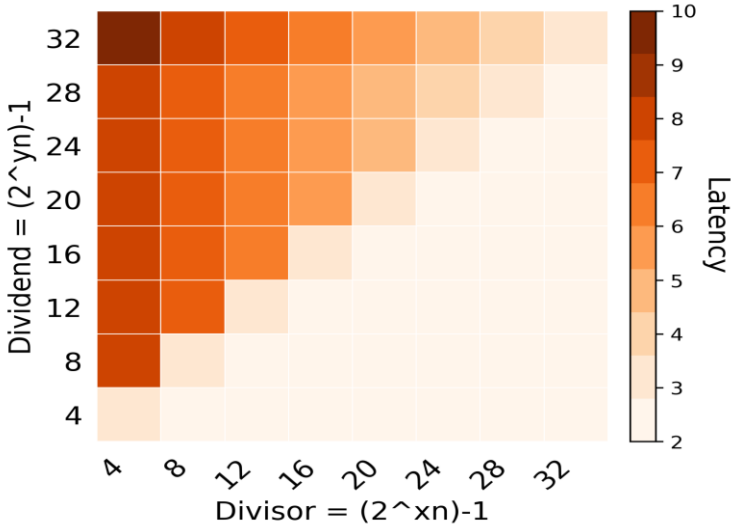


Cache Activity

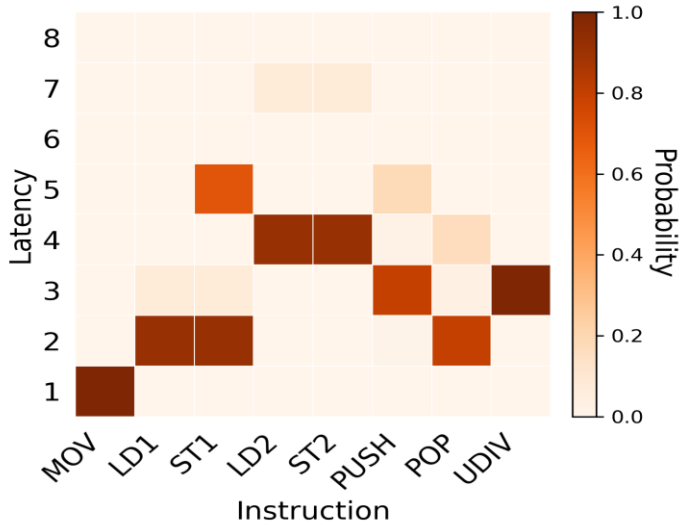


Mstp-BUSted

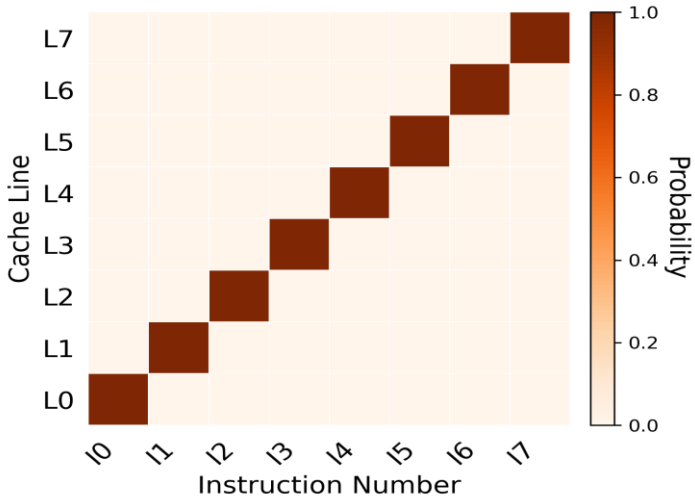
Data-Dependent Divisions



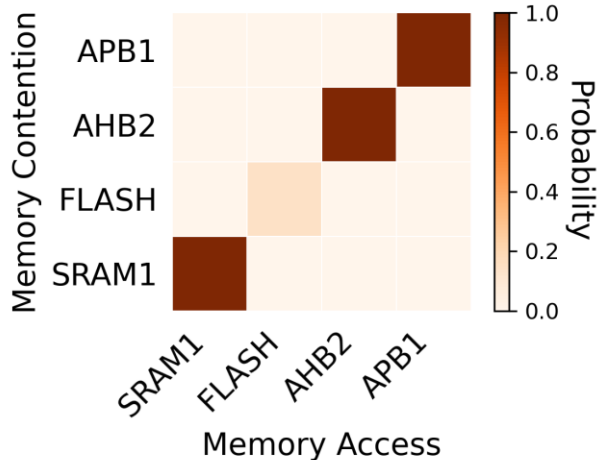
Differentiating Instructions



Cache Activity



Memory Access Activity





M-Step Plugins

M-Step Plugins

Side-Channel

Architectural

Framework

M-Step Plugins

Side-Channel

Mstp-Nemesis

Mstp-Busted

Mstp-Cache

Architectural

Framework

M-Step Plugins

Side-Channel

Mstp-Nemesis

Mstp-Busted

Mstp-Cache

Architectural

Mstp-Zoom

Framework

M-Step Plugins

Side-Channel

Mstp-Nemesis

Mstp-Busted

Mstp-Cache

Architectural

Mstp-Zoom

Framework

Mstp-Production

Mstp-Debug

Mstp-Metrics

Mstp-Emulator

Mstp-Visualizer

Mstp-opDecoder

Mstp-Test

M-Step Plugins

Side-Channel

Mstp-Nemesis

Mstp-Busted

Mstp-Cache

Architectural

Mstp-Zoom

Framework

Mstp-Production

Mstp-Debug

Mstp-Metrics

Mstp-Emulator

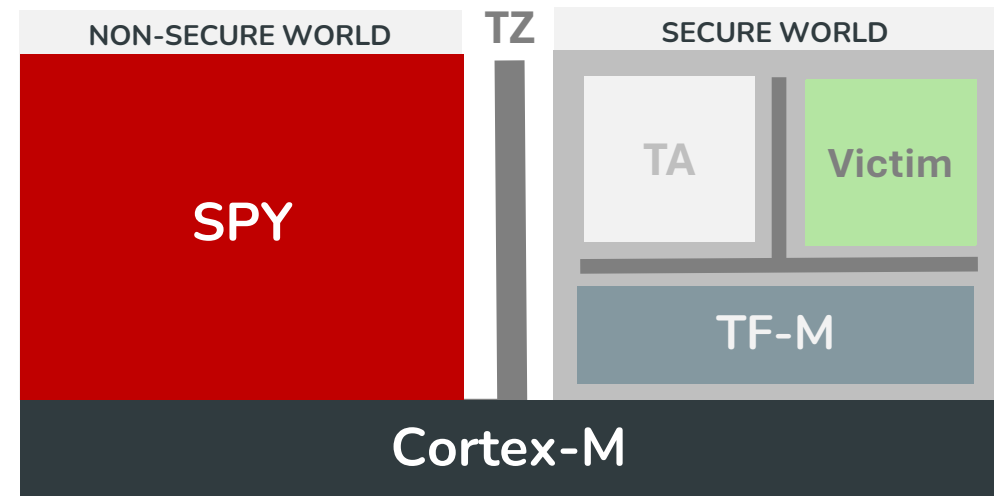
Mstp-Visualizer

Mstp-opDecoder

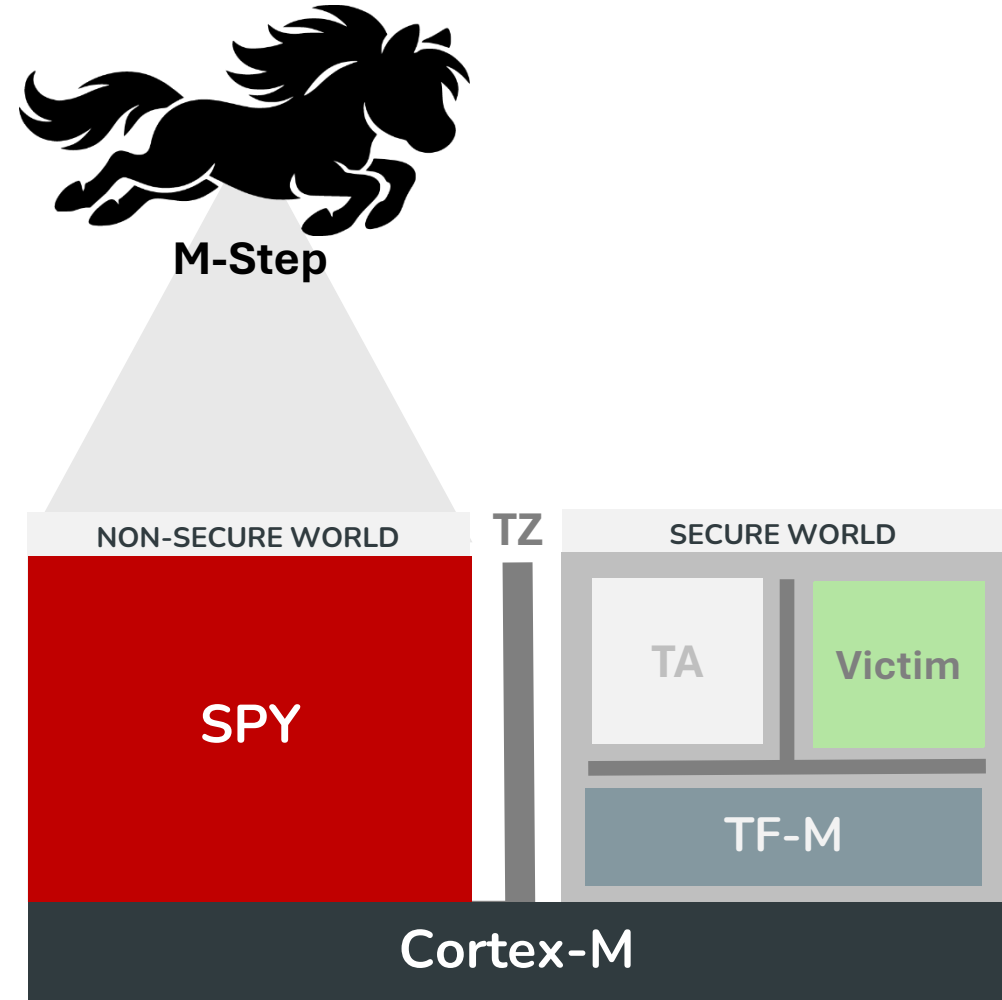
Mstp-Test

BREAKING REAL SOFTWARE

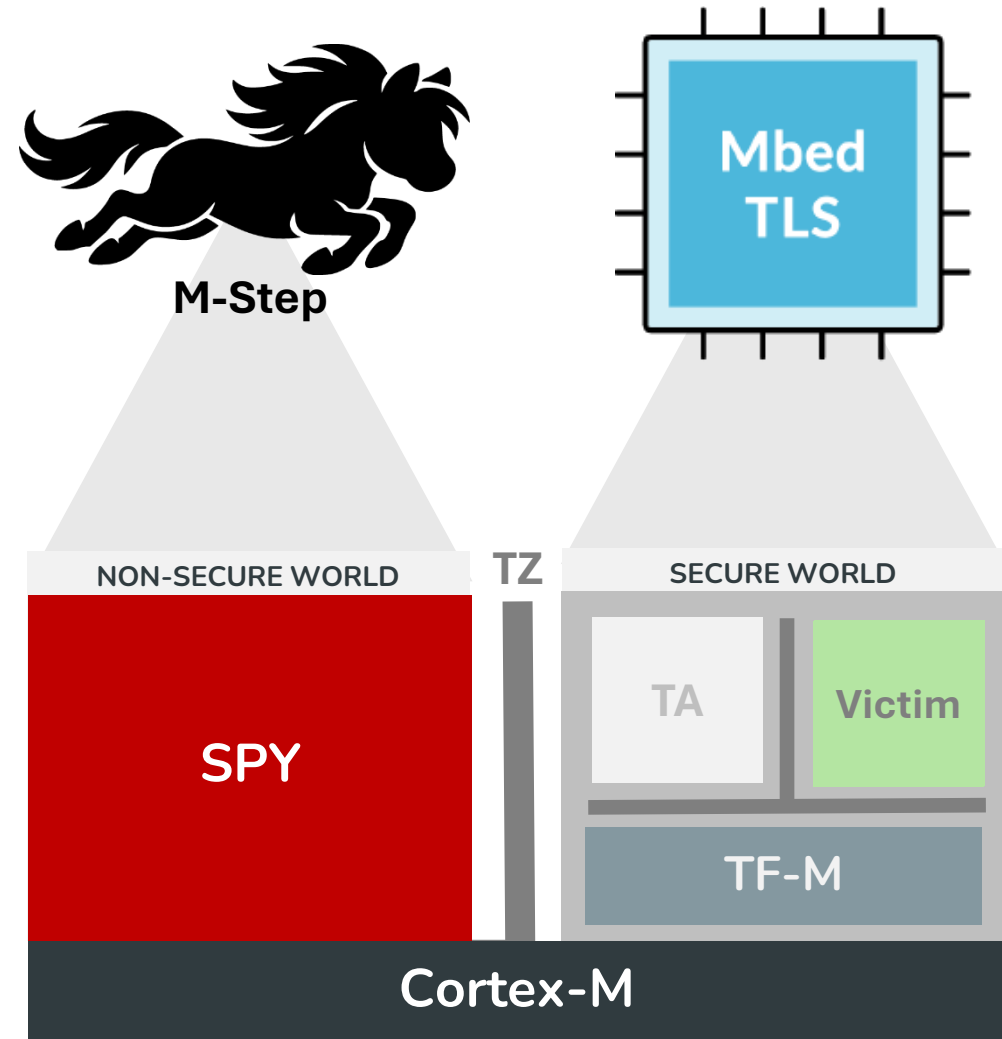
Real World Use Case



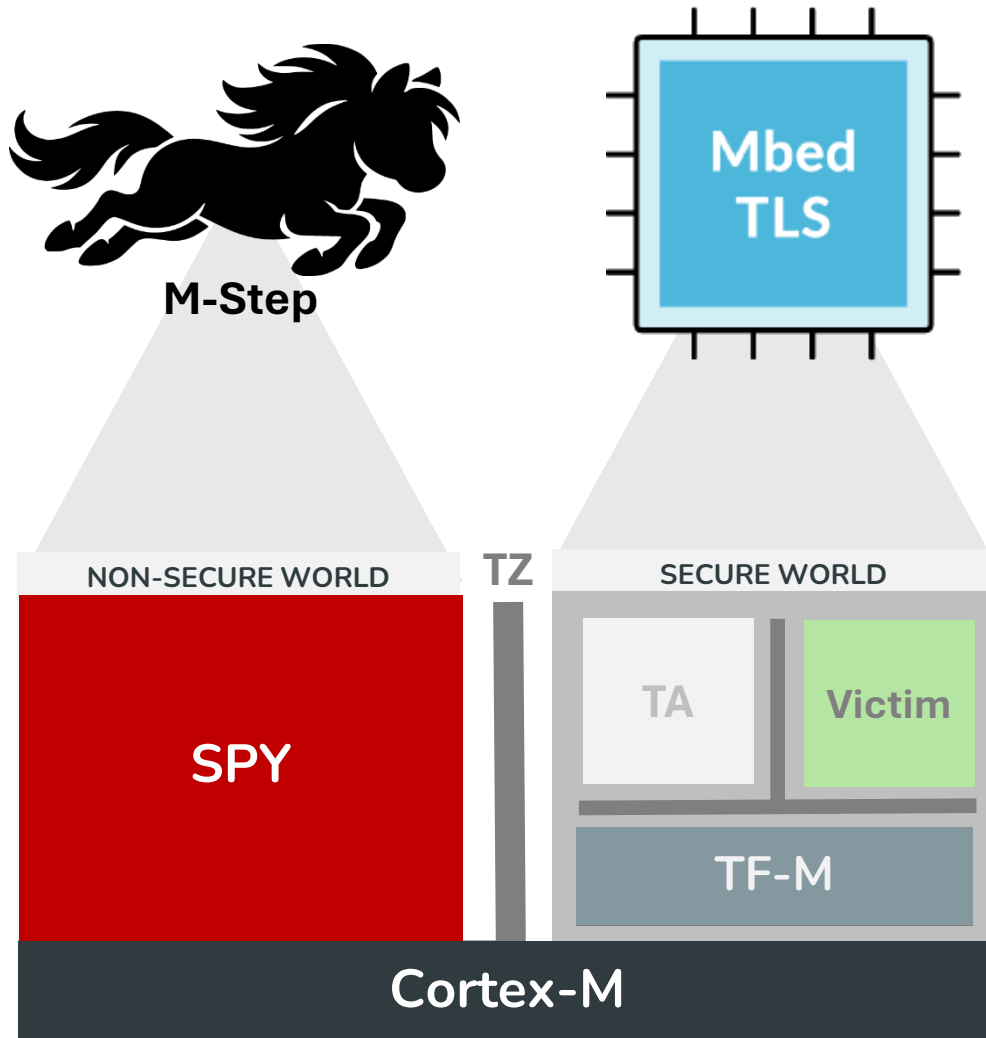
Real World Use Case



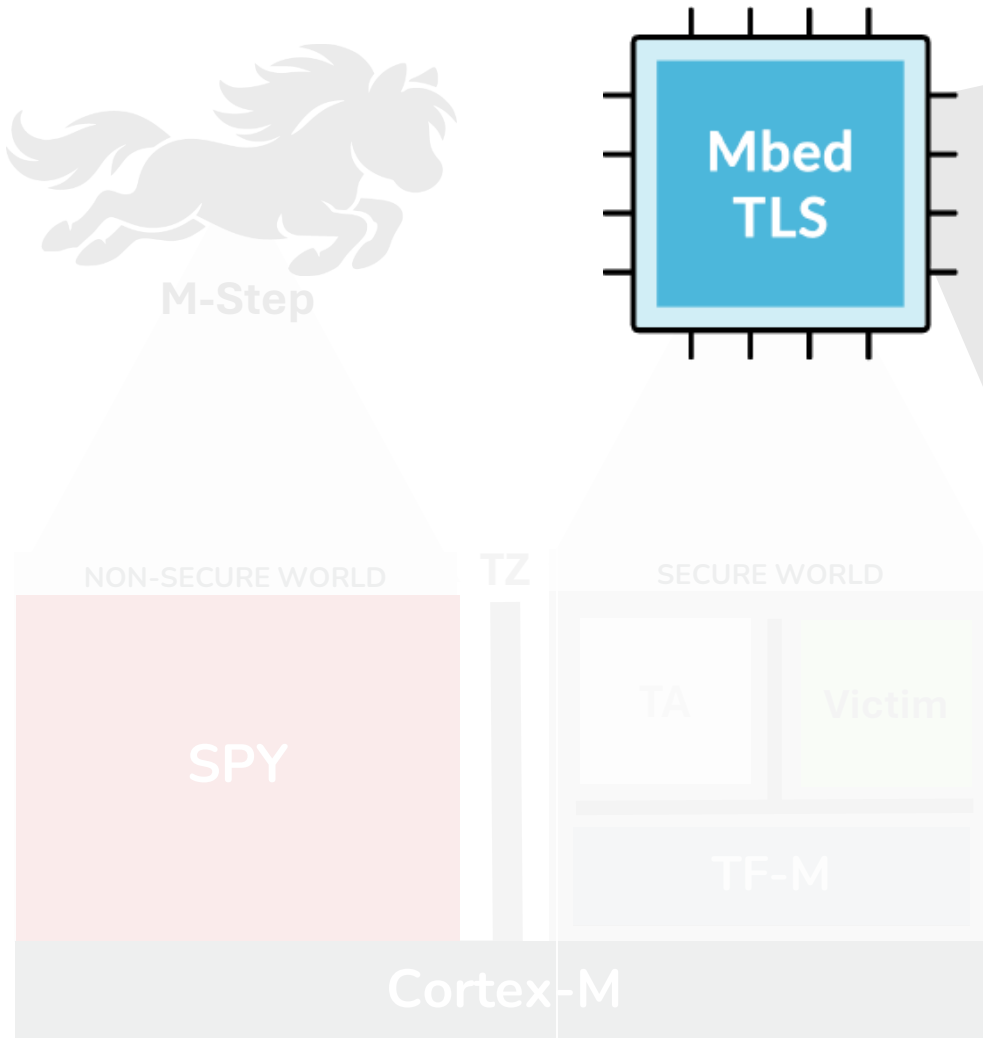
Real World Use Case



Real World Use Case



Real World Use Case

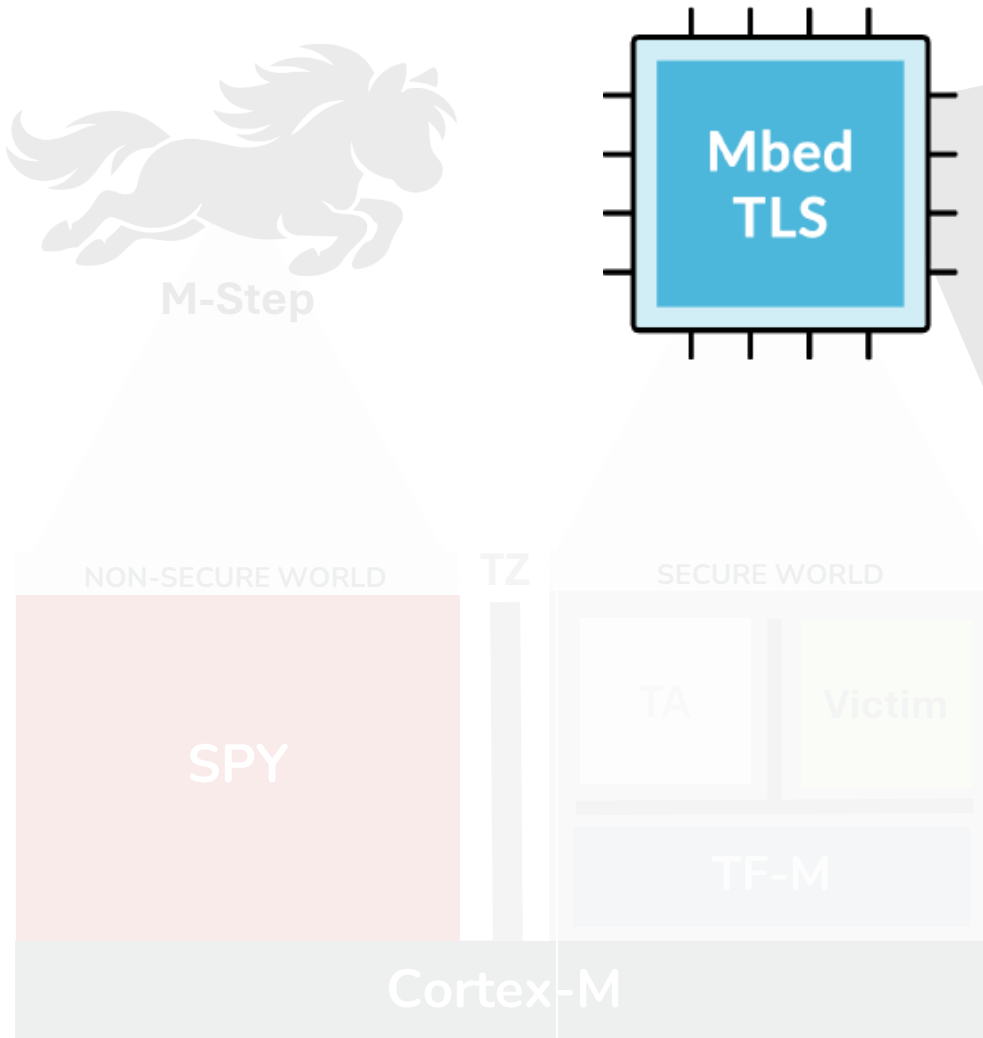


```

1 procedure Mbedtls_BEEA( $X, A, B$ )
2    $TU \leftarrow A, TV \leftarrow B;$ 
3    $U1 \leftarrow 1, U2 \leftarrow 0, V1 \leftarrow 0, V2 \leftarrow 1;$ 
4   while  $TU \neq 0$  do
5     while  $TU$  even do
6        $TU \leftarrow TU/2;$ 
7       if  $U1$  or  $U2$  odd then
8          $U1 \leftarrow U1 + TB; U2 \leftarrow U2 - TA;$ 
9          $U1 \leftarrow U1/2; U2 \leftarrow U2/2;$ 
10      while  $TV$  even do
11         $TV \leftarrow TV/2;$ 
12        if  $V1$  or  $V2$  odd then
13           $V1 \leftarrow V1 + TB; V2 \leftarrow V2 - TA;$ 
14           $V1 \leftarrow V1/2; V2 \leftarrow V2/2;$ 
15      if  $TU \geq TV$  then
16         $TU \leftarrow TU - TV;$ 
17         $U1 \leftarrow U1 - V1; U2 \leftarrow U2 - V2;$ 
18         $TU \leftarrow TU/2;$ 
19      else
20         $TV \leftarrow TV - TU;$ 
21         $V1 \leftarrow V1 - U1; V2 \leftarrow V2 - U2;$ 
22         $TV \leftarrow TV/2;$ 

```

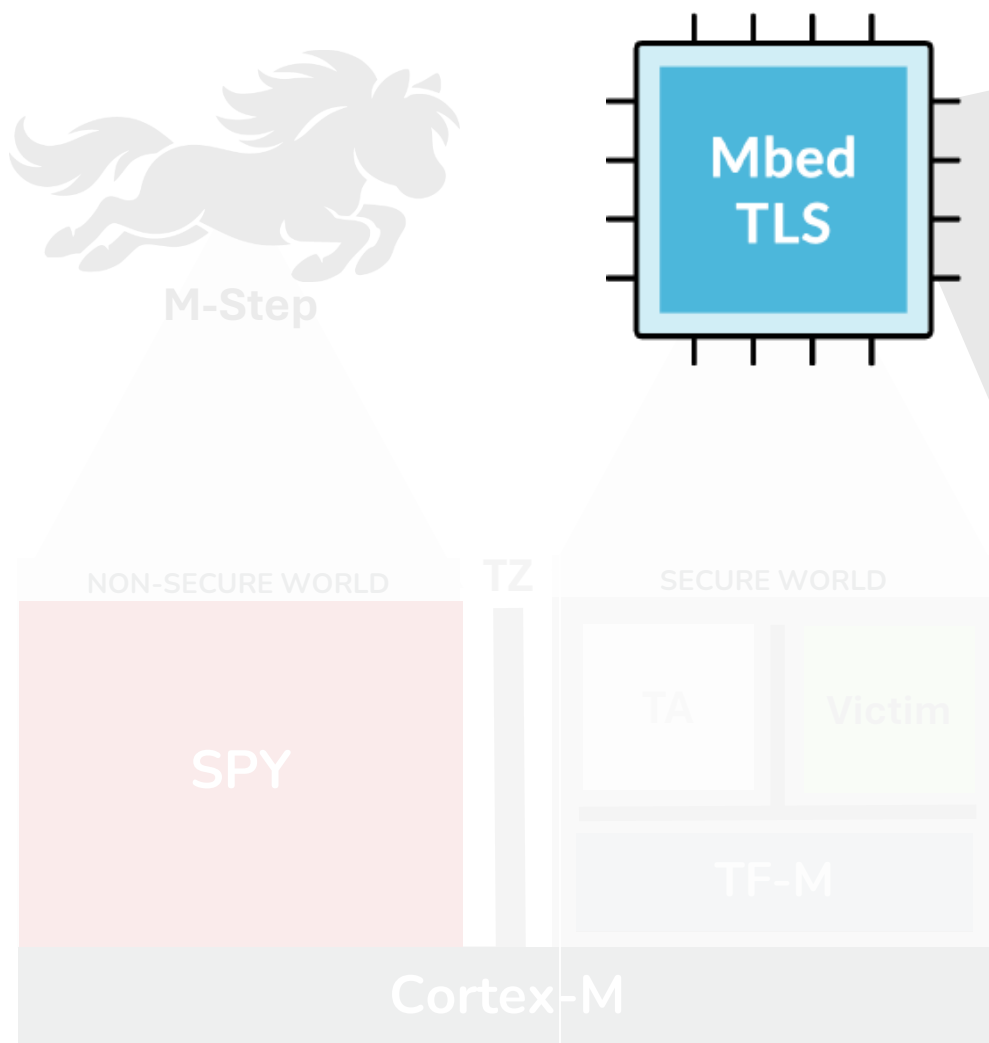
Real World Use Case



```
1 procedure Mbedtls_BEEA(X, A, B)
2    $TU \leftarrow A, TV \leftarrow B;$ 
3    $U1 \leftarrow 1, U2 \leftarrow 0, V1 \leftarrow 0, V2 \leftarrow 1;$ 
4   while  $TU \neq 0$  do
5     while  $TU$  even do
6        $TU \leftarrow TU/2;$ 
7       if  $U1$  or  $U2$  odd then
8          $U1 \leftarrow U1 + TB; U2 \leftarrow U2 - TA;$ 
9          $U1 \leftarrow U1/2; U2 \leftarrow U2/2;$ 
10      while  $TV$  even do
11         $TV \leftarrow TV/2;$ 
12        if  $V1$  or  $V2$  odd then
13           $V1 \leftarrow V1 + TB; V2 \leftarrow V2 - TA;$ 
14           $V1 \leftarrow V1/2; V2 \leftarrow V2/2;$ 
15      if  $TU \geq TV$  then
16         $TU \leftarrow TU - TV;$ 
17         $U1 \leftarrow U1 - V1; U2 \leftarrow U2 - V2;$ 
18         $TU \leftarrow TU/2;$ 
19      else
20         $TV \leftarrow TV - TU;$ 
21         $V1 \leftarrow V1 - U1; V2 \leftarrow V2 - U2;$ 
22         $TV \leftarrow TV/2;$ 
```

OpenSSL
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Real World Use Case



```

1 procedure Mbedtls_BEEA(X, A, B)
2    $TU \leftarrow A, TV \leftarrow B;$ 
3    $U1 \leftarrow 1, U2 \leftarrow 0, V1 \leftarrow 0, V2 \leftarrow 1;$ 
4   while  $TU \neq 0$  do
5     while  $TU$  even do
6        $TU \leftarrow TU/2;$ 
7       if  $U1$  or  $U2$  odd then
8          $U1 \leftarrow U1 + TB; U2 \leftarrow U2 - TA;$ 
9          $U1 \leftarrow U1/2; U2 \leftarrow U2/2;$ 
10      while  $TV$  even do
11         $TV \leftarrow TV/2;$ 
12        if  $V1$  or  $V2$  odd then
13           $V1 \leftarrow V1 + TB; V2 \leftarrow V2 - TA;$ 
14           $V1 \leftarrow V1/2; V2 \leftarrow V2/2;$ 
15      if  $TU \geq TV$  then
16         $TU \leftarrow TU - TV;$ 
17         $U1 \leftarrow U1 - V1; U2 \leftarrow U2 - V2;$ 
18         $TU \leftarrow TU/2;$ 
19      else
20         $TV \leftarrow TV - TU;$ 
21         $V1 \leftarrow V1 - U1; V2 \leftarrow V2 - U2;$ 
22         $TV \leftarrow TV/2;$ 

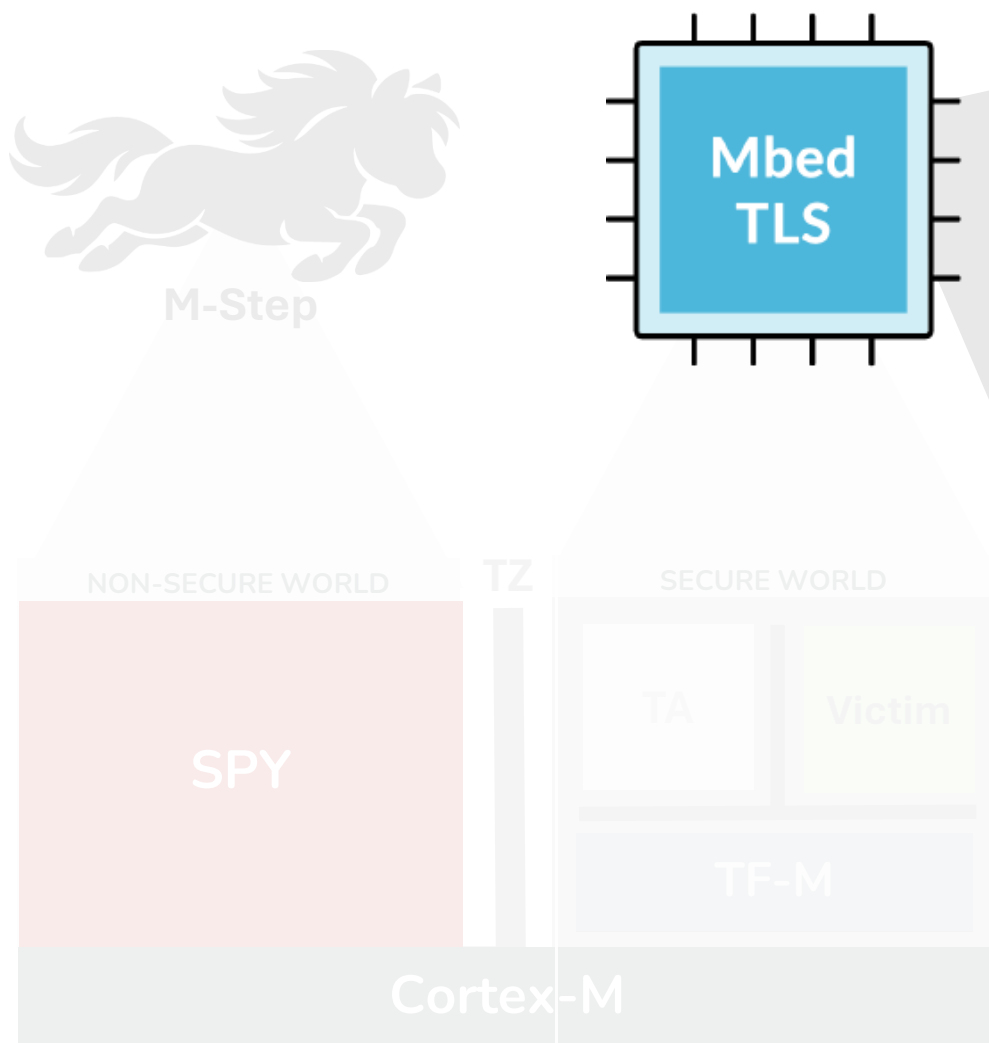
```

OpenSSL
LIBRARY



WolfSSL

Real World Use Case



```

1 procedure Mbedtls_BEEA(X, A, B)
2    $TU \leftarrow A, TV \leftarrow B;$ 
3    $U1 \leftarrow 1, U2 \leftarrow 0, V1 \leftarrow 0, V2 \leftarrow 1;$ 
4   while  $TU \neq 0$  do
5     while  $TU$  even do
6        $TU \leftarrow TU/2;$ 
7       if  $U1$  or  $U2$  odd then
8          $U1 \leftarrow U1 + TB; U2 \leftarrow U2 - TA;$ 
9          $U1 \leftarrow U1/2; U2 \leftarrow U2/2;$ 
10      while  $TV$  even do
11         $TV \leftarrow TV/2;$ 
12        if  $V1$  or  $V2$  odd then
13           $V1 \leftarrow V1 + TB; V2 \leftarrow V2 - TA;$ 
14           $V1 \leftarrow V1/2; V2 \leftarrow V2/2;$ 
15      if  $TU \geq TV$  then
16         $TU \leftarrow TU - TV;$ 
17         $U1 \leftarrow U1 - V1; U2 \leftarrow U2 - V2;$ 
18         $TU \leftarrow TU/2;$ 
19      else
20         $TV \leftarrow TV - TU;$ 
21         $V1 \leftarrow V1 - U1; V2 \leftarrow V2 - U2;$ 
22         $TV \leftarrow TV/2;$ 

```

OpenSSL
LIBRARY



WolfSSL



Libgcrypt

Real World Use Case

```

1 procedure Mbedtis_BEEA(X, A, B)
2   TU ← A, TV ← B;
3   U1 ← 1, U2 ← 0, V1 ← 0, V2 ← 1;

```

	Vuln.	Function	Call Tree (vulnerable path)	Recover Private Key	A	C	!C	P	D
Mod. Inv.	V#1	mb_rsa_gen_key	mb_mpi_inv_mod(D, E, L)	$D = E^{-1} \bmod L$	●	●	●	●	●
	V#2	mb_rsa_gen_key	mb_rsa_deduce_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	●	○	●	●
	V#3	mb_rsa_complete	mb_rsa_deduce_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	●	○	○	●
	V#4	mb_rsa_complete	mb_rsa_deduce_private_exponent → mb_mpi_inv_mod(D, E, K)	$D = E^{-1} \bmod K$	○	○	●	●	●
	V#5	mb_pk_psa_rsa_sign_ext	mb_rsa_write_key → mb_rsa_export_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	○	●	●	●
	V#6	rsa_decrypt_wrap	mb_rsa_write_key → mb_rsa_export_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	○	●	●	●
	V#7	import_pair_into_psa	mb_rsa_write_key → mb_rsa_export_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	○	●	●	●
	V#8	pk_write_rsa_der	mb_rsa_write_key → mb_rsa_export_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	○	●	●	●
	V#9	mb_rsa_export_crt	mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	○	●	●	●
	V#10	mb_psa_rsa_export_key	mb_pk_write_key_der → mb_rsa_export_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	○	●	●	●
	V#11	mb_psa_rsa_export_public_key	mb_psa_rsa_export_key → mb_pk_write_key_der → mb_rsa_export_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	○	●	●	●
	V#12	mb_psa_rsa_import_key	mb_psa_rsa_export_key → mb_pk_write_key_der → mb_rsa_export_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	○	●	●	●
	V#13	mb_psa_rsa_generate_key	mb_psa_rsa_export_key → mb_pk_write_key_der → mb_rsa_export_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	○	●	●	●
Mod. Inv. TF-M	V#14	mb_rsa_complete	mb_rsa_deduce_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	●	○	●	●
	V#15	mb_rsa_parse_key	mb_rsa_complete → mb_rsa_deduce_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	●	○	●	●
	V#16	mb_psa_rsa_load_representation	mb_rsa_parse_key → mb_rsa_complete → mb_rsa_deduce_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	●	○	●	●
	V#17	mb_psa_rsa_sign_hash	mb_psa_rsa_load_representation → mb_rsa_parse_key → mb_rsa_complete → mb_rsa_deduce_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	●	○	●	●
	V#18	mb_psa_rsa_verify_hash	mb_psa_rsa_load_representation → mb_rsa_parse_key → mb_rsa_complete → mb_rsa_deduce_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	●	○	●	●
	V#19	mb_psa_asymmetric_encrypt	mb_psa_rsa_load_representation → mb_rsa_parse_key → mb_rsa_complete → mb_rsa_deduce_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	●	○	●	●
	V#20	mb_psa_asymmetric_decrypt	mb_psa_rsa_load_representation → mb_rsa_parse_key → mb_rsa_complete → mb_rsa_deduce_crt → mb_mpi_inv_mod(QP, Q, P)	$D = E^{-1} \bmod (P-1)(Q-1)$	○	●	○	●	●
GCD	V#21	rsa_gen_key	mb_mpi_gcd(D, E, H)	$D = E^{-1} \bmod H$	●	●	●	●	●
	V#22	rsa_gen_key	mb_mpi_gcd(D, P, Q)	P, Q partial leak following [239]	●	●	●	●	●
	V#23	mb_mpi_inv_mod	mb_mpi_gcd(D, P, Q)	P, Q partial leak following [239]	●	●	●	●	●
	V#24	mb_rsa_deduce_primes	mb_mpi_gcd(D, K, N)	$P = K/N$	●	●	●	●	●

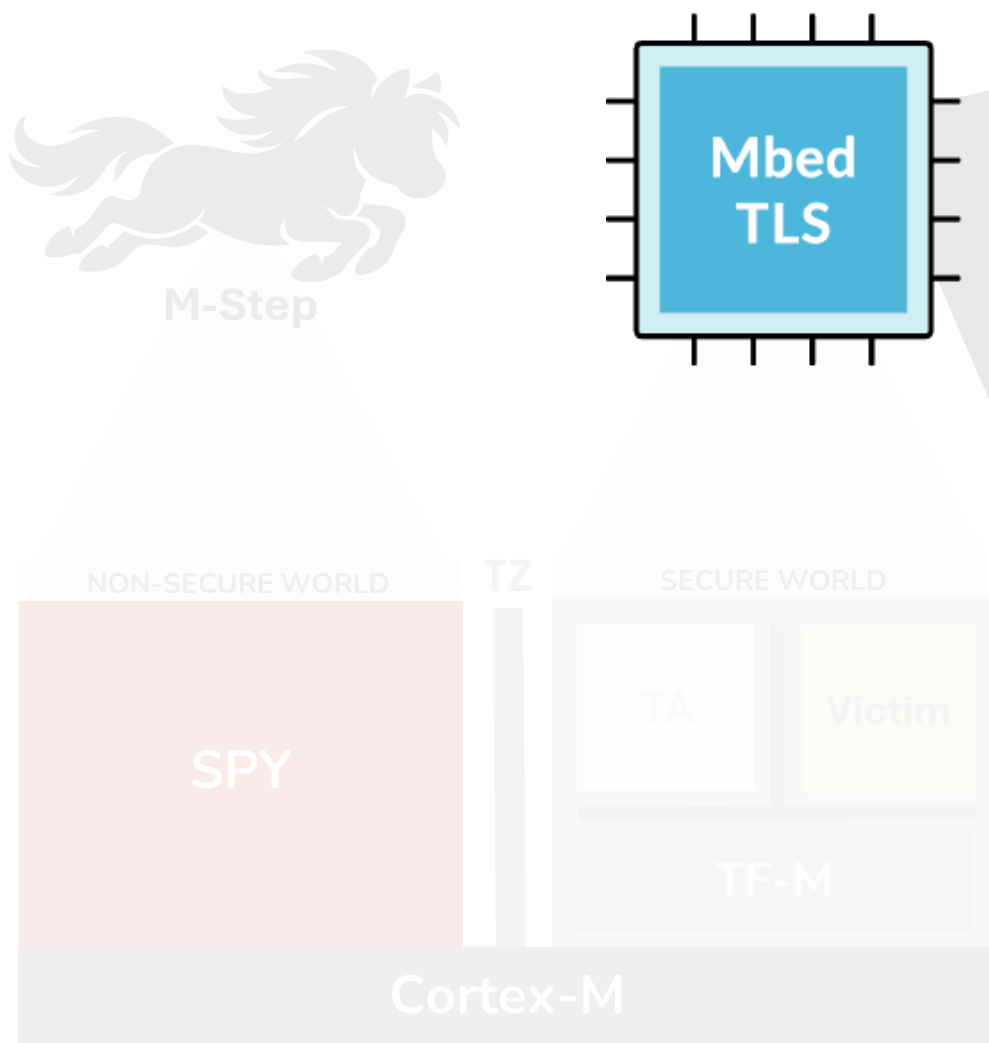
Cortex

22

TV ← TV/2;

Libgcrypt

Real World Use Case



```

1 procedure Mbedtls_BEEA(X, A, B)
2    $TU \leftarrow A, TV \leftarrow B;$ 
3    $U1 \leftarrow 1, U2 \leftarrow 0, V1 \leftarrow 0, V2 \leftarrow 1;$ 
4   while  $TU \neq 0$  do
5     while  $TU$  even do
6        $TU \leftarrow TU/2;$ 
7       if  $U1$  or  $U2$  odd then
8          $U1 \leftarrow U1 + TB; U2 \leftarrow U2 - TA;$ 
9          $U1 \leftarrow U1/2; U2 \leftarrow U2/2;$ 
10      while  $TV$  even do
11         $TV \leftarrow TV/2;$ 
12        if  $V1$  or  $V2$  odd then
13           $V1 \leftarrow V1 + TB; V2 \leftarrow V2 - TA;$ 
14           $V1 \leftarrow V1/2; V2 \leftarrow V2/2;$ 
15      if  $TU \geq TV$  then
16         $TU \leftarrow TU - TV;$ 
17         $U1 \leftarrow U1 - V1; U2 \leftarrow U2 - V2;$ 
18         $TU \leftarrow TU/2;$ 
19      else
20         $TV \leftarrow TV - TU;$ 
21         $V1 \leftarrow V1 - U1; V2 \leftarrow V2 - U2;$ 
22         $TV \leftarrow TV/2;$ 

```

OpenSSL
LIBRARY

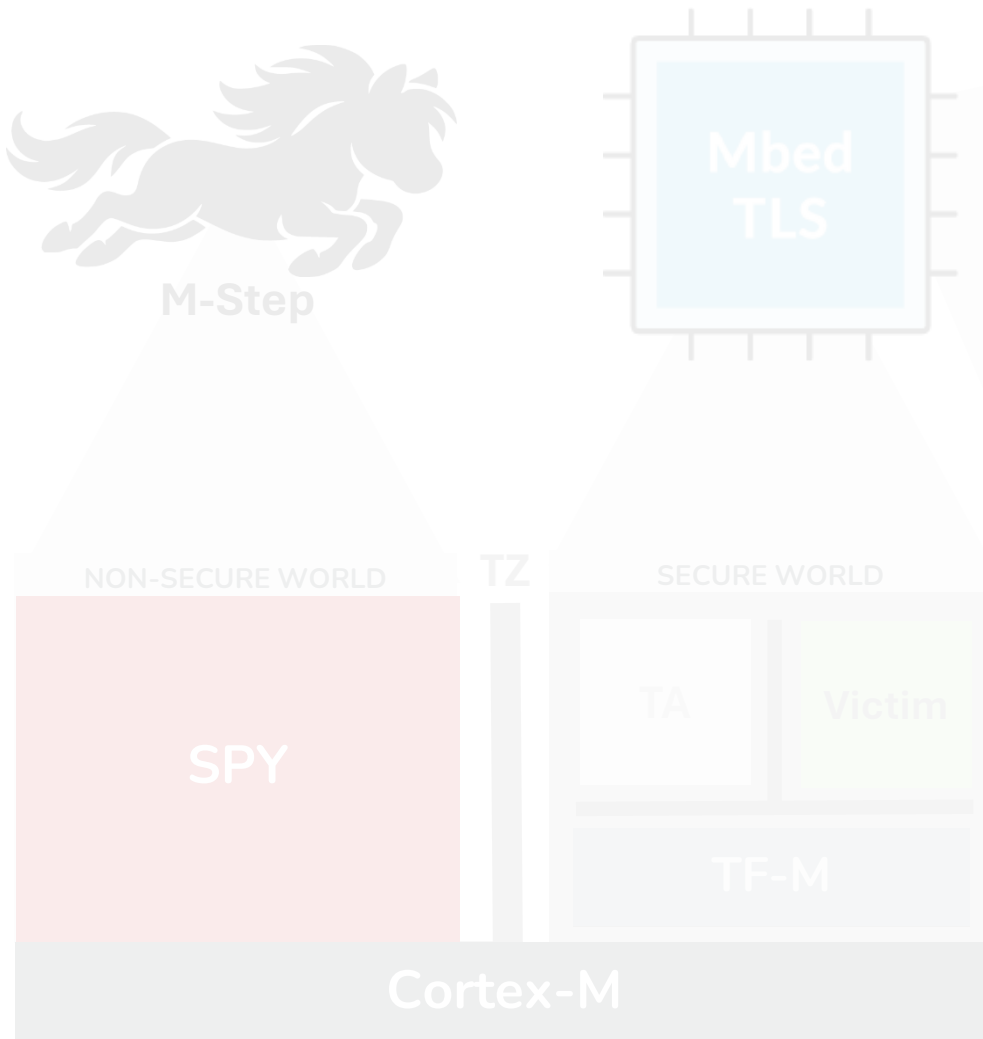


WolfSSL



Libgcrypt

Real World Use Case



```
1 procedure Mbedtls_BEEA(X, A, B)
2    $TU \leftarrow A, TV \leftarrow B;$ 
3    $U1 \leftarrow 1, U2 \leftarrow 0, V1 \leftarrow 0, V2 \leftarrow 1;$ 
4   while  $TU \neq 0$  do
5     while  $TU$  even do
6        $TU \leftarrow TU/2;$ 
7       if  $U1$  or  $U2$  odd then
8          $U1 \leftarrow U1 + TB; U2 \leftarrow U2 - TA;$ 
9          $U1 \leftarrow U1/2; U2 \leftarrow U2/2;$ 
10      while  $TV$  even do
11         $TV \leftarrow TV/2;$ 
12        if  $V1$  or  $V2$  odd then
13           $V1 \leftarrow V1 + TB; V2 \leftarrow V2 - TA;$ 
14           $V1 \leftarrow V1/2; V2 \leftarrow V2/2;$ 
15      if  $TU \geq TV$  then
16         $TU \leftarrow TU - TV;$ 
17         $U1 \leftarrow U1 - V1; U2 \leftarrow U2 - V2;$ 
18         $TU \leftarrow TU/2;$ 
19      else
20         $TV \leftarrow TV - TU;$ 
21         $V1 \leftarrow V1 - U1; V2 \leftarrow V2 - U2;$ 
22         $TV \leftarrow TV/2;$ 
```

OpenSSL
LIBRARY



wolfSSL



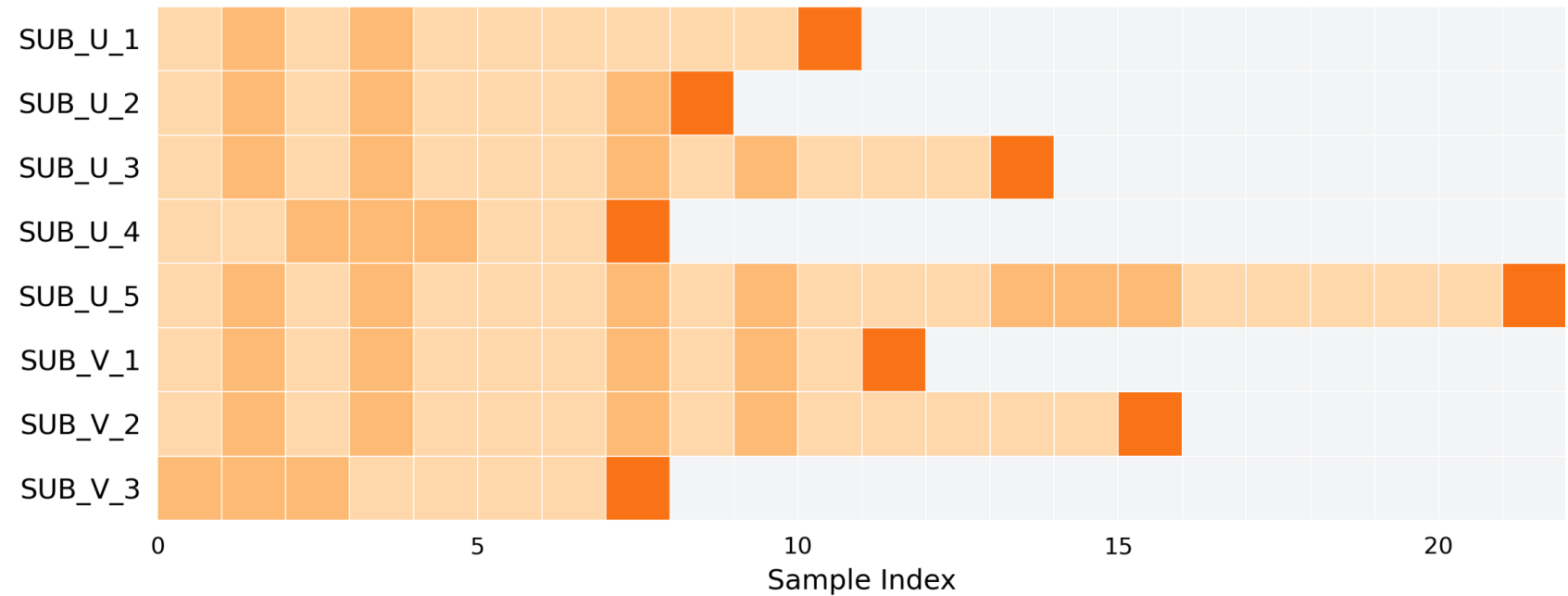
Libgcrypt

BEEA Template Matrices

```

1 procedure Mbedtls_BEEA(X, A, B)
2    $TU \leftarrow A, TV \leftarrow B;$ 
3    $U1 \leftarrow 1, U2 \leftarrow 0, V1 \leftarrow 0, V2 \leftarrow 1;$ 
4   while  $TU \neq 0$  do
5     while  $TU$  even do
6        $TU \leftarrow TU/2;$ 
7       if  $U1$  or  $U2$  odd then
8          $U1 \leftarrow U1 + TB; U2 \leftarrow U2 - TA;$ 
9          $U1 \leftarrow U1/2; U2 \leftarrow U2/2;$ 
10      while  $TV$  even do
11         $TV \leftarrow TV/2;$ 
12        if  $V1$  or  $V2$  odd then
13           $V1 \leftarrow V1 + TB; V2 \leftarrow V2 - TA;$ 
14           $V1 \leftarrow V1/2; V2 \leftarrow V2/2;$ 
15      if  $TU \geq TV$  then
16         $TU \leftarrow TU - TV;$ 
17         $U1 \leftarrow U1 - V1; U2 \leftarrow U2 - V2;$ 
18         $TU \leftarrow TU/2;$ 
19      else
20         $TV \leftarrow TV - TU;$ 
21         $V1 \leftarrow V1 - U1; V2 \leftarrow V2 - U2;$ 
22         $TV \leftarrow TV/2;$ 

```

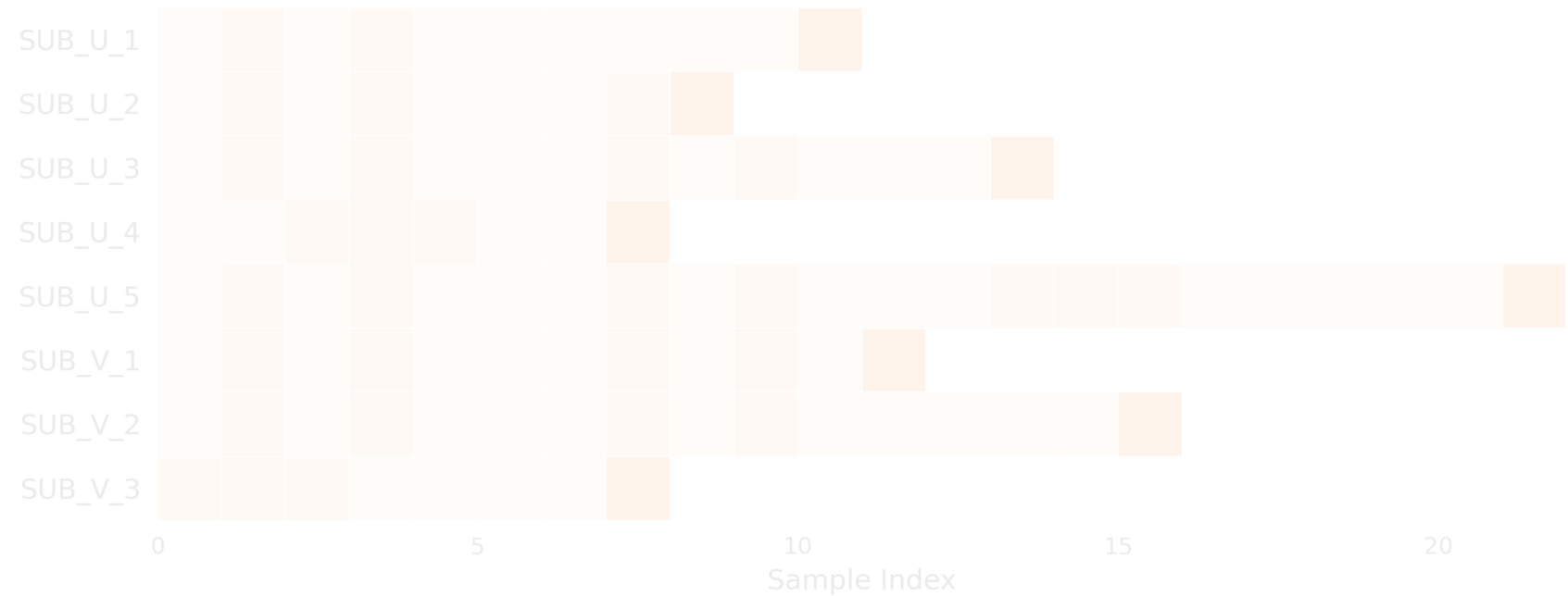
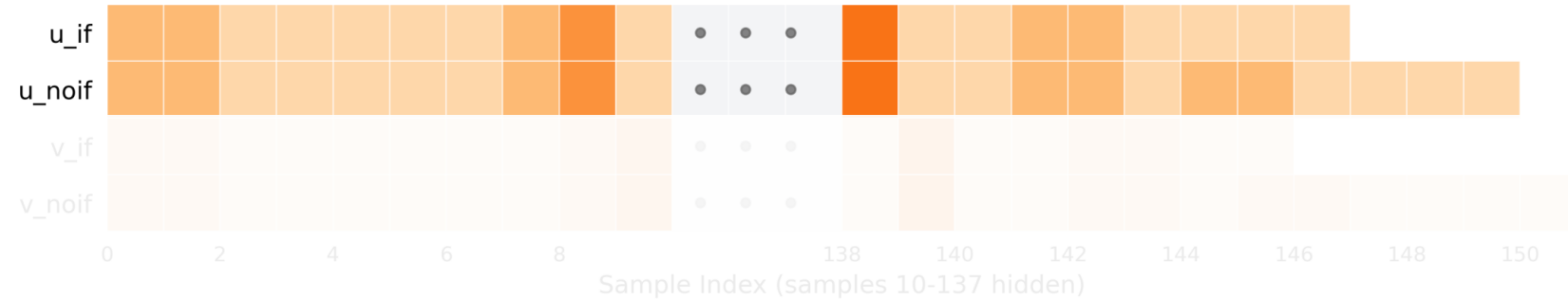


BEEA Template Matrices

```

1 procedure MbedtIs_BEEA(X, A, B)
2    $TU \leftarrow A, TV \leftarrow B;$ 
3    $U1 \leftarrow 1, U2 \leftarrow 0, V1 \leftarrow 0, V2 \leftarrow 1;$ 
4   while  $TU \neq 0$  do
5     while  $TU$  even do
6        $TU \leftarrow TU/2;$ 
7       if  $U1$  or  $U2$  odd then
8          $U1 \leftarrow U1 + TB; U2 \leftarrow U2 - TA;$ 
9          $U1 \leftarrow U1/2; U2 \leftarrow U2/2;$ 
10      while  $TV$  even do
11         $TV \leftarrow TV/2;$ 
12        if  $V1$  or  $V2$  odd then
13           $V1 \leftarrow V1 + TB; V2 \leftarrow V2 - TA;$ 
14           $V1 \leftarrow V1/2; V2 \leftarrow V2/2;$ 
15      if  $TU \geq TV$  then
16         $TU \leftarrow TU - TV;$ 
17         $U1 \leftarrow U1 - V1; U2 \leftarrow U2 - V2;$ 
18         $TU \leftarrow TU/2;$ 
19      else
20         $TV \leftarrow TV - TU;$ 
21         $V1 \leftarrow V1 - U1; V2 \leftarrow V2 - U2;$ 
22         $TV \leftarrow TV/2;$ 

```

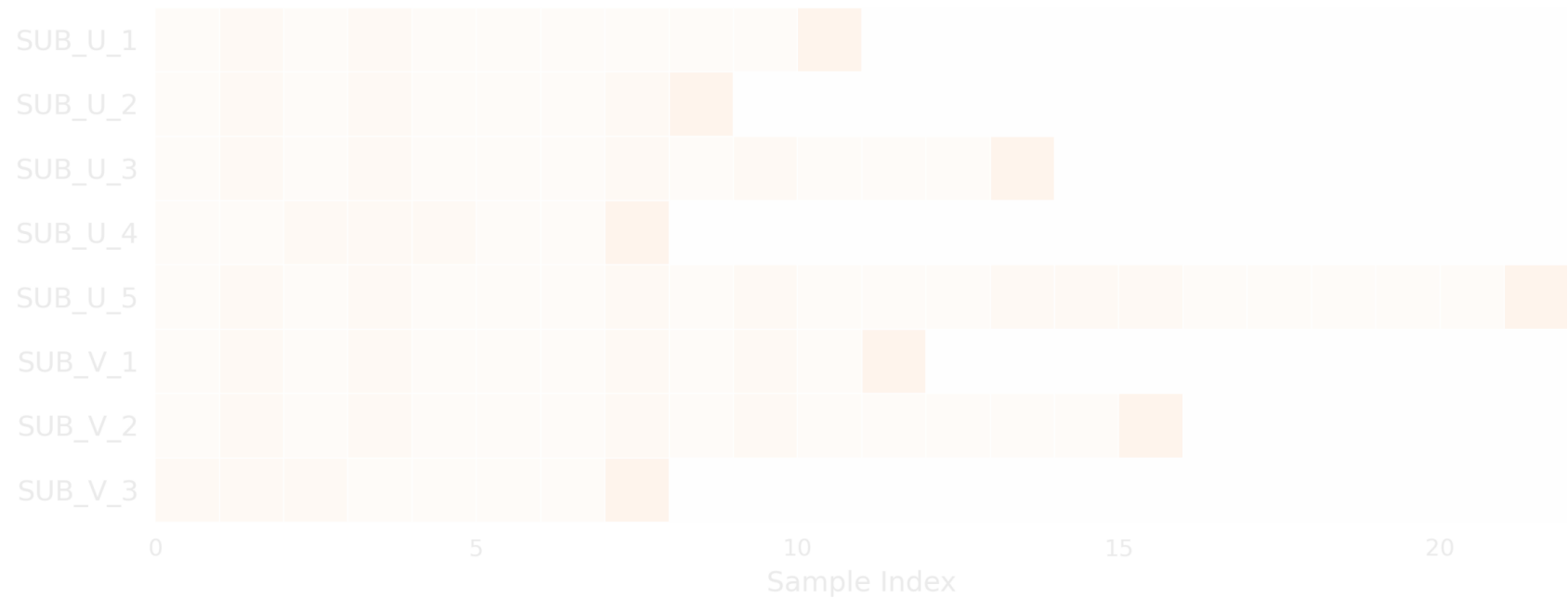
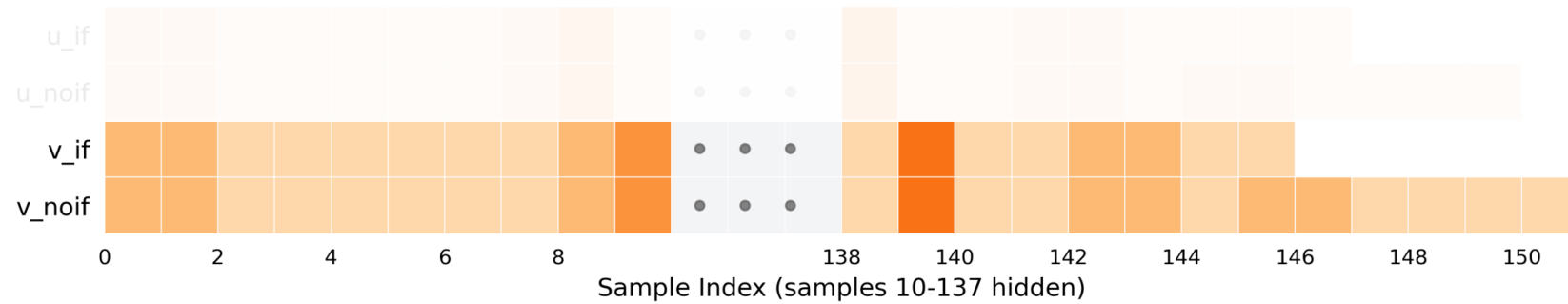


BEEA Template Matrices

```

1 procedure Mbedtls_BEEA(X, A, B)
2    $TU \leftarrow A, TV \leftarrow B;$ 
3    $U1 \leftarrow 1, U2 \leftarrow 0, V1 \leftarrow 0, V2 \leftarrow 1;$ 
4   while  $TU \neq 0$  do
5     while  $TU$  even do
6        $TU \leftarrow TU/2;$ 
7       if  $U1$  or  $U2$  odd then
8          $U1 \leftarrow U1 + TB; U2 \leftarrow U2 - TA;$ 
9          $U1 \leftarrow U1/2; U2 \leftarrow U2/2;$ 
10    while  $TV$  even do
11       $TV \leftarrow TV/2;$ 
12      if  $V1$  or  $V2$  odd then
13         $V1 \leftarrow V1 + TB; V2 \leftarrow V2 - TA;$ 
14         $V1 \leftarrow V1/2; V2 \leftarrow V2/2;$ 
15    if  $TU \geq TV$  then
16       $TU \leftarrow TU - TV;$ 
17       $U1 \leftarrow U1 - V1; U2 \leftarrow U2 - V2;$ 
18       $TU \leftarrow TU/2;$ 
19    else
20       $TV \leftarrow TV - TU;$ 
21       $V1 \leftarrow V1 - U1; V2 \leftarrow V2 - U2;$ 
22       $TV \leftarrow TV/2;$ 

```

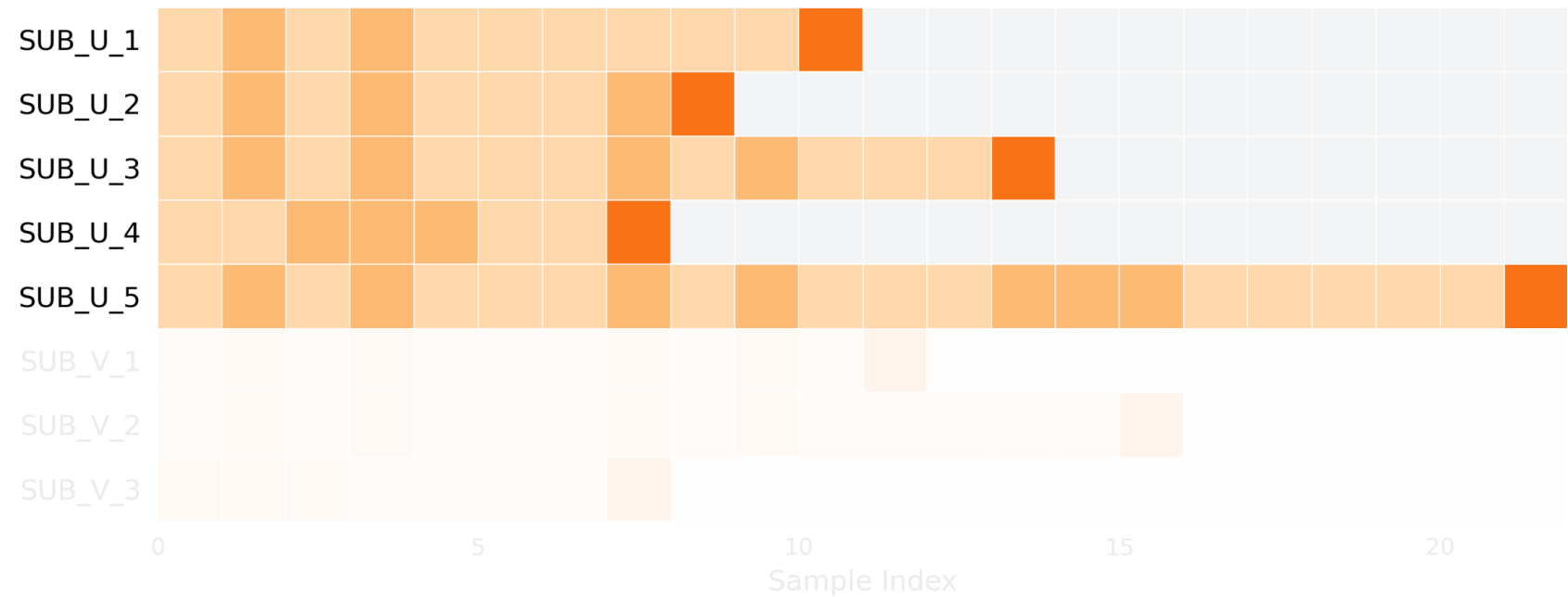
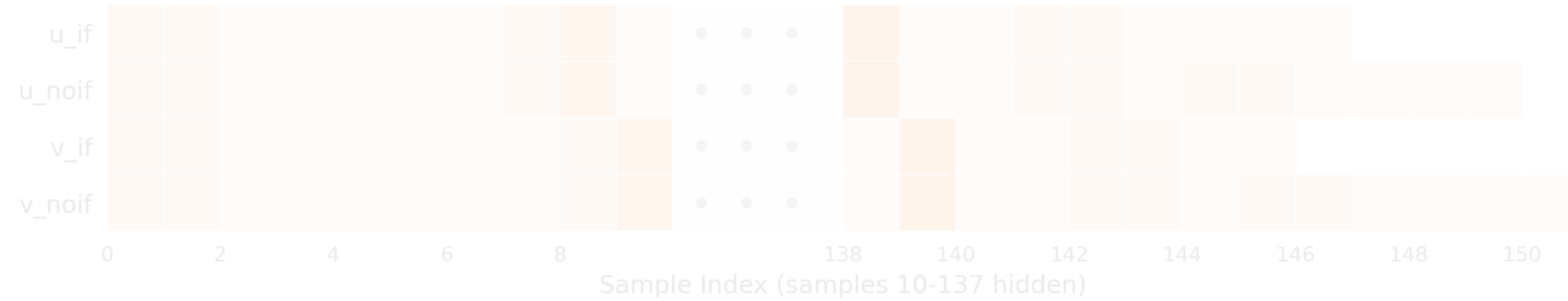


BEEA Template Matrices

```

1 procedure Mbedtls_BEEA(X, A, B)
2    $TU \leftarrow A, TV \leftarrow B;$ 
3    $U1 \leftarrow 1, U2 \leftarrow 0, V1 \leftarrow 0, V2 \leftarrow 1;$ 
4   while  $TU \neq 0$  do
5     while  $TU$  even do
6        $TU \leftarrow TU/2;$ 
7       if  $U1$  or  $U2$  odd then
8          $U1 \leftarrow U1 + TB; U2 \leftarrow U2 - TA;$ 
9          $U1 \leftarrow U1/2; U2 \leftarrow U2/2;$ 
10      while  $TV$  even do
11         $TV \leftarrow TV/2;$ 
12        if  $V1$  or  $V2$  odd then
13           $V1 \leftarrow V1 + TB; V2 \leftarrow V2 - TA;$ 
14           $V1 \leftarrow V1/2; V2 \leftarrow V2/2;$ 
15      if  $TU \geq TV$  then
16         $TU \leftarrow TU - TV;$ 
17         $U1 \leftarrow U1 - V1; U2 \leftarrow U2 - V2;$ 
18         $TU \leftarrow TU/2;$ 
19      else
20         $TV \leftarrow TV - TU;$ 
21         $V1 \leftarrow V1 - U1; V2 \leftarrow V2 - U2;$ 
22         $TV \leftarrow TV/2;$ 

```

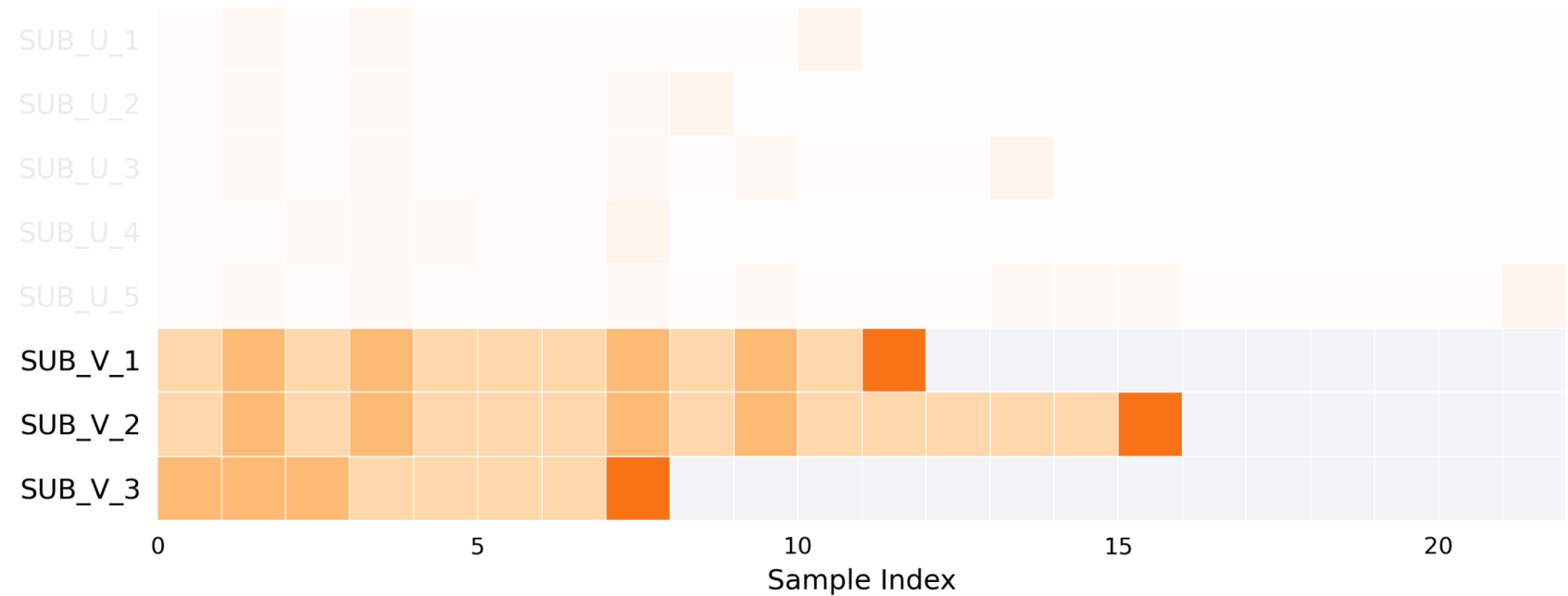
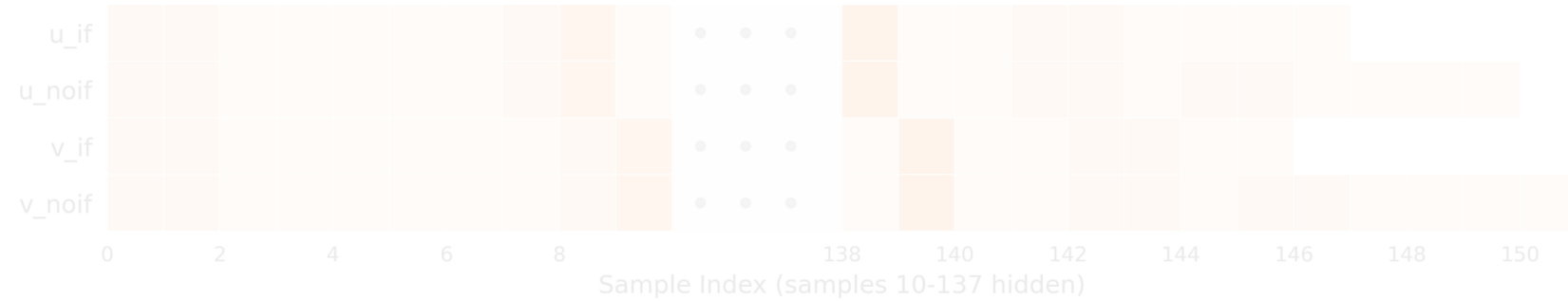


BEEA Template Matrices

```

1 procedure Mbedtls_BEEA(X, A, B)
2    $TU \leftarrow A, TV \leftarrow B;$ 
3    $U1 \leftarrow 1, U2 \leftarrow 0, V1 \leftarrow 0, V2 \leftarrow 1;$ 
4   while  $TU \neq 0$  do
5     while  $TU$  even do
6        $TU \leftarrow TU/2;$ 
7       if  $U1$  or  $U2$  odd then
8          $U1 \leftarrow U1 + TB; U2 \leftarrow U2 - TA;$ 
9          $U1 \leftarrow U1/2; U2 \leftarrow U2/2;$ 
10      while  $TV$  even do
11         $TV \leftarrow TV/2;$ 
12        if  $V1$  or  $V2$  odd then
13           $V1 \leftarrow V1 + TB; V2 \leftarrow V2 - TA;$ 
14           $V1 \leftarrow V1/2; V2 \leftarrow V2/2;$ 
15      if  $TU \geq TV$  then
16         $TU \leftarrow TU - TV;$ 
17         $U1 \leftarrow U1 - V1; U2 \leftarrow U2 - V2;$ 
18         $TU \leftarrow TU/2;$ 
19      else
20         $TV \leftarrow TV - TU;$ 
21         $V1 \leftarrow V1 - U1; V2 \leftarrow V2 - U2;$ 
22         $TV \leftarrow TV/2;$ 

```

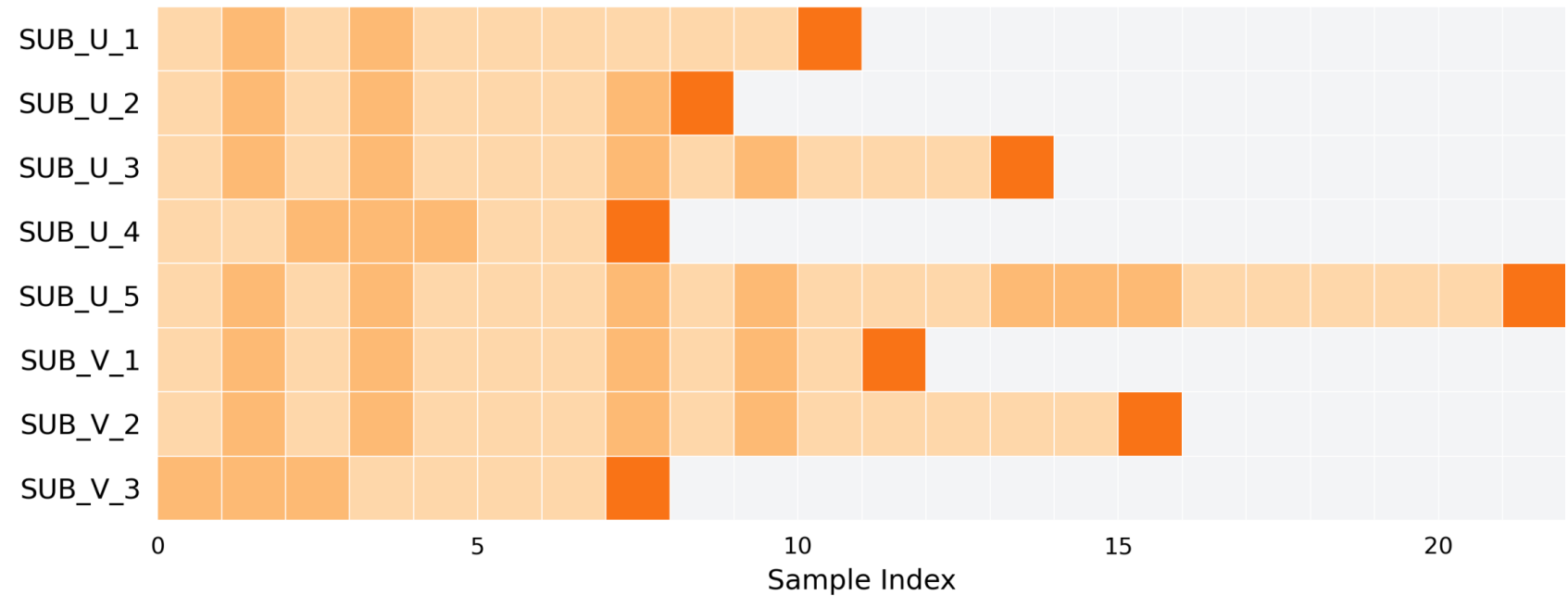


BEEA Template Matrices

```

1 procedure Mbedtls_BEEA(X, A, B)
2    $TU \leftarrow A, TV \leftarrow B;$ 
3    $U1 \leftarrow 1, U2 \leftarrow 0, V1 \leftarrow 0, V2 \leftarrow 1;$ 
4   while  $TU \neq 0$  do
5     while  $TU$  even do
6        $TU \leftarrow TU/2;$ 
7       if  $U1$  or  $U2$  odd then
8          $U1 \leftarrow U1 + TB; U2 \leftarrow U2 - TA;$ 
9          $U1 \leftarrow U1/2; U2 \leftarrow U2/2;$ 
10      while  $TV$  even do
11         $TV \leftarrow TV/2;$ 
12        if  $V1$  or  $V2$  odd then
13           $V1 \leftarrow V1 + TB; V2 \leftarrow V2 - TA;$ 
14           $V1 \leftarrow V1/2; V2 \leftarrow V2/2;$ 
15      if  $TU \geq TV$  then
16         $TU \leftarrow TU - TV;$ 
17         $U1 \leftarrow U1 - V1; U2 \leftarrow U2 - V2;$ 
18         $TU \leftarrow TU/2;$ 
19      else
20         $TV \leftarrow TV - TU;$ 
21         $V1 \leftarrow V1 - U1; V2 \leftarrow V2 - U2;$ 
22         $TV \leftarrow TV/2;$ 

```

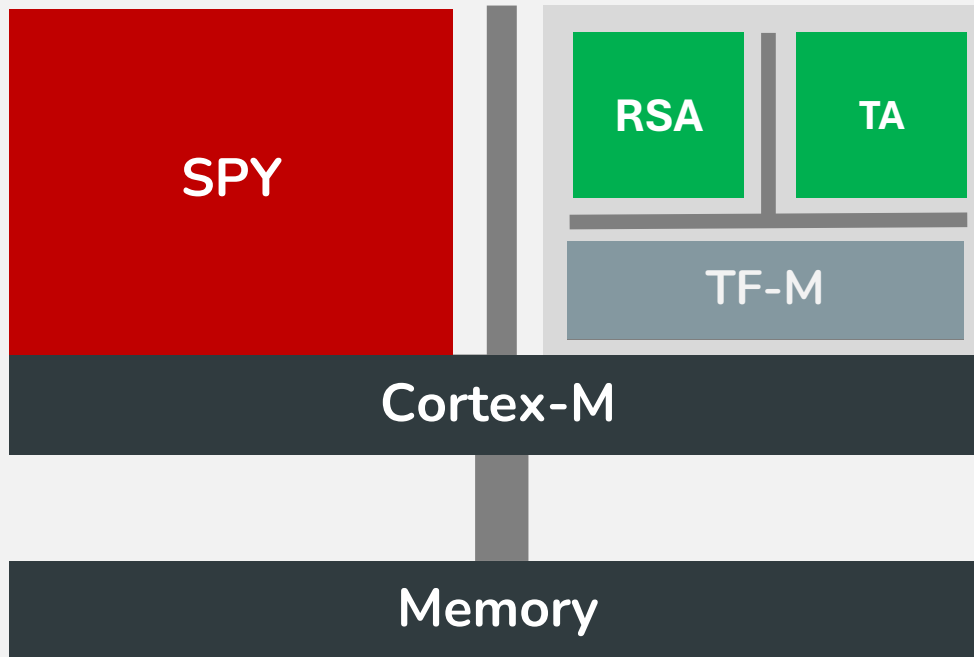


End-to-end Attacks on Mbed TLS

RSA Key Generation

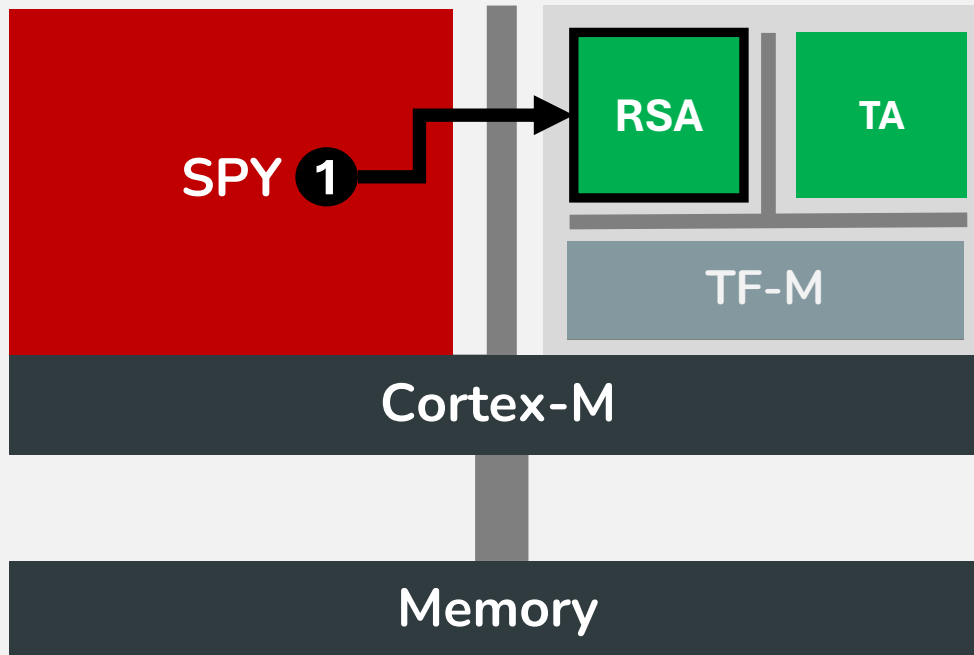
RSA Decryption

RSA Key Generation



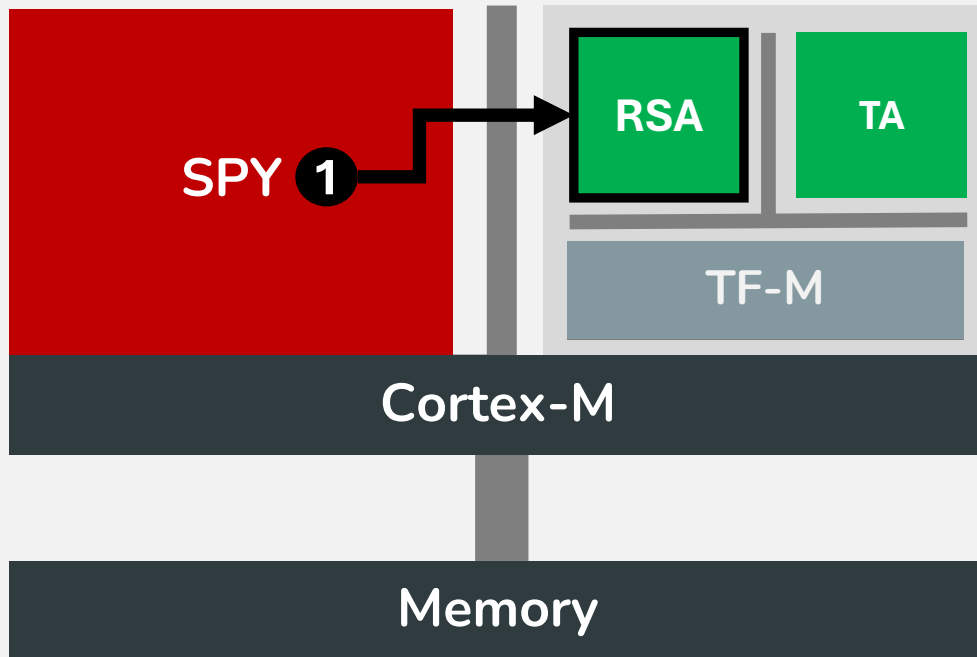
RSA Decryption

RSA Key Generation



RSA Decryption

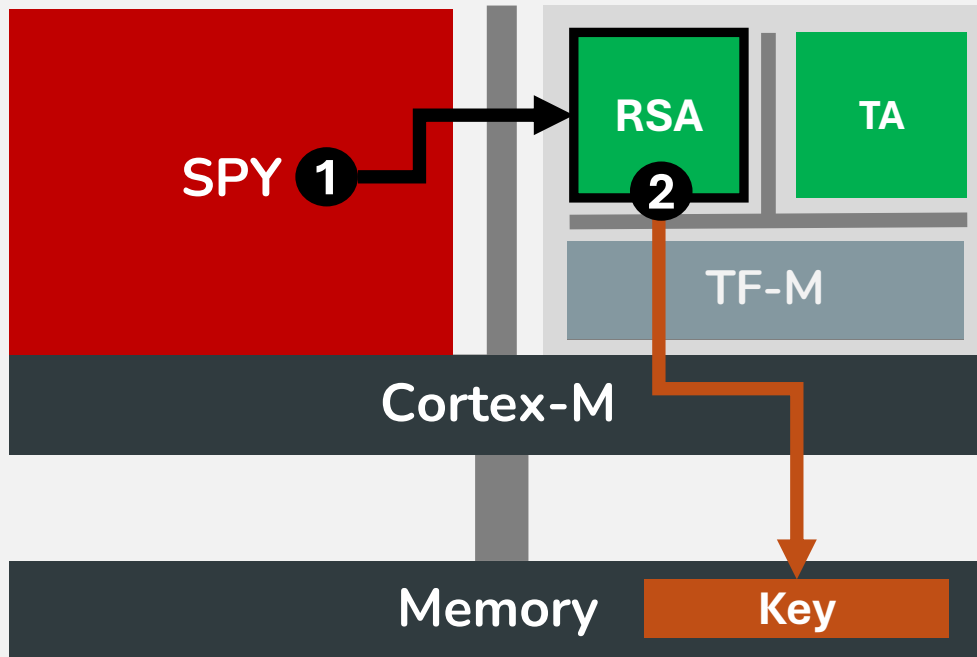
RSA Key Generation



1 Request Public Key

RSA Decryption

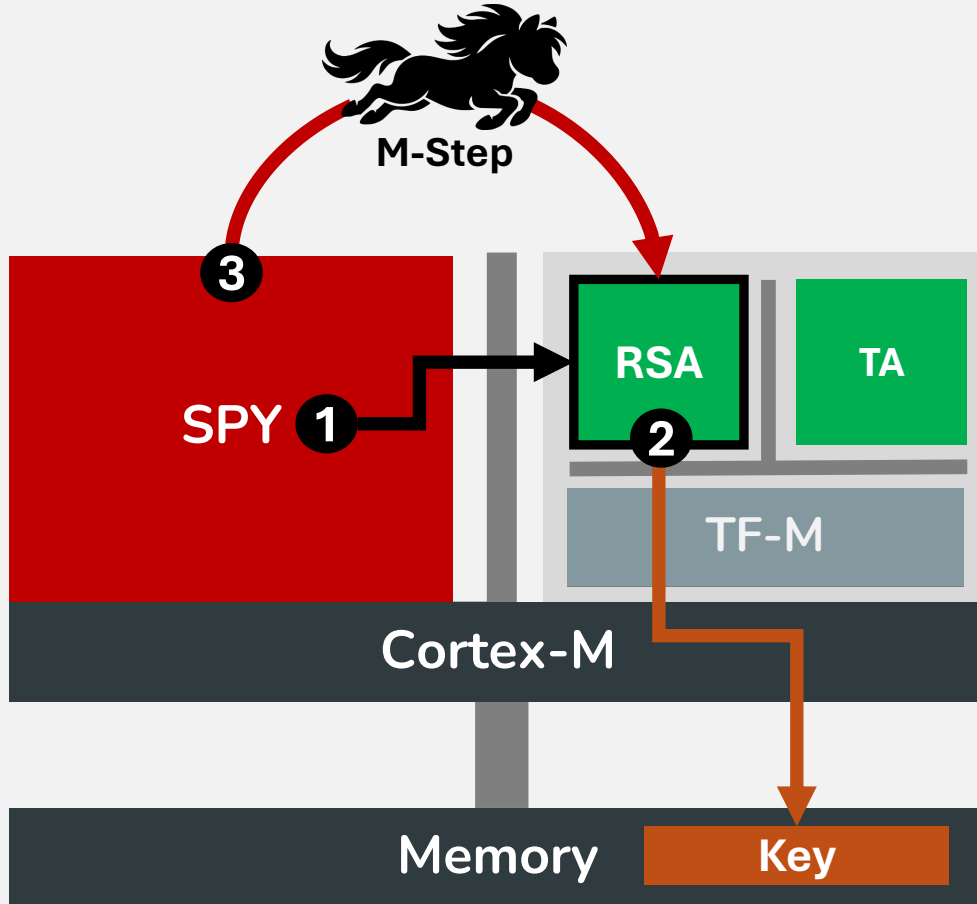
RSA Key Generation



1 Request Public Key

RSA Decryption

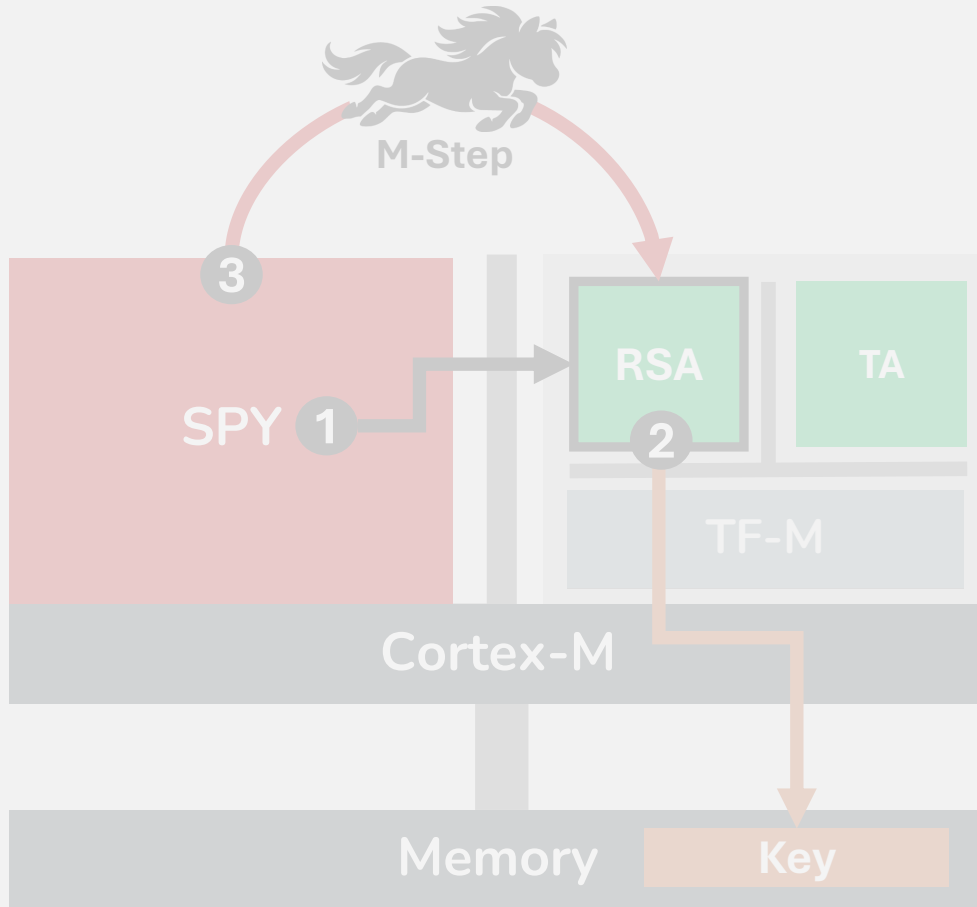
RSA Key Generation



❶ Request Public Key

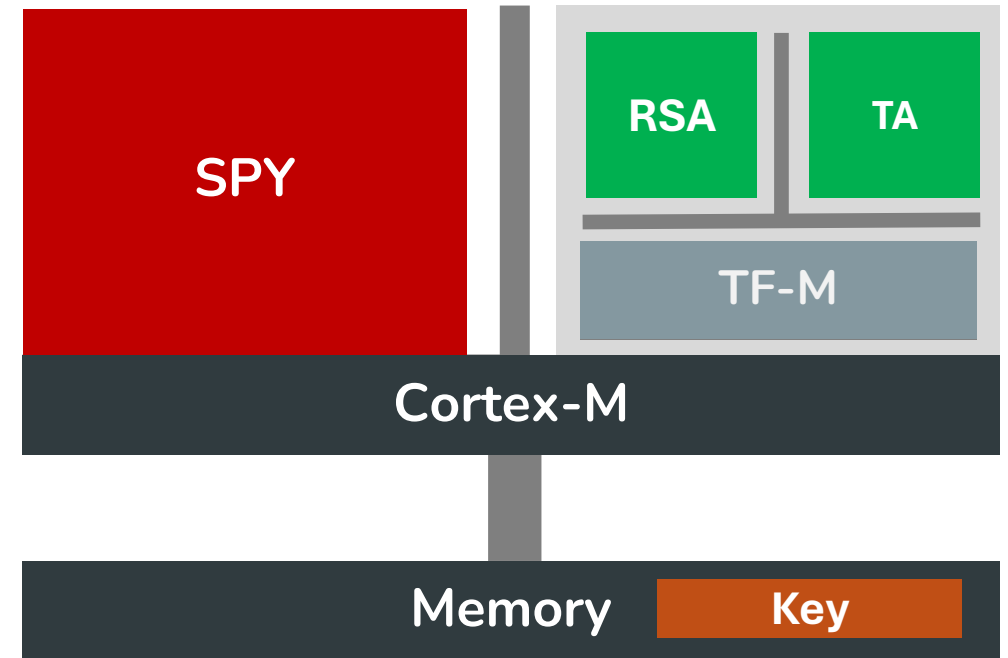
RSA Decryption

RSA Key Generation

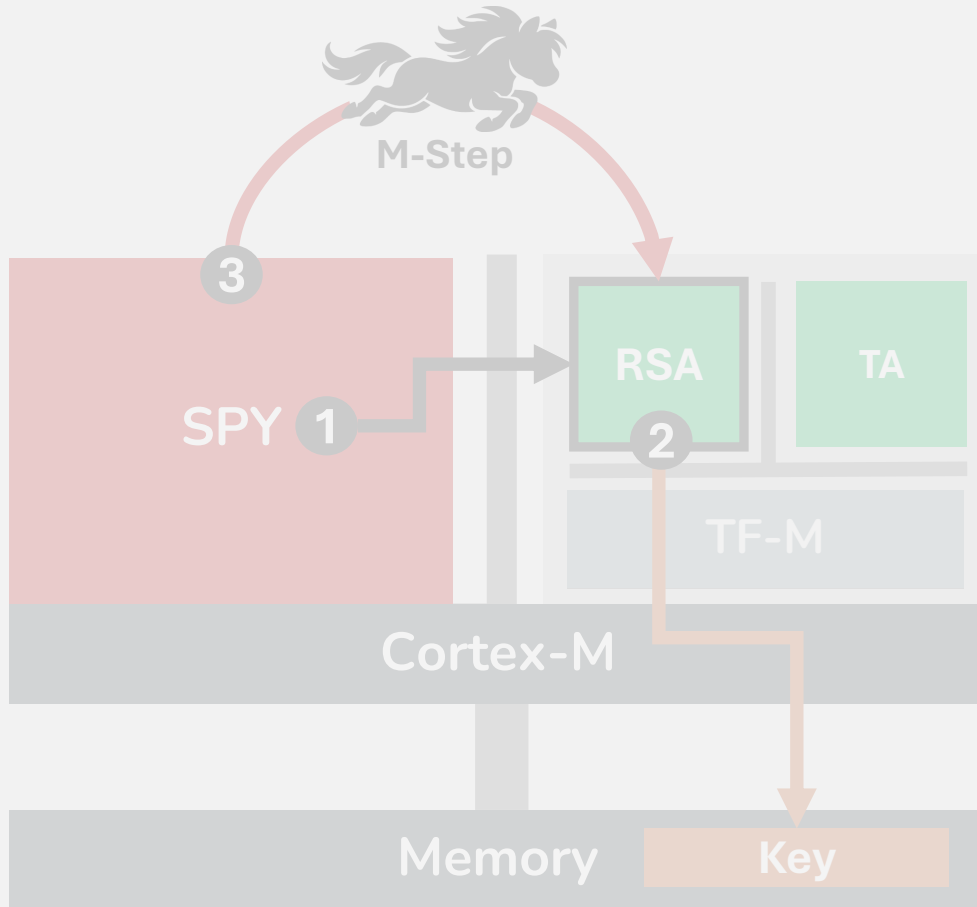


1 Request Public Key

RSA Decryption

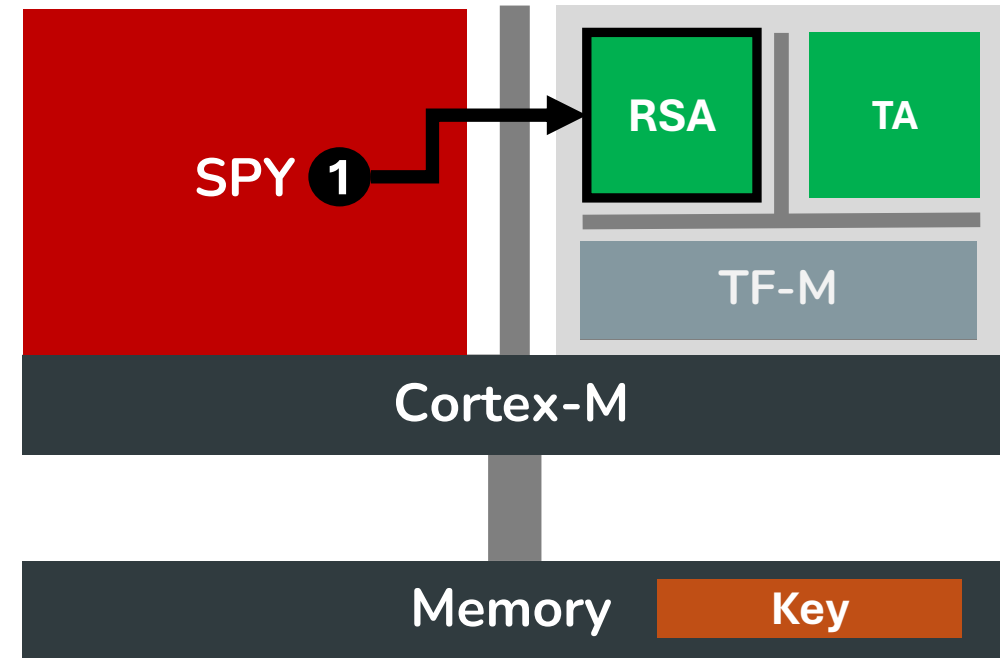


RSA Key Generation

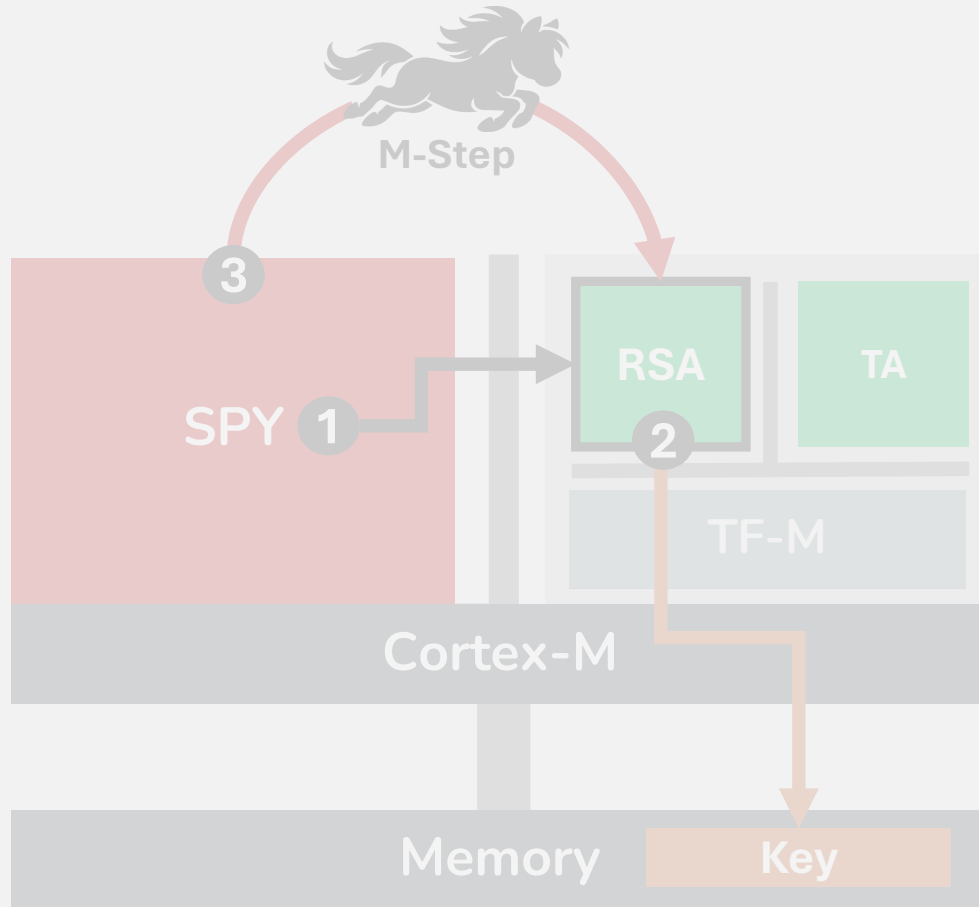


① Request Public Key

RSA Decryption

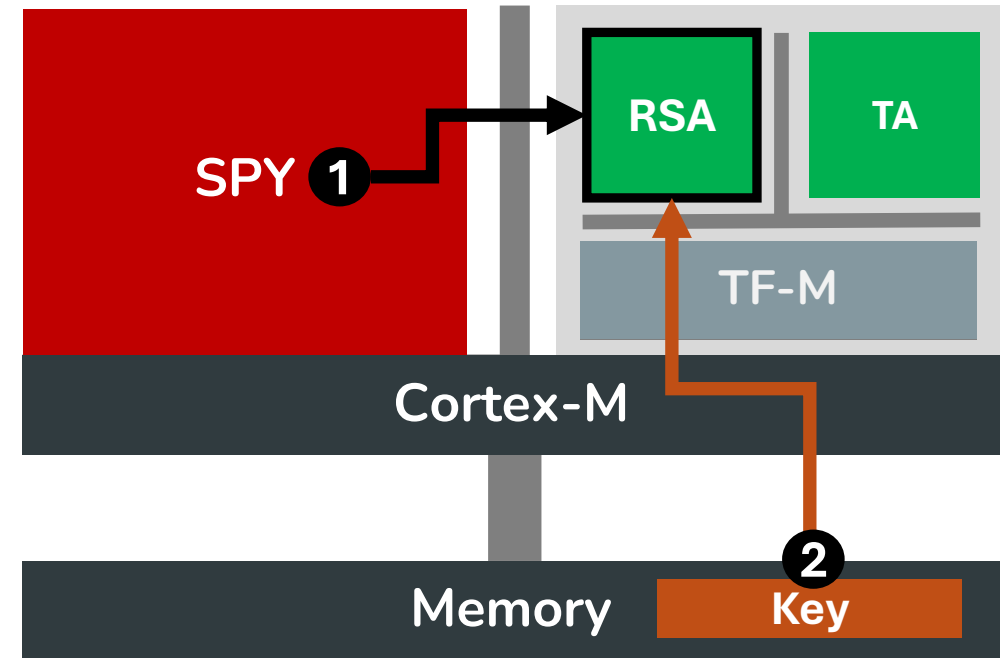


RSA Key Generation

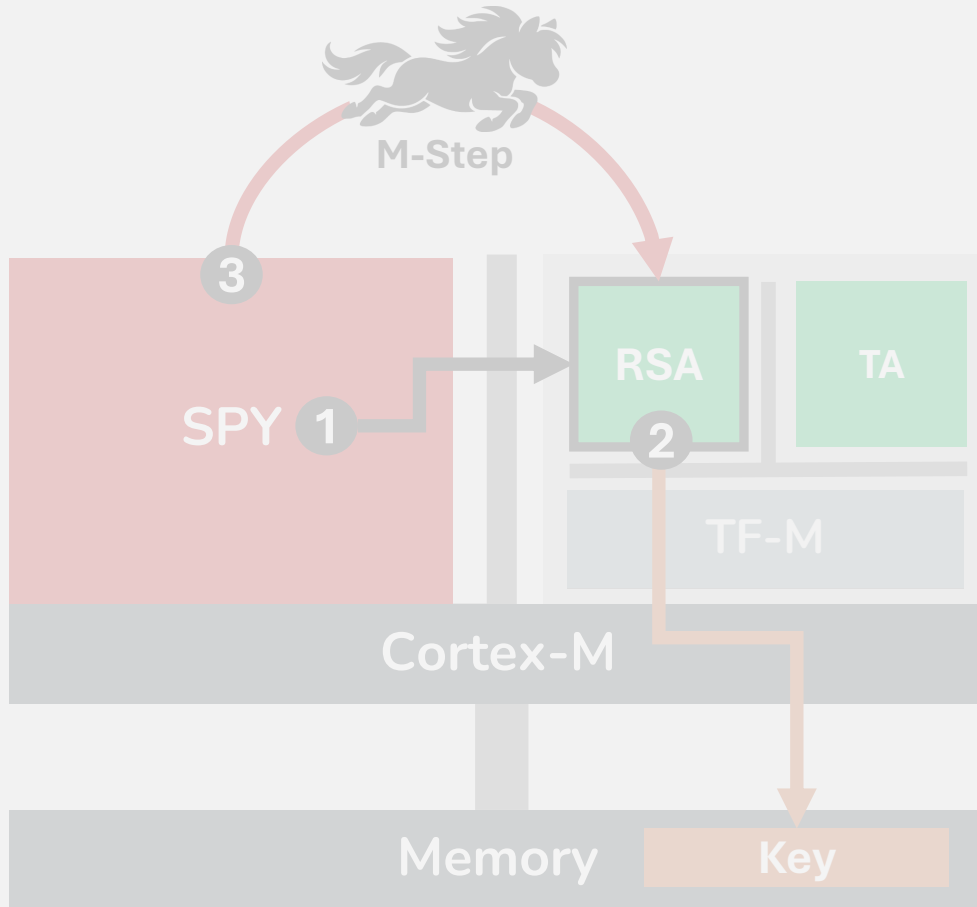


① Request Public Key

RSA Decryption

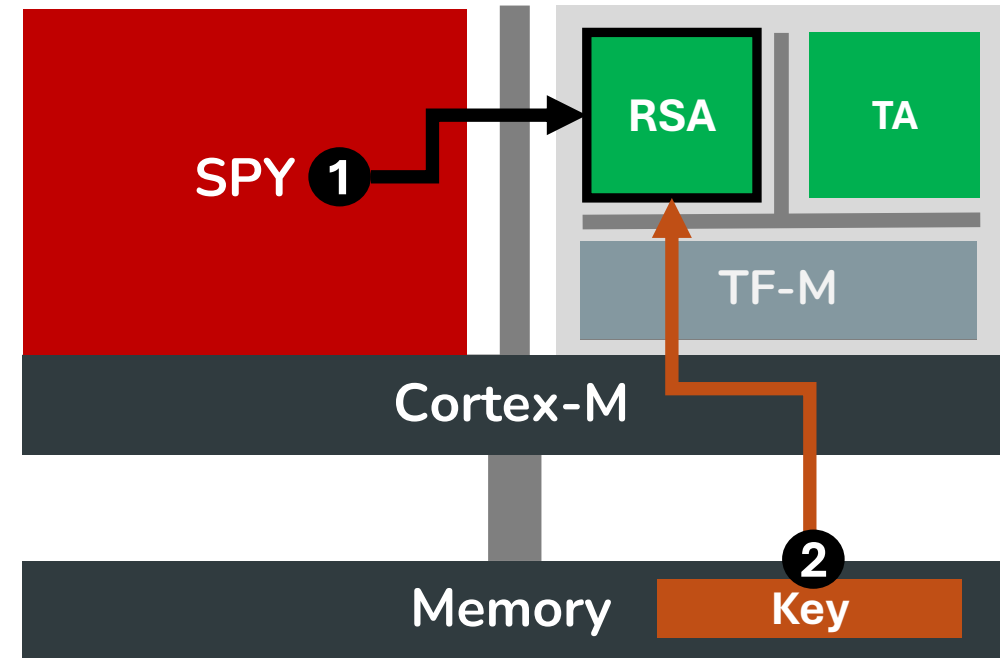


RSA Key Generation



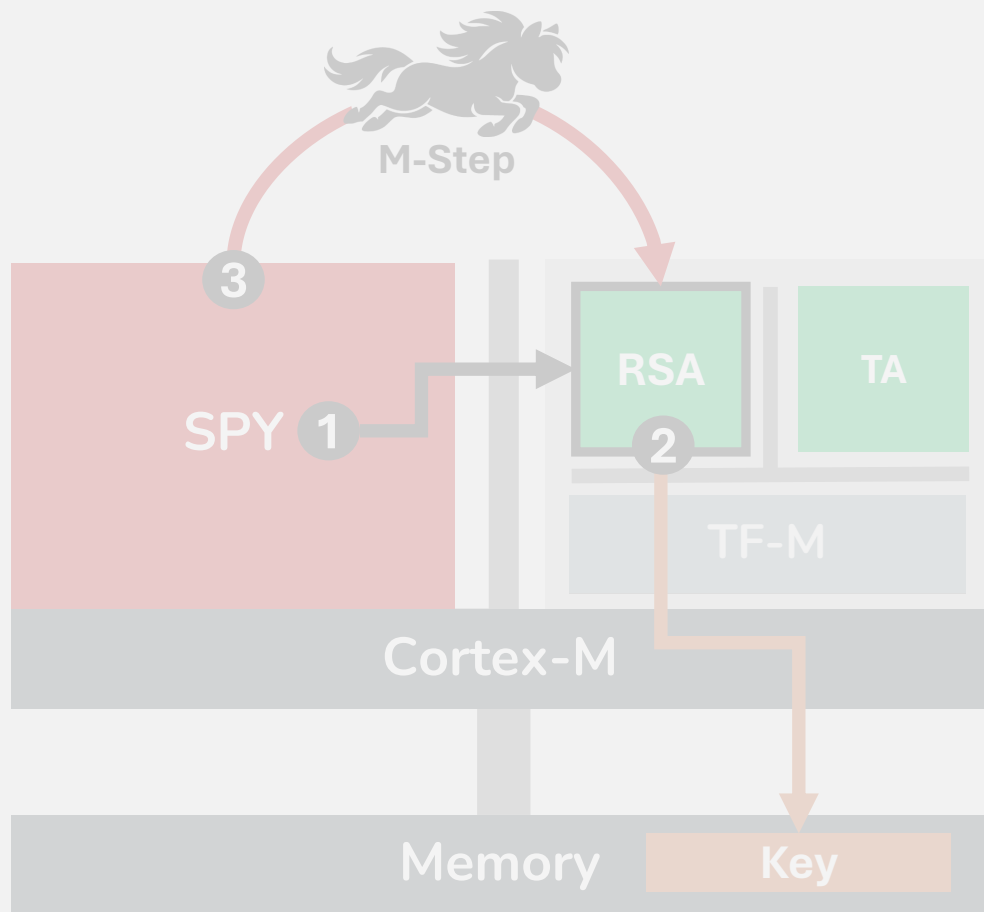
① Request Public Key

RSA Decryption



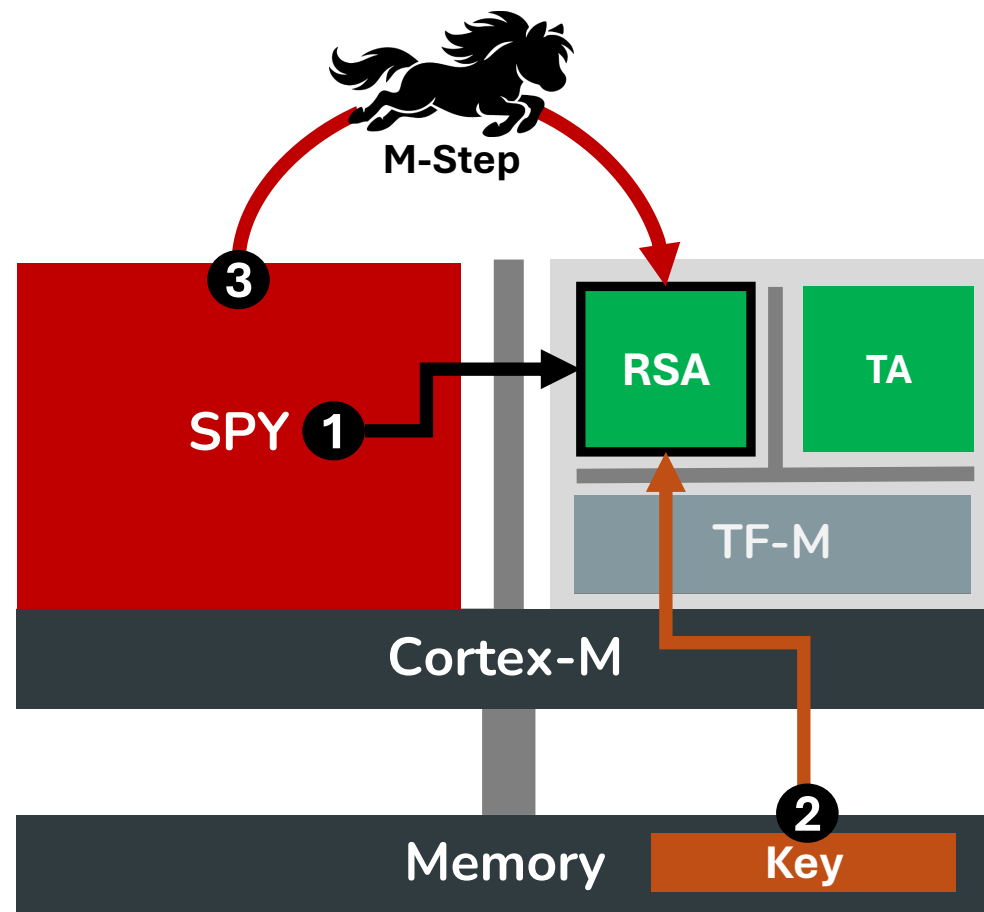
① Request RSA Decryption

RSA Key Generation



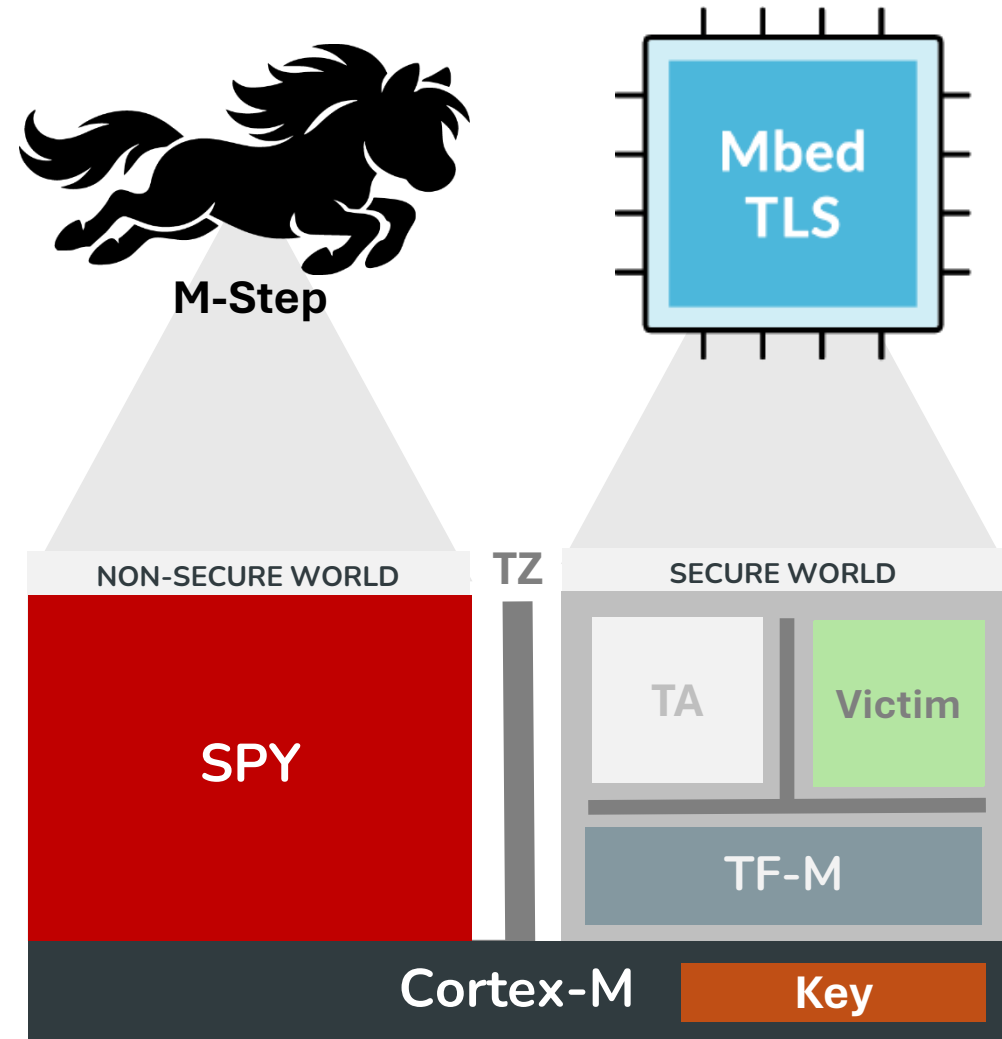
① Request Public Key

RSA Decryption

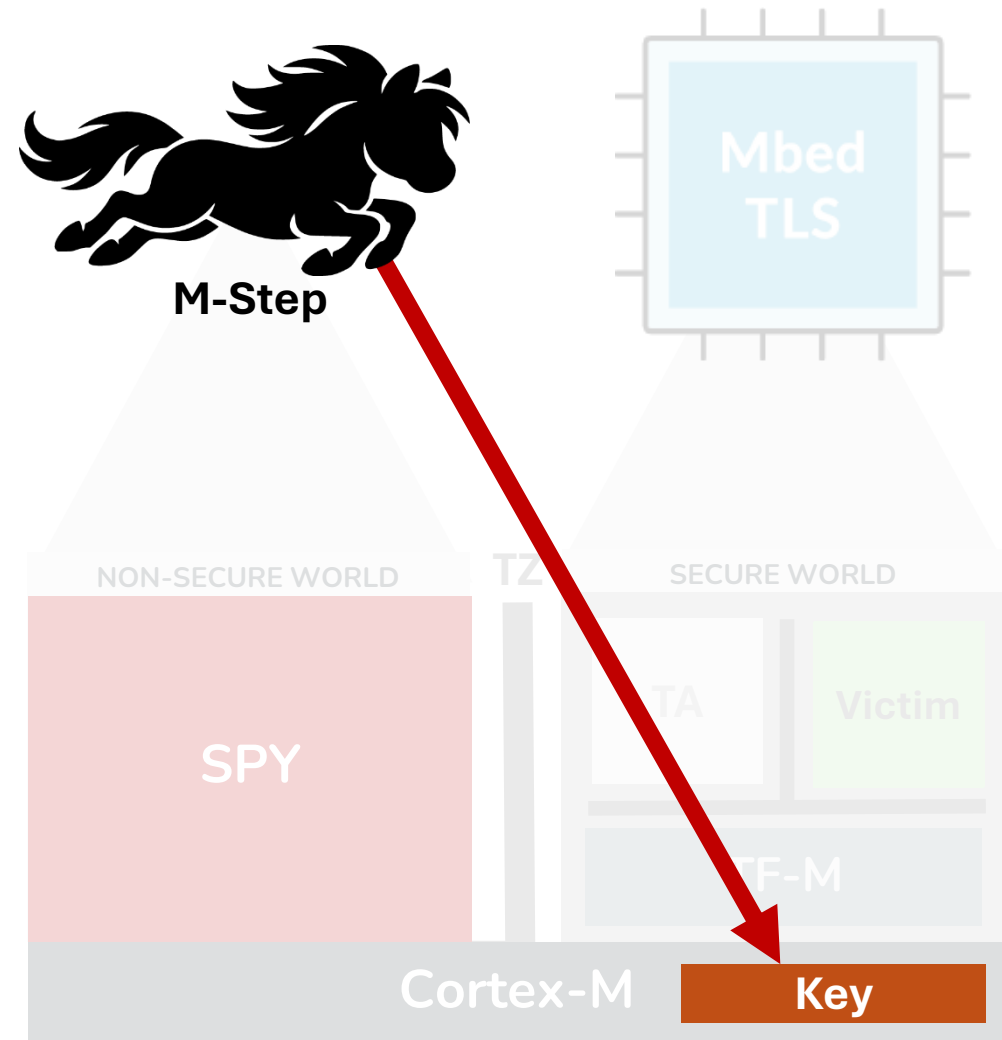


① Request RSA Decryption

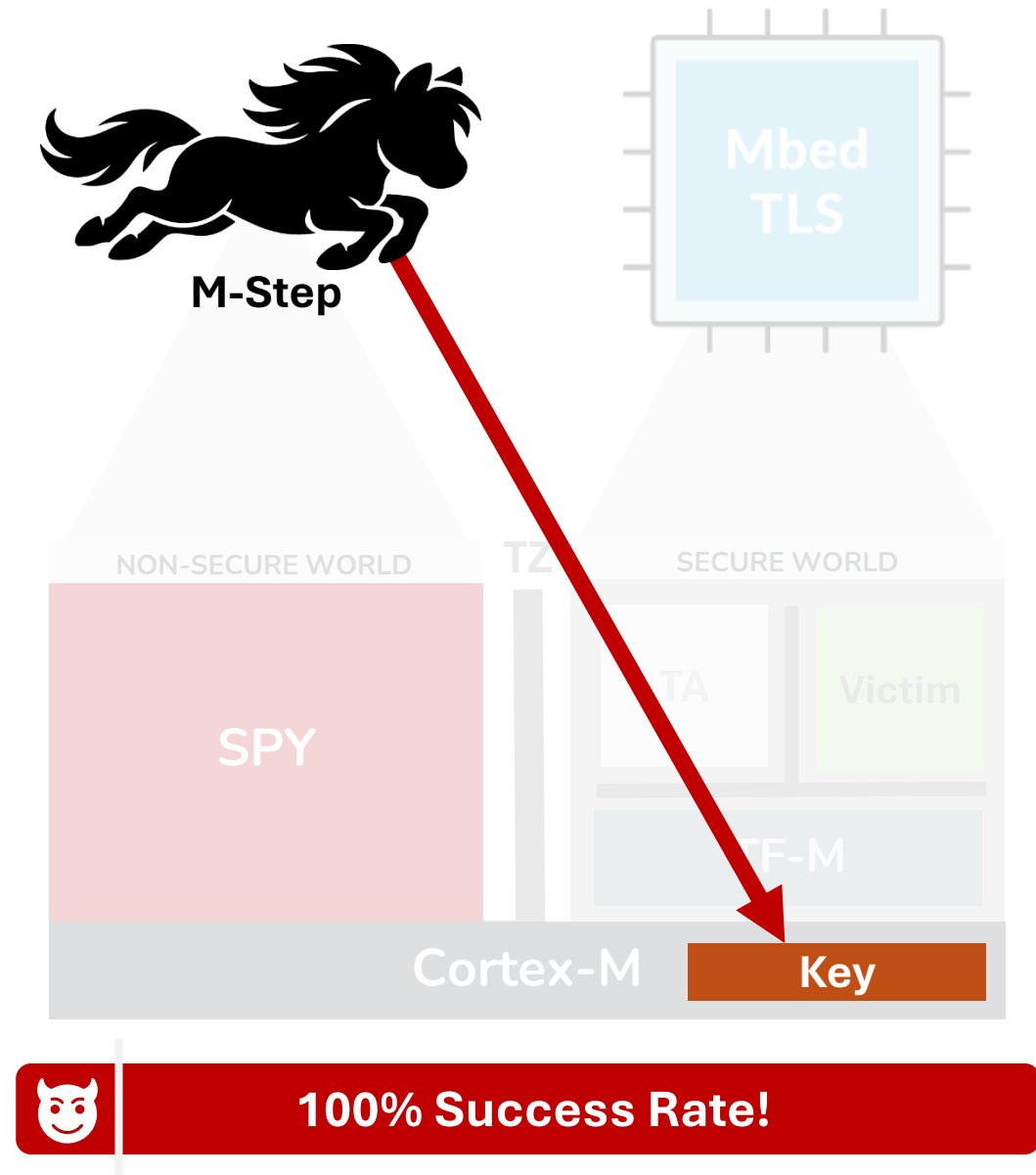
Full Private Key Extraction



Full Private Key Extraction



Full Private Key Extraction

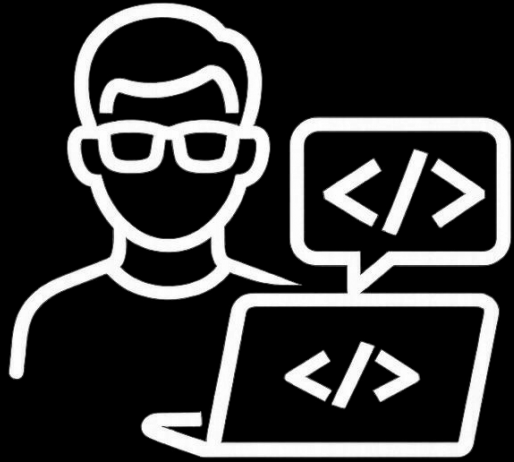


Demo

```
...li/vhs on ☐ exp/cli [x!?] via ☐ impure (tfm-dev-shell-env)  
> ./mstp.sh
```

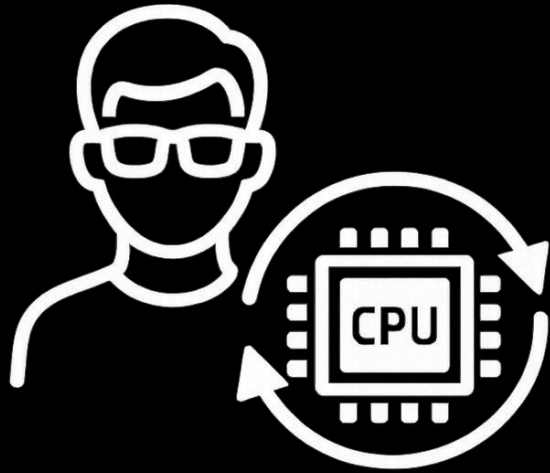
What Changes Now?





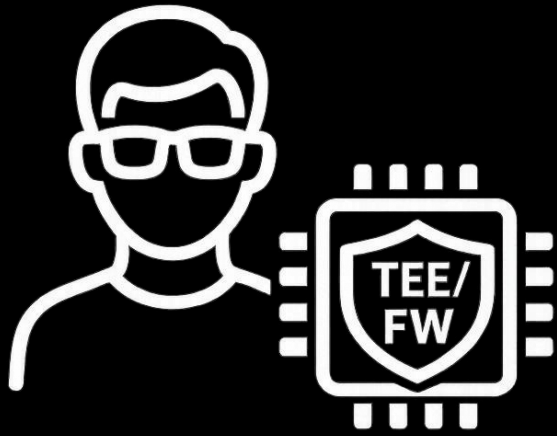
SW Developers

**Constant time code
also matters on MCUs!**



HW Vendors

**There is a need for
secure-by-design
solutions to prevent
leakage!**



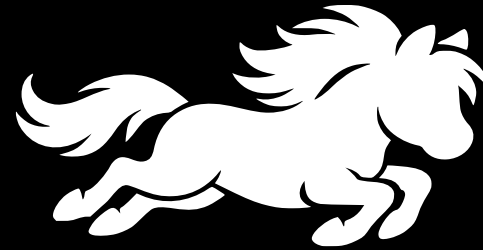
TEE / FW Providers

**Single-stepping Attacks
Should be Consider on The
TZ-M Threat Model!**

Tiny chips...

Tiny chips...

no longer mean tiny leaks!



Thank You!

Cristiano Rodrigues, Sandro Pinto, Marton Bogнар, and Jo Van Bulck
Centro ALGORITMI, University of Minho, DistriNet, KU Leuven





Q&A

Cristiano Rodrigues, Sandro Pinto, Marton Boghar, and Jo Van Bulck
Centro ALGORITMI, University of Minho, DistriNet, KU Leuven



mstep.eu



[Github](#)